迈向长时程智能体：综述

Towards Long-Horizon Agents: A Survey

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核心摘要校订

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随着 LLM 能力迅速提高，人们对 AI 智能体的期待正从简单单轮任务转向现实世界中的长时程任务。本文所称的长时程智能体（Long-Horizon Agents），能够在较长决策范围内规划、与真实环境交互、从自身错误中恢复，并在执行中调整策略。它正成为实用智能体能力的主要瓶颈。然而，领域仍缺少统一定义与分类，“自我演化”和“自主”等相近概念常被混用，也无法清楚归因长时程能力的进步。本文将长时程能力视为外部运行支撑框架（Externalized Harness）与内部模型优化（Internalized Optimization）的协同演化，并按 Foundation、Evolution、Harness、Optimization、Application、Frontier 六个视角组织研究。本文将其形式化为与 Harness 耦合的决策过程，与长时间运行、自主性、自我演化加以区分；梳理从提示控制到运行时系统的发展；按接口组织五类应用并总结评测资源，最后讨论挑战与前沿方向。

Towards Long-Horizon Agents: A Survey

Foundation, Evolution, Harness, Optimization, Application, and Frontier

Guanting Dong[1,*], **Xiaoshuai Song**[1,*], **Yuyang Hu**[1,*], **Jiajie Jin**[1,*], **Chenghao Zhang**[1,*], **Yifei Chen**[1], **Xiaoxi Li**[1], **Huaying Yuan**[1], **Xinyu Yang**[1], **Tongyu Wen**[1], **Jiejun Tan**[1], **Hongjin Qian**[2], **Shijue Huang**[5], **Junting Lu**[2], **Zhenyu Li**[3], **Wanjun Zhong**[4], **Yutao Zhu**[1], **Tat-Seng Chua**[6], **Zhicheng Dou**[1,†], **Ji-Rong Wen**[1]

∗ Core Contributors † Corresponding author

[1]**Renmin University of China**, [2]**Peking University**, [3]**Tsinghua University**, [4]**Sun Yat-Sen University**, [5]**Hong Kong University of Science and Technology**, [6]**National University of Singapore**

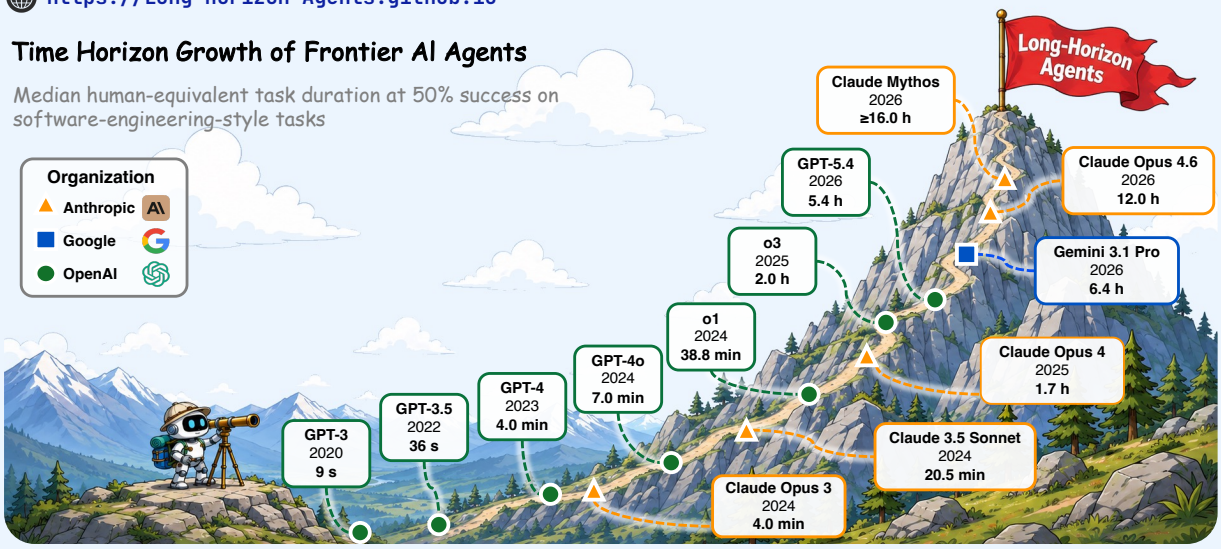
With the rapid advancement of LLM capabilities, expectations for AI agents are shifting from solving simple, single-turn tasks toward carrying out long-horizon tasks in the real world. We refer to such systems as **long-horizon agents**: agents that plan over extended horizons, interact with real-world environments, recover from their own mistakes, and adapt their strategies during execution. This capability is rapidly becoming the central bottleneck for practical agent intelligence. Despite this progress, the field still lacks a shared definition and taxonomy for long-horizon agents. Related concepts such as "self-evolving" or "autonomous" agents are often used interchangeably, none of which clearly captures what it means to build a *long-horizon agent*. This gap leaves the community without a principled way to attribute advances in long-horizon competence.

This paper offers a unified landscape by framing long-horizon agency as the co-evolution of an **externalized harness** and an **internalized optimization**, and organizes the paper around six connected perspectives: **Foundation**, **Evolution**, **Harness**, **Optimization**, **Application**, and **Frontier**. We first formalize long-horizon agency as a harness-coupled decision process and distinguish it from neighboring concepts such as long-running execution, autonomy, and self-evolution. We then trace the field's evolution from prompt-level control to runtime agent systems, classifying existing work through the complementary lenses of externalized harnesses and internalized optimization. Building on this view, we organize five application forms of long-horizon agents by their interfaces, and review the corresponding benchmarks and resources. Finally, we discuss key challenges and frontier directions. Looking ahead, we hope this paper serves not only as a reference for existing work, but also as a foundation for building the next generation of capable, reliable long-horizon agents.

dongguanting@ruc.edu.cn, dou@ruc.edu.cn

https://github.com/RUC-NLPIR/Awesome-Long-Horizon-Agents

https://Long-Horizon-Agents.github.io



*The empirical time-horizon trend of frontier agents: the task-completion horizon has climbed from seconds to over twelve hours, doubling roughly every seven months (https://metr.org/time-horizons).

原文第 1 页 | 中文翻译初稿

迈向长时程智能体：综述

董官廷1_,*、宋小帅1,*、胡玉阳1,*、金家杰1,*、程浩张1,*、陈逸飞1、李晓曦1、袁华英1、杨新宇1、文同宇1、谭杰军1、洪金钱2、黄世爵5、陆俊廷2、李振宇3、钟万军4、朱玉涛1、蔡达胜6、志成窦1,†_、温继荣1

*** 核心贡献者

_†_通讯作者
1 中国人民大学、2 北京大学、3 清华大学、4 中山大学、5 香港科技大学、6 新加坡国立大学

随着LLM能力的快速进步，对AI智能体的期望正在从解决简单的单轮任务转向在现实世界中执行长时程（Long-Horizon）任务。我们将此类系统称为 长时程智能体（Long-Horizon Agents）：智能体可以进行长期规划、与现实世界环境交互、从自己的错误中恢复并在执行过程中调整策略。这种能力正在迅速成为实际智能体智能的中心瓶颈。尽管取得了这些进展，该领域仍然缺乏 长时程智能体（Long-Horizon Agents）的共享定义和分类法。“自我进化”或“自主”智能体等相关概念通常可以互换使用，但没有一个概念清楚地说明了构建 长时程（Long-Horizon） 智能体 的含义。这一差距使得社区没有一种原则性的方法来归因 长时程（Long-Horizon） 能力的进步。

本文通过将 长时程（Long-Horizon） 机构框架为外部化 运行支撑框架（Harness） 和内部化优化的共同进化来提供统一的景观，并围绕六个相互关联的视角组织本文：基础、进化、运行支撑框架（Harness）、优化、应用和前沿。我们首先将 长时程（Long-Horizon） 智能体形式化为 运行支撑框架（Harness） 耦合决策过程，并将其与长期执行、自治和自我进化等邻近概念区分开来。然后，我们追踪该领域从提示级控制到运行时智能体系统的演变，通过外部 运行支撑框架（Harnesses） 和内部优化的互补镜头对现有工作进行分类。基于这个观点，我们按接口整理了 长时程智能体（Long-Horizon Agents） 的五个申请表，并回顾了相应的基准和资源。最后，我们讨论了关键挑战和前沿方向。展望未来，我们希望本文不仅可以作为现有工作的参考，而且可以作为构建下一代有能力、可靠的长时程智能体（Long-Horizon Agents）的基础。

dongguanting@ruc.edu.cn, dou@ruc.edu.cn
https://github.com/RUC-NLPIR/Awesome-长时程（Long-Horizon）-Agents
https://长时程（Long-Horizon）-Agents.github.io

Frontier AI Agents 的时间跨度增长

Claude Mythos
2026 年成功 50% 的人类等效任务持续时间中位数
软件工程类型任务 ≥16.0 小时
组织
人类 GPT-5.4 Claude Opus 4.6
组织 2026 2026
人类 Google 5.4 小时 12.0 h
OpenAI

Google o3
Gemini 3.1 Pro
2025
OpenAI 2.0 h 2026
6.4 h
o1
2024
GPT-4o 38.8 分钟 Claude Opus 4
2025
2024
GPT-4 1.7 h
7.0分钟
GPT-3.5 2023
GPT-3 2020 2022 36 秒 4.0 分钟 Claude 3.5 Sonnet 2024
9 秒 20.5 分钟
Claude Opus 3
2024
4.0 分钟

*边境特工的经验时间范围趋势：任务完成范围已从几秒攀升至十二个小时以上，大约每七个月翻一番（https://metr.org/time-horizons）。

Contents

1 Introduction **4**

2 Foundation: Formalizing Long-Horizon Agents **7**
2.1 Definition . . . 7
2.2 Three Levels of Long-Horizon Tasks and Capabilities . . . 10
2.3 Scaling the Time Horizon of Frontier-Agent Execution . . . 11
2.4 Long-Horizon Agent vs. Neighboring Concepts . . . 13
2.4.1 Long-Horizon Agent vs. Long-Running Agent . . . 13
2.4.2 Long-Horizon Agent vs. Autonomous Agent . . . 13
2.4.3 Long-Horizon Agent vs. Self-Evolving Agent . . . 14

3 Evolution: From Prompting to Runtime **15**
3.1 Stage I: Prompt Engineering (2020–2023) . . . 16
3.2 Stage II: Context Engineering (2023–2025) . . . 17
3.3 Stage III: Runtime Harnesses (2025–Present) . . . 18

4 Harness: Externalizing Long-Horizon Capability **20**
4.1 Loops and Workflows . . . 20
4.1.1 Linear Workflows . . . 21
4.1.2 Plan-Execute Workflows . . . 21
4.1.3 Branching Workflows . . . 22
4.2 Context and Memory . . . 23
4.2.1 Working Context . . . 23
4.2.2 Persistent Memory . . . 24
4.3 Tools, MCP, and Skills . . . 26
4.3.1 Tool Interfaces and Protocols . . . 27
4.3.2 Active Tool Discovery . . . 27
4.3.3 Skill Libraries . . . 28
4.4 Orchestration . . . 29
4.4.1 Decomposition and Roles . . . 29
4.4.2 Coordination Topologies . . . 30
4.4.3 Orchestration Optimization . . . 30
4.4.4 Agent Protocols . . . 30
4.5 Hooks and Middleware . . . 31
4.5.1 Pre-defined Rule-based Hooks . . . 31
4.5.2 Custom User-defined Hooks . . . 32
4.5.3 Runtime-adaptive Hooks . . . 33
4.6 Verification . . . 34
4.6.1 Assessment Targets . . . 34
4.6.2 Verification Levels . . . 36
4.6.3 Verifier Strategies . . . 36

5 Optimization: Internalizing Long-Horizon Capability **37**
5.1 Architectural Substrate . . . 37
5.1.1 Explicit-Context Architectures . . . 38
5.1.2 Compressed-State Architectures . . . 39
5.1.3 Hybrid Architectures . . . 40
5.1.4 High-Throughput Mechanisms . . . 40
5.2 Data and Environment Synthesis . . . 40
5.2.1 Task Synthesis . . . 41
5.2.2 Environment Synthesis . . . 41
5.2.3 Trajectory Synthesis . . . 42
5.3 Pre-training and Mid-training . . . 43
5.3.1 Reasoning Priors . . . 44
5.3.2 Long-Context State . . . 44
5.3.3 Multimodal Perception . . . 44
5.3.4 Data Mixture and Training Strategies . . . 44

原文第 2 页 | 中文翻译初稿

内容

|1
简介	产品	关于	4
2
Fou	ndatio	n：形式化 长时程智能体（Long-Horizon Agents）	7
2.1	定义	定义。。。

18|
|4
工具|性：|外化 长时程（Long-Horizon） 能力|20|
|4.1|循环|和工作流程。。。

29|
||4.4.2|协调拓扑。。。

。 。 。

。 。 。 。 。 。 。 。 。 。 。 。 。 。 。 。 。 。 。 .|. 。 。 。 。 。 。 。 。 。 。 。 .
44|

2

5.4 Fine-tuning . . . 45
5.4.1 Instruction Selection and Mixing . . . 45
5.4.2 Curriculum Learning . . . 46
5.4.3 Distillation . . . 46
5.5 Agentic Reinforcement Learning . . . 47
5.5.1 Credit Assignment . . . 48
5.5.2 Policy Optimization . . . 50
5.5.3 Sampling Strategy . . . 50
5.5.4 Interaction Patterns . . . 51
5.6 On-Policy Distillation . . . 52
5.6.1 Teacher-guided OPD . . . 52
5.6.2 Self-improving OPD . . . 53
5.7 Self-evolution . . . 53
5.7.1 Offline Self-evolution . . . 54
5.7.2 Online Self-evolution . . . 54
5.7.3 Agent–Environment Co-evolution . . . 55

6 Application: Long-Horizon Agents in Practice 55
6.1 Software Engineering . . . 56
6.1.1 Repository Grounding . . . 56
6.1.2 Workflow-level Planning . . . 57
6.1.3 Feedback-driven Repair . . . 58
6.2 Information Seeking . . . 58
6.2.1 Deep Search . . . 58
6.2.2 Wide Search . . . 59
6.2.3 Multimodal Grounding . . . 59
6.2.4 Research Synthesis . . . 59
6.3 Computer Use . . . 60
6.3.1 Browser Agents . . . 60
6.3.2 Desktop GUI Agents . . . 60
6.3.3 Mobile Agents . . . 60
6.4 Multimodal Agents . . . 61
6.4.1 Multimodal Understanding . . . 61
6.4.2 Multimodal Generation . . . 61
6.4.3 Omnimodal Agency . . . 62
6.5 General-purpose Agents . . . 62
6.5.1 Personal Assistants . . . 62
6.5.2 Embodied Agents and World Models . . . 63
6.5.3 Productive Agents . . . 63
6.6 Benchmarks and Resources . . . 64

7 Frontier: Open Challenges and Outlooks 67
7.1 Evolution: Generalization and Learning Over Time . . . 67
7.1.1 Self-evolving Harness and Agents . . . 67
7.1.2 Harness Generalization and Transferability . . . 69
7.1.3 Continual and Lifelong Learning . . . 69
7.2 Effectiveness: Acting in Realistic Environments . . . 70
7.2.1 Real-world Environment Interaction . . . 70
7.2.2 From Digital to Embodied Agents . . . 71
7.3 Efficiency: Spending Compute, Context, and Modality Budgets . . . 71
7.3.1 Cost- and Budget-aware Agency . . . 72
7.3.2 Multimodal and Omni Harnesses . . . 72
7.4 Trustworthiness: Robust and Governed Autonomy . . . 73
7.4.1 Reflection and Error Robustness . . . 73
7.4.2 Safety and Governance . . . 74

8 Conclusion 74

原文第 3 页 | 中文翻译初稿

|5.4|精细
5.4.1|uning。。。

教师指导的 OPD 。 。 。

||6.2.2|广泛搜索
。 。 。

|6.5|通用|通用智能体。。。

具有成本和预算意识的机构
。 。 。

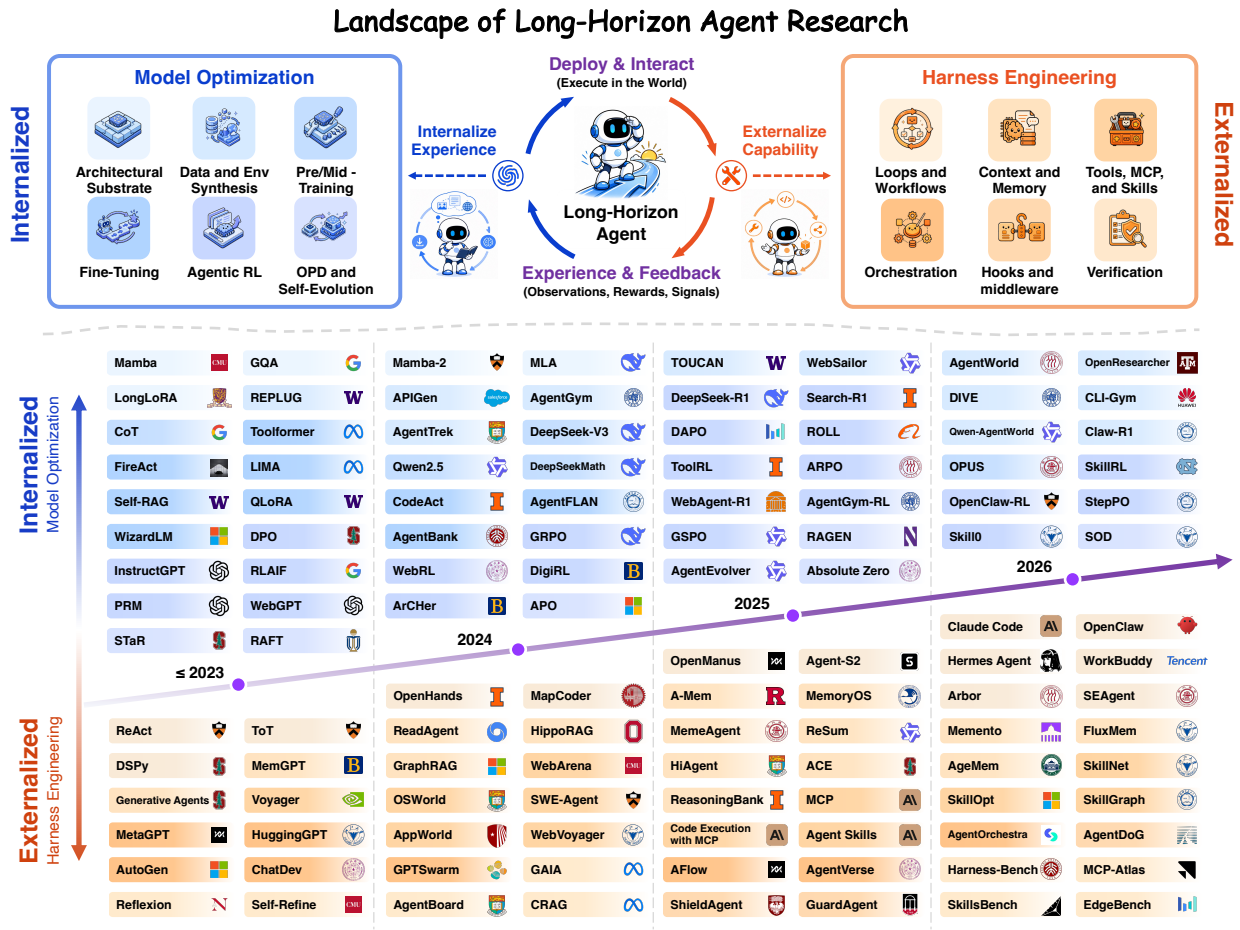


Figure 1 The landscape of long-horizon agent research. This paper adopts a co-evolutionary view of long-horizon agents, organized around *externalized harness engineering* (bottom) and *internalized model optimization* (top).

1 Introduction

Over the past few years, large language models (LLMs) have moved from single-turn chatbots to the decision-making core of autonomous agents in software engineering [260, 747], general-purpose assistance [431], scientific discovery [678], computer use [716, 885], and multimodal interaction [141]. These domains differ on the surface, but they share one decisive requirement that we call *long horizon*: they require persistent iteration across reasoning, tool use, observation, and revision over many interdependent steps—from those completable within a single context window to those that span windows, sessions, or open-ended task streams. To meet this requirement, *long-horizon agents* have emerged, expanding the boundaries of agent applications through continual, sustained interaction with real-world environments.

Long-horizon competence is now advancing rapidly and increasingly marks the practical line for delegating productive work to agents. For instance, frontier coding agents are already able to run for hours on a single project, occasionally close to a full day [14, 456]. Moreover, METR [286] reports that the time horizon of advanced agentic task execution has doubled roughly every seven months over the past several years, accelerating to about every four months for the most recent models. For example, GLM-5.1 is reported to sustain independent execution on a single task for up to eight hours, covering planning, implementation, testing, iterative refinement, and final delivery [179]. Notably, this capability cannot be obtained *simply* by optimizing and scaling the model's internal parameters or enlarging its context window alone; its capability boundary also varies sharply with the *external* harness surrounding the model [123, 378, 468, 511]. Therefore, the central thesis of this paper is that *long-horizon agency is a system-level capability jointly shaped by two*

原文第 4 页 | 中文翻译初稿

长时程（Long-Horizon） 智能体研究景观
部署和交互
模型优化（在世界范围内执行） 运行支撑框架（Harness） 工程
内化外化
经验能力
架构数据和环境前/中循环和上下文和工具、MCP、
基质综合训练工作流程内存和技能
长时程（Long-Horizon）
Agent
微调 Agentic RL OPD 和经验与反馈编排挂钩和验证
自我进化（观察、奖励、信号）中间件
Mamba GQA Mamba-2 MLA TOUCAN WebSailor AgentWorld OpenResearcher
LongLoRA REPLUG APIGen AgentGym DeepSeek-R1 Search-R1 DIVE CLI-Gym
CoT Toolformer AgentTrek DeepSeek-V3 DAPO ROLL Qwen-AgentWorld Claw-R1
FireAct LIMA Qwen2.5 DeepSeekMath ToolRL ARPO OPUS SkillRL
Self-RAG QLoRA CodeAct AgentFLAN WebAgent-R1 AgentGym-RL OpenClaw-RL StepPO
WizardLM DPO AgentBank GRPO GSPO RAGEN Skill0 SOD
InstructGPT RLAIF WebRL DigiRL AgentEvolver Absolute Zero 2026
PRM WebGPT Archer APO 2025
Claude Code OpenClaw
STAR RAFT 2024
OpenManus Agent-S2 Hermes Agent WorkBuddy
≤ 2023
OpenHands MapCoder A-Mem MemoryOS Arbor SEAgent
React ToT ReadAgent HippoRAG MemeAgent ReSum Memento FluxMem
DSPy MemGPT GraphRAG WebArena HiAgent ACE AgeMem SkillNet
生成智能体 Voyager OSWorld SWE-Agent ReasoningBank MCP SkillOpt SkillGraph
MetaGPT HuggingGPT AppWorld WebVoyager 使用 MCP 智能体技能执行代码 AgentOrchestra
AgentDoG
AutoGen ChatDev GPTSwarm GAIA AFlow AgentVerse 运行支撑框架（Harness）-Bench MCP-Atlas
反射自我优化 AgentBoard CRAG ShieldAgent GuardAgent SkillsBench EdgeBench
内部化 外部化
内部化模型优化
外部化运行支撑框架（Harness）工程

图 1 长时程（Long-Horizon） 智能体研究的概况。本文采用 长时程智能体（Long-Horizon Agents） 的协同进化观点，围绕_外部化 运行支撑框架（Harness） 工程_（底部）和_内部化模型优化_（顶部）进行组织。

1 简介

在过去的几年里，大型语言模型（LLM）已经从单轮聊天机器人转变为软件工程 [260, 747]、通用（General-Purpose）辅助 [431]、科学发现 [678]、计算机使用 [716, 885] 和多模式交互 [141] 中自主智能体的决策核心。这些领域在表面上有所不同，但它们有一个决定性的要求，我们称之为“长视野”：它们需要在推理、工具使用、观察和对许多相互依赖的步骤进行修订方面进行持续迭代--从可在单个上下文窗口内完成的步骤到跨窗口、会话或开放式任务流的步骤。为了满足这一要求，长时程智能体（Long-Horizon Agents）应运而生，通过与现实环境持续交互来扩展智能体应用程序的边界。

长时程（Long-Horizon）能力现在正在迅速发展，并日益标志着将生产性工作委托给座席的实用路线。例如，前沿编码智能体已经能够在单个项目上运行数小时，有时接近一整天 [14, 456]

。此外，METR [286] 报告称，在过去几年中，高级智能体任务执行的时间范围大约每七个月翻一番，对于最新模型来说，加速到大约每四个月一次。例如，据报道，GLM-5.1 可维持单个任务的独立执行长达八小时，涵盖规划、实施、测试、迭代细化和最终交付 [179]。值得注意的是，这种能力不能仅仅通过优化和缩放模型的内部参数或扩大其上下文窗口来获得；它的能力边界也随着模型周围的_external_ 运行支撑框架（Harness）而急剧变化[123,378,468,511]。因此，本文的中心论点是_长时程（Long-Horizon）智能体是两者共同塑造的系统级能力_

4

co-evolving components: externalized harness engineering and internalized model optimization.

Why long-horizon task execution is hard for agents. The difficulty of long-horizon task execution does not reduce to per-step accuracy: it arises from the *composition* of many tightly coupled steps into a single trajectory, rather than from any individual decision. Across the systems we organize, three representative failure modes recur throughout long-horizon execution and structure much of the technical discussion.

- **Goal drift and compounding error.** Over thousands of steps an agent can slowly diverge from its original objective, and because no step is perfectly reliable, local errors accumulate along the trajectory and eventually steer it onto a wrong reasoning path [20, 549].
- **Context rot and context-limit pressure.** As the working context fills, an agent tends to degrade abruptly once it passes a certain context-utilization threshold [378, 511]; conversely, an agent that perceives its context limit may terminate prematurely and declare partial progress a success [14].
- **Sparse, delayed rewards and irreversibility.** Many long-horizon tasks return a reward only at the end, leaving the agent with an extremely sparse learning signal over its intermediate decisions [808]; moreover, the longer an agent acts, the more likely it is to take a risky, irreversible action, so monitoring its execution safety and recoverability becomes essential.

Why long-horizon agents need a new taxonomy. Recent works have begun to clarify parts of this landscape, but its taxonomies still exhibit two structural limitations that motivate a new one.

- **Limited and unsystematic coverage.** Existing papers typically examine a *single* technical component of long-horizon agents in isolation, such as memory [694], context engineering [425, 841], agent harnesses [468, 876], self-evolution [165, 709], or agentic reinforcement learning [808]. Each is a genuine piece of the long-horizon agent, but none offers a comprehensive account of what such an agent *is*, nor of how to build an efficient and trustworthy long-horizon agent that stays reliable across a full trajectory.
- **Unclear definitions and overlapping terminology.** As interest in capable agents grows, the concept itself has become broad and blurred. Related notions such as long-context, long-running, autonomous, and self-evolving agents are routinely used interchangeably, even though each emphasizes a different facet, and reported gains entangle changes in the model, the harness, the data pipeline, and the evaluation protocol. Most works cannot say precisely what counts as long-horizon agency or how these techniques relate, which leaves improvements hard to attribute and hard to transfer.

These findings highlight the need for a unified taxonomy for long-horizon agent research.

We therefore organize the long-horizon agent field in this paper through a *co-evolutionary* view (Figure 1). Our organizing principle is that long-horizon agency is realized through the co-evolution of **externalized harness engineering** and **internalized model optimization**, two complementary halves that together constitute a long-horizon agent. Concretely, **externalized harness engineering** supplies runtime structure such as loops and workflows, context and memory, tools, orchestration, hooks and middleware, and verification; **internalized model optimization** improves the agent policy through architecture, data, environments, fine-tuning, reinforcement learning, and self-evolution. Neither half is sufficient on its own; long-horizon agency emerges only through their co-evolution, as harness mechanisms are gradually internalized into the policy and a stronger policy in turn unlocks more advanced harnesses.

Building on this co-evolutionary view, this paper aims to reconcile the fragmented definitions, connect the emerging trends, and articulate the foundational principles of long-horizon agents. Concretely, we organize the discussion around the following key questions:

原文第 5 页 | 中文翻译初稿

共同进化组件：外化运行支撑框架（Harness）工程和内化模型优化。

为什么 长时程（Long-Horizon）任务执行对于智能体来说很难。 长时程（Long-Horizon）任务执行的难度不会降低到每步的准确性：它是由许多紧密耦合的步骤组合成单个轨迹而产生的，而不是来自任何单独的决策。在我们组织的系统中，三种代表性的故障模式在 长时程（Long-Horizon） 的执行过程中反复出现，并构建了大部分技术讨论。

- 目标漂移和复合误差。经过数千个步骤，智能体可能会慢慢偏离其最初的目标，并且由于没有任何步骤是完全可靠的，因此局部错误会沿着轨迹累积并最终将其引导到错误的推理路径上 [20, 549]。

- 上下文腐烂和上下文限制压力。当工作上下文被填满时，一旦超过某个上下文利用率阈值，智能体就会突然降级 [378, 511]；相反，感知到其上下文限制的智能体可能会提前终止并声明部分进度成功[14]。

- 奖励稀疏、延迟且不可逆转。许多 长时程（Long-Horizon）任务仅在最后返回奖励，使智能体在其中间决策中留下极其稀疏的学习信号[808]；此外，智能体行动的时间越长，就越有可能采取有风险的、不可逆转的行动，因此监控其执行安全性和可恢复性变得至关重要。

为什么 长时程智能体（Long-Horizon Agents） 需要新的分类法。最近的工作已经开始阐明这一景观的部分内容，但其分类法仍然表现出两个结构性限制，从而激发了新的结构性限制。

- 覆盖范围有限且不系统。现有论文通常孤立地研究长视野智能体的单个技术组件，例如记忆[694]、上下文工程[425, 841]、智能体运行支撑框架（Harnesses） [468, 876]、自我进化[165, 709]或智能体强化学习（Reinforcement Learning） [808]。每个都是 长时程（Long-Horizon） 智能体的真实部分，但没有一个全面说明此类智能体的含义，也没有提供如何构建在整个轨迹上保持可靠的高效且值得信赖的 长时程（Long-Horizon） 智能体。

- 定义不明确和术语重叠。随着对有能力的智能体人的兴趣不断增长，这个概念本身变得广泛和模糊。长上下文、长时间运行、自主和自我进化智能体等相关概念通常可以互换使用，尽管每个概念都强调不同的方面，并且报告的收益涉及模型、运行支撑框架（Harness）、数据管道和评估协议中的变化。大多数作品无法准确说明什么是 长时程（Long-Horizon） 机构或这些技术如何相互关联，这使得改进难以归因且难以转移。

这些发现强调了 长时程（Long-Horizon） 药剂研究需要统一的分类法。

因此，我们通过_co-evolutionary_视图来组织本文中的长时程（Long-Horizon）智能体字段（图1）。我们的组织原则是，长时程（Long-Horizon）智能体是通过外化运行支撑框架（Harness）工程和内化模型优化的共同进化来实现的，两个互补的部分共同构成一个长时程（Long-Horizon）智能体

。具体来说，外部化的运行支撑框架（Harness）工程提供了运行时结构，例如循环和工作流程、上下文和内存、工具、编排、钩子和中间件以及验证；内部化模型优化通过架构、数据、环境、微调、强化学习（Reinforcement Learning）和自我进化来改进智能体策略。这两半本身都不够。 长时程（Long-Horizon） 机构只能通过它们的共同进化而出现，因为 运行支撑框架（Harness） 机制逐渐内化到策略中，而更强大的策略又会解锁更高级的 运行支撑框架（Harnesses）。

基于这种共同进化的观点，本文旨在协调零散的定义，连接新兴趋势，并阐明 长时程智能体（Long-Horizon Agents） 的基本原则。具体来说，我们围绕以下关键问题组织讨论：

5

Key Questions

- **Q1. Foundation.** How should long-horizon agency be formalized, and how does it differ from long-running execution, autonomy, and self-evolution?
- **Q2. Evolution.** How did the control surface widen from prompts, to context, to runtime agent systems?
- **Q3. Harness.** How does externalized harness engineering keep long-horizon agent execution specifiable, steerable, verifiable, and recoverable at runtime?
- **Q4. Optimization.** How can internalized model optimization build long-horizon competence across the agent's architectural substrate, data and environment synthesis, and full training lifecycle?
- **Q5. Application.** What forms do long-horizon agents take across domains, and how should their long-horizon capability be systematically evaluated?
- **Q6. Frontier.** What open problems define the next stage as horizons grow?

To address these questions, the paper follows this co-evolutionary decomposition through six connected perspectives. **Foundation** formalizes the agent and separates it from adjacent concepts, while **Evolution** traces how the locus of control widened from prompts to context to runtime systems. The two halves of the argument are then treated in turn: **Harness** studies how long-horizon capability is *externalized* into a runtime layer, and **Optimization** studies how it is *internalized* into the policy. Finally, **Application** shows how the same long-horizon pressures recur across domains and how such systems are evaluated, and **Frontier** identifies what remains open. Crucially, these two halves are not independent: capability continually migrates between them, so throughout we treat externalized harness engineering and internalized model optimization as coupled rather than as separate tracks.

Contributions and Roadmap. This paper makes the following contributions, each answering one of the key questions and mapping to one part of the paper. Figure 2 presents an overview of our taxonomy.

1. **Foundation (Section 2).** We formalize long-horizon agency as a coupled model–harness decision process and organize its difficulty into three nested task levels (H1–H3) with matching capabilities (C1–C3): intra-context reasoning, cross-context memory, and cross-task experience. Building on METR's data, we note that the frontier time horizon is scaling with an accelerating doubling time. We further separate long-horizon agency from long-running execution, autonomy, and self-evolution, giving the field a shared vocabulary for attributing where long-horizon competence comes from.

2. **Evolution (Section 3).** We trace how the control surface migrated from prompts, to context, to runtime systems, framing today's long-horizon agents as the current endpoint of a capability-driven trajectory and clarifying why control first moved outward before it could be internalized.

3. **Harness (Section 4).** We organize externalized harness engineering by component: loops and workflows, context and memory, tools, MCP, and skills, orchestration and multi-agent control, hooks and middleware, and verification. Together these components show how a runtime layer keeps long execution specifiable, steerable, verifiable, and recoverable.

4. **Optimization (Section 5).** We organize internalized model optimization along the pipeline, from the architectural substrate and data and environment synthesis to agentic pre/mid-training, fine-tuning, reinforcement learning with long-horizon credit assignment, on-policy distillation, and self-evolution, tracing how capabilities first scaffolded through externalized harness engineering are progressively internalized into the policy.

5. **Application (Section 6).** Organized by the agent–environment interface, we map long-horizon pressures across software engineering, information seeking, computer use, multimodal agents, and general-purpose agents, revealing a common long-horizon core beneath surface-level domain differences, and consolidate horizon-aware evaluation resources.

6. **Frontier (Section 7).** We organize open problems into four axes, namely evolution, effectiveness, efficiency, and trustworthiness, spanning nine concrete directions that range from self-evolving and transferable

原文第 6 页 | 中文翻译初稿

关键问题

- Q1。基础。 长时程（Long-Horizon）智能体应该如何形式化，它与长期运行的执行、自治和自我进化有何不同？

- Q2。进化。控制面如何从提示扩展到上下文，再到运行时智能体系统？

- Q3。 运行支撑框架（Harness）。外部化 运行支撑框架（Harness） 工程如何保持 长时程（Long-Horizon） 智能体执行在运行时可指定、可操纵、可验证和可恢复？

- Q4。优化。内部模型优化如何在智能体的架构基础、数据和环境综合以及完整的训练生命周期中构建 长时程（Long-Horizon） 能力？

- Q5。应用。 长时程智能体（Long-Horizon Agents）跨领域采取什么形式，如何系统评估其长期能力？

- Q6。边境。随着视野的增长，哪些开放性问题定义了下一阶段？

为了解决这些问题，本文通过六个相互关联的视角来遵循这种共同进化分解。 Foundation 形式化了智能体并将其与相邻概念分开，而 Evolution 则追踪控制范围如何从提示扩展到上下文再到运行时系统。然后依次处理参数的两半：运行支撑框架（Harness） 研究 长时程（Long-Horizon） 功能如何“外部化”到运行时层，优化研究如何“内部化”到策略中。最后，应用程序展示了相同的 长时程（Long-Horizon） 压力如何跨域重复出现以及如何评估此类系统，而 Frontier 则确定哪些内容仍然开放。至关重要的是，这两个部分并不是独立的：能力在它们之间不断迁移，因此在整个过程中，我们将外部化的 运行支撑框架（Harness） 工程和内部化的模型优化视为耦合而不是单独的轨道。

贡献和路线图。本文做出以下贡献，每一项贡献都回答了一个关键问题并映射到本文的一部分。图 2 概述了我们的分类法。

1. 基础（第 2 节）。我们将 长时程（Long-Horizon） 智能体形式化为耦合模型 - 运行支撑框架（Harness）决策过程，并将其难度组织为具有匹配能力（C1-C3）的三个嵌套任务级别（H1-H3）：上下文内推理、跨上下文记忆和跨任务体验。根据 METR 的数据，我们注意到前沿时间范围正在随着倍增时间的加速而扩展。我们进一步将 长时程（Long-Horizon） 智能体与长期运行的执行、自治和自我进化分开，为该领域提供了一个共享词汇来归因 长时程（Long-Horizon） 能力的来源。

2. 进化（第 3 节）。我们追踪控制界面如何从提示迁移到上下文，再到运行时系统，将今天的 长时程智能体（Long-Horizon Agents） 视为能力驱动轨迹的当前端点，并阐明为什么控制首先向外移动，然后才能被内化。

3. 运行支撑框架（Harness）（第 4 节）。我们按组件组织外部化 运行支撑框架（Harness） 工程：循环和工作流程、上下文和内存、工具、MCP 和技能、编排和多智能体控制、挂钩和中间件以及验证。这些组件一起展示了运行时层如何保持长时间执行的可指定性、可操纵性、可验证性和可恢复性。

4. 优化（第 5 节）

。我们沿管道组织内化模型优化，从架构基础、数据和环境综合到智能体预/中期训练、微调、强化学习（Reinforcement Learning）与 长时程（Long-Horizon） 贡献归因（Credit Assignment）、策略蒸馏和自我进化，跟踪首先通过外部化 运行支撑框架（Harness） 工程搭建的能力如何逐步内化到策略中。

5. 申请（第 6 节）。通过智能体-环境接口的组织，我们将 长时程（Long-Horizon） 压力映射到软件工程、信息搜索、计算机使用、多模式智能体和 通用（General-Purpose） 智能体中，揭示了表层域差异之下的共同 长时程（Long-Horizon） 核心，并整合了水平感知评估资源。

6. 边境（第 7 节）。我们将开放问题组织成四个轴，即演化、有效性、效率和可信度，跨越从自我演化到可转移的九个具体方向

6

harnesses, to real-world environments and error robustness, to cost- and modality-aware efficiency, and to safety, governance, and embodiment, marking what must be solved as horizons keep growing.

Taken together, these contributions serve three purposes: a *clear definition* of long-horizon agency; a *high-quality taxonomy* (Figure 2) built around the paper's six perspectives; and a *roadmap* sketching the directions toward capable long-horizon agents. Beyond organizing existing work, this co-evolutionary view also offers a vantage point for reasoning about how the balance between externalized harness and internalized optimization may shift as horizons grow, and thus for anticipating the frontier forms that long-horizon agents may take.

Scope and Boundaries. We delimit our scope along four boundaries. **(1)** This is not a general investigation of LLM agents, memory, or reinforcement learning: we treat memory, reinforcement learning, and self-evolution through the co-evolution of externalized harness engineering and internalized model optimization, and only as they bear on long-horizon competence. **(2)** We exclude general ML platform infrastructure (e.g., container orchestration, GPU-cluster scheduling), and consider inference and decoding only where they materially affect long-trajectory execution. **(3)** We address model-architecture internals only for their effect on horizon, i.e., making long trajectories representable, trainable, and affordable to decode, while still treating reasoning and long-context modeling as foundational capabilities. **(4)** We draw on official documentation, open-source implementations, and emerging specifications alongside indexed academic publications, prioritizing documented engineering practice.

2 Foundation: Formalizing Long-Horizon Agents

Overview. Section 1 positioned long-horizon agency as a system-level capability, jointly shaped by the internalized model optimization and the externalized harness engineering that surrounds it. This section makes that claim precise and lays the conceptual groundwork for the rest of the paper, organized around four questions that follow the section's structure:

- **What is a long-horizon agent? (Section 2.1)** We define it by the coupled logical dependencies of a task rather than by elapsed time, and cast the agent as a base policy coupled to a surrounding harness, so long-horizon agency becomes a property of the model–harness system rather than of the foundation model alone.
- **How does its difficulty decompose? (Section 2.2)** We use how far execution stretches beyond one working context as an observable proxy for the underlying dependency burden, giving three nested levels (H1–H3) and their capabilities (C1–C3).
- **How fast is the frontier moving? (Section 2.3)** We quantify the rising time horizon of frontier agents as an empirical scaling trend and compare its full-period and recent-window doubling times.
- **How does it differ from neighboring notions? (Section 2.4)** We separate long-horizon agency from long-running execution, autonomy, and self-evolution, showing that it integrates these rather than reducing to any one.

2.1 Definition

We begin with the two objects of study: the task and the agent that seeks to complete it. As introduced in Section 1, we characterize long-horizon agency not by elapsed time or token count, but by the *coupled logical dependencies* that distribute a task across many interdependent steps. It is this chain of decisions, together with the interaction it entails, rather than duration alone, that determines the length of the horizon.

Long-horizon Tasks. A **long-horizon task** is a goal that may be under-specified at the outset and may require progressive clarification during solving, and whose solution path admits many possibilities and typically couples several complex subtasks together. As a result, it cannot be answered by a single-turn response in isolation, but only by *composing many interdependent decisions* into a single coherent trajectory [286, 811]. Concretely, a long-horizon task requires the agent to reason over many steps while interacting with the environment and, where relevant, the user: it progressively narrows down the solution path and continually tests and revises its reasoning direction until the goal is reached. Such a task may run from minutes to days

原文第 7 页 | 中文翻译初稿

运行支撑框架（Harnesses），到现实环境和错误鲁棒性，到成本和模式感知效率，到安全、治理和实施，标志着随着视野不断增长必须解决的问题。

总的来说，这些贡献有三个目的：长时程（Long-Horizon）机构的_明确定义_；围绕本文的六个视角构建的_高质量分类法_（图 2）；以及一张_路线图_，勾勒出通往有能力的长时程智能体（Long-Horizon Agents）的方向。除了组织现有的工作之外，这种共同进化的观点还提供了一个有利的观点，用于推理外部化 运行支撑框架（Harness）和内部化优化之间的平衡如何随着视野的增长而变化，从而预测 长时程智能体（Long-Horizon Agents）可能采取的前沿形式。

范围和边界。我们沿着四个边界界定我们的范围。(1) 这不是对 LLM 智能体、记忆或 强化学习（Reinforcement Learning）的一般研究：我们通过外化 运行支撑框架（Harness）工程和内化模型优化的共同进化来对待记忆、强化学习（Reinforcement Learning）和自我进化，并且仅当它们与 长时程（Long-Horizon）能力有关时。(2) 我们排除一般的 ML 平台基础设施（例如容器编排、GPU 集群调度），并且仅在对长轨迹执行产生重大影响的情况下考虑推理和解码。(3) 我们仅针对模型架构内部的影响来解决它们对地平线的影响，即使长轨迹可表示、可训练且可负担得起解码，同时仍将推理和长上下文建模视为基础功能。(4) 我们利用官方文档、开源实现和新兴规范以及索引学术出版物，优先考虑记录的工程实践。

2 基础：形式化 长时程智能体（Long-Horizon Agents）

概述。第 1 节将 长时程（Long-Horizon）机构定位为系统级能力，由内部模型优化和围绕它的外部 运行支撑框架（Harness）工程共同塑造。本节使这一主张更加精确，并为本文的其余部分奠定了概念基础，围绕遵循本节结构的四个问题进行组织：

- 长时程（Long-Horizon）智能体是什么？（第 2.1 节）我们通过任务的耦合逻辑依赖性而不是经过时间来定义它，并将智能体作为与周围 运行支撑框架（Harness）耦合的基本策略，因此 长时程（Long-Horizon）智能体成为模型 - 运行支撑框架（Harness）系统的属性，而不是单独的 基础模型（Foundation Model）的属性。

- 它的难度如何分解？（第 2.2 节）我们使用执行超出一个工作上下文的程度作为底层依赖负担的可观察智能体，给出三个嵌套级别 (H1-H3) 及其功能 (C1-C3)。

- 边境移动的速度有多快？（第 2.3 节）我们将前沿主体的上升时间范围量化为经验缩放趋势，并比较其全周期和最近窗口倍增时间。

- 它与邻近的概念有何不同？（第 2.4 节）我们将 长时程（Long-Horizon）智能体与长期运行的执行、自治和自我进化分开，表明它集成了这些而不是简化为任何一个。

2.1 定义

我们从两个研究对象开始：任务和寻求完成任务的智能体

。正如第 1 节中所介绍的，我们不是通过经过的时间或令牌计数来描述 长时程（Long-Horizon）智能体的特征，而是通过将任务分布在许多相互依赖的步骤之间的_耦合逻辑依赖关系_。决定视野长度的正是这一连串的决策，以及它所带来的相互作用，而不仅仅是持续时间。

长时程（Long-Horizon）任务。 长时程（Long-Horizon）任务是一个在一开始可能未明确指定的目标，并且在解决过程中可能需要逐步澄清，其解决方案路径允许多种可能性，并且通常将多个复杂的子任务耦

合在一起。因此，它不能通过孤立的单轮响应来回答，而只能通过“将许多相互依赖的决策”组合成单个连贯的轨迹来回答[286, 811]。具体而言，长时程（Long-Horizon） 任务要求智能体在与环境以及相关用户交互时进行多个步骤的推理：它逐渐缩小解决方案路径，并不断测试和修改其推理方向，直到达到目标。这样的任务可能需要几分钟到几天的时间

7

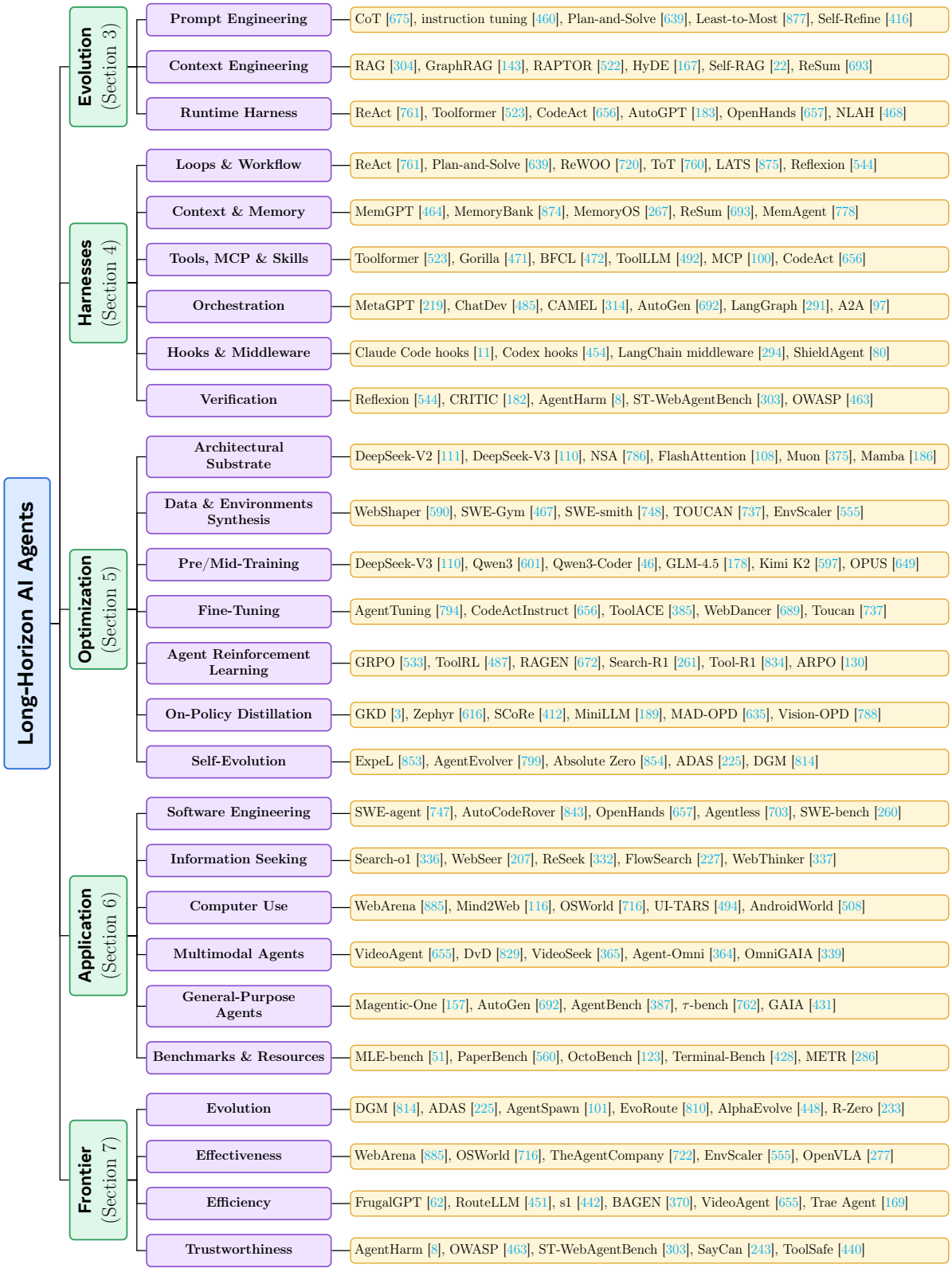


Figure 2 A taxonomy of long-horizon agent research. The paper is structured according to this taxonomy, progressing from **Evolution** (Section 3) through **Harnesses** (Section 4), **Optimization** (Section 5), and **Applications** (Section 6), to open **Frontiers** (Section 7). Each leaf lists representative works or systems discussed in the corresponding part of the paper.

原文第 8 页 | 中文翻译初稿

提示工程 CoT [675]、指令调整 [460]、规划和求解 [639]、从最少到最多 [877]、自我优化 [416]
上下文工程 RAG [304]、GraphRAG [143]、RAPTOR [522]、HyDE [167]、Self-RAG [22]、ReSum [693]
运行时运行支撑框架（Harness） ReAct [761]、Toolformer [523]、CodeAct [656]、AutoGPT [183]、OpenHands [657]、NLAH [468]
循环和工作流程 ReAct [761]、Plan-and-Solve [639]、ReWOO [720]、ToT [760]、LATS [875]、反射 [544]
上下文和内存 MemGPT [464]、MemoryBank [874]、MemoryOS [267]、ReSum [693]、MemAgent [778]
工具、MCP 和技能 Toolformer [523]、Gorilla [471]、BFCL [472]、ToolLLM [492]、MCP [100]、CodeAct [656]
Orchestration MetaGPT [219]、ChatDev [485]、CAMEL [314]、AutoGen [692]、LangGraph [291]、A2A [97]
Hooks 和中间件 Claude Code hooks [11]、Codex hooks [454]、LangChain 中间件 [294]、ShieldAgent [80]
验证反射 [544]、CRITIC [182]、AgentHarm [8]、ST-WebAgentBench [303]、OWASP [463]
架构
Substrate DeepSeek-V2 [111]、DeepSeek-V3 [110]、NSA [786]、FlashAttention [108]、Muon [375]、Mamba [186]
数据和环境
WebShaper [590]、SWE-Gym [467]、SWE-smith [748]、TOUCAN [737]、EnvScaler [555]
合成
训练前/中期 DeepSeek-V3 [110]、Qwen3 [601]、Qwen3-Coder [46]、 GLM-4.5 [178]、Kimi K2 [597]、OPUS [649]
微调 AgentTuning [794]、CodeActInstruct [656]、ToolACE [385]、WebDancer [689]、Toucan [737]
智能体强化
GRPO [533]、ToolRL [487]、RAGEN [672]、Search-R1 [261]、Tool-R1 [834]、ARPO [130]
学习
按策略蒸馏 GKD [3]、Zephyr [616]、SCoRe [412]、MiniLLM [189]、MAD-OPD [635]、Vision-OPD [788]
自我进化 ExpeL [853]、AgentEvolver [799]、绝对零 [854]、ADAS [225]、DGM [814]
软件工程 SWE-agent [747]、AutoCodeRover [843]、OpenHands [657]、Agentless [703]、SWE-bench [260]
信息搜索 Search-o1 [336]、WebSeer [207]、ReSeek [332]、FlowSearch [227]、WebThinker [337]
计算机使用 WebArena [885]、Mind2Web [116]、OSWorld [716]、UI-TARS [494]、AndroidWorld [508]
多模式智能体 VideoAgent [655]、DvD [829]、VideoSeek [365]、 Agent-Omni [364]、OmniGAIA [339]
通用（General-Purpose）
Magentic-One [157]、AutoGen [692]、AgentBench [387]、τ -bench [762]、GAIA [431]
Agents
基准与资源 MLE-bench [51]、PaperBench [560]、 OctoBench [123]、Terminal-Bench [428]、METR [286]
Evolution DGM [814]、ADAS [225]、AgentSpawn [101]、EvoRoute [810]、AlphaEvolve [448]、R-Zero [233]

Effectiveness WebArena [885]、OSWorld [716]、TheAgentCompany [722]、EnvScaler [555]、OpenVLA [277]
效率 FrugalGPT [62]、RouteLLM [451]、s1 [442]、BAGEN [370]、VideoAgent [655]、Trae Agent [169]
可信度 AgentHarm [8]、OWASP [463]、ST-WebAgentBench [303]、SayCan [243]、ToolSafe [440]
进化（第 3 节）
运行支撑框架（Harnesses）（第 4 节）
（第 5 节）
优化
Long-HorizonAIAgents
应用程序（第 6 节）
Frontier
（第 7 节）

图 2 长时程（Long-Horizon）智能体研究的分类。本文根据该分类法进行结构，从进化（第 3 节）到 运行支撑框架（Harnesses）（第 4 节）、优化（第 5 节）和应用（第 6 节），再到开放前沿（第 7 节）。每页都列出了论文相应部分中讨论的代表性作品或系统。

8

and may fit within one context window or stretch far beyond it; its defining feature is not duration itself but this sustained, feedback-driven dependence among steps and the continual interaction required to manage it.

Long-horizon Agents. A **long-horizon agent** is a system designed to solve such tasks in an environment with state, feedback, and side effects [640]. Although the stated intent may be simple, successful execution requires resolving a large and tightly coupled chain of logical dependencies. Unlike a single-turn responder, it does not produce one answer and stop; instead, it must sustain reasoning through repeated cycles of planning, action, observation, and revision, progressively clarifying an under-specified goal as the trajectory unfolds. As argued in Section 1, this competence is not a property of the foundation model alone. It arises from the joint design of an **externalized harness engineering**, which supplies runtime structure such as tools, memory, orchestration, and checks, and an **internalized model optimization** process, which improves the policy through data, training, and experience.

Formalization: Long-Horizon Agent

We model a long-horizon agent as a foundation-model *base policy* π_θ coupled to a surrounding structure $\mathcal{H}$ that closes the loop around it:

$$\boxed{\text{Agent} \;=\; \pi_\theta \;\oplus\; \mathcal{H}} \tag{1}$$

Here π_θ proposes actions, while $\mathcal{H}$ comprises the loops and workflows, system prompts, tools and skills, context and memory, orchestration logic, hooks, and verification mechanisms that govern *when* the model is invoked, *how* its actions are validated and executed, and *what* state persists across steps. The operator $\oplus$ denotes runtime coupling rather than simple addition: $\mathcal{H}$ closes the action–observation–memory loop around π_θ, so long-horizon agency is a property of the coupled system and its training, not of π_θ alone. Equation (1) describes the *deployed* coupling that produces long-horizon behavior at run time; how it *evolves* over time, as π_θ internalizes competence through training and the system accumulates experience, is the co-evolving recipe we develop in Section 5 and Section 3. We refer to these two halves as **Pillar I**, the externalized harness $\mathcal{H}$ (Section 4), and **Pillar II**, the internalized optimization of π_θ (Section 5).

The Long-horizon Agent–Environment Loop. Since a long-horizon agent makes progress precisely by interacting with its environment step after step, its execution is most naturally described as a continual loop in which it acts, observes the consequences, and updates its state before acting again. Following the standard LLM-based agent formulation [228, 640, 761], we place the coupled system in an environment. Let $\mathcal{S}$ be an environment state space, $\mathcal{O}$ an observation space, and $\mathcal{A}$ an action space. The environment evolves by a transition kernel $s_{t+1} \sim \Psi(\cdot \mid s_t, a_t)$ and emits observations through an observation kernel $o_t \sim Z(\cdot \mid s_t)$, which is partial and in general stochastic. We distinguish the base policy π_θ from the *agent policy* $\pi = \pi_\theta \oplus \mathcal{H}$ that the harness closes around it: at each step the harness assembles a contextual signal c_t and the base policy acts on it,

$$c_t = \mathcal{H}(o_{0:t},\, a_{0:t-1},\, \mathcal{Q})\,, \qquad a_t \sim \pi_\theta(\cdot \mid c_t),$$

where $\mathcal{Q}$ is the task specification and c_t is the contextual signal that the harness assembles at step t from the interaction history: the prompt window with retrieved memories, prior tool outputs, reasoning traces, and other state. Execution induces a trajectory $\tau = (s_0, o_0, a_0, s_1, o_1, a_1, \ldots, s_T)$ that runs until termination, i.e. until the first step $T = \min\{t : s_t \in \mathcal{S}_{\text{term}}\}$ at which the state enters a terminal set $\mathcal{S}_{\text{term}}$ (the goal is reached, a budget is exhausted, or the agent halts). Because o_t is only a partial view of s_t and c_t is a lossy summary of history, the coupled system is naturally modeled, following standard POMDP formulations for sequential decision making under partial observability [264], as a partially observable Markov decision process (POMDP). Much of what $\mathcal{H}$ does is the machinery of acting under partial observability. The task assigns each trajectory a utility $R(\tau) \in \mathbb{R}$, typically sparse and delayed and often revealed only at T, so that credit for the intermediate decisions that actually drive success or failure must be inferred from a single terminal signal. This is the long-horizon credit-assignment problem that is central to reinforcement learning methods [579].

原文第 9 页 | 中文翻译初稿

并且可能适合一个上下文窗口或远远超出它；它的定义特征不是持续时间本身，而是步骤之间持续的、反馈驱动的依赖以及管理它所需的持续交互。

长时程智能体（Long-Horizon Agents）。 长时程（Long-Horizon）智能体是一个设计用于在具有状态、反馈和副作用的环境中解决此类任务的系统[640]。尽管所陈述的意图可能很简单，但成功的执行需要解决大量且紧密耦合的逻辑依赖关系链。与单轮响应器不同，它不会产生一个答案并停止；相反，它必须通过重复的计划、行动、观察和修正循环来维持推理，随着轨迹的展开逐步澄清未明确的目标。正如第 1 节所述，这种能力并不是 基础模型（Foundation Model）单独的财产。它源自外部化的 运行支撑框架（Harness）工程的联合设计，该工程提供工具、内存、编排和检查等运行时结构，以及内部化的模型优化流程，该流程通过数据、训练和经验改进策略。
形式化：长时程（Long-Horizon）智能体

我们将 长时程（Long-Horizon）智能体建模为基础模型_基本策略 πθ_，耦合到周围结构 H，从而闭合其周围的循环：

Agent = πθ ⊕H (1)

这里 πθ 提出了操作，而 H 包括循环和工作流程、系统提示、工具和技能、上下文和内存、编排逻辑、挂钩和验证机制，这些机制控制模型何时被调用、如何验证和执行其操作以及跨步骤保持什么状态。运算符 _⊕_表示运行时耦合而不是简单的加法：_H_关闭围绕_πθ_的动作-观察-记忆循环，因此长时程（Long-Horizon）智能体是耦合系统及其训练的属性，而不是_πθ_单独的属性。等式 (1) 描述了在运行时产生 长时程（Long-Horizon）行为的_部署_耦合；随着 πθ 通过训练内化能力以及系统积累经验，它如何随着时间的推移_进化_，是我们在第 5 节和第 3 节中开发的共同进化配方。我们将这两部分称为第一支柱，外化的 运行支撑框架（Harness）H（第 4 节），和第二支柱，πθ 的内部优化（第 5 节）。

长时程（Long-Horizon）智能体-环境循环。由于 长时程（Long-Horizon）智能体通过一步一步与其环境交互来精确地取得进展，因此它的执行最自然地被描述为一个连续循环，在该循环中它执行操作、观察结果并在再次执行操作之前更新其状态。遵循基于 LLM 的标准智能体公式 [228,640,761]，我们将耦合系统放置在一个环境中。令 S 为环境状态空间，O 为观察空间，A 为动作空间。环境通过过渡内核 st +1 ~ Ψ(· | st, at) 演化，并通过观察内核 ot ~ Z (· | st) 发出观测值，该内核是部分随机的。我们将基本策略 πθ 与 运行支撑框架（Harness）围绕其封闭的 agent 策略 π = πθ ⊕H 区分开来：在每一步，运行支撑框架（Harness）都会组装一个上下文信号 ct ，并且基本策略对其进行操作，

ct = H (o 0: t, a 0: t− 1 , Q) , at ~ πθ (· | ct) ,

其中 Q 是任务规范，ct 是 运行支撑框架（Harness）在第 t 步从交互历史记录中组装的上下文信号：带有检索到的内存、先前工具输出、推理跟踪和其他状态的提示窗口。执行产生一条轨迹 τ = (s 0 , o 0 , a 0 , s 1 , o 1 , a 1 , . . . , sT) ，该轨迹一直运行直至终止，即直到第一步 T = min {t : st ∈S term } ，此时状态进入终端集_S_ 期限（达到目标、预算耗尽或智能体停止）。因为 ot 只是 st 的部分视图，而 ct 是历史的有损总结，因此耦合系统自然地建模，遵循部分可观察性下顺序决策的标准 POMDP 公式 [264]，作为部分可观察马尔可夫决策过程 (POMDP)。 H 所做的大部分工作是在部分可观察性下采取行动的机制。该

任务为每个轨迹分配一个效用 R (τ) ε R，通常稀疏且延迟，并且通常仅在 T 处显示，因此必须从单个终端信号推断出实际驱动成功或失败的中间决策的功劳。这是 长时程（Long-Horizon） 信用分配问题，是强化学习（Reinforcement Learning） 方法的核心[579]。

9

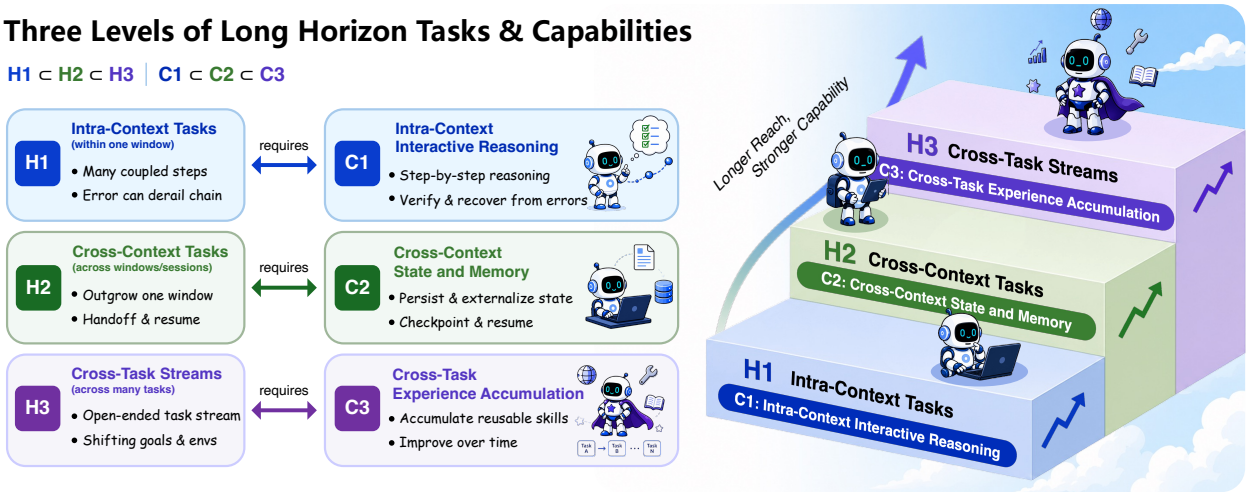


Figure 3 Three levels of long-horizon tasks and the capabilities they demand. As the required horizon widens from intra-context tasks (H1), to cross-context tasks (H2), to cross-task streams (H3), the agent must add intra-context interactive reasoning (C1), cross-context state and memory (C2), and cross-task experience accumulation (C3) capabilities.

2.2 Three Levels of Long-Horizon Tasks and Capabilities

Although the notion of a *long-horizon* agent is by now widely used, it remains conceptually vague, and prior work still lacks a principled way to partition long-horizon competence into distinct levels. Building on the definition above, we organize long-horizon tasks by an operational axis: *how far execution must stretch beyond a single working context*, ranging from intra-context tasks completable within one window (H1), to cross-context tasks that span windows or sessions (H2), to cross-task streams in which an overarching goal unfolds over many episodes (H3). This axis is a proxy for, rather than the definition of, logical dependency: a context boundary matters when coupled decisions and task state must survive it, whereas a long but decomposable batch process need not be long-horizon. The proxy is useful because no step is perfectly reliable and errors compound along the dependency graph, so trajectory-level reliability rather than per-step accuracy governs success [286]. It yields three nested operational levels of increasing reach [14, 228], and we pair each level with the capability an agent needs to succeed at it (Figure 3). Because every level adds a new failure mode on top of the one below, the corresponding demands accumulate. The time ranges below are therefore descriptive of current systems, not definitional thresholds.

Three Levels of Long-Horizon Tasks (H1 ≺ H2 ≺ H3)

H1 — Intra-context, within one window (∼minutes).
A single task can be completed within one working context window, yet it is far from a one-shot query: it requires composing many interdependent decisions into one coherent trajectory, with repeated interaction with the environment and, where relevant, the user to test and revise the reasoning direction until the goal is reached [286, 544].

H2 — Cross-context, across windows/sessions (∼hours to days).
A single task far exceeds one context window or session and must be carried across multiple context windows or by several agents working in concert. The agent(s) must therefore actively compress and externalize historical context, track interaction state in real time, and resume progress without assuming that the full history remains visible at every step [14, 378, 511].

H3 — Cross-task, open-ended task stream (∼lifelong).
Tasks arrive as an open-ended stream rather than a closed episode. A single overarching goal may therefore span many tasks, each of which may itself require multiple context windows to complete. In deployment, such settings often demand persistent, and sometimes lifelong, operation across heterogeneous environments, tools, and sessions [165, 228, 580, 808].

原文第 10 页 | 中文翻译初稿

三个级别的长期任务和功能
H1 ⊂ H2 ⊂ H3 C1 ⊂ C2 ⊂ C3
上下文内任务 上下文内
（在一个窗口内）需要交互式推理
H1 C1
许多耦合步骤 逐步推理
错误可能使链脱轨 验证并从错误中恢复
跨上下文任务跨上下文
（跨窗口/会话）需要状态和内存
H2 C2
扩展一个窗口 持久和外部化状态
切换和恢复 检查点和恢复
跨任务流 跨任务
（跨多个任务）需要经验积累
H3 C3
开放式任务流 积累可重用技能
转移目标和环境 改进时间
触手更远，
能力更强

长期任务和能力的三个层次

图 3 长时程（Long-Horizon）任务的三个级别及其所需的功能。随着所需视野从上下文内任务（H1）扩大到跨上下文任务（H2），再到跨任务流（H3），智能体必须添加上下文内交互推理（C1）、跨上下文状态和记忆（C2）以及跨任务经验积累（C3）能力。
2.2 长时程（Long-Horizon）三个级别的任务和能力

尽管 长时程（Long-Horizon）智能体的概念现在已被广泛使用，但它在概念上仍然模糊，并且先前的工作仍然缺乏将 长时程（Long-Horizon）能力划分为不同级别的原则性方法。基于上述定义，我们通过操作轴组织 长时程（Long-Horizon）任务：执行必须超出单个工作上下文多远，范围从可在一个窗口内完成的上下文内任务 (H1)，到跨窗口或会话的跨上下文任务 (H2)，再到总体目标在多个情节中展开的跨任务流 (H3)。该轴是逻辑依赖关系的智能体，而不是逻辑依赖关系的定义：当耦合决策和任务状态必须在其中生存时，上下文边界很重要，而长但可分解的批处理过程不必是 长时程（Long-Horizon）。智能体很有用，因为没有任何步骤是完全可靠的，并且错误沿着依赖图复合，因此轨迹级可靠性而不是每步骤的准确性决定着成功[286]。它产生三个嵌套的操作级别，其范围不断扩大 [14, 228]，我们将每个级别与智能体成功所需的能力配对（图 3）。因为每一级都会在下面的故障模式之上添加一个新的故障模式，因此相应的需求会不断累积。因此，下面的时间范围是对当前系统的描述，而不是定义阈值。
长时程（Long-Horizon）任务的三个级别 (H1 ≺ H2 ≺ H3)
H1 - 上下文内，在一个窗口内（~ 分钟）。

一项任务可以在一个工作上下文窗口内完成，但它远非一次性查询：它需要将许多相互依赖的决策组合成一个连贯的轨迹，与环境以及相关的用户重复交互，以测试和修改推理方向，直到达到目标[286, 544]。
H2 - 跨上下文、跨窗口/会话（~ 小时到天）。

单个任务远远超出一个上下文窗口或会话，并且必须跨多个上下文窗口或由协同工作的多个智能体来执行

。因此，智能体必须主动压缩和外部化历史上下文，实时跟踪交互状态，并恢复进度，而不假设完整的历史记录在每一步都保持可见[14,378,511]。
H3 - 跨任务、开放式任务流（~ 终身）。

任务以开放式流的形式出现，而不是封闭的片段。因此，一个总体目标可能跨越许多任务，每个任务本身可能需要多个上下文窗口才能完成。在部署中，此类设置通常需要跨异构环境、工具和会话进行持久的、有时甚至是终生的操作[165、228、580、808]。

10

So far we have organized long-horizon *tasks* into three difficulty levels; we now turn from tasks to *agents* and identify, for each level, the capability that its difficulty demands. Each level exposes a distinct insufficiency. H1 shows that per-step accuracy alone is insufficient: the agent must sustain reasoning through continual interaction rather than produce a single response. H2 shows that a larger context window is insufficient: once a task spans multiple windows or agents, state must be compressed, externalized, and tracked even when the full history is not continuously in view. H3 shows that episodic success is insufficient: an agent that completes an individual task may still fail to accumulate reusable competence across an open-ended stream of subsequent tasks. Each level therefore demands a capability of matching reach, denoted C1–C3 respectively, and these capabilities compose cumulatively: C2 presupposes C1, and C3 presupposes both.

Capabilities Demanded by Each Level (C1 $\prec$ C2 $\prec$ C3)

C1 — Intra-context interactive reasoning (for H1).
Within a single context, sustain continuous reasoning through interaction with the environment and the user: take actions, absorb feedback, and revise the trajectory so that many interdependent decisions compose into one coherent path toward the goal [286, 544].

C2 — Cross-context state and memory (for H2).
Building on C1, maintain task state when execution spans windows or agents: compress and externalize historical context, read and write memory, checkpoint progress, and faithfully resume after handoffs [14, 378, 511, 694].

C3 — Cross-task experience accumulation (for H3).
Building on C2, turn experience into lasting competence across an open-ended task stream: switch among complex tasks, each of which may itself span multiple context windows, accumulate reusable skills [630], generalize from past experience [462], and continually improve as goals, environments, and data distributions shift [165, 228, 580].

Because the levels nest, so do the capabilities, in that an agent operating at H3 must still satisfy C1 and C2 within every task it touches. This is where the paper's central question enters. Each capability can be supplied either from *outside* the model, through the harness, or from *inside* it, through training. H1 pressures are often met by C1 within a single context, or absorbed into the model weights; H2 pressures push C2 out of the window into durable external state, namely compressed context, memory, and checkpoints; and H3 pressures demand C3, which turns deployment experience into lasting competence. In many H3 settings, however, neither route is likely to suffice on its own, and the highest level of long-horizon competence is therefore inherently a *system-level* property, jointly constructed by internalized model weights and the externalized harness rather than achievable by either in isolation (Figure 1). We attach each method to the level it primarily serves.

2.3 Scaling the Time Horizon of Frontier-Agent Execution

In the previous subsection, we organized long-horizon difficulty and its corresponding capabilities into a stair-stepped ladder *qualitatively*; here, we quantify how an observable proxy for the frontier scales over time. We build on the measurements of METR [286], which define the *50%-task-completion time horizon* as the human-expert completion time for the class of agentic engineering tasks that a given agent is predicted to complete with 50% reliability. These tasks span software engineering, machine learning engineering, and computer use, providing a more objective proxy for task difficulty than task labels alone.

Figure 4 plots METR's live estimates through May 8, 2026 [286, 429][1]. Because several early models were not re-evaluated under the newer TH1.1 methodology, the full historical series combines TH1.0 estimates for those models with TH1.1 estimates for later models. Under this stitched series, the time horizon of frontier agents rises from seconds for early models (e.g. GPT-2) to approximately twelve hours for Claude Opus 4.6 and at least sixteen hours for an early Claude Mythos Preview by May 2026, spanning nearly four orders of magnitude within only a few years.

[1]METR live dashboard, https://metr.org/time-horizons/, accessed May 8, 2026.

原文第 11 页 | 中文翻译初稿

到目前为止，我们已经将 长时程（Long-Horizon） tasks 组织成三个难度级别；我们现在从任务转向智能体，并为每个级别确定其难度所需的能力。每个级别都暴露出明显的不足。 H1 表明，仅靠每步准确度是不够的：智能体必须通过持续的交互来维持推理，而不是产生单一响应。 H2 表明，较大的上下文窗口是不够的：一旦任务跨越多个窗口或智能体，即使不能连续查看完整历史记录，也必须压缩、外部化和跟踪状态。 H3 表明，情景成功是不够的：完成单个任务的智能体可能仍然无法在开放的后续任务流中积累可重用的能力。因此，每个级别都需要匹配到达的能力，分别表示为C1-C3，并且这些能力累积起来：C2以C1为前提，C3以两者为前提。
每个级别所需的能力（C1 ≺ C2 ≺ C3）
C1--上下文内交互式推理（针对 H1）。

在单一上下文中，通过与环境和用户的交互来维持连续推理：采取行动、吸收反馈并修改轨迹，以便许多相互依赖的决策组成一条通向目标的连贯路径 [286, 544]。
C2 - 跨上下文状态和内存（用于 H2）。

基于 C1 构建，在跨窗口或智能体执行时维护任务状态：压缩和外部化历史上下文、读写内存、检查点进度，并在切换后忠实地恢复 [14, 378, 511, 694]。
C3--跨任务经验积累（针对H3）。

以 C2 为基础，将经验转化为跨开放式任务流的持久能力：在复杂的任务之间切换，每个任务本身可能跨越多个上下文窗口，积累可重用的技能 [630]，从过去的经验中概括 [462]，并随着目标、环境和数据分布的变化而不断改进 [165,228,580]。

因为级别是嵌套的，所以功能也是嵌套的，因为在 H3 上运行的智能体仍然必须在其涉及的每个任务中满足 C1 和 C2。这就是本文的核心问题所在。每个功能都可以通过 运行支撑框架（Harness） 从模型外部提供，也可以通过训练从模型内部提供。 H1 压力通常由 C1 在单一上下文中满足，或吸收到模型权重中；H2 压力将 C2 推出窗外，进入持久的外部状态，即压缩上下文、内存和检查点； H3 压力要求 C3，将部署经验转化为持久的能力。然而，在许多 H3 设置中，这两种途径本身都可能不够，因此 长时程（Long-Horizon） 能力的最高级别本质上是一个_系统级_属性，由内部化模型权重和外部化 运行支撑框架（Harness） 共同构建，而不是通过任何一个单独实现（图 1）。我们将每个方法附加到它主要服务的级别。
2.3 扩展前沿智能体执行的时间范围

在上一小节中，我们将长时程（Long-Horizon）难度及其相应的能力定性地组织成阶梯式的阶梯；在这里，我们量化了边界的可观察智能体如何随着时间的推移而扩展

。我们以 METR [286] 的测量为基础，将_50% 任务完成时间范围_ 定义为预测给定智能体以 50% 可靠性完成的智能体工程任务类的人类专家完成时间。这些任务涵盖软件工程、机器学习工程和计算机使用，为任务难度提供了比单独的任务标签更客观的智能体。

图 4 绘制了 METR 截至 2026 年 5 月 8 日的实时估计 [286, 429]1。由于一些早期模型没有根据较新的 TH1.1 方法进行重新评估，因此完整的历史系列将这些模型的 TH1.0 估计值与后来模型的 TH1.1 估计值结合起来。在这个缝合系列下，前沿特工的时间范围从早期模型（例如 GPT-2）的几秒钟上升到 Claude Opus 4.6 的大约 12 小时，到 2026 年 5 月，早期的 Claude Mythos 预览版至少需要 16 小时，在短短几年内跨越了近四个数量级。

> 1METR 实时仪表板，https://metr.org/time-horizons/，于 2026 年 5 月 8 日访问。

11

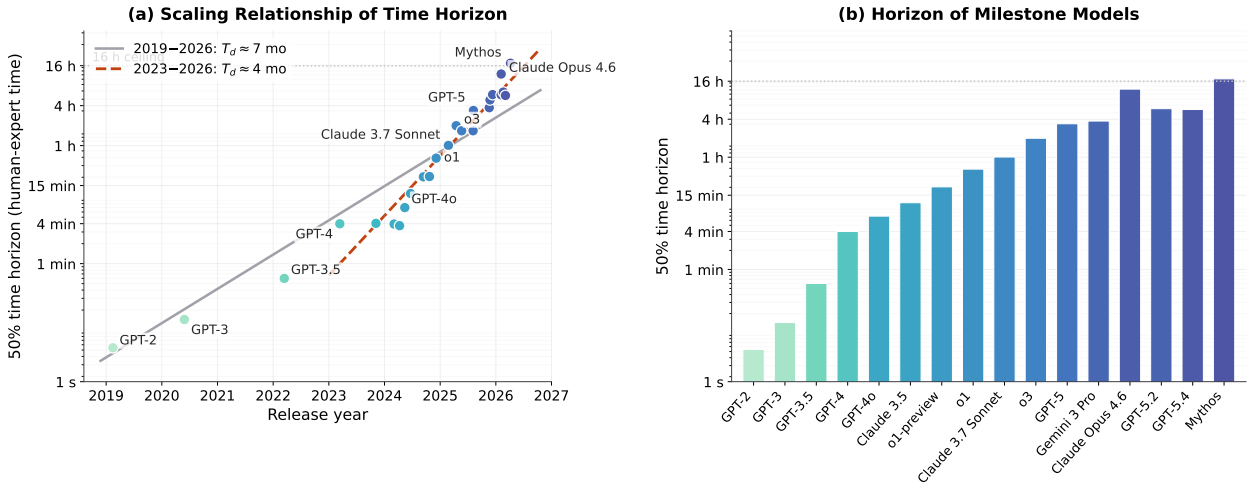


Figure 4 Empirical growth of frontier-agent time horizons. Frontier agents' 50%-task-completion time horizons over model release dates, shown for the full fitted series in **(a)** and selected milestone agents in **(b)**.

The fitted trend in Figure 4 summarizes this rapid expansion. Across the stitched full-period series, the estimated doubling time is approximately 196.5 days, or 6.5 months. Restricting the fit to models released from 2023 onward under TH1.1 yields a shorter doubling time of approximately 130.8 days, or 4.3 months [286, 429, 683]. The dotted line marks the approximately sixteen-hour region above which METR considers estimates unreliable under the current task suite. Consequently, both individual estimates and fitted trends near or beyond this boundary carry substantial uncertainty.

To summarize the trend formally, we borrow the functional spirit of the compute-optimal scaling law of Hoffmann et al. [217] and model the frontier horizon in terms of two quantities: the release date of a frontier agent and the human-expert duration of tasks that the evaluated system can complete. We write this relationship as a compact descriptive summary:

$$\log_2 H(t) \approx \log_2 H_0 + \frac{t - t_0}{T_d}, \qquad \text{equivalently} \qquad H(t) \approx H_0 \cdot 2^{(t-t_0)/T_d}, \tag{2}$$

where $H(t)$ is the estimated time horizon of a frontier agent whose base model was released at date t, t_0 is a reference date, $H_0 = H(t_0)$ is the fitted horizon at that date, and T_d is the *doubling time*. Under this functional form, the full-period and 2023-onward fits correspond to $T_d \approx 196.5$ days and $T_d \approx 130.8$ days, respectively. Although these estimates are derived from self-contained, automatically scorable software-style tasks and become less precise near the upper boundary of the benchmark, these limitations primarily constrain the exact fitted values rather than the overall direction of the trend.

The decisive signal is that the task duration frontier agents can reliably handle has increased approximately exponentially. The shorter doubling time obtained for the 2023-onward subset further suggests that this expansion may have accelerated, although differences in evaluation coverage and methodology prevent treating the two fitted values as definitive evidence of acceleration. The trend nevertheless aligns directly with our capability hierarchy: as completable task horizons expand from minutes to hours and then to tens of hours, frontier agents move beyond intra-context H1 tasks, increasingly address H2 tasks that require cross-context state and memory, and begin to approach the boundary of cross-task, open-ended H3. In this sense, the sustained expansion of the measured time horizon is an observable projection of long-horizon competence climbing the H1→H2→H3 ladder. At the current pace, many multi-hour tasks that remain at the frontier today may become routine in the foreseeable future. That said, the trend characterizes *progress over time*; it does not explain its *source*. In particular, it does not distinguish whether the gains arise from a stronger policy π_θ, a better harness $\mathcal{H}$, or the co-evolution of the two. Attributing observed progress among these factors is the central question of this paper. We therefore turn from measuring the frontier to asking what drives it.

原文第 12 页 | 中文翻译初稿

(a) 时间范围的标度关系 (b) 里程碑模型的范围
16 h 16 h ceilin 2019 2023 g 2026: 2026: T T d d 7 mo 4 mo Mythos Claude Opus 4.6
16 h
GPT-5
4 h
o3 4 h
Claude 3.7 Sonnet
1 h
o1 1 h
15 分钟
GPT-4o 15 分钟
4 分钟 GPT-4 4 分钟
1 分钟 GPT-3.5 1 分钟
GPT-3
GPT-2
1 秒 1 秒
2019 2020 2021 2022 2023 2024 2025 2026 2027
发布年份
GPT-2GPT-3GPT-3.5GPT-4GPT-4oClaude 3.5o1-previewClaude 3.7 Sonneto1 o3GPT-5Gemini 3 ProClaude Opus 4.6GPT-5.2GPT-5.4Mythos
50% 时间地平线
50%时间范围（人类专家时间）

图 4 前沿智能体时间范围的经验增长。前沿智能体在模型发布日期内完成 50% 任务的时间范围，如 (a) 中的完整拟合系列和 (b) 中选定的里程碑智能体所示。

图 4 中的拟合趋势总结了这种快速扩张。在缝合的全周期系列中，预计倍增时间约为 196 __._ 5 天，或 6 ___ 5 个月。将拟合限制在 TH1.1 下从 2023 年起发布的模型会产生更短的倍增时间，约为 130 __._ 8 天，或 4 __._ 3 个月 [286, 429, 683]。虚线标记了大约 16 小时的区域，METR 认为在该区域之上的估计在当前任务套件下不可靠。因此，接近或超出该边界的个人估计和拟合趋势都具有很大的不确定性。

为了正式总结这一趋势，我们借鉴了 Hoffmann 等人的计算最优缩放定律的功能精神。 [217]并根据两个量对前沿视野进行建模：前沿智能体的发布日期和评估系统可以完成的任务的人类专家持续时间。我们将这种关系写成一个紧凑的描述性摘要：

log2 H (t); ≈ ; log2 H 0 + t − t 0 ，相当于 H (t) ； ≈ ; H 0 · 2 , (t−t 0) /Td , (2)
Td

其中 H (t) 是在日期 t 发布基础模型的前沿智能体的估计时间范围， t 0 是参考日期， H 0 = H (t 0) 是该日期的拟合范围， Td 是 倍增时间 。在此函数形式下，全周期和 2023 年以后的拟合对应于 Td ！ ≈ !196 .

5 天，Td！ ≈ !130 . 分别为 8 天。尽管这些估计是从独立的、自动评分的软件式任务中得出的，并且在基准上限附近变得不太精确，但这些限制主要限制了精确的拟合值，而不是趋势的总体方向。

决定性的信号是，边境特工能够可靠处理的任务持续时间几乎呈指数增长。 2023 年以后子集获得的更短的倍增时间进一步表明，这种扩张可能已经加速，尽管评估范围和方法的差异阻止将两个拟合值视为加速的明确证据

。然而，这一趋势与我们的能力层次结构直接一致：随着可完成的任务范围从几分钟扩展到几小时，然后扩展到几十小时，前沿智能体超越了上下文内的 H1 任务，越来越多地解决需要跨上下文状态和内存的 H2 任务，并开始接近跨任务、开放式 H3 的边界。从这个意义上说，测量时间范围的持续扩展是 长时程（Long-Horizon） 攀登 H1 → H2 → H3 阶梯的能力的可观察预测。按照目前的速度，许多目前仍处于前沿的耗时数小时的任务在可预见的未来可能会成为例行公事。也就是说，趋势的特点是_随着时间的推移而取得进展_；它没有解释它的 source 。特别是，它没有区分增益是来自更强的策略 πθ 、更好的 运行支撑框架（Harness） H ，还是两者的共同进化。将观察到的进展归因于这些因素是本文的核心问题。因此，我们从测量边界转向询问驱动边界的因素。

12

2.4 Long-Horizon Agent vs. Neighboring Concepts

Long-horizon agency is often conflated with three adjacent notions: long-running execution, autonomy, and self-evolution. They are related because all appear in modern agent systems, but they answer different questions. Long-running asks how long execution persists, autonomy asks how much human control remains, and self-evolution asks whether the agent improves itself over time. Long-horizon agency asks a different question: can the coupled system sustain interactive reasoning within a context (C1), maintain state and memory across contexts when needed (C2), and accumulate experience across tasks when deployment demands it (C3)? Figure 5 summarizes the overlap and separation.

2.4.1 Long-Horizon Agent vs. Long-Running Agent

A long-running agent is defined by duration: it keeps making progress over hours to weeks across many windows, sessions, or sandboxes, with checkpointing and resumption [14]. It often appears in long-horizon settings, but duration and difficulty are not the same.

Overlap. Many genuinely long-horizon tasks are also long-running, because their trajectories outgrow one context window or one execution session. In these cases the same engineering mechanisms that instantiate C2 matter: compressing and externalizing historical context, durable memory, progress logs, checkpoints, and faithful resumption. Long-running engineering is therefore a common implementation requirement in H2 settings, where task state must survive resets, context boundaries, and hand-offs without assuming that full history remains visible.

Distinctions. The key distinction is that duration and *logical coupling* are independent properties. A task may run for days while consisting mostly of logically decoupled, independently checkpointable steps; a script that loops over a large dataset and applies the same operation runs for a long time, yet its steps carry few interdependencies, so it demands neither dense, sustained interactive reasoning (C1) nor live management of cross-context state (C2). Conversely, a task may complete within a single window while still requiring many interdependent decisions whose direction must be continually tested and revised through interaction with the environment. Long-running engineering thus optimizes persistence, throughput, and cost over time, whereas long-horizon agency concerns whether an agent can sustain reasoning through continual interaction and, when the task demands it, maintain the state and memory needed to carry that reasoning across contexts [286]. Accordingly, execution duration captures temporal persistence, whereas long-horizon agency denotes the capacity to sustain coupled reasoning and state management across the task's dependency structure.

2.4.2 Long-Horizon Agent vs. Autonomous Agent

An autonomous agent is defined by how much it can act without user involvement, often formalized as roles ranging from operator to observer [155]. Autonomy and long-horizon agency are strongly related, but they describe different properties of a system.

Overlap. The two overlap because long-horizon tasks make constant human supervision expensive, so most long-horizon agents are largely automated: they interact with external environments and tools on their own, and autonomously decide when to revise their strategy. If an agent must take many steps, interact with tools and the environment, absorb feedback, and revise its trajectory, then requiring approval at every step can destroy the usefulness of the system, even when the task itself fits within H1. At the same time, autonomy raises the safety stakes of long-horizon execution, since more unattended steps and more coupled side effects mean a larger blast radius, so permission gates and oversight remain central.

Distinctions. The distinction is that autonomy is only a partial, governance-oriented dimension of a system, not a measure of its capability. A strong long-horizon agent can be run with low autonomy behind approval gates, whereas a highly autonomous agent may execute only a short, single-window task, such as a single web action [452]. Moreover, long-horizon agents need not be fully autonomous: their harnesses can introduce permission gates, approval checkpoints, and escalation policies that selectively involve humans at high-impact or uncertain steps. Autonomy therefore asks who remains in the loop, whereas long-horizon agency asks

原文第 13 页 | 中文翻译初稿

2.4 长时程（Long-Horizon） 智能体与邻居概念

长时程（Long-Horizon） 智能体经常与三个相邻的概念混为一谈：长期执行、自治和自我进化。它们是相关的，因为它们都出现在现代智能体系统中，但它们回答不同的问题。长时间运行询问执行持续多长时间，自主性询问人类控制力还剩下多少，自我进化询问智能体是否随着时间的推移而改进自身。 长时程（Long-Horizon） 机构提出了一个不同的问题：耦合系统能否在上下文中维持交互式推理（C1），在需要时跨上下文维护状态和内存（C2），并在部署需要时跨任务积累经验（C3）？图 5 总结了重叠和分离。

2.4.1 长时程（Long-Horizon） 智能体与长时间运行的智能体

长时间运行的智能体由持续时间定义：它在多个窗口、会话或沙箱中持续数小时到数周的时间，并具有检查点和恢复功能 [14]。经常出现在长时程（Long-Horizon）设置中，但持续时间和难度不一样。

重叠。许多真正的 长时程（Long-Horizon） 任务也是长时间运行的，因为它们的轨迹超出了一个上下文窗口或一个执行会话。在这些情况下，实例化 C2 的相同工程机制很重要：压缩和外部化历史上下文、持久内存、进度日志、检查点和忠实恢复。因此，长时间运行的工程是 H2 设置中的常见实现要求，其中任务状态必须能够承受重置、上下文边界和切换，而不假设完整的历史记录仍然可见。

区别。关键区别在于持续时间和_逻辑耦合_是独立的属性。一项任务可能会运行数天，同时主要由逻辑解耦、可独立检查点的步骤组成；循环遍历大型数据集并应用相同操作的脚本会运行很长时间，但其步骤几乎没有相互依赖性，因此它既不需要密集、持续的交互式推理 (C1)，也不需要跨上下文状态的实时管理 (C2)。相反，一项任务可能在单个窗口内完成，同时仍然需要许多相互依赖的决策，这些决策的方向必须通过与环境的交互来不断测试和修改。因此，长期运行的工程随着时间的推移优化了持久性、吞吐量和成本，而 长时程（Long-Horizon） 机构关注智能体是否可以通过持续交互来维持推理，并且在任务需要时维护跨上下文进行推理所需的状态和内存[286]。因此，执行持续时间捕获时间持久性，而 长时程（Long-Horizon） 机构表示跨任务依赖结构维持耦合推理和状态管理的能力。

2.4.2 长时程（Long-Horizon） 智能体与自治智能体

自主智能体的定义是在没有用户参与的情况下它可以采取多少行动，通常形式化为从操作员到观察者的角色[155]。自治和 长时程（Long-Horizon） 智能体密切相关，但它们描述了系统的不同属性。

重叠

。两者重叠是因为 长时程（Long-Horizon） 任务使得持续的人工监督成本高昂，因此大多数 长时程智能体（Long-Horizon Agents） 很大程度上是自动化的：它们自行与外部环境和工具交互，并自主决定何时修改策略。如果智能体必须采取许多步骤，与工具和环境交互，吸收反馈并修改其轨迹，那么每一步都需要批准可能会破坏系统的有用性，即使任务本身适合 H1。与此同时，自主性提高了 长时程（Long-Horizon） 执行的安全风险，因为更多的无人值守步骤和更多的耦合副作用意味着更大的爆炸半径，因此权限门和监督仍然是核心。

区别。区别在于，自治只是系统的部分、以治理为导向的维度，而不是对其能力的衡量标准。强大的 长时程（Long-Horizon） 智能体可以在审批门后以低自治度运行，而高度自治的智能体可能仅执行简短的单窗口任务，例如单个 Web 操作 [452]。此外，长时程智能体（Long-Horizon Agents） 不需要完全自主：它们的 运行支撑框架（Harnesses） 可以引入许可门、批准检查点和升级策略，选择性地让人类参与高影响或不确定的步骤。因此，自治询问谁留在循环中，而 长时程（Long-Horizon） 机构则询问

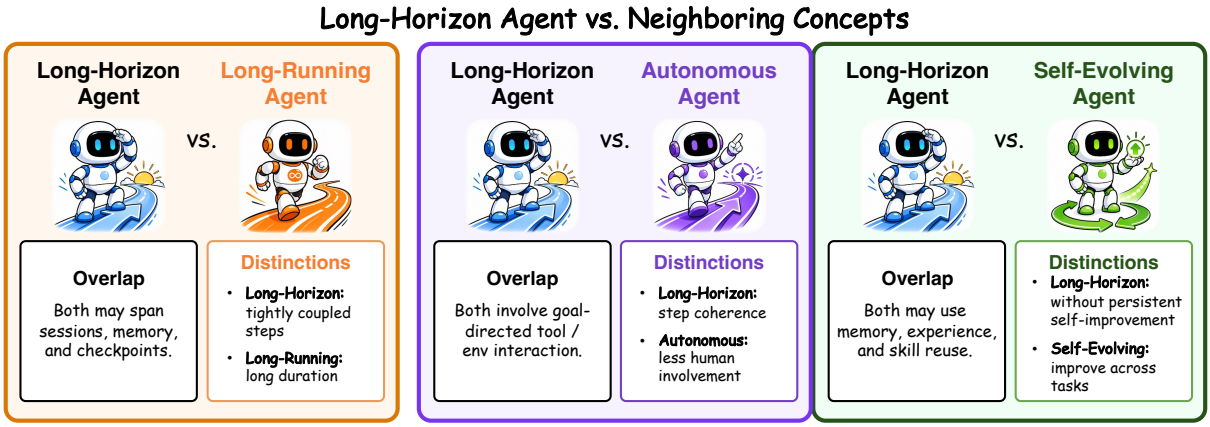


Figure 5 Long-horizon agent and three neighboring concepts. Long-running execution, autonomy, and self-evolution each emphasize a different facet; long-horizon agency asks whether the coupled model–harness system can sustain interactive reasoning (C1), maintain state and memory across contexts when needed (C2), and accumulate experience across tasks when deployment demands it (C3).

whether the coupled system can sustain the reasoning, state management, and, at H3, experience accumulation that the task horizon demands. In short, autonomy controls the degree of human involvement, whereas long-horizon agency measures whether execution can be carried through the required horizon.

2.4.3 Long-Horizon Agent vs. Self-Evolving Agent

A self-evolving agent is defined by experience-driven, *persistent* change to its own parameters, memory, tools, or topology, with the goal of improving future performance. It is the closest neighbor to long-horizon agency because both concern competence beyond a single model call, but they still differ in scope.

Overlap. At the cross-task (H3) scale the two overlap strongly. A long-horizon agent that switches among complex tasks, accumulates reusable skills, consolidates memory, and reuses experience across episodes begins to approach self-evolution. Both also fit the "era of experience" view that agent competence should grow from interaction rather than only from a frozen pretraining corpus [580]. In this paper, self-evolution is one internalized mechanism for realizing C3, namely persistent cross-task experience accumulation.

Distinctions. The distinction is whether experience persistently changes the agent for the future. Self-evolution requires such persistent change and is therefore only one facet of the C3 capability, cross-task experience accumulation, rather than the whole of long-horizon agency, which also rests on core capabilities such as interactive reasoning, tool use, and long-context reasoning. Consequently, many long-horizon agents operating at H1 or H2 complete difficult tasks with a fixed policy and temporary state, finishing a trajectory without updating their weights, tools, or lasting memory. Such agents lack any self-evolution capability, yet they remain long-horizon agents.

Self-evolution therefore concerns persistent improvement *between* tasks, while long-horizon agency additionally concerns reliably carrying each task *through* the extended interaction, state management, tool use, and iterative reasoning that its dependency horizon demands.

Overall, this section has laid the conceptual foundation on which the rest of the paper builds. We defined a long-horizon agent as a coupled model–harness system, $\pi_\theta \oplus \mathcal{H}$, rather than as the foundation model alone (Section 2.1); organized its difficulty and matching capabilities into three nested levels, H1–H3 and C1–C3 (Section 2.2); quantified the empirical growth of a frontier-agent time-horizon proxy and compared alternative fitting windows (Section 2.3); and separated long-horizon agency from long-running execution, autonomy, and self-evolution (Section 2.4). The methodological takeaway is that long-horizon agency is a high-level, system-level capability that integrates these adjacent properties rather than reducing to any one of them. An agent may exhibit sustained running, autonomy, and self-evolution and yet fail an H1 task when it cannot

原文第 14 页 | 中文翻译初稿

长时程（Long-Horizon） 智能体与邻近概念
长时程（Long-Horizon） 长期运行 长时程（Long-Horizon） 自治 长时程（Long-Horizon） 自我演化
智能体 智能体 智能体 智能体 智能体
vs 智能体vs. 与
区别 区别 区别
重叠 • 长时程（Long-Horizon）：重叠 • 长时程（Long-Horizon）：重叠 • 长时程（Long-Horizon）：
两者都可能紧密耦合 两者都涉及目标步骤一致性 两者都可以在没有持久 自我改进（Self-Improvement）
会话、内存和检查站。 • 长时间运行的步骤：环境交互。定向工具/ • 自主：较少的人类记忆、经验和技能
重用。 • 自我进化：
长期参与可以提高跨任务

图 5 长时程（Long-Horizon） 智能体和三个相邻概念。长期执行、自主性和自我进化各自强调不同的方面；长时程（Long-Horizon） 机构询问耦合模型 - 运行支撑框架（Harness） 系统是否可以维持交互式推理（C1），在需要时跨上下文维护状态和内存（C2），并在部署需要时积累跨任务经验（C3）。

耦合系统是否能够维持任务范围所需的推理、状态管理以及 H3 的经验积累。简而言之，自治控制着人类参与的程度，而长时程（Long-Horizon）机构衡量的是执行是否可以在所需的范围内进行。
2.4.3 长时程（Long-Horizon） 智能体与自我进化智能体

自我进化智能体是通过经验驱动的、对其自身参数、内存、工具或拓扑的_持久_更改来定义的，其目标是提高未来的性能

。它是与长期机构最接近的邻居，因为两者都关注超出单一模型调用的能力，但它们的范围仍然不同。

重叠。在跨任务（H3）尺度上，两者有很强的重叠

术语保留：长时程（Long-Horizon）

。一个 长时程（Long-Horizon） 智能体可以在复杂的任务之间切换，积累可重用的技能，巩固记忆，并在不同的场景中重用经验，开始接近自我进化。两者都符合“经验时代”的观点，即智能体能力应该从交互中增长，而不仅仅是从冻结的预训练语料库中增长[580]

。本文中，自我进化是实现C3的一种内在机制，即持久的跨任务经验积累。

区别。区别在于经验是否会持续改变未来的智能体人。自我进化需要这种持续的变化，因此只是C3能力、跨任务经验积累的一个方面，而不是长时程（Long-Horizon）机构的全部，而长时程（Long-Horizon）机构也依赖于交互推理、工具使用和长上下文推理等核心能力。因此，许多在 H1 或 H2 运行的 长时程智能体（Long-Horizon Agents） 以固定的策略和临时状态完成困难的任务，在不更新其权重、工具或持久记忆的情况下完成轨迹

。此类智能体缺乏任何自我进化能力，但它们仍然是 长时程智能体（Long-Horizon Agents）。

因此，自我进化关注任务之间的持续改进，而 长时程（Long-Horizon） 机构还关注通过其依赖范围所需的扩展交互、状态管理、工具使用和迭代推理来可靠地执行每项任务。

总的来说，本节奠定了本文其余部分的概念基础。我们将 长时程（Long-Horizon） 智能体定义为耦合模型 - 运行支撑框架（Harness） 系统 $\pi_\theta \oplus H$ ，而不是单独的 基础模型（Foundation Model）（第 2.1 节）

；将其难度和匹配能力分为三个嵌套级别：H1-H3 和 C1-C3（第 2.2 节）；量化了前沿智能体时间范围智能体的经验增长并比较了替代拟合窗口（第 2.3 节）；并将 长时程（Long-Horizon） 智能体与长期运行的执行、自治和自我演化分开（第 2.4 节）。方法论要点是，长时程（Long-Horizon） 机构是一种高级别的系统级功能，它集成了这些相邻属性，而不是简化为其中任何一个。智能体可能会表现出持续运行、自主性和自我进化，但当它无法完成 H1 任务时就会失败

14

Table 1 Three stages in the co-evolution of long-horizon agents. Each stage widens the *control unit*, opens a new *control surface*, and advances a set of *key technical routes*. The graded *long-horizon profile* on the right scores each stage against the three accumulating capabilities defined in Section 2.2, namely C1 intra-context interactive reasoning, C2 cross-context state and memory, and C3 cross-task experience accumulation (● full, ◗ partial, ○ absent).

Stage (era)	Control unit & surface	Key technical routes	Long-horizon profile		
			C1	C2	C3
I. Prompt engineering (2020–2023)	Single query *language of the prompt*	• Chain-of-thought reasoning • Decomposition, planning, and search • Prompt optimization and internalization	◗	○	○
II. Context engineering (2023–2025)	Conditioned call *information in the context*	• Retrieval-augmented generation • Tool use and function calling • Long context, memory, and context engineering	◗	◗	○
III. Runtime harnesses (2025–Present)	Whole trajectory *runtime around the model*	• Agent loops (reason–act–observe) • Harness engineering • Internalizing the loop	●	●	◗

sustain interactive reasoning through environment feedback, or fail an H2–H3 task when it lacks the state management or experience accumulation that those horizons require. Competence therefore shifts from the model alone to the coupled system $\pi_\theta \oplus \mathcal{H}$ and its training, and evaluation must track task completion, robustness over interaction, cross-context recovery, cross-task transfer, cost, and safety rather than static per-call accuracy. Having framed long-horizon agency as a coupled model–harness property organized along H1–H3 and C1–C3, we now turn to a historical question, namely how the field progressively moved capability from prompts, to context, to persistent runtime systems. Section 3 traces that progression.

3 Evolution: From Prompting to Runtime

As stated in the previous sections, we define a long-horizon agent as the joint product of *internalized model optimization* and *externalized harness engineering*, and reading the field through the lens of evolution shows why this coupling is not a recent design choice but a recurring necessity. Across the history of LLMs, the base model and the engineering structure around it have advanced together, each repeatedly compensating for the other's limits. A base model is a powerful but fragile next-step predictor that, on its own, does not reliably decompose a goal, ground its steps in the world, sustain reasoning through environment interaction, or maintain the cross-context state that longer horizons require (Sections 1 and 2.2), and supplying this missing structure has always been the task of the surrounding engineering. Following this co-evolutionary thread, this section traces the development of long-horizon agents along its two driving forces, model optimization and harness engineering, and organizes the representative work that defines each stage.

As shown in Figure 6, we trace this co-evolution through three nested, chronologically and thematically coherent stages, each opened by an expansion in the base model's own capability, organized around a new control surface exposed by the surrounding harness, and closed by the long-horizon limit that forces the next stage. **(1) Prompt Engineering** (2020–2023) steers behavior through the *language* of a single query, first eliciting multi-step reasoning from a fixed model; **(2) Context Engineering** (2023–2025) extends control to the *information* supplied to the model, grounding the agent in retrieved evidence, tools, and memory; and **(3) Runtime Harnesses** (2025–present) make the unit of control the whole *trajectory*, so that the long-horizon agent proper emerges as a model coupled to a runtime that sustains one goal over many dependent steps. The three are not successive replacements but nested layers, in which each new control surface absorbs the previous one rather than discarding it. Throughout this chapter, the years attached to each stage denote the period in which its control surface became the *dominant* paradigm, not the publication date of any individual work; a stage may therefore rest on foundational methods that predate it and preview ideas that mature later. Table 1 summarizes the three stages along a common set of axes.

原文第 15 页 | 中文翻译初稿

表1 长时程智能体（Long-Horizon Agents）协同进化的三个阶段。每个阶段都拓宽了_控制单元_，开辟了新的_控制面_，并推进了一套_关键技术路线_。右侧的分级_长时程（Long-Horizon） profile_根据2.2节中定义的三种累积能力对每个阶段进行评分，即C1上下文内交互推理、C2跨上下文状态和记忆以及C3跨任务经验积累（●全部，◗部分，_○_缺席）。

舞台（时代）	控制单元及表面	关键技术路线	长时程（Long-Horizon）简介
			C1
C2	C3
我。即时工程
（2020-2023）
II。情境工程
(2023-2025)	单一查询
提示语言
条件调用
情境中的信息	•思路链推理
•分解、规划和搜索
•提示优化和内化
•检索增强生成
•工具使用和函数调用
•长情境、记忆和背景
工程	◗
○
◗
◗	○
○
三．运行时 运行支撑框架（Harnesses）
（2025 年至今）	整个轨迹
模型周围的运行时	• 智能体循环（原因-行动-观察）
• 运行支撑框架（Harness） 工程
• 内部化循环	●
●	◗

通过环境反馈维持交互式推理，或者在缺乏这些视野所需的状态管理或经验积累时失败 H2-H3 任务。因此，能力从单独的模型转移到耦合系统 πθ ⊕H 及其训练和评估，必须跟踪任务完成情况、交互鲁棒性、跨上下文恢复、跨任务转移、成本和安全性，而不是静态的每次调用准确性。将 长时程（Long-Horizon） 机构构建为耦合模型 - 运行支撑框架（Harness） 属性，沿着 H1-H3 和 C1-C3 组织，现在我们转向一个历史问题，即该领域如何逐步将功能从提示转移到上下文，再转移到持久运行时系统。第 3 节追踪了这一进展。

3 演变：从提示到运行时

如前几节所述，我们将 长时程（Long-Horizon） 智能体定义为_内部化模型优化_和_外部化 运行支撑框架（Harness） 工程_的联合产品，并且通过进化的视角来解读该领域，这表明为什么这种耦合不是最近的设计选择，而是反复出现的必要性。纵观法学硕士的历史，基础模型和围绕它的工程结构一起进步，每一个都反复弥补另一个的局限性。基础模型是一个强大但脆弱的下一步预测器，它本身不能可靠地分解目标、在世界中奠定其步骤、通过环境交互维持推理或维持较长视野所需的跨上下文状态（第 1 节和 2.2 节），而提供这种缺失的结构始终是周围工程的任务。沿着这个共同进化的线索，本节沿着模型优化和 运行支撑框架（Harness） 工程这两个驱动力追溯了 长时程智能体（Long-Horizon Agents） 的发展，并组织了定义每个阶段的代表性工作。

如图 6 所示，我们通过三个嵌套的、按时间顺序和主题连贯的阶段来追踪这种协同进化，每个阶段都是通过基础模型自身能力的扩展而打开的，围绕着由周围的 运行支撑框架（Harness） 暴露的新控制面组织起来，并由强制下一阶段的 长时程（Long-Horizon） 限制来关闭。 (1) 即时工程（2020-2023）通过单个查询的_语言_来引导行为，首先从固定模型中引出多步骤推理； (2) 情境工程（2023-2025）将控制扩展到提供给模型的_信息_，落地关联（Grounding） 检索证据、工具和内存中的智能体； (3) 运行时 运行支撑框架（Harnesses）（2025 年至今）使控制单元成为整个_轨迹_，因此 长时程（Long-Horizon） 智能体本身作为一个模型耦合到运行时，在许多相关步骤中维持一个目标。这三个控制面并不是连续的替换，而是嵌套层，其中每个新的控制面都会吸收前一个控制面，而不是丢弃它。在本章中，每个阶段所附的年份表示其控制面成为_主导_范式的时期，而不是任何单独作品的出版日期；因此，一个阶段可能依赖于它之前的基础方法，并预览后来成熟的想法。表 1 总结了沿着一组共同轴的三个阶段。

15

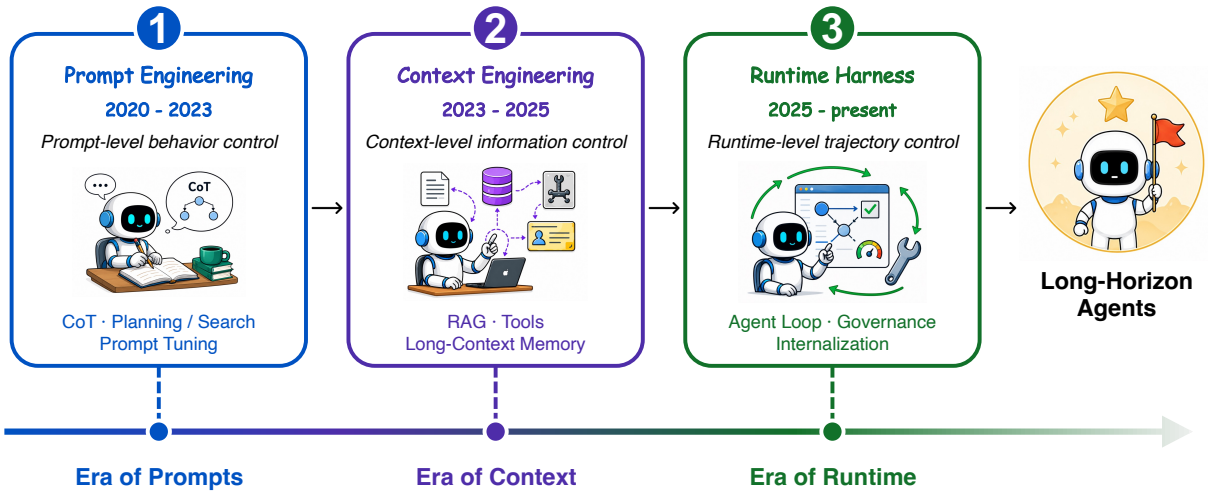


Figure 6 The co-evolution of long-horizon agents across three stages. Control widens from the *language* of a single prompt (2020–2023), to the *information* supplied to the model before it acts, to the whole *trajectory* sustained by a runtime harness (2025–present); each stage absorbs the previous rather than replacing it.

3.1 Stage I: Prompt Engineering (2020–2023)

The prompting stage is the prototype of long-horizon agency, in which the model carries most task capability in its weights and the only surface a builder can touch is the language of the query. The stage opens with *prompt learning*, made widely visible by GPT-3 [40] and consolidated by dedicated papers on prompting and in-context learning [132, 379], which let a pre-trained model perform a new task from a handful of in-prompt examples with no weight update and thereby moved the focus of system building from training the model to shaping its input at inference time. Its central discovery is that the right prompt can *elicit* latent abilities the base model does not show by default, including the embryonic forms of planning and execution that long-horizon tasks demand. Concretely, the prompting stage surfaces this latent competence along three representative threads, which we take up in turn: chain-of-thought reasoning, decomposition and planning, and the optimization and internalization of prompts themselves.

Chain-of-Thought Reasoning. The first thread showed that a fixed model reasons far better when prompted to externalize intermediate steps rather than answer directly. Initially, the scratchpad [450] emitted intermediate computation for multi-step problems; chain-of-thought (CoT) prompting [675] then turned this into a few-shot recipe that elicits a reasoning trace before the answer; and zero-shot CoT [281] further showed that the same behavior could be triggered by a single instruction, confirming that the capacity was latent in the weights and merely needed surfacing. Building on CoT, a family of variants quickly followed that automate demonstration construction, scale reasoning effort with problem complexity, and modularize the prompt into reusable sub-steps [160, 275, 851]. The shared lesson, that the same model exhibits qualitatively different competence depending on how its input is phrased, recurs throughout the paper.

Decomposition, Planning, and Search. A second thread widened the single chain into the structures that prefigure an agent, using prompting not merely to reason but to make a fixed model *decompose* a goal, *plan* a course of action, and *execute* it step by step, which is the earliest recognizable form of an agent. Self-consistency [659] samples many chains and aggregates them by majority vote, trading one fragile path for a more reliable distribution; least-to-most prompting [877] made decomposition explicit by solving ordered subproblems; and Plan-and-Solve prompting [639] split an explicit planning step from execution, producing the first clean "plan then act" template that later agent loops inherit. A parallel line grounded execution in code, with PAL [168] and Program of Thoughts [70] offloading computation to an interpreter rather than the model's own arithmetic, while Tree-of-Thoughts [760] recast reasoning as deliberate search over a tree of partial solutions with look-ahead and backtracking. A closely related line then pushed the

原文第 16 页 | 中文翻译初稿

1 2 3
提示工程上下文工程运行时 运行支撑框架（Harness）
2020 - 2023 2023 - 2025 2025 - 现在
提示级行为控制 上下文级信息控制 运行时级轨迹控制
长时程（Long-Horizon）
CoT · 规划/搜索 RAG · 工具 智能体循环 ·治理智能体
提示调优长上下文内存内化
提示时代上下文时代运行时

图6 长时程智能体（Long-Horizon Agents）跨三个阶段的协同进化。控制范围从单个提示的_语言_（2020-2023），到在模型起作用之前提供给模型的_信息_，再到运行时运行支撑框架（Harness）（2025年至今）维持的整个_轨迹_；每个阶段都会吸收前一个阶段，而不是取代它。
3.1 第一阶段：快速工程（2020-2023）

提示阶段是长时程（Long-Horizon）机构的原型，其中模型在其权重中承载了大部分任务能力，而构建者唯一可以接触的表面是查询的语言。该阶段以_提示学习_开始，通过 GPT-3 [40] 使其广泛可见，并通过有关提示和上下文学习 [132, 379] 的专门论文进行巩固，这让预训练的模型可以在没有权重更新的情况下从少数即时示例中执行新任务，从而将系统构建的重点从训练模型转移到在推理时塑造其输入。它的核心发现是，正确的提示可以引出基本模型默认情况下未显示的潜在能力，包括 长时程（Long-Horizon） 任务所需的规划和执行的雏形。具体来说，提示阶段沿着三个有代表性的线索来展现这种潜在能力，我们依次进行：思想链推理、分解和规划以及提示本身的优化和内化。

思维链推理。第一个线程表明，当提示将中间步骤具体化而不是直接回答时，固定模型的推理效果要好得多。最初，暂存器 [450] 为多步骤问题发出中间计算；思想链（CoT）提示[675]然后将其转化为一个在答案之前引出推理轨迹的几次配方；零样本 CoT [281] 进一步表明，相同的行为可以由单个指令触发，证实容量隐藏在权重中，只需要表面处理。在 CoT 的基础上，很快出现了一系列变体，它们可以自动化演示构建，根据问题复杂性扩展推理工作，并将提示模块化为可重用的子步骤 [160,275,851]。共同的教训是，同一模型根据其输入的措辞方式表现出不同质量的能力，这在整篇论文中反复出现。

分解、规划和搜索。第二个线程将单链扩展为预示智能体的结构，不仅使用提示进行推理，而且使固定模型_分解_目标，_计划_行动过程，并逐步_执行_它，这是最早可识别的智能体形式。自我一致性[659]对许多链进行采样，并通过多数投票聚合它们，用一条脆弱的路径换取更可靠的分布

;从最少到最多的提示[877]通过解决有序子问题使分解变得明确；计划和解决提示[639]将显式计划步骤从执行中分离出来，生成第一个干净的“计划然后行动”模板，稍后智能体循环继承该模板。并行线基于代码执行，PAL [168] 和 Program of Thoughts [70] 将计算卸载给解释器而不是模型自己的算术，而 Tree-of-Thoughts [760] 将推理重新定义为通过前瞻和回溯对部分解决方案树进行有意搜索。随后，一条密切相关的线路推动了

16

same plan-then-act template into *interactive* environments, where SayCan and Inner Monologue grounded language plans in affordances and environment feedback [236, 243], and AdaPlanner, ADaPT, and DEPS closed the loop with adaptive re-planning from execution feedback in embodied and text worlds such as ALFWorld [482, 545, 569, 673]. Read together, this line forms the embryonic agent skeleton, which decomposes a goal, proposes and executes steps, evaluates progress, and revises. Prompt learning alone cannot yet yield a long-horizon agent, since all of this unfolds without a persistent runtime harness, durable state, or robust recovery. What the stage does provide is a soft external constraint on model behavior and an early exploration of the harness-mediated control that later stages systematize, together with the planning-and-execution skeleton here and the elicitation-then-internalization dynamic below.

Prompt Optimization and Internalization. A third thread responded to a property this stage exposed and the rest of the paper inherits, namely that elicited capability is *fragile*, swinging with wording, example order, example choice, or output format [403, 433, 862], which made prompt-based systems hard to reproduce, test, and compose. The community answered by making prompt construction systematic rather than handcrafted. One line recast prompt design as an explicit search or optimization problem [483, 889], while another treated prompts as composable, program-like artifacts [514]. In parallel, what began as a prompting trick started to be *internalized*, as instruction tuning together with reinforcement learning from human feedback let models follow intent without elaborate templates, most visibly in InstructGPT [460], and the same dynamic would later fold full chain-of-thought traces into the weights, which we revisit in Section 5. This is the first concrete instance of the harness→policy migration the paper tracks across all three stages.

Although prompt engineering by itself does not constitute a long-horizon agent, its core idea, shaping a model's reasoning behavior through external constraints rather than weight updates, remains a guiding principle for long-horizon agents to this day. Yet prompting controls only *language* over a fixed knowledge base. A single call can use only the text in the prompt and the knowledge already stored in the parameters [266, 479, 516], so no phrasing can supply information the model never saw. As applications moved toward current events, private data, large repositories, and changing APIs, the binding question shifted from how the instruction is worded to what the model is allowed to see, pushing engineering one layer outward, from the language of the prompt to the information assembled into it.

3.2 Stage II: Context Engineering (2023–2025)

Where Stage I engineered the model's *behavior* through the language of the query, Stage II turns to its *information*, that is, the evidence, tools, and history that a single call is allowed to condition on. Broadening this input, rather than rewording the instruction, becomes the primary means of extending the model's capability boundary. The representative directions of this stage are retrieval-augmented generation, tool use and function calling, and context management and memory, which we detail in turn below.

Retrieval-Augmented Generation. Retrieval-augmented generation (RAG) loosened the boundary between the model and the world by attaching external evidence to the prompt at inference time [304], building on a line of retrieval- and memorization-augmented models that coupled parametric knowledge with an external corpus [38, 200, 246, 247, 270, 273] and organized systematically in dedicated RAG reviews. Its core result is that grounding generation in retrieved text reduces hallucination and improves factuality on knowledge-intensive tasks. Initially a single retrieve-then-read pass, the technique soon evolved into a richer family. HyDE-style hypothetical-document embeddings first improved recall [167], and hierarchical indexing summarized corpora into multi-resolution trees [522]. Retrieval was then interleaved with reasoning, through multi-hop methods that chain retrieval across steps [612] and active retrieval that lets the model decide *when* to look something up [258]. More recently, self-reflective retrieval further lets the model critique and re-retrieve its own evidence [22].

Tool Use and Function Calling. Where retrieval lets a model *read* the world, tool use lets it *query and act on* systems beyond text. Early work established the mechanics. Toolformer [523] showed that a model could teach itself, in a self-supervised way, when and how to call APIs; ToolAlpaca [585] and ToolLLM [492], with the StableToolBench testbed [198], then scaled tool use to thousands of real-world APIs; Gorilla [471] grounded API selection in documentation to curb hallucinated calls; and WebGPT [445] grounded long-form

原文第 17 页 | 中文翻译初稿

将相同的先计划后行动模板引入到交互环境中，其中 SayCan 和 Inner Monologue 在 可供性（Affordances） 和环境反馈中建立了语言计划 [236, 243]，而 AdaPlanner、ADaPT 和 DEPS 通过根据 ALFWorld 等具体世界和文本世界中的执行反馈进行自适应重新规划来闭合循环 [482, 545, 569, 673]。一起阅读，这条线形成了智能体骨架的雏形，它分解目标，提出并执行步骤，评估进度并进行修改。仅靠即时学习还不能产生 长时程（Long-Horizon） 智能体，因为所有这些都是在没有持久运行时 运行支撑框架（Harness）、持久状态或强大恢复的情况下展开的。该阶段提供的是对模型行为的软外部约束，以及对后期阶段系统化的 运行支撑框架（Harness） 介导控制的早期探索，以及此处的规划和执行框架以及下面的引发然后内化动态。

及时优化和内部化。第三个线程响应了本阶段公开的属性，并且论文的其余部分继承了这一属性，即引出的能力是_脆弱_，随措辞、示例顺序、示例选择或输出格式的变化而变化[403,433,862]，这使得基于提示的系统难以重现、测试和组合。社区的回应是使快速施工系统化，而不是手工制作。其中一行将提示设计重新定义为显式搜索或优化问题 [483, 889]，而另一行将提示视为可组合的、类似程序的工件 [514]。与此同时，最初作为提示技巧的东西开始被_内部化_，因为指令与来自人类反馈的 强化学习（Reinforcement Learning） 一起调整，让模型在没有复杂模板的情况下遵循意图，最明显的是 InstructGPT [460]，并且相同的动态随后将完整的思想链痕迹折叠到权重中，我们在第 5 节中重新讨论。这是 运行支撑框架（Harness） 的第一个具体实例_→_ 本文跟踪了所有三个阶段的策略迁移。

尽管即时工程本身并不构成 长时程（Long-Horizon） 智能体，但其核心思想（通过外部约束而不是权重更新来塑造模型的推理行为）至今仍然是 长时程智能体（Long-Horizon Agents） 的指导原则。然而，提示仅控制固定知识库的_语言_。单个调用只能使用提示中的文本和参数中已存储的知识 [266, 479, 516]，因此任何措辞都无法提供模型从未见过的信息。随着应用程序转向时事、私有数据、大型存储库和不断变化的 API，具有约束力的问题从指令的措辞方式转变为允许模型查看的内容，从而将工程向外推一层，从提示语言到组装到其中的信息。

3.2 第二阶段：情境工程（2023-2025）

第一阶段通过查询语言设计模型的_行为_，而第二阶段则转向其_信息_，即允许单个调用作为条件的证据、工具和历史记录。扩大这种输入，而不是重新措辞指令，成为扩展模型能力边界的主要手段

。这一阶段的代表方向是检索增强生成、工具使用和函数调用、上下文管理和记忆，下面我们依次详细介绍。

检索增强生成。检索增强生成（RAG）通过在推理时将外部证据附加到提示上来放松模型与世界之间的界限[304]，建立在将参数知识与外部语料库相结合的检索和记忆增强模型的基础上 [38,200,246,247,270,273]，并在专门的 RAG 评论中系统地组织起来。其核心结果是检索文本中的 落地关联（Grounding） 生成减少了幻觉并提高了知识密集型任务的真实性。最初是一次检索然后读取的过程，该技术很快就发展成为一个更丰富的系列。 HyDE 风格的假设文档嵌入首先提高了召回率 [167]，分层索引将语料库总结为多分辨率树 [522]。然后，检索与推理交织在一起，通过跨步骤链接检索的多跳方法 [612]和主动检索，让模型决定何时查找某些内容[258]。最近，自我反思检索进一步让模型批判并重新检索自己的证据[22]。

工具使用和函数调用。检索让模型“读取”世界，而工具的使用则让模型“查询并作用于”文本以外的系统。早期的工作建立了机制。 Toolformer [523] 表明模型可以以自我监督的方式自学何时以及如何调用 API；ToolAlpaca [585] 和 ToolLLM [492]，使用 StableToolBench 测试平台 [198]，然后将工具使用扩展到数千

个真实世界的 API； Gorilla [471] 在文档中添加了 API 选择，以抑制幻觉调用；和 WebGPT [445] 接地长格式

17 号

answering in live browsing. As the practice consolidated, function calling and structured outputs standardized the interface so that model decisions could be parsed and dispatched reliably [385, 472], a maturation charted by dedicated tool-learning papers [493, 497]. The decisive bridge is ReAct [761], which interleaves reasoning with tool calls so that each observation feeds the next thought, while HuggingGPT [537] further showed how an LLM could route tasks across specialized models in a tool ecosystem. Both sit at the boundary of two stages, turning one-shot tool use into the beginnings of the loop that Stage III makes central.

Long Context, Memory, and Context Engineering. The third development enlarged the container itself. Efficient-attention kernels such as FlashAttention [108] improved the practicality of long-sequence serving, while sparse-attention architectures such as Longformer [32] and BigBird [792] explored algorithmic routes to longer inputs; later frontier systems exposed hundred-thousand- to million-token windows [593], and state-space architectures offered near-linear alternatives to attention [186]. This made it physically possible to condition on whole documents, codebases, and histories rather than summarizing them away, which shifted the question from whether the model can reach information to whether it can use the information well once present. LongBench [28] was among the first benchmarks to quantify how poorly models handle genuinely long-context tasks, a gap that dedicated long-context suites continue to make precise [7, 220, 833]. Because no window is unbounded, a complementary line externalized state into persistent memory, with MemGPT treating context as a paging hierarchy over external storage [464] and generative agents maintaining a long-running memory stream with retrieval and reflection over remembered experience [470]. This abundance, however, created its own problem, since more context is not better context. Long inputs are read unevenly, with material in the middle systematically ignored [378]; irrelevant evidence actively distracts reasoning [541]; and utilization past a threshold degrades sharply, a failure measured as *context rot* [511]. The methodological response, named by practitioners *context engineering*, is the systematic selection, organization, compression, and refresh of the model's input under hard token budgets [13, 425], with a toolkit spanning prompt compression [254, 351, 773], working-memory mechanisms [224, 268, 778], and rolling summarization for long runs [693, 890]. More recently, this curation is itself becoming agentic, as *agentic context engineering* lets the agent treat its own context as an evolving object that it revises and improves over successive steps [824]. Together, retrieval, tools, longer windows, memory, and context engineering shifted the relevant capacity from how much knowledge sits inside the model to how effectively the surrounding structure can selectively surface, organize, and use relevant information at each step; here too capability was partly internalized, as retrieval, tool use, and long-context handling increasingly became native, trainable model skills rather than bolted-on calls.

For all its reach, context engineering keeps the structural assumption it inherited from prompting, in that it optimizes the input to a *single* model call. The unit is wider, but it remains a one-shot, open-loop conditioning step that shapes what the model sees *before* it acts, not what happens after. Once deployments began chaining many such calls into consequential actions on tools, files, browsers, and operating systems, the dominant failures became properties of the *trajectory*, as errors compounded across steps, belief state drifted from the world, and side effects became irreversible. Closing that loop requires control over the runtime itself, which is the surface Stage III opens.

3.3 Stage III: Runtime Harnesses (2025–Present)

In Stage III, capability expansion manifests as a shift from answering questions to acting in environments, and the long-horizon agent takes its recognizable form. This also marks the transition of agents from chatbots toward genuinely productive settings, in which they carry out consequential work rather than merely converse. The transition was made concrete by executable environments and benchmarks such as WebShop [759], WebArena [885], SWE-bench [260], OSWorld [716], and Terminal-Bench [428], which score completed actions in web, code, desktop, and terminal settings rather than static answers, and thereby supply the trajectory-level feedback long-horizon agents are optimized against. The unit of engineering is no longer a prompt or a single conditioned call but a *trajectory*, comprising repeated cycles of reasoning, action, observation, verification, and recovery. We summarize the stage in three movements, namely the loop that makes the agent act, the harness that governs the loop, and the internalization that absorbs harness capabilities back into the model.

Agent Loops. The core engineering unit is a *trajectory* that unfolds repeatedly in an environment, in which the agent reasons, acts, reads feedback, and decides what to do next, following the widely adopted

原文第 18 页 | 中文翻译初稿

在实时浏览中回答。随着实践的巩固，函数调用和结构化输出标准化了接口，以便可以可靠地解析和分派模型决策 [385, 472]，这是专用工具学习论文 [493, 497] 所描绘的成熟过程。决定性的桥梁是 ReAct [761]，它将推理与工具调用交织在一起，以便每次观察都能满足下一个想法，而 HuggingGPT [537] 进一步展示了法学硕士如何在工具生态系统中的专门模型之间路由任务。两者都位于两个阶段的边界，将一次性工具的使用变成了第三阶段的核心循环的开始。

长上下文、记忆和上下文工程。第三项发展扩大了容器本身。 FlashAttention [108] 等高效注意力内核提高了长序列服务的实用性，而 Longformer [32] 和 BigBird [792] 等稀疏注意力架构则探索了通往更长输入的算法路线；后来的前沿系统暴露了数十万到数百万个令牌窗口[593]，状态空间架构提供了接近线性的注意力替代方案[186]。这使得在物理上可以以整个文档、代码库和历史为条件，而不是对它们进行总结，这将问题从模型是否可以获取信息转移到是否可以很好地使用一旦出现的信息。 LongBench [28] 是最早量化模型处理真正长上下文任务的能力的基准之一，专用的长上下文套件继续缩小这一差距[7,220,833]。因为没有窗口是无限的，所以互补的行外部化状态进入持久内存，MemGPT 将上下文视为外部存储上的分页层次结构 [464]，并且生成智能体通过对记忆的经验进行检索和反射来维护长期运行的内存流 [470]。然而，这种丰富性也产生了自己的问题，因为更多的背景并不意味着更好的背景。长输入的读取不均匀，中间的材料被系统地忽略[378]；不相关的证据会严重干扰推理[541]；超过阈值后利用率急剧下降，失败被衡量为_context rot_ [511]。由实践者_context Engineering_命名的方法响应是在硬代币预算下对模型输入进行系统选择、组织、压缩和刷新[13, 425]，并使用涵盖即时压缩[254, 351, 773]、工作记忆机制[224, 268, 778]和长期运行滚动总结[693, 890]的工具包。最近，这种管理本身正在变得智能体化，因为“智能体上下文工程”让智能体将自己的上下文视为一个不断发展的对象，并通过连续的步骤进行修改和改进[824]。检索、工具、更长的窗口、内存和上下文工程一起将相关能力从模型内部有多少知识转移到周围结构如何有效地在每个步骤中选择性地呈现、组织和使用相关信息；在这里，能力也被部分内化，因为检索、工具使用和长上下文处理逐渐成为本地的、可训练的模型技能，而不是附加的调用。

就其影响范围而言，上下文工程保留了它从提示中继承的结构假设，因为它优化了单个模型调用的输入。该单元更宽，但它仍然是一次性的开环调节步骤，它塑造模型在其行动之前看到的内容，而不是之后发生的情况。一旦部署开始将许多此类调用链接到工具、文件、浏览器和操作系统上的后续操作，主要的故障就成为_轨迹_的属性，因为错误会跨步骤复合，信念状态会偏离世界，副作用变得不可逆转。关闭该循环需要控制运行时本身，这是第三阶段打开的表面。

3.3 第三阶段：运行时 运行支撑框架（Harnesses）（2025 年至今）

在第三阶段，能力扩展表现为从回答问题到在环境中行动的转变，长时程（Long-Horizon）智能体采取了可识别的形式。这也标志着智能体从聊天机器人向真正高效的环境的转变，在这种环境中，他们执行重要的工作而不仅仅是交谈。这种转变是通过可执行环境和基准测试实现的，例如 WebShop [759]、WebArena [885]、SWE-bench [260]、OSWorld [716] 和 Terminal-Bench [428]，它们对 Web、代码、桌面和终端设置中完成的操作而不是静态答案进行评分，从而提供针对 长时程智能体（Long-Horizon Agents）进行优化的轨迹级反馈。工程单元不再是一个提示或单个条件调用，而是一个_轨迹_，包括推理、行动、观察、验证和恢复的重复循环。我们将这个阶段总结为三个动作，即使智能体行动的循环、控制循环的 运行支撑框架（Harness）以及将 运行支撑框架（Harness） 功能吸收回模型的内化。

智能体循环。核心工程单元是一个在环境中反复展开的“轨迹”，在该环境中，智能体根据广泛采用的规则进行推理、行动、读取反馈并决定下一步该做什么。

18

thought–action–observation cycle. Early work such as ReAct, Reflexion, and Self-Refine established the basic pattern of interleaving reasoning with tool use and verbal reflection after failure [416, 544, 761], while language-agent tree search wrapped the loop in value-guided search with backtracking [875]; around the same time, open-source autonomous agents such as AutoGPT and BabyAGI popularized the self-directed loop for a broad audience [183, 444]. Since 2025, the loop has tightened along three axes rather than inventing a new paradigm. First, *action representation* became more executable and modular, as CodeAct unified step-level actions as code [656] and ReWOO separated planning from tool execution [720]. Second, *reliability* moved inward to the loop itself, as ReflAct grounds decisions in goal-state reflection [276] and test-and-repair loops such as CRITIC introduce tool-interactive verify-then-correct steps that catch and revise errors mid-trajectory [182], complementing the reflect-and-retry pattern of Reflexion and the value-guided search of LATS above. Third, *task form* widened from QA-style interaction to open-ended, multi-tool trajectories in benchmarks such as GAIA and AssistantBench [431, 774], to vision–language control in SeeAct [864], and to multi-stage research pipelines that interleave planning, retrieval, and writing [258, 613, 871]. Alongside these free-form loops, a complementary line re-emphasizes structured *agent workflows*, in which recurring procedures are captured as reusable graphs or memories and replayed across tasks, as in Agent Workflow Memory [674] and automated workflow generation [353]. The loop has thus become a runtime primitive that sustain one goal across many coupled steps while continuously changing external state, and the remaining question is how to govern it.

Harness Engineering. Once a loop stretches over a horizon, failures become properties of the *execution process* rather than of any single input, as errors compound, goals drift, success is declared too early, state distorts as context fills, and side effects escape authorization. The engineering response must therefore enter the runtime itself. *Harness engineering* names the organization of prompts, context, tools, state, permissions, verification, and recovery into a system-level control surface. The field's first response was to make multi-step execution explicit. Communicative multi-agent efforts led by ChatDev and MetaGPT cast long-horizon software work as role-specialized message passing [219, 485], AutoGen turned that pattern into a programmable conversation runtime [692], and Magentic-One added dual ledgers that separate outer replanning from inner dispatch [157]. As trajectories grew longer and more inspectable, control migrated from ad hoc dialogue to graph-structured runtimes exemplified by LangGraph and packaged into deep-research systems such as Deep Agents and Flash-Searcher, where state can be resumed, branched, and audited [291, 293, 491]. In parallel, the interface to the outside world was standardized, as MCP unified heterogeneous tools behind a discoverable, authorizable service layer [99], while AGENTS.md and Agent Skills externalized project rules as portable harness configuration [16, 98]. Long-running deployment then forced a third concern, what an agent is allowed to do, into the foreground through hooks, middleware, pre-action gates, sandboxes, and policy layers [17, 18, 41, 463, 617]. These layers are what product systems ultimately integrate, including repository-oriented harnesses such as OpenHands and SWE-agent, whose Agent–Computer Interface showed that interface design alone can determine long-horizon success [657, 747], local personal-agent runtimes such as Claude Code and OpenClaw [10, 457, 513], and GUI-facing systems such as Operator [12, 452]. A later turn makes the harness itself an object of search and improvement, as in workflow discovery by AFlow and ADAS [225, 383, 662, 816, 897] and self-editing harness code in the Darwin Gödel Machine [814]. Reliable long-horizon agency is therefore a system property to be engineered and learned. Practitioners increasingly describe this outward shift as *loop engineering*, a move from writing one-shot prompts to designing loops that dispatch, monitor, verify, and recover agents over a trajectory.

Internalizing the Loop. Stage III is also where the paper's two halves become visible together, as behaviors first organized by the harness are increasingly absorbed into model weights through trajectory-level SFT and RL. Post-2024 work no longer pursues a single generic agent-tuning recipe; instead, internalization *tracks harness type*, because each runtime exposes a different action grammar, feedback channel, and failure mode. Search-centric harnesses are distilled by agentic-search RL such as Search-R1 [261]; repository harnesses by software-engineering RL such as SWE-RL, increasingly trained inside neural environments like SWE-World [572, 680]; terminal harnesses by systems such as RLAnything that unify shell, tool-use, and coding rollouts [663]; GUI harnesses by UI-TARS-2 [632]; and personal runtimes by Claw-R1, which routes rich OpenClaw trajectories into RL backends through step-level middleware [457, 627]. Agent-World pushes this further by coupling environment–task discovery with continual self-evolving training, so that internalization is no longer confined to fixed benchmarks [131]. Evaluation moves in the same direction,

原文第 19 页 | 中文翻译初稿

思想-行动-观察的循环。 ReAct、Reflexion 和 Self-Refine 等早期工作建立了交叉推理与工具使用和失败后言语反思的基本模式 [416, 544, 761]，而语言智能体树搜索将循环包裹在带有回溯的价值引导搜索中 [875]；大约在同一时间，AutoGPT 和 BabyAGI 等开源自主智能体向广大受众普及了自主循环 [183, 444]。自 2025 年以来，循环沿着三个轴收紧，而不是发明一个新的范式。首先，“动作表示”变得更加可执行和模块化，因为 CodeAct 将步骤级动作统一为代码 [656]，而 ReWOO 将规划与工具执行分开 [720]。其次，_可靠性_向内移动到循环本身，因为 ReflAct 在目标状态反射 [276] 中做出决策，而测试和修复循环（例如 CRITIC）引入了工具交互式验证然后纠正步骤，捕获并修正轨迹中间的错误 [182]，补充了 Reflexion 的反射和重试模式以及上面 LATS 的价值引导搜索。第三，任务形式从 QA 式交互扩展到 GAIA 和 AssistantBench 等基准测试中的开放式多工具轨迹 [431, 774]，再到 SeeAct 中的视觉语言控制 [864]，以及交叉规划、检索和写作的多阶段研究流程 [258, 613, 871]。除了这些自由形式的循环之外，补充线重新强调结构化的_智能体工作流程_，其中重复的过程被捕获为可重用的图形或内存，并在任务之间重放，如智能体工作流程内存[674]和自动工作流程生成[353]。因此，循环已成为一种运行时原语，它在多个耦合步骤中维持一个目标，同时不断改变外部状态，剩下的问题是如何管理它。

运行支撑框架（Harness）工程。一旦循环延伸到一定范围，失败就会成为“执行过程”的属性，而不是任何单个输入的属性，因为错误会复合，目标会漂移，成功宣布得太早，状态会随着上下文填充而扭曲，副作用会逃避授权。因此，工程响应必须进入运行时间本身。 运行支撑框架（Harness）Engineering 将提示、上下文、工具、状态、权限、验证和恢复的组织命名为系统级控制界面。该领域的第一个反应是明确多步骤执行。 ChatDev 和 MetaGPT 领导的通信多智能体工作将 长时程（Long-Horizon）软件作为角色专用消息传递工作 [219, 485]，AutoGen 将这种模式转变为可编程对话运行时 [692]，Magentic-One 添加了双账本，将外部重新规划与内部调度分开 [157]。随着轨迹变得更长、更易于检查，控制从临时对话迁移到以 LangGraph 为代表的图结构运行时，并打包到 Deep Agents 和 Flash-Searcher 等深度研究系统中，其中状态可以恢复、分支和审核 [291, 293, 491]。与此同时，与外界的接口也实现了标准化，因为 MCP 在可发现、可授权的服务层后面统一了异构工具 [99]，而 AGENTS.md 和 Agent Skills 将项目规则外部化为可移植的 运行支撑框架（Harness）配置 [16, 98]

。然后，长时间运行的部署迫使第三个关注点，即允许智能体做什么，通过钩子、中间件、预操作门、沙箱和策略层进入前台[17、18、41、463、617]。这些层是产品系统最终集成的内容，包括面向存储库的运行支撑框架（Harnesses），例如 OpenHands 和 SWE-agent，其智能体计算机接口表明，仅界面设计就可以决定 长时程（Long-Horizon）的成功 [657, 747]、本地个人智能体运行时，例如 Claude Code 和 OpenClaw [10, 457, 513]，以及面向 GUI 的系统，例如 Operator [12, 452]。后来的转变使 运行支撑框架（Harness）本身成为搜索和改进的对象，例如 AFlow 和 ADAS [225,383,662,816,897] 的工作流程发现以及达尔文哥德尔机 [814] 中的自编辑 运行支撑框架（Harness）代码。因此，可靠的 长时程（Long-Horizon）机构是一个需要设计和学习的系统属性。从业者越来越多地将这种向外的转变描述为“循环工程”，这是一种从编写一次性提示到设计在轨迹上调度、监控、验证和恢复智能体的循环的转变。

内部化循环。第三阶段也是论文的两半同时可见的地方，因为最初由 运行支撑框架（Harness）组织的行为越来越多地通过轨迹级 SFT 和 RL 被吸收到模型权重中。 2024 年后的工作不再追求单一的通用智能体调整方法；相反，内部化_tracks 运行支撑框架（Harness）type_，因为每个运行时都会公开不同的操作语法、反馈通道和故障模式。以搜索为中心的 运行支撑框架（Harnesses）由智能体搜索 RL 提炼而成，例如 Search-R1 [261]；由 SWE-RL 等软件工程 RL 构建的存储库 运行支撑框架（Harnesses），在 SWE-World 等神经环境中进行越来越多的训练 [572, 680]；通过 RLAnything 等统一 shell、工具使用和编

码 展开轨迹（Rollouts） [663] 的系统来终端 运行支撑框架（Harnesses）； GUI 运行支撑框架（Harnesses），由 UI-TARS-2 [632] 提供；以及 Claw-R1 的个人运行时，它通过步骤级中间件将丰富的 OpenClaw 轨迹路由到 RL 后端 [457, 627]。 Agent-World 通过将环境-任务发现与持续的自我进化训练相结合，进一步推动了这一点，使内化不再局限于固定的基准[131]。评价朝着同一个方向发展，

19

being less concerned with a single end-to-end score under one hand-built harness and more with whether a model natively carries the capability a harness would otherwise supply. Recent benchmarks accordingly probe cross-industry computer-use workflows [575], long-horizon tool and MCP use [322, 700], local control and multi-day collaboration [426, 764], and productive office reasoning [458]. Internalization does not make the harness disappear; it shifts the field from hand-assembling a runtime for every model toward distilling runtime experience into weights and validating transfer with harness-native benchmarks, a theme developed systematically in Section 5.

This is the co-evolution at the heart of the paper. Long-horizon capability is supplied *jointly* by an externalized harness and an internalized optimization process, with experience traces carrying capability between them. The next two chapters take up these halves in turn, treating the externalized harness in Section 4 and the internalized optimization in Section 5.

4 Harness: Externalizing Long-Horizon Capability

To analyze long-horizon capability, we start from the externalization perspective. Long-horizon capabilities need not originate in weights: before a model learns to plan, remember, or recover through training, these capacities can be *externalized*, that is, engineered into the runtime system through the harness that wraps the model at inference time and mediates every interaction it has with the world. It contains the machinery of loops, memory, tools, coordination, and oversight that extends a model into a long-horizon agent without changing what the model knows. Where Section 5 asks what training can absorb, this chapter asks what harness can supply. As shown in Table 2, we organize the harness into six components that recur across essentially every contemporary agent, ordered from the control loop outward to the systems that govern it:

The Six Components of an Agent Harness

1. **Loops and Workflows** (Section 4.1): specifies how execution moves across states, including when the model is called, how actions are taken, and how control returns after observations.
2. **Context and Memory** (Section 4.2): manages the state available to the model, from the active context of the current run to persistent memory across windows, tasks, and sessions.
3. **Tools, MCP, and Skills** (Section 4.3): defines how the agent accesses external capabilities, represents actions, and reuses acquired procedures.
4. **Orchestration** (Section 4.4): organizes work beyond a single agent loop, including task decomposition, role assignment, routing, and inter-agent coordination.
5. **Hooks and Middleware** (Section 4.5): provides predefined interception points in the runtime where execution can be inspected, modified, paused, or handed off.
6. **Verification** (Section 4.6): checks whether states, actions, outputs, or side effects satisfy task, safety, and quality constraints before execution continues or results are accepted.

4.1 Loops and Workflows

The innermost layer of a long-horizon agent harness is the workflow that advances execution from one state to the next. At this layer, the harness specifies the control pattern by which model calls, tool calls, state updates, checks, and feedback are sequenced. Recent workflow papers describe agent workflows as execution structures in which agents perceive state, reason, decide, act, receive feedback, and adapt under a workflow engine [777]. An example of a long-horizon run can be found in Figure 7.

For long-horizon agents, the central design question is how much of the trajectory is governed by this external control pattern rather than by the model's next-token policy alone. A workflow may simply repeat an observe–act loop, maintain an explicit plan across steps, or branch over multiple candidate trajectories before committing. We therefore classify harness workflows into three structures: *linear workflows*, which advance through a single trajectory; *plan-execute workflows*, which externalize task decomposition and progress tracking; and *branching workflows*, which search over alternatives using comparison, pruning, or backtracking.

原文第 20 页 | 中文翻译初稿

不太关心手工构建的 运行支撑框架（Harness） 下的单个端到端分数，而更关心模型本身是否具有 运行支撑框架（Harness） 本应提供的功能。最近的基准相应地探讨了跨行业计算机使用工作流程 [575]、长时程（Long-Horizon） 工具和 MCP 使用 [322, 700]、本地控制和多日协作 [426, 764] 以及高效办公推理 [458]。内化不会使运行支撑框架（Harness）消失；它将该领域从为每个模型手动组装运行时转向将运行时经验提炼为权重并使用 运行支撑框架（Harness） 原生基准测试验证传输，这是第 5 节中系统开发的主题。

这是本文核心的共同进化。 长时程（Long-Horizon） 能力由外部化的 运行支撑框架（Harness） 和内部化的优化流程共同提供，经验痕迹在它们之间承载能力。接下来的两章依次讨论这两个部分，在第 4 节中处理外部化的 运行支撑框架（Harness） ，在第 5 节中处理内部化的优化。

4 运行支撑框架（Harness）：外部化长时程（Long-Horizon）能力

为了分析长时程（Long-Horizon）的能力，我们从外部化的角度出发。 长时程（Long-Horizon） 能力不需要源自权重：在模型学会通过训练来计划、记忆或恢复之前，这些能力可以被_外部化_，即通过 运行支撑框架（Harness） 设计到运行时系统中，运行支撑框架（Harness） 在推理时包装模型并调解它与世界的每一次交互。它包含循环、内存、工具、协调和监督机制，将模型扩展为 长时程（Long-Horizon） 智能体，而无需更改模型所知道的内容。第 5 节询问训练可以吸收什么，而本章询问 运行支撑框架（Harness） 可以提供什么。如表 2 所示，我们将 运行支撑框架（Harness） 组织成六个组件，这些组件基本上在每个当代智能体中重复出现，从控制循环向外到管理它的系统排序：

智能体的六个组成部分 运行支撑框架（Harness）

1. 循环和工作流（第 4.1 节）：指定执行如何跨状态移动，包括何时调用模型、如何采取操作以及观察后控制如何返回。

2. 上下文和内存（第 4.2 节）：管理模型可用的状态，从当前运行的活动上下文到跨窗口、任务和会话的持久内存。

3. 工具、MCP 和技能（第 4.3 节）：定义智能体如何访问外部功能、表示操作以及重用已获取的过程。

4. 编排（第 4.4 节）：组织单个智能体循环之外的工作，包括任务分解、角色分配、路由和智能体间协调。

5. 挂钩和中间件（第 4.5 节）：在运行时提供预定义的拦截点，可以在其中检查、修改、暂停或切换执行。

6. 验证（第 4.6 节）：在继续执行或接受结果之前检查状态、操作、输出或副作用是否满足任务、安全和质量约束。

4.1 循环和工作流程

长时程（Long-Horizon） 智能体 运行支撑框架（Harness） 的最内层是将执行从一种状态推进到下一种状态的工作流

。在这一层，运行支撑框架（Harness） 指定模型调用、工具调用、状态更新、检查和反馈排序的控制模式。最近的工作流论文将智能体工作流描述为执行结构，其中智能体在工作流引擎下感知状态、推理、决定、行动、接收反馈和适应[777]。 长时程（Long-Horizon） 运行的示例如图 7 所示。

对于 长时程智能体（Long-Horizon Agents） 来说，核心设计问题是轨迹有多少是由这种外部控制模式控制的，而不是仅由模型的下一个代币策略控制的。工作流程可以简单地重复观察-行动循环，跨步骤维护明确的计划，或者在提交之前在多个候选轨迹上进行分支。因此，我们将 运行支撑框架（Harness） 工作流

程分为三种结构： 线性工作流程，通过单个轨迹前进； 计划-执行工作流程，将任务分解和进度跟踪具体化；和_分支工作流程_，它使用比较、修剪或回溯来搜索替代方案。

20

Table 2 The agent harness by component. *Harness responsibility* states what the component must supply at runtime; *representative mechanisms* lists the component's sub-mechanisms, and *representative systems* pairs each sub-mechanism with two concrete systems on the same line.

Component (§)	Harness responsibility	Representative mechanisms	Representative systems
Loop & Workflow (§4.1)	Schedules model calls, action execution, and observation feedback across steps	• Linear workflows • Plan–execute • Branching	ReAct [761], Self-Refine [416] ReWOO [720], Arbor [262] Tree-of-Thoughts [760], LATS [875]
Context & Memory (§4.2)	Bounds in-run state and persists what must survive across windows and sessions	• Working context • Factual memory	ReSum [693], MemAgent [778] Mem0 [88], MemoryBank [874] ExpeL [853], ReasoningBank [462]
Tools, MCP & Skills (§4.3)	Interfaces model decisions with external capabilities and reusable procedures	• Tool interfaces & MCP • Active discovery • Skill libraries	Toolformer [523], MCP [100] AnyTool [140], ToolGen [647] Voyager [630], SkillNet [360]
Orchestration (§4.4)	Decomposes goals, assigns roles, and coordinates work beyond a single loop	• Decomposition • Topologies • Routing & protocols	MetaGPT [219], CAMEL [314] Magentic-One [157], LangGraph [291] GPTSwarm [897], A2A [97]
Hooks & Middleware (§4.5)	Enforces allow, block, or hand-off at predefined points on the action path	• Rule-based hooks • User-defined policies • Adaptive guards	AEGIS [783], IsolateGPT [698] AgentSpec [633], ShieldAgent [80] AdaptiveGuard [754], AGrail [408]
Verification (§4.6)	Scores states and outputs for correctness, safety, and task fit before continuing or accepting results	• Assessment targets • Verification levels • Verifier strategies	SelfCheckGPT [418], VeriGuard [432] PRMs [363], LLM-as-a-Judge [867] Math-Shepherd [643], RLV [521]

4.1.1 Linear Workflows

Linear workflows are the simplest way to keep an agent moving: the harness repeatedly renders the current state to the model, executes the selected action, records the resulting observation, and feeds the updated state into the next call. The workflow follows one evolving trace rather than maintaining a separate plan or search tree. This makes it suitable for interactive tasks where the agent must react to fresh evidence, such as web navigation, debugging, tool use, and exploratory search. ReAct [761] is the canonical instance of this pattern, where reasoning and acting are interleaved and each observation conditions the next model invocation.

The strength of a linear workflow is responsiveness, but its weakness is that all progress depends on a single trace. Early mistakes can contaminate later steps, repeated reasoning can exhaust the context window, and the model may loop, drift, or stop prematurely. Many variants therefore wrap feedback, critique, retrieval, memory, or self-correction around the basic chain [22, 416, 544, 567]. In this sense, a bare linear workflow is only the inner loop, while the emerging practice of *loop engineering* designs the recurring outer system that automatically prompts, monitors, verifies, persists, and resumes the agent. This outer system adds external stop conditions, persistent state, verification gates, cost limits, and recovery policies, turning the workflow into an unattended long-horizon run in which a human need not decide every next prompt.

4.1.2 Plan-Execute Workflows

Plan-execute workflows address the short-sightedness of purely reactive loops by making task structure explicit. The harness first creates or maintains a plan, then uses it to guide later actions through subgoals, dependencies, progress markers, budgets, and failure conditions. Early forms ask the model to plan before solving, while more structured variants plan tool calls, decompose tasks recursively, or revise the plan during execution [482, 639, 720]. Across these systems, the plan becomes an execution object that the harness can inspect, update, resume, invalidate, or assign to different modules.

This structure is especially useful for long-horizon loops because an unattended agent cannot safely run on a vague instruction such as "improve the project". It needs a goal that can be decomposed, monitored, and verified. A maintained plan provides this control surface: it translates an underspecified objective into concrete checkpoints and failure conditions, while allowing the model to make local decisions under persistent task commitments. Recent long-horizon systems therefore couple planning with execution monitoring, verification, replanning, or learning-based policy improvement [763, 832]. Arbor [262] extends this paradigm to autonomous research by maintaining a persistent hypothesis tree across coordinator–executor cycles, where each branch

原文第 21 页 | 中文翻译初稿

表 2 按组件划分的智能体 运行支撑框架（Harness）。 _运行支撑框架（Harness）职责_说明组件在运行时必须提供什么； _代表性机制_列出了组件的子机制，并且_代表性系统_将每个子机制与同一行上的两个具体系统配对。

组成部分(§)	运行支撑框架（Harness） 责任	代表机制	代表制度
循环和工作流程
(§4.1)	跨步骤安排模型调用、操作
执行和观察反馈
• 线性工作流程
• 计划-执行
• 分支	ReAct [761]、Self-Refine [416]
ReWOO [720]、Arbor [262]
Tree-of-Thoughts [760]、LATS 【875】
上下文和内存
(§4.2)	限制运行中状态并持续存在
必须跨窗口
和会话生存的内容	• 工作上下文
• 事实内存	ReSum [693]、MemAgent [778]
Mem0 [88]、MemoryBank [874]
ExpeL [853]、 ReasoningBank [462]
工具、MCP 和
技能 (§4.3)	将模型决策与
外部功能和可重用
过程进行接口	• 工具接口和 MCP
• 主动发现
• 技能库	Toolformer [523]、MCP [100]
AnyTool [140]、ToolGen [647]
Voyager [630]、SkillNet [360]
编排
(§4.4)	分解目标、分配角色并
在单个
循环之外协调工作	• 分解
• 拓扑
• 路由和协议	MetaGPT [219]、CAMEL [314]
Magentic-One [157]、LangGraph [291]
GPTSwarm [897]、A2A [97]
挂钩 &	强制允许、阻止或移交	• 基于规则的挂钩	AEGIS [783]、IsolateGPT [698]
中间件 (§4.5)	操作路径上的预定义点	• 用户定义的策略
• 自适应防护	AgentSpec [633]、ShieldAgent [80]
AdaptiveGuard [754]、AGrail [408]

|验证 (§4.6)|在继续或接受结果之前
正确性、安全性和任务适合度
对状态和输出进行评分|• 评估目标
• 验证级别
• 验证者策略|SelfCheckGPT [418]、VeriGuard [432]
PRM [363]、LLM-as-a-Judge [867]
Math-Shepherd [643]、 RLV [521]|

4.1.1 线性工作流程

线性工作流程是保持智能体移动的最简单方法：运行支撑框架（Harness） 重复将当前状态呈现给模型，执行选定的操作，记录结果观察，并将更新的状态输入到下一个调用中。工作流程遵循一条不断发展的轨迹，而不是维护单独的计划或搜索树。这使得它适合智能体必须对新证据做出反应的交互式任务，例如网络导航、调试、工具使用和探索性搜索。 ReAct [761] 是这种模式的典型实例，其中推理和行动是交错的，并且每个观察都影响下一个模型调用。

线性工作流程的优点是响应能力，但其缺点是所有进度都依赖于单个跟踪。早期的错误可能会影响后续步骤，重复推理可能会耗尽上下文窗口，并且模型可能会循环、漂移或过早停止。因此，许多变体围绕基本链包裹反馈、批评、检索、记忆或自我纠正[22、416、544、567]

。从这个意义上说，裸露的线性工作流程只是内部循环，而新兴的循环工程实践则设计了自动提示、监视、验证、持久和恢复智能体的循环外部系统。该外部系统添加了外部停止条件、持久状态、验证门、成本限制和恢复策略，将工作流程转变为无人值守的 长时程（Long-Horizon） 运行，其中人类无需决定每个下一个提示。

4.1.2 计划-执行工作流程

计划执行工作流通过使任务结构明确来解决纯粹反应性循环的短视问题。 运行支撑框架（Harness） 首先创建或维护计划，然后使用它通过子目标、依赖性、进度标记、预算和失败条件来指导后续行动。早期形式要求模型在求解之前进行计划，而更结构化的变体则计划工具调用、递归分解任务或在执行期间修改计划[482、639、720]。在这些系统中，计划成为 运行支撑框架（Harness） 可以检查、更新、恢复、失效或分配给不同模块的执行对象。

此结构对于 长时程（Long-Horizon） 循环特别有用，因为无人值守的智能体无法安全地运行诸如“改进项目”之类的模糊指令。它需要一个可以分解、监控和验证的目标。维护计划提供了这样的控制面：它将未指定的目标转化为具体的检查点和故障条件，同时允许模型在持久的任务承诺下做出本地决策。因此，最近的 长时程（Long-Horizon） 系统将规划与执行监控、验证、重新规划或基于学习的策略改进结合起来[763, 832]。 Arbor [262] 通过在协调器-执行器周期中维护持久的假设树，将这种范式扩展到自主研究，其中每个分支

21

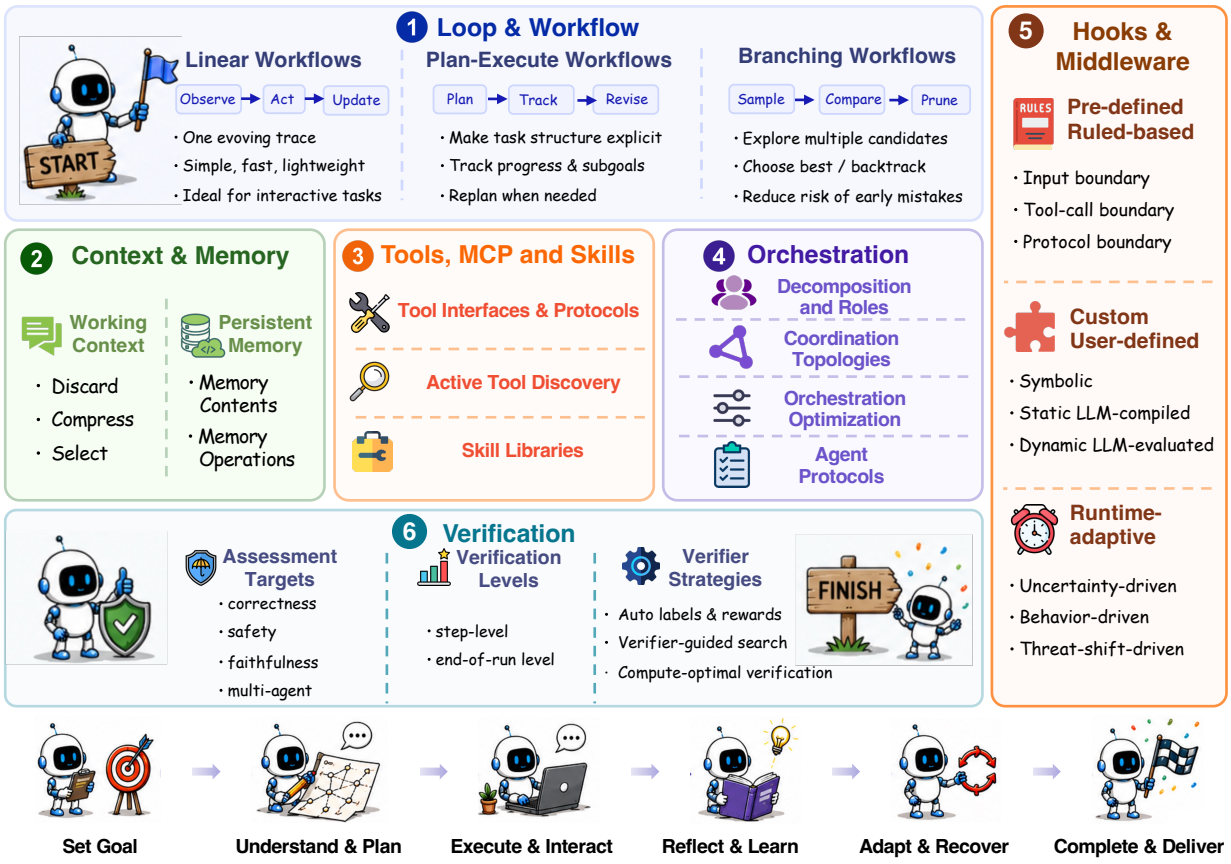


Figure 7 Overview of the agent harness. Six core harness components support the end-to-end agent lifecycle, from goal formulation and planning to execution, adaptation, verification, and delivery, enabling coherent long-horizon execution across interdependent steps.

records hypotheses, experimental artifacts, evidence, and distilled insights that propagate across iterations, turning isolated attempts into a cumulative strategy-driven search.

4.1.3 Branching Workflows

Branching workflows are useful when one wrong commitment can derail the whole task. Instead of immediately following the first plausible action, plan, or answer, the harness generates several alternatives and evaluates them before deciding which one to pursue. This pattern appears in reasoning trees, tree-search-based acting, graph-based thought, code-generation search, and memory-augmented search variants [33, 91, 760, 875]. The defining feature is evaluation before commitment: the harness controls the branching factor, scoring function, pruning rule, backtracking policy, and final selection procedure [319].

For long-horizon agents, branching reduces the risk of spending a large trajectory budget on a weak early decision [280]. It is also closely related to loop engineering, where unattended loops need reliable verification rather than repeated self-approval. A branching workflow can separate generation and checking: one branch proposes a patch or plan, another critiques it, a verifier tests constraints, and the harness decides whether to continue, revise, or stop. This maker-checker separation helps reduce self-evaluation bias and Goodhart-style failures, where the agent satisfies the visible completion condition while violating the user's underlying intent.

Branching is most useful when partial states can be evaluated cheaply and meaningfully. It is less attractive when feedback is weak, delayed, or expensive, because search can multiply token and tool costs, create incompatible partial worlds, and make provenance harder to audit [515]. Thus, branching workflows are best

原文第 22 页 | 中文翻译初稿

循环和工作流程 5 个钩子和
线性工作流程 计划执行工作流程 分支工作流程 中间件
观察行动 更新 计划跟踪 修订样本 比较 修剪 预定义
· 一个演变跟踪 · 使任务结构明确 · 探索多个候选对象 基于规则
· 简单、快速、轻量级 · 跟踪进度和子目标 · 选择最佳/回溯
· 输入边界
· 交互式任务的理想选择 · 需要时重新计划 · 降低风险早期错误
· 工具调用边界
· 协议边界
上下文和内存工具、MCP 和技能编排
分解
工作持久工具接口、协议和角色自定义
上下文内存协调用户定义
拓扑
· 丢弃· 内存活动工具发现· 符号
· 压缩内容编排优化· 静态LLM 编译
· 选择· 操作内存技能库智能体·动态 LLM 评估
协议
运行时-
6 验证自适应
评估验证验证器
目标级别策略·不确定性驱动
·正确性
·自动标签和奖励·行为驱动
·安全性·步骤级别
·验证者引导搜索·威胁转移驱动
·忠诚度·运行结束级别
·计算最优验证
·多智能体
设定目标 理解和计划 执行和交互 反思和学习 适应和恢复 完成和交付

图 7 智能体 运行支撑框架（Harness） 概述。六个核心 运行支撑框架（Harness） 组件支持端到端智能体生命周期，从目标制定和规划到执行、适应、验证和交付，从而实现跨相互依赖的步骤的一致 长时程（Long-Horizon） 执行。

记录假设、实验工件、证据和在迭代中传播的提炼见解，将孤立的尝试转变为累积的策略驱动的搜索。
4.1.3 分支工作流程

当一个错误的承诺可能使整个任务脱轨时，分支工作流程非常有用。 运行支撑框架（Harness） 不会立即遵循第一个看似合理的行动、计划或答案，而是会生成多个替代方案并对其进行评估，然后再决定采用哪一个。这种模式出现在推理树、基于树搜索的行动、基于图的思维、代码生成搜索和记忆增强搜索变体中

[33,91,760,875]。定义特征是承诺前的评估：运行支撑框架（Harness） 控制分支因子、评分函数、修剪规则、回溯策略和最终选择程序[319]。

对于 长时程智能体（Long-Horizon Agents），分支降低了在较弱的早期决策上花费大量轨迹预算的风险[280]。它还与回路工程密切相关，其中无人值守的回路需要可靠的验证而不是重复的自我批准。分支工作流程可以将生成和检查分开：一个分支提出补丁或计划，另一个分支对其进行批评，验证者测试约束，然后 运行支撑框架（Harness） 决定是否继续、修改或停止。这种制造者与检查者的分离有助于减少自我评估偏差和古德哈特式失败，即智能体满足可见的完成条件，同时违反用户的潜在意图。

当可以廉价且有意义地评估部分状态时，分支是最有用的

。当反馈微弱、延迟或昂贵时，它的吸引力就不那么大了，因为搜索会增加令牌和工具成本，创建不兼容的部分世界，并使出处更难审计[515]。因此，分支工作流程是最好的

22

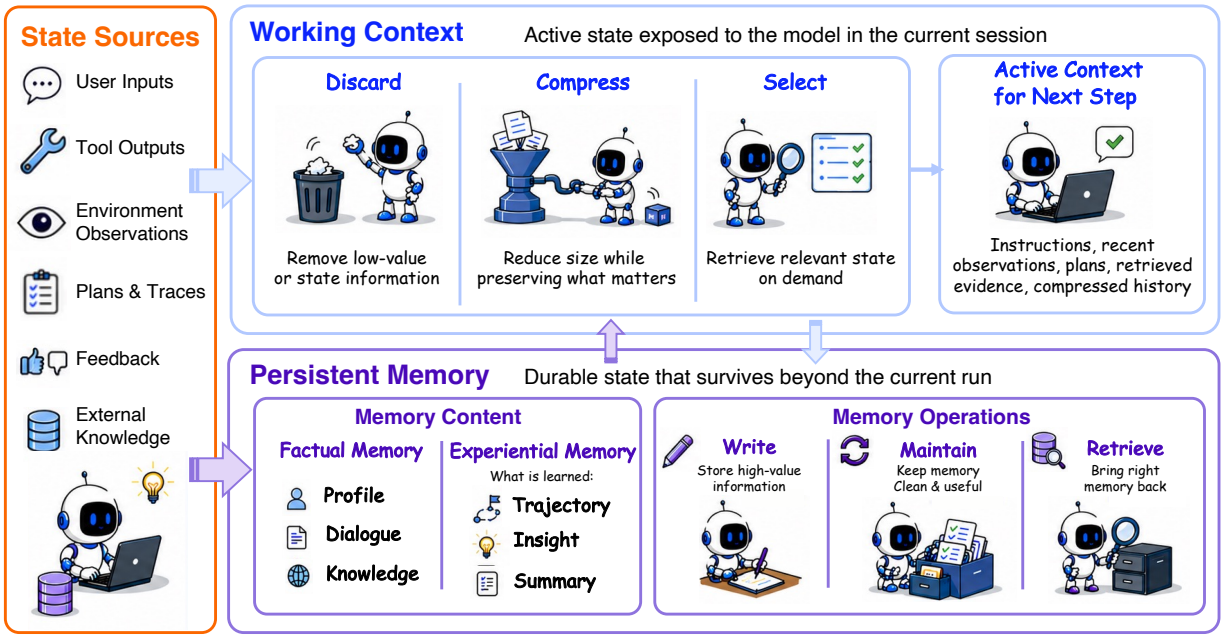


Figure 8 Context and memory in an agent harness. The harness governs the state interface between the workflow and the model: it selects, compresses, or discards active context under a token budget, and reads from and writes to persistent memory that outlives a single window, task, or session.

viewed as selective search mechanisms invoked when the cost of a bad commitment is higher than the cost of evaluating alternatives [343]. Across linear, plan-execute, and branching workflows, the common requirement is that the harness must maintain the state on which these workflows operate. We therefore turn next to the state machinery that supports long-horizon execution.

4.2 Context and Memory

If workflows define how an agent proceeds, context and memory define what state it proceeds with [228]. For long-horizon agents, the harness cannot simply pass along an ever-growing transcript. It must decide what the next model call can see, what should be compressed or discarded when the context window fills, and what information should persist beyond the current window, task, or session [876]. In this sense, context and memory are the state interface between the workflow and the model: the harness reads from prior observations, tool outputs, plans, traces and stored knowledge, then writes the selected state back into the next step.

We distinguish context and memory by their time horizon and role at runtime. *Context* is the active state exposed to the model in the current run, including instructions, recent observations, intermediate reasoning, retrieved evidence, and compressed history. *Memory* is the durable state that survives beyond the immediate window and can be retrieved or updated over time. We organize this component accordingly. Section 4.2.1 covers working state, namely the in-window or near-window state maintained within a run; Section 4.2.2 covers persistent state, namely the durable memory that survives across windows, tasks, and sessions. In Table 3, we list representative works for each sub-section. The overview of this section is shown in Figure 8.

4.2.1 Working Context

Every step of an agent run produces new state: model outputs, tool returns, environment observations, partial plans, and intermediate artifacts. Runtime context maintenance is the harness process that decides which of these states should condition the next model call. Some work refers to this in-window or near-window state as the agent's *working memory*, emphasizing its role as a limited workspace for keeping task-relevant information available within a single episode [228]. For long-horizon agents, the central challenge is not only that the context window is finite, but that useful, stale, and distracting information accumulate together over time.

原文第 23 页 | 中文翻译初稿

状态源 工作上下文 在当前会话中向模型公开的活动状态
活动上下文
用户输入 放弃压缩 选择
下一步
工具输出
环境
观察
指令，最近
删除低值 减小大小，同时检索相关状态观察、计划、检索
或状态信息保留按需重要内容
计划和跟踪证据，压缩历史
反馈
持久内存 在当前运行后仍然存在的持久状态
外部内存内容内存操作
知识事实内存经验内存写入维护检索
学到了什么：存储高价值保持内存右移
配置文件信息干净且有用的内存返回
轨迹
对话洞察
知识摘要

图 8 智能体 运行支撑框架（Harness） 中的上下文和内存。 运行支撑框架（Harness） 管理工作流和模型之间的状态接口：它在令牌预算下选择、压缩或丢弃活动上下文，并读取和写入比单个窗口、任务或会话寿命更长的持久内存。

被视为当不良承诺的成本高于评估替代方案的成本时调用的选择性搜索机制[343]。在线性、计划执行和分支工作流中，共同的要求是 运行支撑框架（Harness） 必须维护这些工作流运行的状态。因此，我们接下来转向支持 长时程（Long-Horizon） 执行的状态机。

4.2 上下文和记忆

如果工作流定义了智能体如何进行，那么上下文和内存定义了它进行的状态[228]。对于 长时程智能体（Long-Horizon Agents），运行支撑框架（Harness） 不能简单地传递不断增长的转录本。它必须决定下一个模型调用可以看到什么，当上下文窗口填充时应该压缩或丢弃什么，以及哪些信息应该在当前窗口、任务或会话之外保留[876]。从这个意义上说，上下文和内存是工作流和模型之间的状态接口：运行支撑框架（Harness）从先前的观察、工具输出、计划、跟踪和存储的知识中读取，然后将所选状态写回下一步。

我们通过时间范围和运行时的角色来区分上下文和内存。 Context 是当前运行中暴露给模型的活动状态，包括指令、最近的观察结果、中间推理、检索到的证据和压缩的历史记录。 Memory 是持久状态，它在即时窗口之外仍然存在，并且可以随着时间的推移进行检索或更新。我们相应地组织这个组件。 4.2.1节涵盖了工作状态，即运行中保持的窗口内或接近窗口状态； 4.2.2 节介绍了持久状态，即跨窗口、任务和会话保存的持久内存。在表3中，我们列出了每个小节的代表性作品。本节的概述如图 8 所示。

4.2.1 工作环境

智能体运行的每一步都会产生新的状态：模型输出、工具返回、环境观察、部分计划和中间工件。运行时上下文维护是 运行支撑框架（Harness） 过程，它决定这些状态中的哪一个应该作为下一个模型调用的条件

。一些工作将这种窗口内或近窗口状态称为智能体的_工作记忆_，强调其作为有限工作空间的作用，用于在单个情节中保持任务相关信息可用[228]。对于 长时程智能体（Long-Horizon Agents） 来说，核心挑战不仅在于上下文窗口是有限的，而且有用的、陈旧的和分散注意力的信息随着时间的推移而积累在一起。

23

Too much context can degrade performance through *context rot* and *lost-in-the-middle* effects, while too little context removes constraints, evidence, or partial progress that the agent still needs [378, 511, 541]. The harness must therefore turn an expanding trajectory into a bounded, high-signal workspace. In practice, long-running agents combine three operations: *discard*, *compress*, and *select* [13, 14, 293].

Context Discard. Discarding removes information from the active window [113]. Common policies include resetting context at task boundaries, dropping bulky tool responses after they have been parsed or summarized, and keeping only the most recent turns [597]. For short interactions, discard mainly saves tokens; for long-horizon runs, it is also a reliability mechanism. Stale observations, obsolete plans, verbose logs, and irrelevant tool outputs can continue to bias later decisions if they remain visible.

Discard can also happen before information enters the trajectory. Tool or schema pruning, for example, prevents rarely useful capabilities or verbose specifications from occupying the prompt in the first place [27, 622]. This matters because noise compounds over long runs: a small amount of irrelevant state repeated across many steps can become a first-order source of drift. Discard is therefore the most direct harness-level way to enforce forgetting, rather than relying on the model to ignore low-value context.

Context Compression. Compression preserves task-relevant information while reducing its token and attention footprint. A useful operational distinction is between compressing the current payload and compressing the accumulated trajectory. The first case handles large immediate inputs, such as retrieved passages, tool outputs, web pages, logs, or DOM trees [57, 254, 773]. In agent harnesses, similar operations selectively rewrite verbose observations into compact, task-oriented state descriptions.

The second case is trajectory consolidation: replacing a growing interaction history with a maintained state. This state may be a summary, a list of open constraints, a table of facts, a plan with completion markers, or a graph of entities and dependencies. For long-horizon agents, the goal is not just to shorten the transcript, but to preserve what still matters for future action: completed decisions, remaining subgoals, unresolved failures, and produced artifacts. Recent methods explore learned, task-aware, budget-aware, and hierarchical forms of such compression [224, 268, 573, 693, 696, 768, 778, 824, 890]. This also marks a shift from hand-written context rules toward learned policies for when and how to fold state.

Context Selection. Selection keeps richer state outside the active window and retrieves only what is relevant to the current decision. The indexed state may include documents, events, artifacts, plans, tool outputs, or previous observations. Unlike ordinary RAG, selection in an agent harness often retrieves from the agent's own trajectory, not only from an external knowledge base [202, 669]. Typically, CL-bench is designed to target and evaluate a model's long-context comprehension and selection capabilities [133]. The goal is to keep the current context small while preserving access to the evidence trail.

From the harness perspective, selection involves four choices: when to retrieve, how to form the query, which store to search, and how to filter or aggregate the results before injection. Retrieving too often increases cost and noise; retrieving too rarely risks losing dependencies that were established many steps earlier. Long-running agents therefore usually combine selection with discard and compression: bulky outputs are removed, completed subtasks are folded, and relevant past state is retrieved on demand [13, 230, 231, 824, 846]. These operations keep the active context bounded without forcing the agent to forget the full task history.

4.2.2 Persistent Memory

Working context keeps a single run usable; persistent memory lets the harness carry state beyond the current window, trajectory, or session [228, 583, 694, 848]. This is crucial for long-horizon agents because useful information often outlives the step in which it was produced. A failed attempt may become a lesson for later recovery, a completed subtask may define constraints for downstream actions, and a discovered procedure may be reused in future tasks. Without persistent memory, the agent must repeatedly rediscover the same facts, strategies, and skills as the context window turns over. We use a simple harness-oriented view. *Memory Contents* asks what persistent state is stored; *Memory Operations* asks how that state is written, maintained, retrieved, and evaluated.

原文第 24 页 | 中文翻译初稿

太多的上下文会通过“上下文腐烂”和“中间丢失”效应降低性能，而太少的上下文会消除智能体仍然需要的约束、证据或部分进展[378,511,541]。因此，运行支撑框架（Harness） 必须将扩展轨迹转变为有界的高信号工作空间。在实践中，长时间运行的智能体结合了三个操作：discard、_compress_和_select_[13,14,293]。

上下文丢弃。丢弃从活动窗口中删除信息[113]。常见的策略包括在任务边界重置上下文、在解析或总结后丢弃大量的工具响应以及仅保留最近的轮次[597]。对于较短的交互，discard主要是节省token；对于长期运行来说，它也是一种可靠性机制。陈旧的观察结果、过时的计划、冗长的日志和不相关的工具输出如果仍然可见，可能会继续影响以后的决策。

丢弃也可能发生在信息进入轨迹之前。例如，工具或模式修剪可以防止很少有用的功能或冗长的规范首先占据提示[27, 622]。这很重要，因为噪声会在长期运行中复合：在许多步骤中重复的少量不相关状态可能成为漂移的一阶源。因此，丢弃是强制遗忘最直接的 运行支撑框架（Harness） 级方法，而不是依赖模型来忽略低价值上下文。

上下文压缩。压缩保留了任务相关信息，同时减少了其标记和注意力足迹。一个有用的操作区别是压缩当前有效负载和压缩累积轨迹。第一种情况处理大量即时输入，例如检索的段落、工具输出、网页、日志或 DOM 树 [57,254,773]。在智能体 运行支撑框架（Harnesses） 中，类似的操作有选择地将详细的观察重写为紧凑的、面向任务的状态描述。

第二种情况是轨迹整合：用维护的状态替换不断增长的交互历史。该状态可以是摘要、开放约束列表、事实表、带有完成标记的计划或者实体和依赖关系图。对于 长时程智能体（Long-Horizon Agents），目标不仅仅是缩短脚本，而是保留对未来行动仍然重要的内容：已完成的决策、剩余的子目标、未解决的故障和产生的工件。最近的方法探索了这种压缩的学习、任务感知、预算感知和分层形式[224、268、573、693、696、768、778、824、890]。这也标志着从手写上下文规则到何时以及如何折叠状态的学习策略的转变。

上下文选择。选择在活动窗口之外保留更丰富的状态，并仅检索与当前决策相关的内容。索引状态可以包括文档、事件、工件、计划、工具输出或先前的观察结果。与普通的 RAG 不同，智能体 运行支撑框架（Harness） 中的选择通常从智能体自己的轨迹中检索，而不仅仅是从外部知识库中检索 [202, 669]。通常，CL-bench 旨在瞄准和评估模型的长上下文理解和选择能力 [133]

。目标是保持当前上下文较小，同时保留对证据线索的访问。

从运行支撑框架（Harness）的角度来看，选择涉及四个选择：何时检索、如何形成查询、搜索哪个存储以及在注入之前如何过滤或聚合结果。过于频繁地检索会增加成本和噪音；检索太少可能会丢失之前许多步骤建立的依赖关系。因此，长时间运行的智能体通常将选择与丢弃和压缩结合起来：删除大量输出，折叠已完成的子任务，并根据需要检索相关的过去状态[13,230,231,824,846]。这些操作保持活动上下文有界，而不强迫智能体忘记完整的任务历史记录。

4.2.2 持久内存

工作环境保持单次运行可用；持久内存允许 运行支撑框架（Harness） 携带当前窗口、轨迹或会话之外的状态 [228、583、694、848]。这对于 长时程智能体（Long-Horizon Agents） 至关重要，因为有用的信息通常比生成它的步骤还要长久。失败的尝试可能成为以后恢复的教训，完成的子任务可以定义下游操作的约束，并且发现的过程可以在未来的任务中重用。如果没有持久记忆，智能体必须在上下文窗口翻转时

反复重新发现相同的事实、策略和技能。我们使用一个简单的面向 运行支撑框架（Harness） 的视图。_Memory Contents_询问存储了什么持久状态； _内存操作_询问如何写入、维护、检索和评估该状态。

24

Table 3 Representative context and memory systems in the agent harness. Rows follow the split of this subsection: *working context* (Section 4.2.1) is bounded within a single run, while *persistent memory* (Section 4.2.2) survives across windows, tasks, and sessions. **Type** names what the method does with state. **State form** is the representation it maintains. **Train-free** (✔) marks methods that plug into a frozen model as pure harness machinery, versus (✘) those whose policy must be trained or RL-tuned. **Trigger** is when the operation fires. **Key mechanism** briefly describes the core operation.

Method	Type	State form	Train-free	Trigger	Key mechanism
Working Context					
Effective Harnesses [14]	Discard	Turns	✔	Boundary	Reset context, drop parsed tool output
Deep Agents [293]	Discard	Files	✔	Boundary	Offload state to files
Tool pruning [622]	Discard	Schemas	✔	Start-of-run	Prune tools/schemas before prompt
LLMLingua [254]	Compress	Summary	✘	Per-payload	Token-level prompt compression
CompAct [773]	Compress	Summary	✘	Per-payload	Actively compress retrieved docs
ReSum [693]	Compress	Summary	✔	Boundary	Learned trajectory summarization
MEM1 [890]	Compress	Summary	✘	Per-step	RL fuses memory and reasoning
MemAgent [778]	Compress	Summary	✘	Per-step	RL over fixed-size memory
ACON [268]	Compress	Summary	✔	Boundary	Optimizes context-compression policy
ContextBudget [696]	Compress	Summary	✘	Per-step	Budget-aware context allocation
HiAgent [224]	Compress	Plan	✔	Boundary	Subgoal-scoped working memory
Context-Folding [573]	Compress	Summary	✘	Per-step	Folds history under a budget
AgentFold [768]	Compress	Summary	✘	Per-step	Proactive history folding
ACE [824]	Select	NL-note	✔	On-demand	Retrieves, evolves context entries
LEGOMem [202]	Select	Summary	✔	On-demand	Modular per-subtask procedural memory
Memex(RL) [669]	Select	Vector	✘	On-demand	RL over indexed experience memory
Memory-as-Action [846]	Mixed	Summary	✘	Per-step	Memory read/write as actions
Persistent Memory					
CLAUDE.md [10]	Factual	NL-note	✔	Start-of-run	Project-rule file loaded per run
Cursor Rules [104]	Factual	NL-note	✔	Start-of-run	Persistent project rules per run
AGENTS.md [98]	Factual	NL-note	✔	Start-of-run	Portable project-knowledge spec
MemoryBank [874]	Factual	Vector	✔	On-demand	Store with forgetting-curve updates
Mem0 [88]	Factual	Graph	✔	On-demand	Linked long-term fact store
MemoryOS [267]	Factual	Mixed	✔	On-demand	OS-style tiered memory
HippoRAG [199]	Factual	Graph	✔	On-demand	Graph memory for associative recall
A-MEM [731]	Factual	Graph	✔	On-demand	Self-organizing memory notes
Reflexion [544]	Experiential	NL-note	✔	After-failure	Verbal lessons from failures
ExpeL [853]	Experiential	NL-note	✔	On-demand	Reuses insights from trajectories
Synapse [868]	Experiential	Summary	✔	On-demand	Trajectory-as-exemplar prompting
SAGE [359]	Experiential	NL-note	✔	On-demand	Reflective memory-augmented evolution
Buffer of Thoughts [750]	Experiential	NL-note	✔	On-demand	Reusable thought templates
ReasoningBank [462]	Experiential	NL-note	✔	On-demand	Distills reusable reasoning strategies
AWM [674]	Experiential	Code-skill	✔	Boundary	Induces workflows from traces
Agent KB [587]	Experiential	NL-note	✔	On-demand	Cross-domain experience knowledge base
Voyager [630]	Experiential	Code-skill	✔	On-demand	Reusable executable skill library
MemGPT [464]	Mixed	Mixed	✔	On-demand	OS-style context/memory paging
Generative Agents [470]	Mixed	Summary	✔	On-demand	Memory stream with reflection

原文第 25 页 | 中文翻译初稿

表 3 智能体 运行支撑框架（Harness） 中的代表性上下文和内存系统。行遵循本小节的分割：工作上下文（第 4.2.1 节）仅限于一次运行，而_持久内存_（第 4.2.2 节）跨窗口、任务和会话保存。类型命名了该方法对状态执行的操作。状态形式是它所维护的表征。免训练 (✓) 标记将插入冻结模型作为纯 运行支撑框架（Harness） 机器的方法，而不是 (✗) 那些必须对其策略进行训练或 RL 调整的方法。触发器是操作触发的时间。按键机制简要描述了核心操作。

方法	类型	状态形式	免训练	触发	关键机制
工作环境					
有效 运行支撑框架（Harnesses） [14]	丢弃	转弯	✓	边界	重置上下文，删除已解析的工具输出
深度智能体 [293]	丢弃	文件	✓	边界	将状态卸载到文件
工具修剪 [622]	丢弃	模式	✓	运行开始	提示前修剪工具/模式
LLMLingua [254]	压缩	摘要	✗	每个有效负载	令牌级提示压缩
CompAct [773]	压缩	摘要	✗	每个有效负载	主动压缩检索到的文档
ReSum [693]	压缩	摘要	✓	边界	学习轨迹摘要
MEM1 [890]	压缩	总结	✗	每步	RL 融合记忆和推理
MemAgent [778]	压缩	摘要	✗	每步	固定大小内存上的强化学习
ACON [268]	压缩	摘要	✓	边界	优化上下文压缩策略
ContextBudget [696]	压缩	摘要	✗	每步	预算感知上下文分配
HiAgent [224]	压缩	计划	✓	边界	子目标范围的工作记忆
上下文折叠 [573]	压缩	摘要	✗	每步	在预算范围内折叠历史记录
AgentFold [768]	压缩	摘要	✗	每步	主动历史记录折叠
ACE [824]	选择	NL-note	✓	按需	检索、演变上下文条目
LEGOMem [202]	选择	摘要	✓	按需	模块化每子任务程序内存
Memex(RL) [669]	选择	矢量	✗	按需	基于索引经验记忆的强化学习
内存即操作 [846]	混合	摘要	✗	每步	内存读/写作为操作
持久内存					
CLAUDE.md [10]	事实	NL-note	✓	运行开始	每次运行加载的项目规则文件
光标规则 [104]	事实	NL-note	✓	运行开始	每次运行的持久项目规则
AGENTS.md [98]	事实	NL-note	✓	运行开始	便携式项目知识规范
MemoryBank [874]	事实	矢量	✓	按需	存储遗忘曲线更新
Mem0 [88]	事实	图表	✓	按需	链接的长期事实存储
MemoryOS [267]	事实	混合	✓	按需	操作系统风格分层内存
HippoRAG [199]	事实	图表	✓	按需	用于联想回忆的图形存储器
A-MEM [731]	事实	图表	✓	点播	自组织记忆笔记
反思[544]	体验	NL-note	✓	失败后	失败的口头教训
ExpeL [853]	体验式	NL-note	✓	按需	重用来自轨迹的见解
Synapse [868]	体验式	总结	✓	按需	轨迹作为范例提示
SAGE [359]	体验式	NL-note	✓	点播	反思性记忆增强进化
思想缓冲区 [750]	体验式	NL-note	✓	按需	可重复使用的思想模板
ReasoningBank [462]	体验式	NL-note	✓	按需	提炼可重复使用的推理策略

|AWM [674]|体验|代码技能|✓|边界|从跟踪中引入工作流程|
|智能体知识库

[587]|体验式|NL-note|✓|点播|跨域体验知识库|
Voyager [630]	体验式	代码技能	✓	按需	可重复使用的可执行技能库
MemGPT [464]	混合	混合	✓	按需	操作系统样式上下文/内存分页
生成智能体 [470]	混合	摘要	✓	按需	带反射的内存流

25

Memory Contents. Persistent memory mainly stores two kinds of state. *Factual memory* stores declarative information that the agent should not have to rediscover: user preferences, project rules, commitments, environment state, document or codebase facts, and the capabilities of external systems [228]. For a long-horizon harness, these facts support continuity: user-level facts keep behavior consistent across sessions, while environment-level facts prevent the agent from repeatedly re-inferring stable constraints. Practical examples include project-rule files such as `CLAUDE.md`, Cursor `Rules`, and `AGENTS.md`, which provide stable project knowledge outside the immediate prompt [10, 98, 104]. User memory systems and graph-structured stores further extend this idea by maintaining profiles, preferences, and linked factual records that can be consolidated and audited over time [88, 199, 229, 267, 507, 731, 874].

Experiential memory stores what the agent has learned from prior trajectories. It may contain failure lessons, successful cases, state-action exemplars, reusable reasoning patterns, plans, workflows, or executable procedures. The common purpose is to reduce long-horizon cost by replacing repeated rediscovery with retrieval of prior structure. Some systems store verbal reflections or lessons from past failures; others maintain experience pools, reusable reasoning templates, workflow memories, or cross-domain solution paths [44, 262, 359, 462, 544, 587, 674, 750, 807, 853, 868]. At the most operational end, experiential memory can be distilled into skills, scripts, API wrappers, or tool-like procedures, as in skill-library agents such as Voyager [446, 630]. Once stored in this form, memory begins to overlap with the tool layer: the harness can retrieve, version, validate, reuse, or retire a learned procedure without changing model weights.

Memory Operations. A persistent store is useful only if the harness manages how memory is written, maintained, and read back into context. *Memory writing* decides what should be saved from a long run. Tool outputs, partial plans, reasoning traces, user feedback, observations, and intermediate artifacts are often too noisy to store as-is. The harness therefore extracts higher-value units such as facts, summaries, cases, lessons, workflows, or skills, using summarization, distillation, or structured construction [221, 407, 464, 470, 674]. The key question is what level of experience should survive: an event, a workflow, or an executable procedure.

Memory maintenance keeps the store usable as new information arrives. Over long horizons, memories can become stale, repeated, or inconsistent. The harness must merge related entries, update facts when users or environments change, and remove outdated or low-value material before retrieval becomes noisy. Existing systems use memory streams, graphs, linked notes, or other structured stores to support this process [199, 470, 731]. The model may suggest what to remember, but the harness decides whether a memory is inserted, merged, linked, archived, or forgotten.

Memory retrieval decides how stored state re-enters the active context. Some memories are loaded at the start of a run, such as project rules or user preferences [664]; others are retrieved at subtask boundaries, after failures, before tool use, or when the model asks for them. Retrieval requires choices about the query, the store to search, the ranking or filtering method, and how much content to inject [488, 568, 610, 694]. Too little retrieval makes the agent repeat mistakes or miss long-range dependencies; too much retrieval turns memory back into context overload. Recent work therefore treats memory read/write as a policy problem, learning when to remember, update, or retrieve during inference [778, 846].

Persistent memory is also hard to evaluate. End-task success often mixes memory quality with model strength, tool reliability, and benchmark variance. New benchmarks test long-term recall, cross-session consistency, and tasks that depend on earlier sessions, but recent analyses still find saturation, judge sensitivity, and limited coverage of update cost and memory maintenance [235, 253, 686]. For long-horizon harnesses, the key question is not whether an agent can recall an isolated fact, but whether it can save, update, retrieve, and inject the right memory when it changes the next action. We collect evaluation resources in Section 6.6.

4.3 Tools, MCP, and Skills

Where context and memory settle what the agent *knows*, this component settles what it can *do*. Without tools that return observations, even a capable model can only emit text; tools are what let it query a search engine, run code, or read and write a database. We organize this capability interface in three parts: the tools exposed by the runtime and the protocol layer, most prominently the Model Context Protocol (MCP), that standardizes them (Section 4.3.1); the discovery mechanisms that keep a large tool surface usable (Section 4.3.2); and the skills an agent accumulates on top of them (Section 4.3.3). This interface matters more as the horizon grows,

原文第 26 页 | 中文翻译初稿

内存内容。持久内存主要存储两种状态。 _事实内存_存储智能体不应重新发现的声明性信息：用户偏好、项目规则、承诺、环境状态、文档或代码库事实以及外部系统的功能[228]。对于长期 运行支撑框架（Harness），这些事实支持连续性：用户级事实使跨会话的行为保持一致，而环境级事实阻止智能体重复重新推断稳定约束。实际示例包括 CLAUDE.md、Cursor Rules 和 AGENTS.md 等项目规则文件，它们在即时提示之外提供稳定的项目知识 [10,98,104]。用户记忆系统和图形结构存储通过维护配置文件、偏好和链接的事实记录进一步扩展了这一想法，这些记录可以随着时间的推移进行整合和审计 [88,199,229,267,507,731,874]。

_经验记忆_存储智能体从先前轨迹中学到的知识。它可能包含失败教训、成功案例、状态动作范例、可重用推理模式、计划、工作流程或可执行程序。共同的目的是通过检索先前结构代替重复的重新发现来降低长时程（Long-Horizon）成本。有些系统会存储口头反思或过去失败的教训；其他人维护经验池、可重用推理模板、工作流程记忆或跨域解决方案路径[44、262、359、462、544、587、674、750、807、853、868]。在最具操作性的一端，经验记忆可以被提炼成技能、脚本、API 包装器或类似工具的程序，如 Voyager 等技能库智能体 [446, 630]。一旦以这种形式存储，内存就开始与工具层重叠：运行支撑框架（Harness）可以检索、版本、验证、重用或废弃学习的过程，而无需更改模型权重。

内存操作。仅当 运行支撑框架（Harness）管理内存的写入、维护和读回上下文的方式时，持久存储才有用。 _内存写入_决定长期运行时应该保存什么。工具输出、部分计划、推理轨迹、用户反馈、观察结果和中间工件通常噪音太大，无法按原样存储。因此，运行支撑框架（Harness）使用总结、提炼或结构化构造来提取更高价值的单元，例如事实、摘要、案例、课程、工作流程或技能[221、407、464、470、674]。关键问题是应该保留什么级别的经验：事件、工作流程或可执行过程。

_内存维护_使存储在新信息到达时保持可用。随着时间的推移，记忆可能会变得陈旧、重复或不一致。 运行支撑框架（Harness）必须合并相关条目，在用户或环境发生变化时更新事实，并在检索变得嘈杂之前删除过时或低价值的材料。现有系统使用内存流、图表、链接笔记或其他结构化存储来支持此过程 [199,470,731]。该模型可能会建议要记住什么，但 运行支撑框架（Harness）决定是否插入、合并、链接、存档或忘记记忆。

_内存检索_决定存储的状态如何重新进入活动上下文。一些记忆在运行开始时加载，例如项目规则或用户首选项[664]

;其他的则在子任务边界、失败之后、工具使用之前或模型需要时检索。检索需要选择查询、要搜索的存储、排名或过滤方法以及要注入的内容量[488、568、610、694]。检索太少会使智能体重复错误或错过远程依赖关系；过多的检索会使记忆重新陷入上下文过载。因此，最近的工作将内存读/写视为一个策略问题，学习在推理过程中何时记住、更新或检索[778, 846]。

持久记忆也很难评估。最终任务的成功通常将内存质量与模型强度、工具可靠性和基准方差混合在一起。新的基准测试长期召回、跨会话一致性以及依赖于早期会话的任务，但最近的分析仍然发现饱和、判断敏感性以及更新成本和内存维护的有限覆盖范围[235,253,686]。对于 长时程（Long-Horizon）运行支撑框架（Harnesses）来说，关键问题不是智能体是否能够回忆起一个孤立的事实，而是当它改变下一个动作时是否能够保存、更新、检索和注入正确的内存。我们在第 6.6 节中收集评估资源。

4.3 工具、MCP 和技能

上下文和内存决定了智能体_知道_的内容，而该组件决定了它可以_做什么_。如果没有返回观察结果的工具，即使有能力的模型也只能发出文本；工具是让它查询搜索引擎、运行代码或读写数据库的工具。我们

将这个功能接口分为三个部分：运行时和协议层公开的工具，最突出的是模型上下文协议（MCP），它对它们进行了标准化（第 4.3.1 节）；保持大型工具表面可用的发现机制（第 4.3.2 节）；以及智能体在其之上积累的技能（第 4.3.3 节）。随着地平线的增长，这个界面变得更加重要，

26

because extended execution becomes a long chain of interdependent calls whose state must be carried, verified, and recovered across steps.

4.3.1 Tool Interfaces and Protocols

Function Calling. Tool interfaces let a language model act beyond its parametric knowledge. Foundational work established when, what, and how to call: Toolformer self-supervises where API calls should be inserted, Gorilla maps requests onto large API surfaces while suppressing hallucinated calls, and ToolLLM scales instruction-tuned tool use to thousands of real-world REST APIs [471, 492, 523]. Papers consolidate this line into a workflow of task planning, tool selection, tool calling, and response generation within a controller–tools–environment view [497, 724]. Single-call competence is by now well covered by mature benchmarks (BFCL, StableToolBench) and data-synthesis pipelines (APIGen, ToolACE) [198, 385, 392, 472].

Tool Protocols and MCP. The dominant engineering response to a growing, heterogeneous tool surface is protocolization. The Model Context Protocol (MCP) defines a host–client–server lifecycle with capability negotiation, authorization, and tool annotations, turning ad hoc integrations into a uniform, discoverable layer; its ecosystem, architecture, and security are covered in dedicated and broader protocol papers [100, 758]. Because it lets agents compose independently maintained servers and exposes the tool space uniformly, MCP also sets up the scaling techniques of Section 4.3.2. MCP-grounded benchmarks such as MCP-Universe reveal what single-call leaderboards miss: once tasks span many servers and tools in stateful sandboxes, even the strongest models succeed on only a minority of them, with input tokens rising rapidly over steps and failures becoming increasingly cognitive rather than syntactic [31, 350, 411, 700]. Beyond MCP, a broader wave of tool-use benchmarks probes schema discipline, state mutation, conversational tool legality, and messy real-world user behavior, from constrained tool-legality suites such as τ-bench to long-horizon, cross-application, and in-the-wild benchmarks such as Toolathlon [213, 322, 399, 539, 548, 762, 779].

Multi-turn Tool Use. Function calling and MCP standardize the *interface* of an individual call, but tool use at scale is inherently multi-turn: an agent chains many dependent calls, each conditioned on the previous observation. As these sequences grow to tens or hundreds of invocations, the bottleneck migrates from isolated-call correctness to sustaining the whole chain [724], and even strong agents complete only a fraction of realistic tasks, degrading through context saturation and memory decay as the turn count rises [322, 325, 406]. The limiting constraint is increasingly the state and context accumulated across turns rather than the mechanics of any single call. This is a property of the execution process rather than the call interface; the control-flow responses to it, namely decomposing, scheduling, and coordinating many turns, are orchestration, which we treat separately, so this subsection stays on the tool surface itself.

4.3.2 Active Tool Discovery

A fixed, prompt-resident tool list does not scale. Once catalogs reach thousands of tools, their descriptions crowd out the task, token cost tracks registry size rather than step demand, and selection accuracy degrades; the tool-retrieval saturation diagnosed above motivates active discovery [350, 411]. The remedy is to stop placing the entire surface in context and instead let the agent actively discover the few tools each step needs.

Tool Retrieval. The first move is retrieval-based discovery: index tool metadata and inject only the top-ranked schemas at each step. RAG-MCP, AnyTool, and MCP-Zero develop this idea with plain, hierarchical, and agent-initiated retrievers respectively, sharply reducing token use on large toolsets [140, 153, 162]. A complementary, supply-side approach generates MCP servers directly from OpenAPI specifications, and Anthropic's Tool Search Tool demonstrates the same retrieval principle in production, scaling to thousands of tools with large token savings and higher tool-selection accuracy [9, 423]. Pushing this further, DeepAgent makes tool search an action inside a single reasoning process, dynamically discovering tools from registries of tens of thousands as the task unfolds, while ToolGen removes the separate retriever altogether by representing each tool as a vocabulary token the model emits during generation [338, 647].

Programmatic Tool Use. A second move treats the tool surface itself as something the agent can read and program against. CodeAct treats executable code as the action space, letting a single step compose

原文第 27 页 | 中文翻译初稿

因为扩展执行变成了一长串相互依赖的调用，其状态必须跨步骤进行、验证和恢复。

4.3.1 工具接口和协议

函数调用。工具接口让语言模型的行为超越其参数知识。基础工作确定了何时调用、调用什么以及如何调用：Toolformer 自我监督应在何处插入 API 调用，Gorilla 将请求映射到大型 API 表面，同时抑制幻觉调用，而 ToolLLM 将指令调整工具的使用扩展到数千个现实世界的 REST API [471,492,523]。论文将这条线整合到控制器-工具-环境视图中的任务规划、工具选择、工具调用和响应生成的工作流程中[497, 724]。目前，成熟的基准测试（BFCL、StableToolBench）和数据合成管道（APIGen、ToolACE）已经很好地涵盖了单次调用能力[198、385、392、472]。

工具协议和 MCP。对不断增长的异构工具表面的主要工程响应是协议化。模型上下文协议 (MCP) 通过功能协商、授权和工具注释定义了主机-客户端-服务器生命周期，将临时集成转变为统一的、可发现的层；其生态系统、架构和安全性包含在专门且更广泛的协议文件中 [100, 758]。因为它允许智能体组成独立维护的服务器并统一公开工具空间，所以 MCP 还设置了第 4.3.2 节的扩展技术。 MCP-Universe 等基于 MCP 的基准测试揭示了单次调用排行榜所遗漏的内容：一旦任务跨越有状态沙箱中的许多服务器和工具，即使是最强大的模型也只能在其中的一小部分上取得成功，输入标记随着步骤和失败的增加而迅速上升，变得越来越认知而不是句法 [31,350,411,700]。除了 MCP 之外，更广泛的工具使用基准测试还探讨了模式规则、状态突变、会话工具合法性和混乱的现实世界用户行为，从受约束的工具合法性套件（如 τ -bench 到 长时程（Long-Horizon））、跨应用程序和野外基准测试（如 Toolathlon [213、322、399、539、548、762、 779]。

多回转工具的使用。函数调用和 MCP 标准化了单个调用的_接口_，但大规模工具的使用本质上是多轮的：一个智能体链接许多依赖的调用，每个调用都以先前的观察为条件。随着这些序列增长到数十或数百个调用，瓶颈从孤立调用的正确性迁移到维持整个链[724]，即使强大的智能体也只能完成现实任务的一小部分，随着轮数的增加，通过上下文饱和和记忆衰退而降级[322,325,406]。限制约束越来越多地是各个轮次累积的状态和上下文，而不是任何单个调用的机制。这是执行过程的属性而不是调用接口的属性；控制流对其的响应，即分解、调度和协调许多回合，是编排，我们单独处理，因此本小节保留在工具表面本身。

4.3.2 主动工具发现

固定的、提示驻留的工具列表无法扩展

。一旦目录达到数千个工具，它们的描述就会挤占任务，令牌成本跟踪注册表大小而不是步骤需求，并且选择准确性会降低；上面诊断出的工具检索饱和度激发了主动发现[350, 411]。补救措施是停止将整个表面置于上下文中，而是让智能体主动发现每个步骤所需的少数工具。

工具检索。第一步是基于检索的发现：索引工具元数据并在每一步仅注入排名靠前的模式。 RAG-MCP、AnyTool 和 MCP-Zero 分别使用简单的、分层的和智能体启动的检索器开发了这个想法，大大减少了大型工具集上的令牌使用[140,153,162]。一种补充性的供应方方法直接从 OpenAPI 规范生成 MCP 服务器，而 Anthropic 的工具搜索工具在生产中展示了相同的检索原理，可扩展到数千种工具，并具有大量令牌节省和更高的工具选择准确性 [9, 423]。进一步推动这一点，DeepAgent 使工具搜索成为单个推理过程中的一个动作，随着任务的展开，动态地从数以万计的注册表中发现工具，而 ToolGen 通过将每个工具表示为模型在生成过程中发出的词汇标记来完全删除单独的检索器 [338, 647]。

编程工具的使用。第二步将工具表面本身视为智能体可以读取和编程的东西。 CodeAct 将可执行代码视为操作空间，让单个步骤组成

27

calls, branch, and loop [656]; in MCP settings, code-execution patterns present servers as importable APIs and coordinate many calls inside one code block, cutting token consumption by orders of magnitude [9, 15], and Code-Execution MCP formalizes the shift from tool-by-tool orchestration to a single model-generated program, keeping context consumption nearly constant in the call count and tool-ecosystem size at the cost of a larger executable attack surface [154].

As coding agents strengthen, tool *selection* itself becomes a workspace activity: rather than picking from a static registry, the agent inspects tool definitions, documentation, or source on demand, using progressive disclosure to keep a large ecosystem cheap in context [730]. The same insight drives agentic search. Direct corpus interaction has a strong agent search a raw corpus with terminal tools (grep, find, read) instead of a top-k retriever, beating sparse, dense, and reranking baselines [355], and agentic keyword search with only grep-style tools matches vector-database RAG [566]. Applied to code, a terminal-only RL-trained agent localizes code as well as far larger models [578], turning discovery from a static input choice into an agent-driven search. Together, code execution and workspace-mediated discovery bound a tool surface that would otherwise grow with every call.

4.3.3 Skill Libraries

Once tools and workspace state are in place, the next layer of externalized capability is the *skill*: a reusable, parameterizable procedure that packages a validated combination of tools, control flow, and domain knowledge. Recent papers cast this as the step from episodic tool use to persistent, colleague-like agents under a "Workspace + Skill" paradigm, where a persistent workspace supplies state and a curated skill library supplies reusable operational knowledge [841]. We unpack it along three axes: what a skill is, how skills are acquired and reused, and how the library evolves.

Procedural Skills. On top of tool interfaces, the *skill* abstraction packages validated combinations of tools, control flow, and domain knowledge into named, reusable procedures. Rather than re-deriving a tool sequence each time, the agent recalls a procedure that already works, shifting the interface from tools to skills [888]. Papers frame skills as procedural artifacts with a representation, acquisition, retrieval, and evolution lifecycle, and as a layer orthogonal to MCP: where MCP supplies connectivity, skills supply the procedural knowledge for interpreting outputs and recovering from failure. The current landscape converges on three representation families: natural-language or Markdown procedures that condition a frozen agent, realized as a progressively disclosed file tree by Anthropic's Agent Skills [16]; executable code the agent can call, inspect, and edit, as in Voyager [630]; and structured relational libraries, where SkillNet weaves over 200,000 skills into a typed skill graph (depend_on, compose_with, ...) [360]. The shared move is from flat, similarity-retrieved entries toward curated, relationally organized procedures owned by the harness as first-class assets.

Skill Acquisition. Work on skill acquisition and reuse shows the payoff. Voyager, the canonical instance, discovers executable Minecraft skills and reloads them on later tasks, and later systems extend the idea to inference-time tool synthesis, cumulative scientific-reasoning skills, and skills for unfamiliar software [4, 237, 400, 577, 630]. Related systems evaluate higher-order tool-composition skills and convert episodic narratives into executable skills with explicit activation and termination conditions [65, 430]. Across these systems the recurring benefit is lower token cost, higher reliability, and shorter effective horizons through reuse.

Skill Library Evolution. What distinguishes a skill from a tool is that the library is a first-class object the harness owns and curates: skills can be versioned, retrieved, refined, retired, and reused, so where tools supply flexibility, skills supply efficiency and reliability [340]. Because the library lives outside the weights, it can evolve without gradient updates: Memento treats skill acquisition as memory-based reinforcement learning, Alita generates and reuses MCP servers on demand, and observability-driven harness evolution finds most of its gains in rewriting the tool, middleware, and memory layers rather than the system prompt [456, 495, 881]. A first line optimizes the skill text directly: SkillOpt trains a frozen agent by editing its natural-language skill document, accepting a change only when it improves a held-out validation score [757]. A second structures the library as a graph rather than a flat list: the prerequisite-skill graph originates in data-efficient training (Skill-it) [63], and is imported into agent libraries by SkillGraph, which co-evolves a typed skill graph with the policy via reinforcement learning [341], by GraSP, which compiles retrieved skills into an executable

原文第 28 页 | 中文翻译初稿

调用、分支和循环[656]；在 MCP 设置中，代码执行模式将服务器呈现为可导入的 API，并协调一个代码块内的许多调用，从而将令牌消耗减少了几个数量级 [9, 15]，而代码执行 MCP 正式实现了从逐个工具编排到单个模型生成的程序的转变，以更大的可执行攻击面为代价，使上下文消耗在调用计数和工具生态系统大小中保持几乎恒定[154]。

随着编码智能体的增强，工具_选择_本身就变成了一项工作空间活动：智能体不是从静态注册表中进行选择，而是按需检查工具定义、文档或源，使用渐进式披露来保持大型生态系统在上下文中的廉价性[730]。同样的洞察力推动着智能体搜索。直接语料库交互具有强大的智能体搜索原始语料库的终端工具（grep、find、read）而不是 top-k 检索器，击败稀疏、密集和重新排名基线 [355]，并且仅使用 grep 式工具匹配矢量数据库 RAG [566] 的智能体关键字搜索。应用于代码时，仅终端 RL 训练的智能体可以本地化代码以及更大的模型 [578]，将发现从静态输入选择转变为智能体驱动的搜索。代码执行和工作区介导的发现一起限制了工具表面，否则该工具表面将随着每次调用而增长。

4.3.3 技能库

一旦工具和工作区状态就位，下一层外部化功能就是_skill_：一个可重用、可参数化的过程，它将工具、控制流和领域知识的经过验证的组合打包。最近的论文将其视为“工作空间+技能”范式下从情景工具使用到持久的、类似同事的智能体的一步，其中持久的工作空间提供状态，而策划的技能库提供可重用的操作知识[841]。我们沿着三个轴展开它：什么是技能、如何获取和重用技能以及库如何发展。

程序技能。在工具接口之上，skill 抽象将经过验证的工具、控制流和领域知识的组合打包到命名的、可重用的过程中。智能体不是每次都重新派生工具序列，而是调用已经有效的程序，将界面从工具转移到技能[888]。论文将技能描述为具有表示、获取、检索和演化生命周期的程序工件，并作为与 MCP 正交的层：MCP 提供连接性，技能则提供用于解释输出和从故障中恢复的程序知识。当前的情况集中在三个表示系列上：自然语言或 Markdown 程序，用于调节冻结智能体，由 Anthropic 的智能体技能实现为逐步公开的文件树 [16]；智能体可以调用、检查和编辑的可执行代码，如 Voyager [630] 中所示；以及结构化关系库，其中 SkillNet 将超过 200,000 种技能编织成类型化技能图（depend_on、compose_with、……）[360]。共同的转变是从扁平的、相似性检索的条目转向 运行支撑框架（Harness） 作为一流资产拥有的策划的、相关组织的程序。

技能获取。技能获取和重用方面的工作显示出回报

。 Voyager 是典型的实例，它发现可执行的 Minecraft 技能并在以后的任务中重新加载它们，后来的系统将这一想法扩展到推理时间工具合成、累积科学推理技能以及不熟悉软件的技能 [4, 237, 400, 577, 630]。相关系统评估高阶工具组合技能，并将情景叙述转换为具有明确激活和终止条件的可执行技能[65, 430]。在这些系统中，经常性的好处是更低的代币成本、更高的可靠性以及通过重用缩短有效期限。

技能库的演变。技能与工具的区别在于，库是 运行支撑框架（Harness） 拥有和管理的一流对象：技能可以进行版本控制、检索、细化、退役和重用，因此工具提供灵活性，技能则提供效率和可靠性[340]。因为该库位于权重之外，所以它可以在没有梯度更新的情况下进化：Memento 将技能获取视为基于内存的 强化学习（Reinforcement Learning），Alita 按需生成和重用 MCP 服务器，而可观察性驱动的 运行支撑框架（Harness） 进化发现其大部分收益来自于重写工具、中间件和内存层，而不是系统提示符[456,495,881]。第一行直接优化技能文本：SkillOpt 通过编辑其自然语言技能文档来训练冻结智能体，仅当它提高了保留的验证分数时才接受更改[757]。第二个将库构建为图表而不是平面列表：先决条件技能图源自数据高效训练（Skill-it）[63]，并通过 SkillGraph 导入到智能体库中，SkillGraph 通过 GraSP 通过 强

化学习（Reinforcement Learning） [341] 与策略共同演化类型化技能图，GraSP 将检索到的技能编译为可执行文件

28

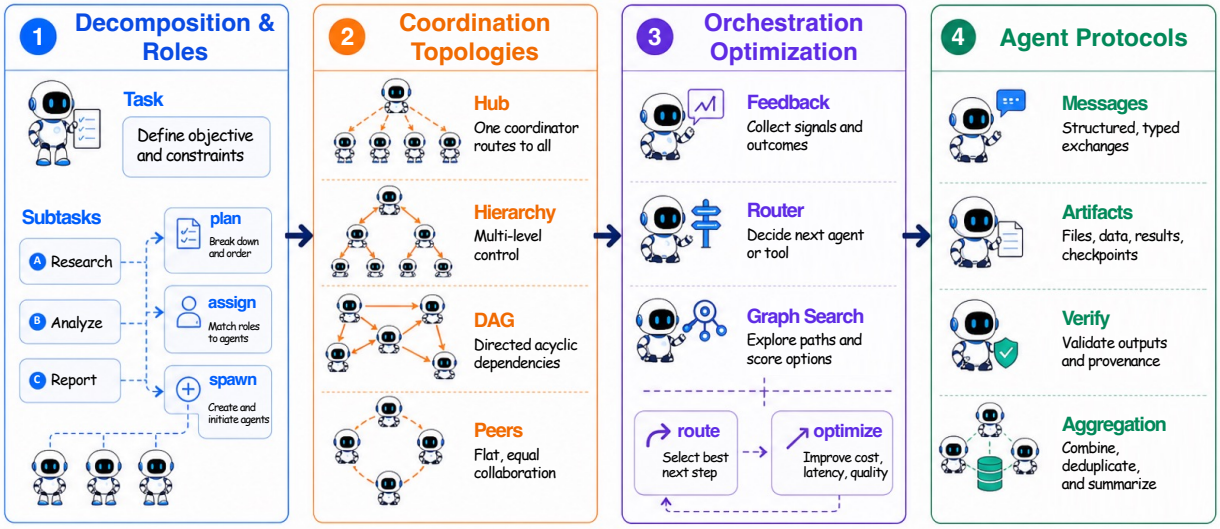


Figure 9 Overview of agent orchestration. Tasks are decomposed into roles, coordinated through different topologies, optimized via feedback-driven routing, and integrated through communication and aggregation protocols.

DAG with locality-bounded repair [707]. A growing line couples skill libraries directly with reinforcement learning: Skill0, for instance, trains with an in-context curriculum that progressively withdraws skill files until the policy runs skill-free, beginning to fold the library into the weights themselves [405]. This library-level self-improvement is the externalized form of evolution; its internalized counterpart is taken up in Section 5.

4.4 Orchestration

A single agent loop can attend to only one thing at a time; many long-horizon tasks are better served by composing several loops, whether as subtasks of one agent or as a team of specialized agents. Multi-agent collaboration can be described along a few dimensions: the participating agents, the interaction type (cooperation, competition, or a mix), the communication structure, the coordination strategy, and whether the arrangement is static or dynamic [197, 611]. We adopt a pragmatic version of this taxonomy and ask four questions: how work is decomposed and assigned, what topology the resulting computation takes, whether that topology is fixed or learned, and how the agents communicate. Orchestration is the externalized form of long-horizon control structure [468]: once a task outgrows what a single loop can sustain, how the work is split and coordinated often affects end-to-end success more than swapping the underlying model does. Composition is not free, however. A systematic study of multi-agent failures attributes most breakdowns to under-specified roles, inter-agent misalignment, and missing verification rather than to weak base models, a reminder that orchestration must be designed rather than merely assembled [48]. The overall framework of agent orchestration is shown in Figure 9.

4.4.1 Decomposition and Roles

The first decision is how a long task is split into units and assigned to roles. The earliest harnesses fix this split statically, hard-coding the roles and their interaction pattern, whether as software-team standard operating procedures, planner–executor role-play, staged role-conditioned dialogues, or a general programmable substrate in which roles and their conversations are first-class, as in frameworks such as software-team SOPs [219], role-play collaboration [314], staged dialogues [485], and a programmable substrate [692]. A predefined subtask graph is simple but brittle, since an early failure propagates downstream and a hand-coded role set scales poorly; this motivates dynamic decomposition that revises the split from feedback and can spawn a subtask-tailored subagent on the fly, as in TDAG [660]. Either way the harness must keep the split well-formed: agent-oriented planning, for example, holds the decomposing meta-agent to solvability, completeness, and non-redundancy, re-planning unresolvable or duplicated subtasks before any agent runs [305]. How fine to

原文第 29 页 | 中文翻译初稿

分解和协调编排
1 2 3 4 智能体协议
角色拓扑优化
任务中心反馈消息
定义目标和约束一个协调器路由到所有收集信号和结果结构化、类型化交换
子任务计划层次结构路由器工件
分解多级决定下一个智能体文件、数据、结果
研究和订单控制或工具检查点
分配
B 分析 匹配角色 DAG 图搜索 验证
到智能体 定向非循环 探索路径并验证输出
依赖项评分选项和出处
C 报告生成
创建和
启动智能体 对等路由优化 聚合
扁平、平等协作 选择最佳下一步 改进成本、延迟、质量 合并、重复数据删除
和总结

图 9 智能体编排概述。任务被分解为角色，通过不同的拓扑进行协调，通过反馈驱动的路由进行优化，并通过通信和聚合协议进行集成。

具有局部限制修复的 DAG [707]。不断增长的产品线将技能库直接与 强化学习（Reinforcement Learning） 结合起来：例如，Skill0 使用上下文课程进行训练，该课程逐渐撤回技能文件，直到策略无技能运行，开始将库折叠到权重本身中 [405]。这个库级的自我改进（Self-Improvement）就是进化的外化形式；第 5 节讨论了其内化对应部分。

4.4 编排

单个智能体循环一次只能处理一件事；许多 长时程（Long-Horizon） 任务通过组合多个循环可以更好地服务，无论是作为一个智能体的子任务还是作为一组专门智能体。多主体协作可以从几个维度进行描述：参与主体、交互类型（合作、竞争或混合）、通信结构、协调策略以及安排是静态还是动态[197, 611]。我们采用这种分类法的实用版本，并提出四个问题：如何分解和分配工作、计算结果采用什么拓扑、该拓扑是固定的还是学习的，以及智能体如何通信。编排是 长时程（Long-Horizon） 控制结构 [468] 的外化形式：一旦任务超出了单个循环的承受能力，工作的拆分和协调方式通常比交换底层模型更能影响端到端的成功。然而，构图并不是免费的。对多智能体故障的系统研究将大多数故障归因于角色指定不足、智能体间错位和验证缺失，而不是基础模型薄弱，这提醒我们必须设计编排而不是仅仅组装[48]。 Agent编排的总体框架如图9所示。

4.4.1 分解与角色

第一个决定是如何将长任务分成单元并分配给角色

。最早的 运行支撑框架（Harnesses） 静态地解决了这种分裂，对角色及其交互模式进行硬编码，无论是作为软件团队标准操作程序、计划者-执行者角色扮演、分阶段角色条件对话，还是角色及其对话是一流的

通用可编程基底，如软件团队 SOP [219]、角色扮演协作 [314]、分阶段对话 [485] 和可编程基底等框架中 [692]。预定义的子任务图简单但脆弱，因为早期故障会向下游传播，并且手工编码的角色集扩展性很差；这会激发动态分解，修正反馈的分割，并可以动态生成子任务定制的子智能体，如 TDAG [660] 中所示。无论哪种方式，运行支撑框架（Harness） 都必须保持分割的良好形式：例如，面向智能体的规划使分解的元智能体保持可解决性、完整性和非冗余性，在任何智能体运行之前重新规划无法解决或重复的子任务 [305]。多好

29

decompose is then a granularity trade-off: too coarse and a subtask overwhelms its agent, too fine and coordination overhead dominates, and that overhead grows with the horizon. Much of the benefit usually attributed to role personas in fact comes from the structural decomposition they induce, namely parallelism, context isolation, and tool specialization, rather than from the personas themselves [48].

4.4.2 Coordination Topologies

Once units and roles are set, the way agents are wired together fixes the shape of the computation. These topologies fall into centralized forms built around a hub or orchestrator, decentralized or distributed forms such as peer debate and dynamic graphs, and hierarchical forms [611]. Centralized designs, exemplified by Magentic-One, place a hub or orchestrator that drives a team, in some variants stacking layers that successively refine one another or dynamically recruiting collaborators [69, 157, 638]. Decentralized designs instead let peers select and prune one another at inference time or scale collaboration over large network structures, as in DyLAN and MacNet [390, 486]. The dominant programmable abstraction across both is the directed acyclic graph: LangGraph defines workflows as DAGs whose nodes are agents or functions and whose edges carry typed state, while Agint and Flash-Searcher compile instructions into typed DAGs and recast web research as a parallel DAG to cut step counts [90, 491]. On long-horizon coding tasks this design space can matter more than the choice of model: on SWE-bench Verified, a fixed frontier model varies far more across scaffolds than different models do under a single fixed scaffold [468].

The wirings above coordinate agents toward a shared objective, but a complementary family makes the interaction itself *competitive*, pitting two roles in a GAN-like loop where one proposes ever-harder problems and the other strives to solve them, so the pair co-evolves an automatic curriculum with no external data. R-Zero instantiates a Challenger rewarded for posing tasks at the edge of a Solver's competence and a Solver rewarded for cracking them, while Absolute Zero folds both roles into a single model that proposes and solves verifiable tasks in one self-play loop [233, 854]; earlier, SPAG showed that setting a model against a copy of itself in a two-player adversarial language game steadily sharpens reasoning [84]. Because this adversarial coordination improves the participants through training rather than routing a fixed roster of models, its payoff is ultimately internalized (Section 5); yet the arrangement itself remains a distinctly competitive orchestration pattern, the counterpart to the cooperative composition that dominates the rest of this section.

4.4.3 Orchestration Optimization

A topology can be hand-designed or searched. AFlow treats workflow design as program-space search with Monte-Carlo tree search, and GPTSwarm represents a multi-agent system as an optimizable graph and learns both node prompts and edges [816, 897]. A related line learns who executes each unit. At the model level, routers learn to select among models under performance, cost, latency, and congestion constraints [447, 810, 880]; at the agent level, MasRouter jointly chooses collaboration mode, roles, and the model per role, while other systems spawn subagents on the fly or pair central planning with a tool–environment–agent protocol [519, 791, 828]. Taken to its limit, the coordination structure itself is allowed to emerge: a central orchestrator can be trained with reinforcement learning to induce compact coordination at lower cost, and related systems refine functionality and coordination from scratch or couple shared memory with asynchronous execution [107, 496, 836]; at scale, mixed self-organizing coordination can even outperform hand-designed hierarchies [127]. Like skill-library growth (Section 4.3.3), these black-box optimizations of the orchestration graph are an externalized form of evolution.

4.4.4 Agent Protocols

Finally, composed agents must exchange messages reliably. The A2A protocol foregrounds Agent Cards, Tasks, and Artifacts over JSON-RPC with streaming, IBM's Agent Communication Protocol offers a lighter REST-native alternative, and MCP, ACP, A2A, and ANP have been compared across architecture, discovery, communication, and security [97, 144, 512, 758]. Framework-level messaging conventions coexist with richer schemes that add decentralized identity and verifiable credentials, switch between structured protocols and natural language according to communication frequency, or perform decentralized discovery over encrypted peer-to-peer channels [219, 314, 422, 501, 504, 692]. A complementary harness duty is aggregation: when parallel trials or subagents run, their outputs must be reduced to a single result, through best-of-N or

原文第 30 页 | 中文翻译初稿

分解是一种粒度权衡：太粗，子任务会压垮它的智能体，太细，协调开销占主导地位，而且开销会随着时间的推移而增长。通常归因于角色角色的大部分好处实际上来自它们引发的结构分解，即并行性、上下文隔离和工具专业化，而不是来自角色本身[48]。

4.4.2 协调拓扑

一旦设置了单元和角色，智能体连接在一起的方式就确定了计算的形状。这些拓扑分为围绕中心或协调器构建的集中式形式、分散式或分布式形式（例如同行辩论和动态图）以及分层形式[611]。以 Magentic-One 为代表的集中式设计放置了一个驱动团队的中心或协调器，在某些变体中，堆叠层不断完善彼此或动态招募协作者 [69,157,638]。相反，去中心化设计让同行在推理时相互选择和修剪，或者在大型网络结构上进行大规模协作，如 DyLAN 和 MacNet [390, 486]。两者中占主导地位的可编程抽象是有向无环图：LangGraph 将工作流程定义为 DAG，其节点是智能体或函数，其边缘携带类型化状态，而 Agint 和 Flash-Searcher 将指令编译为类型化 DAG，并将 Web 研究重新构建为并行 DAG 以减少步骤计数 [90, 491]。在 长时程（Long-Horizon） 编码任务中，这个设计空间比模型的选择更重要：在 SWE-bench Verified 上，固定前沿模型在支架之间的变化比单个固定支架下不同模型的变化要大得多 [468]。

上面的连线协调智能体实现共同的目标，但互补的家庭使交互本身具有竞争性，在类似 GAN 的循环中设置两个角色，一个提出更难的问题，另一个努力解决这些问题，因此两人共同开发了一个没有外部数据的自动课程。 R-Zero 实例化了一个因在解决者能力边缘提出任务而获得奖励的挑战者和因破解这些任务而获得奖励的解决者，而绝对零将这两个角色折叠成一个模型，在一个自我游戏循环中提出并解决可验证的任务 [233, 854]；早些时候，SPAG 表明，在两人对抗性语言游戏中针对自身副本设置模型可以稳定地提高推理能力 [84]。因为这种对抗性协调通过训练而不是路由固定的模型名册来改善参与者，所以它的回报最终是内部化的（第 5 节）；然而，编曲本身仍然是一种明显竞争性的编曲模式，与主导本节其余部分的合作作曲相对应。

4.4.3 编排优化

拓扑可以手工设计或搜索。 AFlow 将工作流程设计视为使用蒙特卡罗树搜索的程序空间搜索，而 GPTSwarm 将多智能体系统表示为可优化的图，并学习节点提示和边缘 [816, 897]。相关行了解谁执行每个单元。在模型级别，路由器学习在性能、成本、延迟和拥塞约束下的模型之间进行选择[447,810,880]

;在智能体级别，MasRouter 共同选择协作模式、角色和每个角色的模型，而其他系统则动态生成子智能体或将中央规划与工具-环境-智能体协议配对[519,791,828]。考虑到其极限，协调结构本身可以出现：可以使用 强化学习（Reinforcement Learning） 来训练中央编排器，以较低的成本诱导紧凑的协调，相关系统从头开始完善功能和协调，或者将共享内存与异步执行结合起来[107,496,836]；在规模上，混合自组织协调甚至可以胜过手工设计的层次结构[127]。与技能库增长（第 4.3.3 节）一样，编排图的这些黑盒优化是进化的一种外化形式。

4.4.4 智能体协议

最后，组成的智能体必须可靠地交换消息。 A2A 协议通过流式处理 JSON-RPC 来实现智能体卡、任务和工件，IBM 的智能体通信协议提供了更轻量级的 REST 本机替代方案，并且在架构、发现、通信和安全性方面对 MCP、ACP、A2A 和 ANP 进行了比较 [97,144,512,758]。框架级消息传递约定与更丰富的方案共存，这些方案添加去中心化身份和可验证凭证，根据通信频率在结构化协议和自然语言之间切换，或者通过加密的点对点通道执行去中心化发现[219、314、422、501、504、692]。 运行支撑框架（Harness）

的补充职责是聚合：当并行试验或子智能体运行时，它们的输出必须通过 best-of-N 或

30

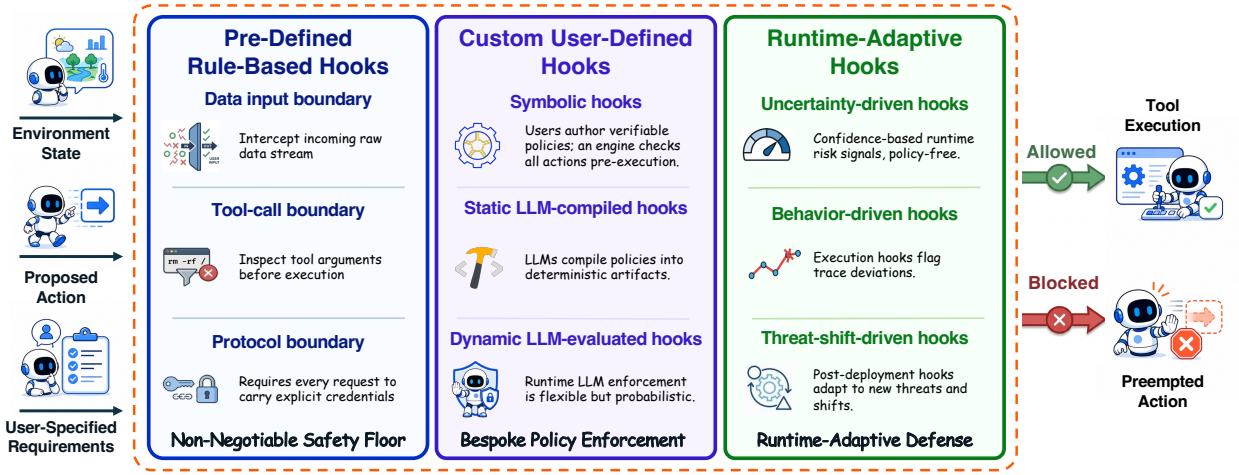


Figure 10 Overview of hooks and middleware. We categorize hooks into three paradigms based on the origin and adaptiveness of their safety criteria: pre-defined rules, user-defined policies, and runtime-adaptive signals.

voting for verifiable tasks [323, 442, 659] and through confidence-aware agentic aggregation for open-ended ones [296, 298]. This shift from prescribed hierarchies toward protocol-mediated coordination stands in productive tension with the deterministic governance we turn to next, and remains one of open problems.

4.5 Hooks and Middleware

The preceding subsections establish the full execution infrastructure of an agent harness: loops govern control flow, context and memory manage information state, tools extend capabilities, and orchestration composes agents into collaborative systems. Yet the very expressiveness that makes these mechanisms powerful also amplifies the risk surface: an agent equipped with file-system access, API credentials, and multi-step autonomy can cause irreversible harm through hallucinated commands or adversarial jailbreaks [1]. Prompt-level instructions alone are insufficient to constrain such behavior, as they remain vulnerable to the same failure modes they aim to prevent. Recent architectures [421, 529] have therefore shifted toward programmable boundaries enforced at the harness level, which we collectively term *hooks*. Hooks perform enforcement, gating the action path with allow, block, or hand-off decisions, whether through rule-based evaluation or model judgment. As illustrated in Figure 10, based on who defines the safety criterion and how the trigger condition is determined, we identify three complementary paradigms: (1) pre-defined rule-based hooks enforce universal safety invariants hard-coded by framework designers; (2) custom user-defined hooks enforce domain-specific policies authored by deployers; and (3) runtime-adaptive hooks derive their trigger conditions from runtime signals such as model uncertainty and behavioral patterns rather than from any predetermined specification.

4.5.1 Pre-defined Rule-based Hooks

At the foundation of agent safety, harnesses employ immutable, hard-coded middleware to intercept or block actions that violate universal safety invariants without requiring user configuration. These hooks are triggered at fixed lifecycle points such as immediately before a shell command or API call, and because they operate on the action path rather than the reasoning path, they run outside the LLM's context window and are therefore less directly exposed to prompt-level manipulation; their coverage, however, still depends on policy completeness and adversarial robustness, since encoded payloads, obfuscated arguments, or gaps in the policy can still slip past them. Concretely, pre-defined hooks manifest at three successive boundaries along the agent execution pipeline:

- **Input Boundary.** Operating before data ingestion by the core language model, this mechanism intercepts incoming raw data streams, including user instructions, prompt templates, and retrieved external context. Its primary objective is to filter out adversarial prompt injections, jailbreaks, and malformed

原文第 31 页 | 中文翻译初稿

预定义自定义用户定义运行时自适应
基于规则的钩子钩子钩子
数据输入边界符号钩子不确定性驱动的钩子工具
环境状态拦截传入的原始数据流用户编写可验证的策略；引擎检查基于置信度的运行时风险信号，与策略无关。允许执行
执行前的所有操作。
工具调用边界静态 LLM 编译的挂钩行为驱动的挂钩
检查工具参数 LLM 将策略编译为执行挂钩标志
在执行确定性工件之前建议。跟踪偏差。被阻止
操作
协议边界动态LLM评估挂钩威胁转移驱动挂钩
要求每个请求携带显式凭据运行时LLM执行是灵活的但具有概率性。部署后挂钩适应新的威胁和变化。
PreemptedAction
用户指定
要求 不可协商的安全底线 定制策略执行 运行时自适应防御

图 10 挂钩和中间件概述。我们根据钩子安全标准的起源和适应性将钩子分为三种范式：预定义规则、用户定义策略和运行时自适应信号。

对可验证的任务进行投票 [323, 442, 659]，并通过对开放式任务进行信任感知的智能体聚合 [296, 298]。这种从规定的层次结构向协议介导的协调的转变与我们下一步转向的确定性治理存在着富有成效的紧张关系，并且仍然是一个悬而未决的问题。

4.5 钩子和中间件

前面的小节建立了智能体 运行支撑框架（Harness） 的完整执行基础架构：循环控制控制流，上下文和内存管理信息状态，工具扩展功能，编排将智能体组成协作系统。然而，使这些机制变得强大的表现力也放大了风险面：配备文件系统访问、API 凭证和多步骤自主权的智能体可能会通过幻觉命令或对抗性越狱造成不可逆转的伤害 [1]。仅提示级指令不足以限制此类行为，因为它们仍然容易受到其旨在防止的相同故障模式的影响。因此，最近的架构 [421, 529] 已转向在 运行支撑框架（Harness） 级别强制实施的可编程边界，我们统称为 hooks 。钩子执行强制执行，通过允许、阻止或移交决策来控制操作路径，无论是通过基于规则的评估还是模型判断。如图 10 所示，根据谁定义了安全标准以及如何确定触发条件，我们确定了三种互补范例：（1）预定义的基于规则的钩子强制框架设计者硬编码的通用安全不变量；(2) 自定义用户定义的钩子强制执行部署者编写的特定于域的策略；(3)运行时自适应钩子从运行时信号（例如模型不确定性和行为模式）而不是从任何预定的规范中得出触发条件。

4.5.1 预定义的基于规则的钩子

在智能体安全的基础上，运行支撑框架（Harnesses） 采用不可变的硬编码中间件来拦截或阻止违反通用安全不变量的操作，而无需用户配置。这些钩子在固定的生命周期点触发，例如在 shell 命令或 API 调用之前，并且因为它们在操作路径而不是推理路径上操作，所以它们在 LLM 的上下文窗口之外运行，因此较少直接暴露于提示级操作；然而，它们的覆盖范围仍然取决于策略的完整性和对抗性的鲁棒性，因为编码的有效负载、模糊的论点或策略中的漏洞仍然可能会被忽略。具体来说，预定义的钩子出现在智能体执行管道的三个连续边界处：

- 输入边界。该机制在核心语言模型摄取数据之前进行操作，拦截传入的原始数据流，包括用户指令、提示模板和检索的外部上下文。其主要目标是过滤掉对抗性提示注入、越狱和格式错误的攻击

31

system directives designed to compromise the model's behavioral alignment. Systems implementing this boundary [529] deploy independent, token-level or semantic static filters to reject non-compliant inputs before they can influence the model's internal attention state.

- **Tool-call Boundary.** Acting as a pre-execution firewall, this mechanism inspects tool arguments before execution to detect dangerous options (e.g., `rm -rf`) and common attack patterns such as code injection or unauthorized file access. AEGIS [783] instantiates this design by interposing a policy engine on every tool call, validating arguments against a curated set of known vulnerability patterns, including path traversal, with negligible latency overhead.
- **Protocol Boundary.** Enforced at the inter-component communication layer, this boundary requires every request to carry explicit credentials and receive per-action authorization, ensuring no message is implicitly trusted regardless of its origin. Representative systems include Authenticated Workflows [503], which attaches cryptographic tokens to protocol messages for authenticity verification, and MagenticUI [441], which mandates explicit user approval before dispatching agent actions.

These three boundaries form independent, complementary defense layers: because each targets a distinct attack surface, the failure of any single layer does not compromise the others [421]. This defense-in-depth property is further strengthened by architectural isolation techniques such as IsolateGPT [698], whose hub-and-spoke design routes all tool interactions through an isolated mediator to prevent cross-tool information leakage. Together, these pre-defined mechanisms establish a non-negotiable safety floor that holds irrespective of the model's reasoning quality, the user's configuration choices, or the particular task being executed.

4.5.2 Custom User-defined Hooks

Beyond immutable rules, modern harnesses expose programmable interfaces that allow developers to define domain-specific policies enforced at runtime. While pre-defined hooks target universally dangerous operations, custom hooks address the far broader space of application-specific requirements: **workflow constraints, regulatory compliance, access control scoping, and safety boundaries that vary across deployment contexts.** The defining characteristic is the locus of policy authorship: in every approach below, it is the user or developer, not the framework designer, who specifies the substantive safety criterion. "User-defined" thus refers to who writes the policy, not how it is enforced; the enforcement mechanism ranges from deterministic symbolic engines to probabilistic model evaluators, but the policy content originates with the deployer in all cases.

Existing approaches vary along two orthogonal dimensions: (i) how the user specifies a policy (formal language vs. natural language), and (ii) what enforces it at runtime (a deterministic symbolic engine vs. a probabilistic model). The interplay of these two dimensions yields three distinct categories. Formal-language policies pair naturally with symbolic enforcement. Natural-language policies split by whether an LLM translates them into executable artifacts offline and exits the enforcement path, or remains on the critical path to evaluate compliance at every step. The resulting spectrum trades specification effort against enforcement strength.

(1) Symbolic Hooks. Users author policies in a formal, machine-verifiable language, and a deterministic engine evaluates every action against the policy before execution. Because the LLM is entirely absent from the enforcement path, compliance guarantees are hard: a compliant action is provably within policy, and no prompt injection can subvert the check.

Several systems design event-triggered DSLs whose predicate guards and enforcement actions are executed by lightweight symbolic runtimes [509, 633]. Others go further by translating temporal-logic or JSON-schema specifications into first-order constraints and using SMT solvers or constrained token generation to make non-compliant tool calls physically impossible to emit [265, 542]. For multi-agent settings, PCAS [466] models inter-agent state as a causal dependency graph and enforces Datalog-derived policies via Differential Datalog with a reference monitor intercepting every action. A complementary line draws on operating-system security primitives such as capability labels and cryptographic audit records to enforce information-flow control and provenance tracking at the tool-call boundary [109, 617]. Across all variants, the enforcement engine is a formally deterministic program whose correctness can be audited independently of the LLM it guards.

(2) Static LLM-compiled Hooks. A second paradigm accepts policies written in natural language or semi-structured documents but compiles them into executable artifacts that are then enforced deterministically

原文第 32 页 | 中文翻译初稿

旨在损害模型行为一致性的系统指令。实现此边界的系统[529]部署独立的、令牌级或语义静态过滤器，以在不合规的输入影响模型的内部注意力状态之前拒绝它们。

- 工具调用边界。作为预执行防火墙，该机制在执行前检查工具参数，以检测危险选项（例如 rm -rf）和常见攻击模式，例如代码注入或未经授权的文件访问。 AEGIS [783] 通过在每个工具调用上插入一个策略引擎来实例化这种设计，根据一组精心设计的已知漏洞模式验证参数，包括路径遍历，延迟开销可以忽略不计。

- 协议边界。该边界在组件间通信层强制执行，要求每个请求都携带显式凭据并接收每个操作授权，确保任何消息都不会被隐式信任，无论其来源如何。代表性系统包括 Authenticated Workflows [503]，它将加密令牌附加到协议消息以进行真实性验证；以及 MagenticUI [441]，它在分派智能体操作之前要求明确的用户批准。

这三个边界形成独立、互补的防御层：因为每个层都针对不同的攻击面，所以任何单个层的失败都不会损害其他层[421]。这种纵深防御特性通过 IsolateGPT [698] 等架构隔离技术得到进一步加强，其中心辐射型设计通过隔离中介路由所有工具交互，以防止跨工具信息泄漏。这些预定义的机制共同建立了一个不可协商的安全底线，无论模型的推理质量、用户的配置选择或正在执行的特定任务如何。

4.5.2 自定义用户定义的Hook

除了不可变的规则之外，现代 运行支撑框架（Harnesses） 还公开了可编程接口，允许开发人员定义在运行时强制执行的特定于域的策略。预定义的钩子针对的是普遍危险的操作，而自定义钩子则解决了更广泛的特定于应用程序的需求：工作流约束、法规遵从性、访问控制范围以及因部署环境而异的安全边界。定义特征是策略作者的所在地：在下面的每种方法中，指定实质性安全标准的是用户或开发人员，而不是框架设计者。因此，“用户定义”是指由谁编写策略，而不是如何执行策略；执行机制的范围从确定性符号引擎到概率模型评估器，但在所有情况下策略内容均源自部署者。

现有方法在两个正交维度上有所不同：（i）用户如何指定策略（形式语言与自然语言），以及（ii）在运行时执行策略的内容（确定性符号引擎与概率模型）。这两个维度的相互作用产生了三个不同的类别。正式语言策略与象征性执行自然配对

。自然语言策略的划分取决于法学硕士是否将其离线翻译为可执行工件并退出执行路径，或者保留在关键路径上以评估每一步的合规性。由此产生的频谱将规范工作与执行力度进行交换。

(1) 符号钩子。用户以正式的、机器可验证的语言编写策略，确定性引擎在执行之前根据策略评估每个操作。由于 LLM 完全不存在于执行路径中，因此合规性保证很困难：合规性操作可证明在策略范围内，并且任何即时注入都无法破坏检查。

一些系统设计了事件触发的 DSL，其谓词保护和强制执行操作由轻量级符号运行时执行 [509, 633]。其他人则更进一步，将时间逻辑或 JSON 模式规范转换为一阶约束，并使用 SMT 求解器或约束令牌生成来使不合规的工具调用在物理上不可能发出 [265, 542]。对于多智能体设置，PCAS [466]将智能体间状态建模为因果依赖图，并通过差异数据记录强制执行数据记录派生策略，并使用拦截每个操作的参考监视器。一条补充线利用操作系统安全原语（例如功能标签和加密审计记录）来在工具调用边界实施信息流控制和来源跟踪[109, 617]。在所有变体中，执行引擎都是一个形式确定性程序，其正确性可以独立于它所保护的法学硕士进行审核。

(2) 静态 LLM 编译的 Hook。第二种范式接受用自然语言或半结构化文档编写的策略，但将它们编译成可执行的工件，然后确定性地强制执行

32

or quasi-deterministically. The LLM participates at compile time but is absent from the enforcement path, yielding guarantees bounded by compilation quality but independent of runtime model reasoning.

The compilation targets range from executable guardrail code that runs deterministically to admit or block each action [708], to probabilistic rule circuits derived from regulatory documents and verified by formal model checkers [80], to formally verified policy code generated from Hoare-triple specifications and validated by an offline verifier before deployment [432]. The common thread is a two-phase architecture: an LLM-powered compilation step that transforms human-readable intent into machine-executable policy, followed by a symbolic or quasi-symbolic enforcement step. Errors can be introduced at compile time, but once compiled, enforcement proceeds without further LLM involvement.

(3) Dynamic LLM-evaluated Hooks. A third paradigm also starts from user-authored natural-language policies but, unlike static compilation, keeps the LLM on the enforcement critical path: the model directly evaluates each action or trajectory against the user-supplied policy at runtime. The deployer must provide domain-specific safety criteria without which the system has nothing to enforce, so the "user-defined" property holds in the same sense as the preceding categories. This maximizes expressiveness and minimizes specification effort at the cost of replacing hard guarantees with probabilistic ones.

One approach deploys a guardian LLM that interprets natural-language policies with contextual reasoning across multiple control points [285]. A classifier-based alternative fine-tunes or prompts language models as configurable safety classifiers whose risk taxonomies or policy definitions can be adapted without retraining [244, 274]. A third variant composes multiple heterogeneous scanners into a layered pipeline that users can extend with custom scanners tailored to their threat model [87]. Because the enforcer is itself a language model, these approaches can handle nuanced, context-dependent policies that resist formalization, but compliance is probabilistic and susceptible to the same failure modes that motivate guardrails in the first place.

Collectively, these three categories trace a design-space frontier between specification cost and enforcement strength: symbolic hooks demand the most expertise to author but provide the hardest guarantees; LLM-compiled hooks offer a middle ground by lowering the authoring barrier while keeping the LLM off the enforcement path; and LLM-evaluated hooks are the most accessible but the least formally bounded, trading deterministic compliance for the ability to express nuanced, context-dependent policies. In practice, production deployments increasingly layer multiple paradigms (e.g., symbolic access-control checks composed with model-based content classifiers) to exploit the complementary strengths of each.

4.5.3 Runtime-adaptive Hooks

While rule-based and user-defined hooks operate on predetermined conditions, a third, runtime-adaptive paradigm derives dynamic, context-sensitive triggers from runtime signals rather than fixed rules. The defining characteristic is that neither the trigger condition nor the enforcement threshold is fully specified before deployment; instead, these hooks continuously monitor the agent's internal confidence, task progress, and external behavioral patterns in real time, activating timely protective measures when the system detects that the agent is operating outside its reliable competence range. We distinguish runtime-adaptive hooks from user-defined hooks by the locus of criterion definition: user-defined hooks enforce human-authored policies, whereas runtime-adaptive hooks let the system itself discover what to check from various data.

Existing approaches can be organized by the type of runtime signal to which the hook responds: the model's own uncertainty, deviations in the agent's behavioral patterns, or shifts in the external threat distribution.

(1) Uncertainty-driven Hooks. The most direct form of runtime adaptation monitors the model's own confidence in its reasoning. These hooks extract calibrated risk scores from the inference process and surface them to downstream systems for intervention decisions, without themselves enforcing any fixed policy. Representative methods estimate semantic entropy over sampled completions to detect confabulations [151], propagate verbalized confidence scores across reasoning steps and trigger targeted recomputation when accumulated uncertainty exceeds a threshold [815], or model situational weights via hidden Markov models to extend uncertainty propagation to multi-step agentic reasoning [858]. Because these signals are derived from the model's internal state at each inference step, the effective trigger condition adapts continuously without requiring any external training data or human-authored rules.

原文第 33 页 | 中文翻译初稿

或准确定性。 LLM 在编译时参与，但不参与执行路径，产生受编译质量限制但独立于运行时模型推理的保证。

编译目标范围从确定性运行以允许或阻止每个操作的可执行护栏代码[708]，到从监管文档导出并由正式模型检查器验证的概率规则电路[80]，再到从霍尔三重规范生成并在部署前由离线验证器验证的正式验证策略代码[432]。公共线程是一个两阶段架构：一个由 LLM 支持的编译步骤，将人类可读的意图转换为机器可执行的策略，然后是符号或准符号执行步骤。错误可能会在编译时引入，但一旦编译，就会继续执行，而无需法学硕士进一步参与。

(3) 动态法学硕士评估挂钩。第三种范式也从用户编写的自然语言策略开始，但与静态编译不同，它使 LLM 保持在执行关键路径上：模型在运行时根据用户提供的策略直接评估每个操作或轨迹。部署者必须提供特定于域的安全标准，否则系统无法执行任何操作，因此“用户定义”属性与前面的类别具有相同的含义。这最大化了表达能力并最小化了规范工作，但代价是用概率保证取代了硬保证。

一种方法部署了一名监护人法学硕士，通过跨多个控制点的上下文推理来解释自然语言策略[285]。基于分类器的替代方案将语言模型微调或提示为可配置的安全分类器，其风险分类或策略定义可以在无需重新训练的情况下进行调整[244, 274]。第三种变体将多个异构扫描器组合成一个分层管道，用户可以使用适合其威胁模型的自定义扫描器进行扩展[87]。因为执行者本身就是一个语言模型，所以这些方法可以处理微妙的、依赖于上下文的、抵制形式化的策略，但合规性是概率性的，并且容易受到最初激发护栏的相同故障模式的影响。

总的来说，这三个类别追踪了规范成本和执行强度之间的设计空间边界：符号挂钩需要最多的专业知识来编写，但提供最难的保证； LLM编译的钩子通过降低创作障碍，同时使LLM远离执行路径，提供了一个中间立场；和 LLM 评估的钩子是最容易获得但最不受正式限制的，用确定性合规性换取表达微妙的、依赖于上下文的策略的能力。在实践中，生产部署越来越多地分层多个范例（例如，由基于模型的内容分类器组成的符号访问控制检查）以利用每个范例的互补优势。

4.5.3 运行时自适应钩子

虽然基于规则和用户定义的钩子在预定条件下运行，但第三种运行时自适应范例从运行时信号而不是固定规则中派生出动态的上下文敏感触发器

。定义特征是在部署之前既没有完全指定触发条件也没有执行阈值；相反，这些钩子实时持续监控智能体的内部信心、任务进度和外部行为模式，当系统检测到智能体在其可靠能力范围之外运行时，及时激活保护措施。我们通过标准定义的轨迹来区分运行时自适应钩子和用户定义的钩子：用户定义的钩子强制执行人类编写的策略，而运行时自适应钩子让系统本身发现要从各种数据中检查什么。

现有的方法可以根据钩子响应的运行时信号类型进行组织：模型自身的不确定性、智能体行为模式的偏差或外部威胁分布的变化。

(1) 不确定性驱动的钩子。运行时适应的最直接形式是监控模型对其推理的信心。这些钩子从推理过程中提取经过校准的风险评分，并将其提供给下游系统以进行干预决策，而无需执行任何固定策略。代表性方法估计采样完成的语义熵以检测虚构[151]，在推理步骤中传播言语置信度分数并在累积的不确定性超过阈值时触发有针对性的重新计算[815]，或通过隐马尔可夫模型对情境权重进行建模以将不确定性传播扩展到多步骤智能体推理[858]。由于这些信号是从模型在每个推理步骤的内部状态导出的，因此有效的触发条件会不断适应，而不需要任何外部训练数据或人工编写的规则。

33

(2) Behavior-driven Hooks. A second class of hooks monitors the agent's observable execution behavior and intervenes when action patterns deviate from learned expectations. Unlike uncertainty-driven hooks that read internal model states, these approaches build structural models of normal agent behavior from execution traces and use them to detect anomalies at runtime. One line of work learns formal behavioral models from historical traces and applies model checking to anticipate violations before they occur, enabling proactive intervention [634]. Another line operates at the trajectory level, training compact verifiers on synthetically perturbed trajectories for step-level error localization and targeted rollback [388], or constructing dynamic execution graphs from distributed traces to detect structural irregularities such as infinite loops, prompt injections, and collusive delegation across multi-agent interactions [214]. The common thread is that the definition of "normal behavior" is learned from data rather than specified by a human, making these hooks adaptive to the particular agent and task at hand.

(3) Threat-shift-driven Hooks. The preceding two categories respond to signals that arise within a fixed threat model; a third category addresses the scenario where the threat landscape itself evolves after deployment. These hooks detect that previously unseen attack types or domain shifts have rendered the current safety model inadequate, and update their own detection criteria accordingly. AdaptiveGuard [754] identifies novel attack types as out-of-distribution inputs on guardrail-model activations and rapidly adapts via continual learning, maintaining high detection performance where static guardrails degrade sharply. AGrail [408] takes a lifelong-learning approach: two cooperative LLMs generate and iteratively refine safety checks per action type via test-time adaptation, accumulating them in a memory module that generalizes across tasks. DuoGuard [119] employs adversarial co-evolution between a generator and a classifier during training to produce a robust safety classifier; notably, its adaptiveness is a training-time property and the deployed model is static, placing it at the boundary between this category and fixed guardrail classifiers.

Benchmarks such as Agent-SafetyBench [849] and ToolEmu [520], which reveal that even frontier agents struggle to achieve satisfactory safety scores, underscore the necessity of these signal-driven mechanisms. By deriving their trigger conditions from learned models rather than human-authored rules, runtime-adaptive hooks complement user-defined hooks: the former discovers what to watch for, while the latter enforces what the developer already knows.

4.6 Verification

Hooks establish deterministic boundaries that intercept or block actions, but interception alone is not sufficient: the system must also *evaluate* the quality, correctness, and safety of agent behavior at every stage. Verification mechanisms serve this complementary role, acting as the harness's assessment engine that judges whether an intercepted action should proceed, whether intermediate reasoning is on track, and whether final outputs meet task requirements. Together with hooks, they close the governance loop by shifting the harness from a passive execution wrapper to an active controller. Table 4 provides a structured overview of representative verification and evaluation methods discussed in this subsection, organized by assessment target, verification level, and verifier strategy.

4.6.1 Assessment Targets

Verification mechanisms can be taxonomized by the property they certify: the factual correctness of outputs, the operational safety of actions, faithfulness to the user's instructions, or agreement across verifiers.

(1) Correctness Verification. Correctness verification certifies the factual accuracy of agent outputs. SelfCheck-GPT [418] detects hallucinations through sampling consistency alone, providing a zero-resource baseline for factual self-assessment without any external knowledge source.

(2) Safety Verification. Safety verification certifies that agent actions do not cause harm before they take effect. VeriGuard [432] generates both functional code and formal verification artifacts for each action, achieving near-zero attack success. A rapidly growing line builds dedicated safety verifiers and diagnostic guardrails: AgentDoG [287] organizes moderation around a three-dimensional risk taxonomy (source, failure mode, consequence) for fine-grained, trajectory-level judgments with root-cause diagnosis beyond binary labels, and its successor AgentDoG 1.5 [288] reports strong moderation from a small amount of training data while supporting real-time online deployment. ToolSafe [440] targets tool-invocation safety with a step-level guardrail

(2) 行为驱动的钩子。第二类钩子监视智能体的可观察执行行为，并在操作模式偏离学习期望时进行干预。与读取内部模型状态的不确定性驱动的钩子不同，这些方法从执行跟踪构建正常智能体行为的结构模型，并使用它们在运行时检测异常。其中一项工作是从历史痕迹中学习正式的行为模型，并应用模型检查来预测违规行为，从而实现主动干预[634]。另一条线路在轨迹级别运行，在综合扰动轨迹上训练紧凑验证器，以实现步骤级错误定位和有针对性的回滚[388]，或从分布式轨迹构建动态执行图以检测结构不规则性，例如无限循环、提示注入和跨多智能体交互的共谋委托[214]。共同点是“正常行为”的定义是从数据中学习的，而不是由人类指定的，使得这些钩子能够适应特定的智能体和手头的任务。

(3) 威胁转移驱动的钩子。前两类响应固定威胁模型中出现的信号；第三类涉及部署后威胁形势本身演变的场景。这些钩子检测到以前未见过的攻击类型或域转移导致当前的安全模型不足，并相应地更新自己的检测标准。 AdaptiveGuard [754] 将新颖的攻击类型识别为护栏模型激活的分布外输入，并通过 持续学习（Continual Learning） 快速适应，在静态护栏急剧下降的情况下保持高检测性能。 AGrail [408] 采用终身学习方法：两个合作的 LLM 通过测试时适应生成并迭代地完善每个操作类型的安全检查，将它们累积在跨任务泛化的内存模块中。 DuoGuard [119] 在训练过程中采用生成器和分类器之间的对抗性共同进化来产生鲁棒的安全分类器；值得注意的是，它的适应性是训练时的属性，并且部署的模型是静态的，将其置于该类别和固定护栏分类器之间的边界。

Agent-SafetyBench [849] 和 ToolEmu [520] 等基准表明，即使是前沿智能体也很难获得令人满意的安全分数，这强调了这些信号驱动机制的必要性。通过从学习模型而不是人类编写的规则中导出触发条件，运行时自适应挂钩补充了用户定义的挂钩：前者发现要监视的内容，而后者则强制执行开发人员已经知道的内容。

4.6 验证

挂钩建立拦截或阻止操作的确定性边界，但仅拦截是不够的：系统还必须在每个阶段评估智能体行为的质量、正确性和安全性。验证机制起到了这种补充作用，充当运行支撑框架（Harness）的评估引擎，判断拦截的动作是否应该继续，中间推理是否正常，以及最终输出是否满足任务要求

。它们与钩子一起，通过将 运行支撑框架（Harness） 从被动执行包装器转变为主动控制器来关闭治理循环。表 4 提供了本小节中讨论的代表性验证和评估方法的结构化概述，按评估目标、验证级别和验证者策略进行组织。

4.6.1 评估目标

验证机制可以根据其所证明的属性进行分类：输出的事实正确性、操作的操作安全性、对用户指令的忠实度或验证者之间的协议。

(1)正确性验证。正确性验证证明智能体输出的事实准确性。 SelfCheckGPT [418]仅通过采样一致性来检测幻觉，为事实自我评估提供零资源基线，无需任何外部知识源。

(2)安全验证。安全验证证明智能体操作在生效之前不会造成伤害。 VeriGuard [432] 为每个操作生成功能代码和形式验证工件，实现近乎零的攻击成功率。快速增长的产品线构建了专用的安全验证器和诊断护栏：AgentDoG [287] 围绕三维风险分类（源、故障模式、后果）组织审核，以进行细粒度的轨迹级判断，并提供超越二进制标签的根本原因诊断，其后继者 AgentDoG 1.5 [288] 报告了对少量训练数据的强大审核，同时支持实时在线部署。 ToolSafe [440] 通过阶梯级护栏来实现工具调用安全

Table 4 Representative verification and evaluation methods for LLM agents. Rows follow the three dimensions of this subsection: assessment targets, verification levels, and verifier strategies. **Train-free** (✔) marks methods that work with a frozen model, versus (✘) those requiring dedicated training.

Method	Train-free	Key mechanism
Assessment Targets		
SelfCheckGPT [418]	✔	Sampling consistency for factuality verification
VeriGuard [432]	✔	Verified code generation for action safety
ToolSafe [440]	✘	RL-trained proactive tool-invocation safety guardrail
NSVIF [563]	✔	Neuro-symbolic constraint checking for instruction faithfulness
Multiagent Debate [139]	✔	Inter-model deliberation for factual agreement
Multi-Agent Verification [362]	✔	Independent aspect verifiers for cross-verifier agreement
Verification Levels		
Let's Verify Step by Step [363]	✘	Human-supervised process rewards for intermediate steps
AgentPRM [702]	✘	Per-decision process rewards for agent trajectories
CRITIC [182]	✔	Tool-interactive stepwise verify-then-correct loop
LLM-as-a-Judge [867]	✔	Model-based evaluation of final outputs
Agent-as-a-Judge [898]	✔	Tool-augmented assessment after task completion
Reflexion [544]	✔	End-of-trial verbal reflection with episodic memory
Verifier Strategies		
Math-Shepherd [643]	✘	Monte Carlo process labels for verifier training
Implicit PRM [787]	✘	Process rewards from policy logits without extra training
Self-Consistency [659]	✔	Majority voting over diverse reasoning paths
LATS [875]	✔	Tree search with LLM-generated value estimates
ReST-MCTS* [805]	✘	PRM-guided self-training via iterative tree search
Compute-optimal scaling [550]	✘	Difficulty-adaptive compute allocation with process verifiers

(TS-Guard) trained via multi-task reinforcement learning that flags unsafe calls before execution, substantially reducing harmful invocations. These verifiers are increasingly stress-tested by dedicated safety benchmarks. The need for them is underscored by AgentHarm [8], a motivating benchmark showing that leading LLMs remain surprisingly compliant with explicitly malicious multi-step tasks; complementary benchmarks then probe specific settings: R-Judge [790] measures safety-risk awareness over multi-turn interaction records, RedCode [193] probes risky code execution and generation for code agents, ST-WebAgentBench [303] evaluates policy-constrained actions for web agents, and OS-Sentinel [571] (distinct from the SentinelAgent runtime monitor of §4.5) combines formal and neural validation for mobile GUI agents in realistic workflows.

(3) Faithfulness Verification. Faithfulness verification certifies that an agent follows the given instructions and stays on task, rather than drifting toward a superficially plausible but misaligned objective. Neuro-symbolic checkers translate natural-language instructions into formal constraints that an output can be mechanically checked against [563], while instruction-following evaluations for multi-turn, tool-using web agents test whether constraints stated early in a session are still honored many steps later [118]. This concern is sharpened by evidence of *goal drift*: agents can gradually abandon their original objective under accumulated context or contextual pressure, so faithfulness verification must track adherence across the whole trajectory rather than at a single step [20, 427]. In agentic settings it is thus the natural counterpart to the maker–checker separation of Section 4.1.3: an independent checker guards against completions that satisfy the visible goal while violating the user's underlying intent.

(4) Multi-agent Cross-verification. Multi-agent cross-verification employs several models for mutual critique. Multiagent debate [139] shows that iterative inter-model deliberation substantially reduces factual errors, and Multi-Agent Verification [362] demonstrates that scaling the number of independent aspect verifiers yields weak-to-strong generalization.

原文第 35 页 | 中文翻译初稿

表4 LLM智能体人的代表性验证和评估方法。行遵循本小节的三个维度：评估目标、验证级别和验证者策略。免训练 (✓) 标记可与冻结模型配合使用的方法，而 (✗) 则标记需要专门训练的方法。

方法	免火车	关键机制
评估目标		
SelfCheckGPT [418]	✓	用于事实验证的抽样一致性
VeriGuard [432]	✓	经过验证的代码生成，确保操作安全
ToolSafe [440]	✗	经过强化学习训练的主动工具调用安全护栏
NSVIF [563]	✓	指令忠实度的神经符号约束检查
多智能体辩论 [139]	✓	模型间审议以达成事实一致
多智能体验证[362]	✓	用于交叉验证者协议的独立方面验证者
验证级别		
让我们逐步验证[363]	✗	中间步骤的人工监督过程奖励
AgentPRM [702]	✗	智能体轨迹的每决策过程奖励
CRITIC [182]	✓	工具交互式逐步验证然后纠正循环
法学硕士作为法官 [867]	✓	最终输出的基于模型的评估
智能体人作为法官 [898]	✓	任务完成后的工具增强评估
反思 [544]	✓	审判结束时的口头反思与情景记忆
验证者策略		
Math-Shepherd [643]	✗	用于验证者训练的蒙特卡洛过程标签
隐式 PRM [787]	✗	无需额外训练即可处理来自策略 logits 的奖励
自洽[659]	✓	对不同推理路径的多数投票
LATS [875]	✓	使用 LLM 生成的价值估计进行树搜索
ReST-MCTS* [805]	✗	通过迭代树搜索进行 PRM 引导的自我训练
计算最优扩展 [550]	✗	使用流程验证器进行难度自适应计算分配

(TS-Guard) 通过多任务 强化学习（ Reinforcement Learning ） 进行训练，在执行前标记不安全调用，从而大大减少有害调用。这些验证者越来越多地接受专用安全基准的压力测试。 AgentHarm [8] 强调了对它们的需求，这是一个激励性的基准，表明领先的法学硕士仍然令人惊讶地符合明确的恶意多步骤任务；补充基准然后探测特定设置：R-Judge [790] 测量多轮交互记录的安全风险意识，RedCode [193] 探测代码智能体的危险代码执行和生成，ST-WebAgentBench [303] 评估 Web 智能体的策略约束操作，OS-Sentinel [571] (与第 4.5 节的 SentinelAgent 运行时监视器不同) 结合了实际工作流程中移动 GUI 智能体的形式验证和神经验证。

(3) 真实性验证。忠诚度验证证明智能体遵循给定的指示并继续执行任务，而不是朝着表面上合理但不一致的目标发展。神经符号检查器将自然语言指令翻译成形式约束，可以对输出进行机械检查[563]，而多轮、使用工具的网络智能体的指令跟踪评估则测试在会话早期声明的约束是否在许多步骤后仍然得到遵守[118]

。“目标漂移”的证据加剧了这种担忧：在累积的背景或背景压力下，智能体可能会逐渐放弃其最初的目标，因此忠诚度验证必须跟踪整个轨迹的遵守情况，而不是单个步骤 [20, 427]。因此，在智能体设置中，它是第 4.1.3 节中的制造者-检查者分离的自然对应物：独立的检查者防止在违反用户潜在意图的同时满足可见目标的完成。

(4)多智能体交叉验证。多智能体交叉验证采用多种模型进行相互批评。多智能体辩论[139]表明迭代模型间审议大大减少了事实错误，多智能体验证[362]表明扩大独立方面验证者的数量会产生从弱到强的泛化。

35

4.6.2 Verification Levels

Verification mechanisms can also be characterized by the stage at which they intervene during the agent's execution lifecycle.

(1) Step-level Verification. Step-level verification evaluates intermediate reasoning quality at each step, complementing hooks that enforce binary allow-or-block decisions via deterministic rules. Where a hook asks "is this action permitted?", a step-level verifier asks "is this reasoning step productive toward the goal?", producing scalar quality signals that guide search and correction rather than gating execution. Building on earlier evidence that training answer verifiers and combining process- with outcome-based feedback improves math reasoning [96, 618], process reward models (PRMs) [363] show that per-step supervision substantially outperforms outcome-only evaluation for complex reasoning, AgentPRM [702] extends this to agentic tasks by scoring each decision on goal proximity and incremental progress, and CRITIC [182] introduces a tool-interactive verify-then-correct loop that establishes external feedback as essential for reliable self-correction. Recent work advances step-level verifiers along three axes. *Generative verification*, which recasts reward modeling as next-token prediction [822], lets GenPRM [857] and ThinkPRM [272] reason step-by-step before scoring, showing that generative verifiers trained on minimal labels can match or outperform discriminative PRMs and rival strong proprietary judges on process benchmarks. *Domain-specialized PRMs* tailor verifiers to concrete agent domains: Web-Shepherd [49] builds structured subgoal checklists for web navigation at substantially lower cost than GPT-4o-based verification, while SWE-PRM [163] and SWE-TRACE [203] deliver trajectory-level course-correction and rubric-based dense feedback for software-engineering agents. *Efficient supervision* concentrates labels where they matter: CSO [329] restricts preference learning to verified critical steps that demonstrably flip task outcomes, requiring supervision at only a small fraction of trajectory steps, and ToolPRMBench [311] contributes the first large-scale PRM benchmark for tool-using settings, showing that reinforcement learning substantially improves robustness and generalization in tool-specialized verifiers. Most recently, step-level verification is moving toward fully *agentic* verifiers that themselves reason and invoke tools while scoring each decision, rather than emitting a single scalar in one pass, as in AgentV-RL [818].

(2) End-of-run Verification. End-of-run verification evaluates agent outputs after task completion. The LLM-as-a-Judge paradigm [867] uses strong models to score final outputs, and its evolution toward agentic judges has proceeded along two lines. *Agent-as-a-Judge* augments the judge with tools and multi-step reasoning: Agent-as-a-Judge [898] equips the evaluator with tool-augmented assessment for substantially higher human agreement, and a recent paper [775] systematizes how agentic judges combine planning, tool-augmented verification, and persistent memory. *Multi-agent collaborative evaluation* leverages inter-model deliberation: CollabEval [489] runs a three-phase initial-evaluation, discussion, and consensus process that outperforms competitive debate; a complementary route instead scales test-time compute for judging itself, which we discuss under verifier strategies below. *Self-correction and environment grounding* close the loop after execution: Reflexion [544] performs verbal self-reflection with episodic memory across trials, though CorrectBench [605] finds that self-correction yields only limited gains on reasoning models at significant cost; environment-based frameworks such as WebArena [885] and SWE-bench [260] verify success through functional correctness of final states, and SWE-RM [546] complements test-suite checks with an execution-free reward model that measurably lifts SWE-bench Verified accuracy. Chain-of-Verification [121] decomposes a response into atomic claims for independent post-hoc checking. Moving beyond static post-hoc scoring, agentic verifiers increasingly probe the environment after a run: verifying a GUI agent through online, proactive interaction lets the judge actively confirm claimed outcomes rather than trust the final transcript alone [102].

4.6.3 Verifier Strategies

Orthogonal to temporal granularity, verifier quality can be improved through training-free architectural designs and learning-based methods, which we group into three families.

(1) Automated labels and implicit rewards. Math-Shepherd [643] automates process-label generation via Monte Carlo estimation, eliminating costly human annotation, and Implicit PRM [787] shows that process rewards can be extracted from outcome-trained models at no additional cost. Self-consistency [659] offers a complementary training-free signal through majority voting across diverse reasoning paths.

(2) Tree search with verifier guidance. A family of methods couples Monte Carlo tree search with LLM self-

原文第 36 页 | 中文翻译初稿

4.6.2 验证级别

验证机制还可以通过它们在智能体执行生命周期期间进行干预的阶段来表征。

(1) 分级验证。步骤级验证评估每个步骤的中间推理质量，补充通过确定性规则强制执行二元允许或阻止决策的挂钩。当钩子问“这个动作是否被允许？”时，步骤级验证器问“这个推理步骤对目标有效吗？”，产生指导搜索和纠正而不是限制执行的标量质量信号。早期证据表明，训练答案验证者并将过程与基于结果的反馈相结合可以改善数学推理 [96, 618]，过程奖励模型 (PRM) [363] 表明，对于复杂推理，每步监督大大优于仅结果评估，AgentPRM [702] 通过对目标接近度和增量进度的每个决策进行评分，将其扩展到智能体任务，CRITIC [182] 引入了一个工具交互式验证然后纠正循环，该循环建立外部反馈对于可靠的自我纠正至关重要。最近的工作沿着三个轴推进了阶梯级验证器。 _生成验证_将奖励模型重新定义为下一个令牌预测[822]，让 GenPRM [857] 和 ThinkPRM [272] 在评分之前逐步推理，表明在最小标签上训练的生成验证器可以匹配或优于判别性 PRM，并在流程基准上与强大的专有法官相媲美。 _领域专用 PRM_针对具体智能体域定制验证器：Web-Shepherd [49] 以比基于 GPT-4o 的验证低得多的成本构建用于 Web 导航的结构化子目标清单，而 SWE-PRM [163] 和 SWE-TRACE [203] 为软件工程智能体提供轨迹级课程校正和基于标题的密集反馈。 _高效监督_将标签集中在重要的地方：CSO [329] 将偏好学习限制在可明显翻转任务结果的已验证关键步骤，仅需要对轨迹步骤的一小部分进行监督，而 ToolPRMBench [311] 为工具使用设置提供了第一个大规模 PRM 基准，表明 强化学习（Reinforcement Learning） 大大提高了工具专用验证器的鲁棒性和泛化性。最近，步骤级验证正在朝着完全智能体验证者的方向发展，验证者在对每个决策进行评分时本身进行推理并调用工具，而不是像 AgentV-RL [818] 那样一次性发出单个标量。

(2) 运行结束验证。运行结束验证在任务完成后评估智能体输出。 LLM-asa-Judge 范式 [867] 使用强大的模型来对最终输出进行评分，其向智能体法官的演变沿着两条路线进行。 _智能体作为法官_通过工具和多步骤推理来增强法官：智能体作为法官[898]为评估者配备了工具增强评估，以显着提高人类一致性，最近的一篇论文[775]系统化了智能体法官如何将计划、工具增强验证和持久记忆结合起来

。 _多智能体协作评估_利用模型间审议：CollabEval [489] 运行一个三阶段的初始评估、讨论和共识过程，其性能优于竞争性辩论；相反，补充路线可以扩展测试时间计算来判断自身，我们将在下面的验证者策略下讨论这一点。 自我纠正和环境 落地关联（Grounding） 执行后关闭循环：反射 [544] 在试验中利用情景记忆进行口头自我反思，尽管 CorrectBench [605] 发现自我纠正在推理模型上仅产生有限的收益，但代价高昂；基于环境的框架，如 WebArena [885] 和 SWE-bench [260] 通过最终状态的功能正确性来验证成功，SWE-RM [546] 通过免执行奖励模型补充测试套件检查，可显着提高 SWE-bench 验证的准确性。验证链[121]将响应分解为原子声明，以进行独立的事后检查。超越静态事后评分，智能体验证者越来越多地在运行后探测环境：通过在线、主动交互验证 GUI 智能体，让法官主动确认所声称的结果，而不是仅仅信任最终的成绩单 [102]。

4.6.3 验证者策略

与时间粒度正交，验证者的质量可以通过免训练的架构设计和基于学习的方法来提高，我们将其分为三个系列。

(1) 自动化标签和隐性奖励。 Math-Shepherd [643] 通过蒙特卡洛估计自动生成流程标签，消除了昂贵的人工注释，而隐式 PRM [787] 表明可以从结果训练模型中提取流程奖励，无需额外成本。自我一致性[659]通过跨不同推理路径的多数投票提供了补充的免训练信号。

(2) 在验证者指导下进行树搜索。一系列方法将蒙特卡罗树搜索与法学硕士自我结合起来

36

evaluation for structured exploration. LATS [875] and rStar-Math [191] combine MCTS with model-generated value estimates, AB-MCTS [436] adds adaptive branching that decides whether to "go wider" or "go deeper," and ReST-MCTS* [805] develops reinforced self-training that harvests higher-quality reasoning traces through iterative process-reward-guided tree search.

(3) Compute-optimal Verification Scaling. Test-time compute scaling [550] shows that verifier-guided search lets smaller models match substantially larger ones, and DeepSeek-R1 [112] demonstrates that self-verification can emerge from reinforcement learning with rule-based rewards alone. RLV [521] unifies LLM reasoners with learned value functions, improving MATH accuracy while enabling markedly more efficient scaling, while MCTS-Judge [665] carries test-time scaling into the LLM-as-a-Judge setting, markedly improving code-correctness accuracy through System-2 reasoning. Extending this scaling to specialized domains, tool-integrated verification interleaves reinforcement learning with external tools so that the verifier can certify multi-step reasoning that pure model judgment cannot [812].

5 Optimization: Internalizing Long-Horizon Capability

Overview. In Section 4, we describe how long-horizon capability can be supplied by the harness that specifies, steers, and consolidates each run. This section turns to *optimization*: how such capabilities can be progressively internalized into model parameters. We organize the section along the *training pipeline*. We begin with the *architectural substrate* (Section 5.1), which determines whether long-horizon training and inference are computationally feasible in the first place: a policy that cannot represent and efficiently process a long trajectory cannot be effectively trained on one. We then examine six stages of capability development: data and environment synthesis (Section 5.2), pre-training and mid-training (Section 5.3), fine-tuning (Section 5.4), reinforcement learning (Section 5.5), on-policy distillation (Section 5.6), and self-evolution (Section 5.7). For each stage, we identify the long-horizon capabilities it develops, the mechanisms through which they are acquired, and the limitations that remain. Figure 11 provides an overview of the pipeline, while Table 5 summarizes the characteristics, mechanisms, and representative works associated with each stage.

The Architectural Substrate and Six Training Stages

1. **Architectural Substrate** (Section 5.1): makes long trajectories efficient to train on and decode from.
2. **Data and Environment Synthesis** (Section 5.2): constructs verifiable long-horizon experience, including tasks, environments, and trajectories, that later stages can use for training and feedback.
3. **Pre-/Mid-training** (Section 5.3): builds action–observation and reasoning priors, so agentic behavior becomes native rather than prompted.
4. **Fine-tuning** (Section 5.4): instills behavioral and tool-use discipline from curated agentic trajectories.
5. **Reinforcement Learning** (Section 5.5): optimizes whole trajectories and confronts long-horizon credit assignment under sparse, delayed reward.
6. **On-policy Distillation** (Section 5.6): consolidates the policy on its own state distribution, reducing distribution shift and stabilizing earlier learned behaviors.
7. **Self-evolution** (Section 5.7): turns self-generated experience, feedback, and task-environment updates into continuing policy improvement.

5.1 Architectural Substrate

A long-horizon agent can internalize only what its architecture can represent: a policy that cannot ingest, retain, and efficiently use a long interaction history, such as observations, tool results, and constraints that accrue across many steps, cannot be trained or served on long trajectories. The architectural substrate therefore sets the ceiling for every later training stage, which is why we begin here. From an architecture view, the central question is how the model keeps and uses a long history. We group the main approaches into three representational classes: explicit-context architectures, compressed-state architectures, and hybrid

原文第 37 页 | 中文翻译初稿

结构化探索的评估。 LATS [875] 和 rStar-Math [191] 将 MCTS 与模型生成的值估计相结合，AB-MCTS [436] 添加了自适应分支，决定是“更广泛”还是“更深”，而 ReST-MCTS* [805] 开发了强化自我训练，通过迭代过程奖励引导的树搜索来收获更高质量的推理轨迹。

(3) 计算最优验证缩放。测试时计算扩展[550]表明，验证者引导的搜索可以让较小的模型匹配更大的模型，而DeepSeek-R1[112]表明，自我验证可以从仅基于规则的奖励的强化学习（Reinforcement Learning）中出现。 RLV [521] 将 LLM 推理器与学习值函数统一起来，提高了数学准确性，同时实现了显着更高效的缩放，而 MCTS-Judge [665] 将测试时间缩放纳入 LLM-as-a-Judge 设置中，通过 System-2 推理显着提高了代码正确性准确性。将这种扩展扩展到专门领域，工具集成验证将 强化学习（Reinforcement Learning） 与外部工具交织在一起，以便验证者可以验证纯模型判断无法验证的多步骤推理[812]。
5 优化：内化长时程（Long-Horizon）能力

概述。在第 4 节中，我们描述了 运行支撑框架（Harness） 如何提供 长时程（Long-Horizon） 功能，以指定、引导和合并每次运行。本节转向_优化_：如何将这些功能逐步内化到模型参数中。我们沿着_训练管道_组织这一部分。我们从_架构基底_（第 5.1 节）开始，它首先决定了 长时程（Long-Horizon） 训练和推理在计算上是否可行：无法表示和有效处理长轨迹的策略无法对其进行有效训练。然后，我们研究了能力开发的六个阶段：数据和环境综合（第 5.2 节）、预训练和训练中期（第 5.3 节）、微调（第 5.4 节）、强化学习（Reinforcement Learning）（第 5.5 节）、on-policy 蒸馏（第 5.6 节）和自我进化（第 5.7 节）。对于每个阶段，我们都会确定其开发的 长时程（Long-Horizon） 功能、获取这些功能的机制以及仍然存在的限制。图 11 提供了该流程的概述，而表 5 总结了与每个阶段相关的特征、机制和代表性工作。

架构基础和六个训练阶段

1. Architectural Substrate（第 5.1 节）：使长轨迹能够有效地进行训练和解码。

2. 数据和环境综合（第 5.2 节）：构建可验证的 长时程（Long-Horizon） 经验，包括任务、环境和轨迹，后续阶段可用于训练和反馈。

3. 预/中期训练（第 5.3 节）：建立行动观察和推理先验，因此智能体行为变得自然而不是提示。

4. 微调（第 5.4 节）：从策划的智能体轨迹中灌输行为和工具使用纪律。

5. 强化学习（Reinforcement Learning）（第5.5节）：优化整个轨迹并在稀疏、延迟奖励下对抗长时程（Long-Horizon） 贡献归因（Credit Assignment）。

6

。策略蒸馏（第 5.6 节）：将策略巩固在其自身的状态分布上，减少分布转移并稳定早期学习的行为。

7.自我进化（第5.7节）：将自我产生的经验、反馈和任务环境更新转化为持续的策略改进。
5.1 建筑基底

长时程（Long-Horizon） 智能体只能内化其架构所代表的内容：无法摄取、保留和有效使用长交互历史记录（例如跨多个步骤产生的观察结果、工具结果和约束）的策略，无法在长轨迹上进行训练或服务。因此，建筑基底为以后的每个训练阶段设定了上限，这就是我们从这里开始的原因。从架构的角度来看，核心问题是模型如何保存和使用悠久的历史。我们将主要方法分为三个表示类别：显式上下文架构、压缩状态架构和混合架构

37

Table 5 The agentic training pipeline by stage. Each row is the architectural substrate or one of the training stages. **Representative mechanisms** lists the stage's sub-mechanisms (one per line, following its subsections), and **representative works** pairs each sub-mechanism with two concrete works on the same line.

Stage (§)	Internalized capability	Representative mechanisms	Representative works
Architectural Substrate (§5.1)	Affordably representing and serving long histories	• Explicit-context • Compressed-state • Hybrid memory • High-throughput	Longformer [32], Big Bird [792] Mamba [186], RWKV [476] Jamba [300], Kimi Linear [842] EAGLE [352], GQA [5]
Data & Environment Synthesis (§5.2)	Executable, verifiable long-horizon experience	• Task synthesis • Environment synthesis • Trajectory synthesis	TaskCraft [540], WebShaper [590] SWE-Gym [467], EnvScaler [555] TOUCAN [737], AgentGym-RL [701]
Pre-/Mid-training (§5.3)	Reasoning and perception priors	• Reasoning priors • Long-context priors • Multimodal priors • Data-mixture design	DeepSeek-V3 [110], Qwen3-Coder [46] YaRN [477], LongLoRA [78] Qwen2-VL [644], InternVL3 [892] DoReMi [715], OPUS [649]
Fine-tuning (§5.4)	Behavioral and tool-use discipline	• Selection & mixing • Curriculum • Distillation	AgentTuning [794], LIMI [711] ScalingInter-RL [701], E2H Reasoner [469] Agent Distillation [269], Agent-R [789]
Reinforcement Learning (§5.5)	Feedback-driven long-horizon decisions	• Credit assignment • Policy optimization • Sampling strategy • Interaction patterns	ToolRL [487], RuscaRL [887] DAPO [780], GiGPO [156] Tree-GRPO [250], TCOD [636] GLIDER [232], Memory-R1 [743]
On-policy Distillation (§5.6)	Consolidating behavior on the policy's own states	• Teacher-guided • Self-improving	GKD [3], DAgger-LLM [309] π-Play [838], Self-Distillation Zero [216]
Self-evolution (§5.7)	Cross-task experience accumulation	• Offline bootstrap • Online interaction • Co-evolution	STaR [793], rStar-Math [191] R-Zero [233], Agent0 [704] Env Tuning [401], Agent-World [131]

architectures. These are not strict model families, but different ways to represent past information within the model. We discuss high-throughput mechanisms separately as a serving-oriented axis rather than as a fourth representation class. This view follows the main split in efficient Transformer and state-space papers [591]. We do not treat Mixture-of-Experts (MoE) as a fourth representational class. MoE routes tokens to different feed-forward experts, as in GShard and Switch Transformer [152, 301]. It primarily changes model capacity, but it does not by itself decide how the sequence history is stored. Accordingly, MoE models can still use explicit attention, latent attention, or hybrid blocks [111, 300, 601].

5.1.1 Explicit-Context Architectures

Explicit-context architectures keep history as a token sequence. The model then uses attention to route information across these tokens. Full attention gives the cleanest form of this design, because any token can attend to any other token [620]. Long-context Transformer variants reduce the cost with local attention, sliding windows, block-sparse attention, or global tokens. Longformer and Big Bird are classic examples of this direction [32, 792]. Mistral 7B uses sliding-window attention and grouped-query attention (GQA) for lower-cost decoding [252]. Gemma 2 and Gemma 3 use local-global attention patterns. Gemma 3 uses more local attention layers to reduce KV-cache cost at long context [595, 596]. Efficient Transformer papers give a broader view of this design space [591]. More recent work makes sparsity *natively trainable* rather than an inference-time approximation, making long-horizon training affordable: DeepSeek NSA couples a hardware-aligned, natively trainable sparse-attention design with efficient kernels [786], and MoBA applies a mixture-of-block-attention routing that lets each query attend to a learned subset of context blocks [398].

This class is useful for long-horizon agents because earlier states remain explicitly represented. The model can, in principle, look back at an old observation, a tool result, or a user constraint. These objects are common in reasoning-and-acting agents and web agents [761, 885]. The cost is the main weakness. As the context grows, attention compute and key-value (KV) cache size also grow [5, 620]. If the model uses a sparse pattern, it may also miss a remote token that matters. Long-context benchmarks and analyses show that locating useful evidence inside long inputs remains hard [28, 378]. Many open base model reports still fit this class. Qwen2.5 and Qwen3 mainly use decoder-only Transformer architectures; Qwen3 adds both dense and MoE variants,

原文第 38 页 | 中文翻译初稿

表 5 各阶段的智能体训练流程。每一行都是建筑基底或训练阶段之一。代表性机制列出了该阶段的子机制（每条线一个，在其子部分之后），代表性作品将每个子机制与同一线上的两个具体作品配对。

阶段（§）	内化能力	代表性机制	代表性作品
架构
Substrate (§5.1)	经济实惠地表示
并服务于悠久的历史	• 显式上下文
• 压缩状态
• 混合内存
• 高吞吐量	Longformer [32]、Big Bird [792]
Mamba [186]、RWKV [476]
Jamba [300]、Kimi Linear [842]
鹰[352]、GQA [5]
数据与环境	可执行、可验证	• 任务综合	TaskCraft [540]、WebShaper [590]
综合（§5.2）	长时程（Long-Horizon）经验	• 环境综合
• 轨迹综合	SWE-Gym [467]、EnvScaler [555]
TOUCAN [737]、AgentGym-RL [701]
训练前/中期	推理与感知	• 推理先验	DeepSeek-V3 [110]、Qwen3-Coder [46]
(§5.3)	先验	• 长上下文先验
• 多模态先验
• 数据混合设计	YaRN [477]、LongLoRA [78]
Qwen2-VL [644]、InternVL3 [892]
DoReMi [715]、OPUS [649]
微调 (§5.4)	行为和工具使用
规则	• 选择与混合
• 课程
• 蒸馏	AgentTuning [794]、LIMI [711]
ScalingInter-RL [701]、E2H Reasoner [469]
Agent Distillation [269]、Agent-R [789]
强化	反馈驱动	• 贡献归因（Credit Assignment）	ToolRL [487]、RuscaRL [887]
学习（§5.5）	长时程（Long-Horizon）决策	• 策略优化
• 采样策略
• 交互模式	DAPO [780]、GiGPO [156]
Tree-GRPO [250]、TCOD [636]
GLIDER [232]、Memory-R1 [743]
策略蒸馏
(§5.6)	巩固策略自身状态的行为	• 教师指导
• 自我改进（Self-Improving）	GKD [3]，DAgger-LLM [309]
π-Play [838]，自蒸馏零[216]
自我进化（§5.7）	跨任务经验

积累|• 离线引导
• 在线互动
• 共同进化| STaR [793]、rStar-Math [191]
R-Zero [233]、Agent0 [704]
Env Tuning [401]、Agent-World [131]|

架构。这些不是严格的模型系列，而是在模型中表示过去信息的不同方式。我们将高吞吐量机制作为面向服务的轴而不是第四个表示类单独讨论。这种观点遵循高效 Transformer 和状态空间论文 [591] 中的主要分歧。我们不将混合专家 (MoE) 视为第四个代表性类别。 MoE 将令牌路由到不同的前馈专家，如 GShard 和 Switch Transformer [152, 301]。它主要改变模型容量，但它本身并不决定序列历史记录的存储方式。因此，MoE 模型仍然可以使用显式注意力、潜在注意力或混合块 [111,300,601]。

5.1.1 显式上下文架构

显式上下文架构将历史记录保存为令牌序列。然后，该模型使用注意力在这些标记之间路由信息。充分关注给出了这种设计最简洁的形式，因为任何令牌都可以关注任何其他令牌[620]

。长上下文 Transformer 变体通过局部注意力、滑动窗口、块稀疏注意力或全局令牌来降低成本。 Longformer 和 Big Bird 是这个方向的经典例子 [32, 792]。 Mistral 7B 使用滑动窗口注意力和分组查询注意力（GQA）来实现较低成本的解码[252]。 Gemma 2 和 Gemma 3 使用局部-全局注意力模式。 Gemma 3 使用更多的局部注意力层来减少长上下文中的 KV 缓存成本 [595, 596]。 Efficient Transformer 论文对这个设计空间提供了更广阔的视野 [591]。最近的工作使稀疏性“本地可训练”而不是推理时间近似，从而使 长时程（Long-Horizon） 训练变得经济实惠：DeepSeek NSA 将硬件对齐、本地可训练的稀疏注意力设计与高效内核结合起来[786]，而 MoBA 应用了块注意力混合路由，让每个查询关注上下文块的学习子集[398]。

此类对于 长时程智能体（Long-Horizon Agents） 很有用，因为早期状态仍显式表示。原则上，该模型可以回顾旧的观察结果、工具结果或用户约束。这些对象在推理与行动智能体和网络智能体中很常见[761, 885]。成本是主要弱点。随着上下文的增长，注意力计算和键值 (KV) 缓存大小也会增长 [5, 620]。如果模型使用稀疏模式，它也可能会错过重要的远程令牌。长上下文基准和分析表明，在长输入中找到有用的证据仍然很困难 [28, 378]。许多开放基础模型报告仍然适合此类。 Qwen2.5和Qwen3主要使用仅解码器的 Transformer架构； Qwen3 添加了密集和 MoE 变体，

38

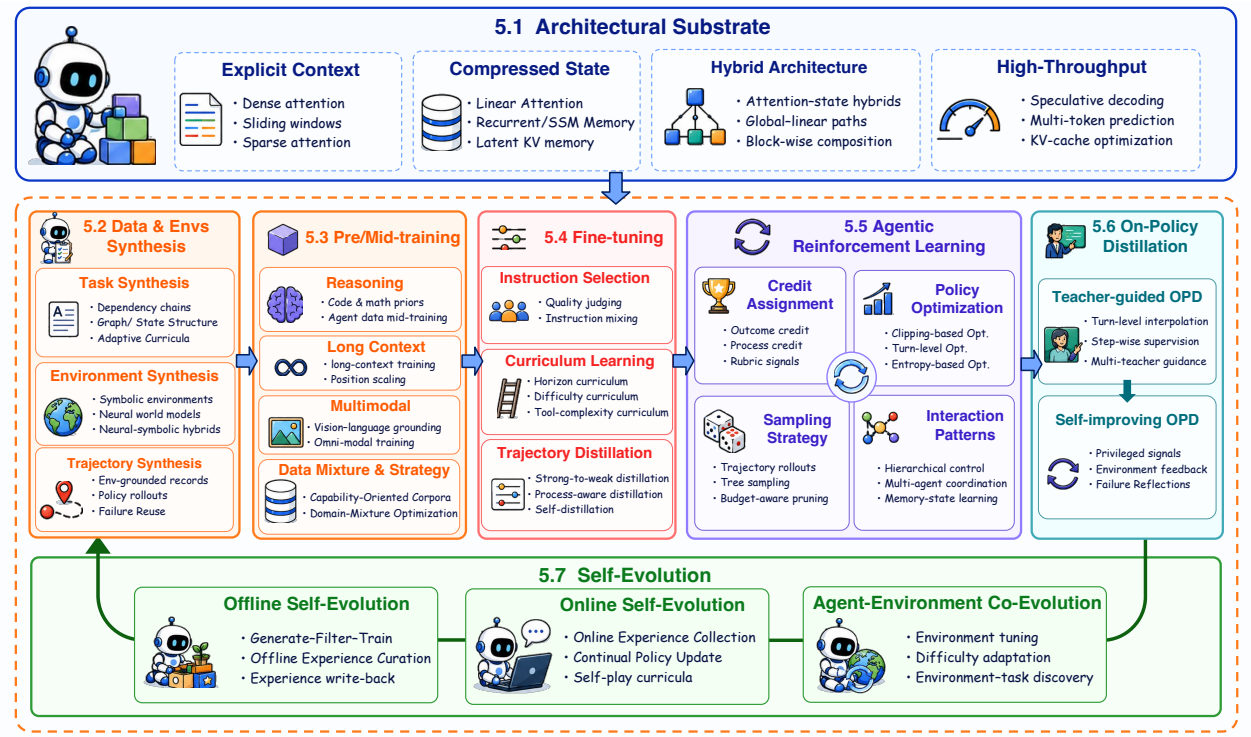


Figure 11 The agentic training pipeline for internalizing long-horizon capabilities. As the internalized counterpart to the externalized harness described in Section 4, the pipeline builds on an architectural substrate that supports long-sequence representation and proceeds through six stages. Each stage either internalizes a capability previously provided at runtime or generates the experience required for its subsequent internalization.

but the history is still accessed through attention [601, 744]. The GLM series follows the same pattern. The ChatGLM report describes the path from GLM-130B to GLM-4. GLM-4.5 is an MoE foundation model for agentic, reasoning, and coding tasks, but it is still an attention-based Transformer family [178, 795].

5.1.2 Compressed-State Architectures

Compressed-state architectures do not keep all pairwise token interactions. They instead encode history into a more compact internal representation. Some methods compress the attention map or the token set, as in low-rank attention and kernelized linear attention [92, 651]. Other methods use a recurrent or state-space form. This includes Retentive Network (RetNet), Receptance Weighted Key Value (RWKV), structured State Space Models (SSMs), and selective SSMs such as Mamba [186, 187, 476, 576].

This class fits the stream-like nature of agent work. In many agent settings, a model observes, reasons, acts, and observes again [761, 885]. A compressed state can be updated step by step, so the runtime memory can stay small [186, 187, 476, 576]. The weakness is precision. Once old tokens are folded into a state, the model may not recover a rare fact or an exact old string. This is a core trade-off between efficient recurrence and content-level access [473, 606]. This is a problem when an agent must obey a past constraint exactly, as in tool-use and web tasks that carry instructions across steps [761, 885]. A partial, and importantly different, base-model version of this idea is Multi-head Latent Attention (MLA), which compresses the KV cache into a latent form. DeepSeek-V2 introduced MLA together with DeepSeekMoE, and DeepSeek-V3 kept this design at larger scale [110, 111]; Kimi-K2 likewise pairs a large MoE model with MLA [597]. Unlike recurrent or state-space compression, however, MLA keeps a per-token latent and still attends over all past tokens, so it preserves per-token addressability and does not incur the exact-recall loss described above. It is therefore best read not as a recurrent compressed state but as a low-rank KV compression *of* explicit context, sitting between the two classes and doubling as a throughput mechanism (Section 5.1.4).

原文第 39 页 | 中文翻译初稿

5.1 架构底层
显式上下文压缩状态混合架构高吞吐量
• 密集注意力• 线性注意力• 注意力-状态混合• 推测性解码
• 滑动窗口• 循环/SSM 内存• 全局线性路径• 多令牌预测
• 稀疏注意力• 潜在KV 内存• 逐块组合• KV 缓存优化
5.2 数据和环境5.5 智能体 5.6 策略
综合 5.3 训练前/训练中 5.4 微调 强化学习（Reinforcement Learning） 蒸馏
任务综合 • 依赖链• 图/状态结构• 自适应课程• 代码和数学先验• 智能体数据训练中期推理长上下文指令选择• 质量判断• 指令混合• 结果学分• 过程学分分配学分 • 基于剪辑• 转弯水平选项。优化策略选项教师指导的 OPD • 轮级插值 • 逐步监督
• 长上下文训练 课程学习 • 红字信号 • 基于熵的选项。 • 多教师指导
环境综合 • 位置缩放 • 地平线课程
• 符号环境• 神经世界模型（World Models）• 神经符号混合• 视觉-语言• 全模态训练 多模态落地关联（Grounding） • 难度课程• 工具复杂性课程 采样策略 交互模式 自我改进（Self-Improving） OPD
轨迹蒸馏 • 特权信号
轨迹综合 • 基于环境的记录 数据混合与策略 • 从强到弱的蒸馏 • 轨迹 展开轨迹（Rollouts）• 树采样 • 分层控制• 多智能体协调 • 环境反馈• 故障反思
• 策略 展开轨迹（Rollouts） • 面向能力的语料库 • 流程感知蒸馏 • 预算感知修剪 •记忆状态学习
• 失败重用• 域混合优化• 自蒸馏
5.7 自进化
离线自进化在线自进化智能体-环境协同进化
• 生成-过滤-训练• 在线经验收集• 环境调优
• 离线体验管理• 持续策略更新• 难度适应
• 经验回写• 自玩课程•环境--任务发现

图 11 用于内化 长时程（Long-Horizon） 功能的智能体训练管道。作为第 4 节中描述的外部化 运行支撑框架（Harness） 的内部化对应物，该管道构建在支持长序列表示的架构基底上，并通过六个阶段进行。每个阶段要么内化先前在运行时提供的功能，要么生成其后续内化所需的经验。

但历史仍然可以通过注意力来获取[601, 744]。 GLM系列也遵循同样的模式。 ChatGLM 报告描述了从 GLM-130B 到 GLM-4 的路径。 GLM-4.5 是一个用于智能体、推理和编码任务的 MoE 基础模型（Foundation Model），但它仍然是一个基于注意力的 Transformer 家族 [178, 795]。
5.1.2 压缩状态架构

压缩状态架构不保留所有成对令牌交互。相反，它们将历史编码为更紧凑的内部表示。一些方法压缩注意力图或标记集，如低秩注意力和核化线性注意力 [92, 651]

。其他方法使用循环或状态空间形式。这包括保持网络（RetNet）、接收加权键值（RWKV）、结构化状态空间模型（SSM）和选择性 SSM，例如 Mamba [186,187,476,576]。

该类适合智能体工作的流式性质。在许多智能体设置中，模型观察、推理、行动，然后再次观察 [761, 885]。压缩状态可以逐步更新，因此运行时内存可以保持较小[186,187,476,576]。弱点是精度。一旦旧的

令牌被折叠成状态，模型可能无法恢复罕见的事实或确切的旧字符串。这是高效循环和内容级访问之间的核心权衡[473, 606]。当智能体必须严格遵守过去的约束时，这是一个问题，就像在工具使用和跨步骤携带指令的网络任务中一样[761, 885]。这个想法的一个部分且重要不同的基本模型版本是多头潜在注意力（MLA），它将 KV 缓存压缩为潜在形式。 DeepSeek-V2 与 DeepSeekMoE 一起引入了 MLA，而 DeepSeek-V3 则在更大范围内保留了这种设计 [110, 111]； Kimi-K2 同样将大型 MoE 模型与 MLA [597] 配对。然而，与循环或状态空间压缩不同，MLA 保持每个令牌的潜在状态，并且仍然关注所有过去的令牌，因此它保留了每个令牌的可寻址性，并且不会导致上述的精确召回损失。因此，最好不要将其视为循环压缩状态，而是将其视为显式上下文的低等级 KV 压缩，位于两个类之间并作为吞吐量机制加倍（第5.1.4节）。

39

5.1.3 Hybrid Architectures

Hybrid architectures combine explicit attention with compressed or recurrent state [300, 842]. The goal is to combine precise content access with efficient long-range state tracking. Attention can keep precise access to important tokens [378, 620]. A compressed state can carry long-range signals at low cost [186, 606]. In this view, attention is used for exact evidence, while the state module is used for long-term continuity [300, 842]. Jamba is a direct example: it mixes Transformer layers, Mamba layers, and MoE layers [300]. Kimi Linear is another clear example. It combines Kimi Delta Attention units with a global MLA layer. The model therefore keeps both a linear-attention path and an explicit global attention path [842].

This makes hybrid design a plausible direction for long-horizon agents. Agent runs need both exact recall and cheap long-term tracking, as shown by web-agent and long-context tasks [28, 885]. Pure attention gives direct access but can be expensive at long context [591, 620]. Pure compressed state is cheaper but may lose details [473, 606]. A hybrid model can assign these roles to different layers, heads, or blocks, as in Jamba and Kimi Linear [300, 842]. The reports above show the same direction at system scale: strong models often combine attention, KV-cache reduction, and MoE routing [110, 597, 601]. These parts solve different problems and should not be collapsed into one label [5, 152, 301].

5.1.4 High-Throughput Mechanisms

High-throughput mechanisms form a second axis under architecture [5, 111, 180, 302, 535]. They do not decide how the model stores long history; they reduce the serial work or cache traffic needed to decode from that history [5, 111, 180, 302, 535]. A complementary, kernel-level route keeps attention *exact* but IO-aware: FlashAttention tiles the computation to avoid materializing the full attention matrix in high-bandwidth memory, cutting memory traffic and enabling longer sequences without altering the model's outputs [108]. One group reduces serial dependence in the output path through parallel generation with diffusion language models [559] or through autoregressive lookahead [43, 180, 302, 352]. Multi-token prediction trains the model to predict several future tokens at each position, so the model learns a longer local prediction horizon [180]. Speculative decoding uses a draft model to propose several tokens and lets the target model verify them in parallel [302]. Medusa keeps the draft process inside one model by adding multiple decoding heads [43]. EAGLE moves the draft signal into feature space and treats feature uncertainty as central to speculative sampling [352]. DeepSeek-V3 uses multi-token prediction as an auxiliary training objective and also presents it as a path to speculative decoding [110].

The other group reduces the cost of reading and writing the KV cache [5, 111, 535]. Multi-query attention shares key and value projections across query heads to speed decoding [535]. Grouped-query attention keeps more key-value groups than multi-query attention, but still reduces the cache cost relative to full multi-head attention [5]. MLA compresses the KV cache into a latent form, so it is both a compressed-state design and a throughput design [110, 111]. These mechanisms matter for long horizon agents because long runs often produce many intermediate tokens after a large context has already been built [180, 302, 761, 885]. They reduce the serving cost of long decoding, but they do not remove the need for explicit recall, compressed state, or hybrid memory when old evidence must be found accurately [28, 180, 302, 378].

5.2 Data and Environment Synthesis

Long-horizon behavior is learned from interaction experience that is executable, feedback-bearing, and reusable. Unlike static instruction–response pairs, such experience records how an agent decides, observes feedback, corrects errors, and accumulates information over a long trajectory. Deployment logs, demonstrations, and benchmark rollouts offer natural sources of such data, but their direct use is constrained by acquisition cost, privacy, and limited reproducibility. Controlled data synthesis has therefore become an important route for constructing long-horizon training experience. It centers on three coupled objects: *tasks* define goals and capability pressures, *environments* instantiate execution and verification, and *trajectories* turn interaction histories into trainable signals.

原文第 40 页 | 中文翻译初稿

5.1.3 混合架构

混合架构将显式注意力与压缩或循环状态结合起来[300, 842]。目标是将精确的内容访问与高效的远程状态跟踪结合起来。注意力可以保持对重要标记的精确访问[378, 620]。压缩状态可以以低成本承载远程信号[186, 606]。在这个观点中，注意力用于精确证据，而状态模块用于长期连续性[300, 842]。 Jamba 就是一个直接的例子：它混合了 Transformer 层、Mamba 层和 MoE 层 [300]。 Kimi Linear 是另一个明显的例子。它将 Kimi Delta Attention 单元与全局 MLA 层相结合。因此，该模型保留了线性注意力路径和显式全局注意力路径[842]。

这使得混合设计成为 长时程智能体（Long-Horizon Agents） 的一个合理方向。智能体运行需要精确的召回和廉价的长期跟踪，如网络智能体和长上下文任务所示[28, 885]。纯粹的注意力可以直接访问，但在长上下文中可能会很昂贵 [591, 620]。纯压缩状态更便宜，但可能会丢失细节[473, 606]。混合模型可以将这些角色分配给不同的层、头或块，如 Jamba 和 Kimi Linear [300, 842] 中所示。上面的报告在系统规模上显示了相同的方向：强模型通常结合注意力、KV 缓存减少和 MoE 路由 [110, 597, 601]。这些部分解决不同的问题，不应合并为一个标签 [5, 152, 301]。

5.1.4 高通量机制

高吞吐量机制形成架构下的第二个轴[5,111,180,302,535]。他们不决定模型如何存储悠久的历史；它们减少了从历史记录中解码所需的串行工作或缓存流量 [5, 111, 180, 302, 535]。互补的内核级路线既保持注意力精确，又保持 IO 感知：FlashAttention 平铺计算以避免在高带宽内存中实现完整的注意力矩阵，从而减少内存流量并在不改变模型输出的情况下实现更长的序列[108]。一组通过扩散语言模型的并行生成[559]或通过自回归前瞻[43,180,302,352]减少输出路径中的串行依赖性。多令牌预测训练模型预测每个位置的多个未来令牌，因此模型学习更长的局部预测范围[180]。推测性解码使用草稿模型提出多个标记，并让目标模型并行验证它们[302]。 Medusa 通过添加多个解码头将草稿过程保留在一个模型内 [43]。 EAGLE 将草稿信号移入特征空间，并将特征不确定性视为推测采样的核心[352]。 DeepSeek-V3 使用多标记预测作为辅助训练目标，并将其作为推测解码的路径[110]。

另一组降低了读写KV缓存的成本[5,111,535]。多查询注意力在查询头之间共享键和值投影以加速解码[535]。分组查询注意力比多查询注意力保留更多的键值组，但相对于完整的多头注意力仍然降低了缓存成本[5]

。 MLA 将 KV 缓存压缩为潜在形式，因此它既是压缩状态设计，也是吞吐量设计 [110, 111]。这些机制对于长期智能体很重要，因为在构建大型上下文之后，长期运行通常会产生许多中间令牌[180,302,761,885]。它们降低了长解码的服务成本，但当必须准确找到旧证据时，它们并没有消除对显式回忆、压缩状态或混合记忆的需要[28,180,302,378]。

5.2 数据与环境综合

长时程（Long-Horizon） 行为是从可执行、反馈且可重用的交互体验中学习的。与静态指令-响应对不同，这种经验记录了智能体如何决策、观察反馈、纠正错误以及在长轨迹上积累信息。部署日志、演示和基准 展开轨迹（Rollouts） 提供了此类数据的自然来源，但它们的直接使用受到获取成本、隐私和有限的可重复性的限制。受控数据合成因此成为构建长时程（Long-Horizon）训练经验的重要途径。它以三个耦合对象为中心：_任务_定义目标和能力压力，_环境_实例化执行和验证，_轨迹_将交互历史转化为可训练信号。

40

5.2.1 Task Synthesis

Task synthesis determines the objective distribution for long-horizon agents. A task defines the target behavior through goals, constraints, initial and terminal states, and success criteria. Task generation sits at the entry of interactive experience collection, since a task definition largely fixes horizon, dependency structure, and failure modes [238, 808]. Existing methods mainly construct tasks through dependency chains, structured information spaces, state transitions, and capability-adaptive curricula.

Dependency-chain Construction. The most direct way to lengthen horizon is to chain subgoals so that early decisions constrain the actions later available. Long-horizon diagnostics relate failures to intrinsic horizon and compositional depth [658]; horizon-length studies further show that longer tasks aggravate optimization instability and credit assignment [279]. AgentSynth composes easy-to-generate subtasks into longer computer-use tasks, while TaskCraft expands atomic tasks into multi-hop, multi-tool objectives [540, 713]. The horizon here grows through necessary intermediate dependencies rather than through mere interaction length.

Graph-structured Information Construction. When horizon arises from evidence rather than from actions, tasks are drawn as paths through web pages, documents, tools, or evidence graphs, and the difficulty shifts to selecting relevant evidence under branching, incomplete, or masked observations. WebWalker and WebDancer construct multi-step information-seeking tasks from traversal, crawling, and backward question evolution [689, 690]. WebSailor uses knowledge-graph walks and information masking, while WebShaper formulates information seeking through set and knowledge projection to control evidence structure, answer space, and verifiability [327, 590]. WebSailor-V2 and WebWeaver further extend this line to more open-ended information-seeking tasks [326, 356].

State-transition Construction. A third view frames a task as the gap between an initial and a target state, or as recovery from an erroneous one. In software settings, SWE-Gym, R2E, and SWE-smith reconstruct tasks from repositories, issues, commits, and tests [248, 467, 748]. Reverse synthesis makes this state-transition view explicit: DIVE generates tasks from executed evidence traces, TASTE evolves valid tool-sequence skeletons, and CLI-Gym derives repair tasks from abnormal command-line states and historical errors [53, 271, 369]. Execution traces and state gaps are thereby recast as tasks with verifiable outcomes.

Curriculum-driven Construction. The routes above fix a task distribution in advance; curricula instead let difficulty track the current policy, reducing the mismatch between static task sets and evolving capability. WebRL generates new web tasks from failed attempts, and Learn-by-interact reconstructs instructions and supervision from environment documents and interaction histories [484, 562]. AgentFrontier targets the model's zone of proximal development, while CuES uses exploration and curiosity signals to synthesize environment-grounded RL tasks [75, 417]. Self-evolving agent papers view such mechanisms as part of joint agent–task evolution [165].

These routes impose complementary capability pressures on plan maintenance, evidence selection, and state recovery, whereas curricula act orthogonally, tuning how fast difficulty grows rather than what it tests. Which pressure to emphasize depends on the target deployment, since a distribution poorly matched to the policy tends to either saturate or destabilize training.

5.2.2 Environment Synthesis

Environment synthesis supplies the training ground where long-horizon agents accumulate the interaction experience that static corpora cannot provide. Whereas tasks specify what to solve, environments determine how actions are executed, observed, and rewarded, closing the action–observation–feedback loop that turns a passive predictor into an active decision maker. A synthesized environment makes this loop reproducible, resettable, parallelizable, and verifiable, so that agents can act, fail safely, and receive checkable feedback at scale. Real repositories, browsers, APIs, databases, desktop systems, and game simulators provide fidelity [145], but only controlled synthesis makes such interaction cheap and reliable enough for repeated rollout. We organize environment synthesis into three families according to how their transitions and feedback are produced: symbolic environments, neural environments, and neural–symbolic hybrids.

5.2.1 任务综合

任务综合确定 长时程智能体（Long-Horizon Agents） 的目标分布。任务通过目标、约束、初始和最终状态以及成功标准来定义目标行为。任务生成位于交互式体验收集的入口，因为任务定义很大程度上修复了范围、依赖结构和故障模式[238, 808]。现有方法主要通过依赖链、结构化信息空间、状态转换和能力自适应课程来构造任务。

依赖链构建。延长视野的最直接方法是将子目标链接起来，以便早期的决策限制以后可用的行动。 长时程（Long-Horizon） 诊断将故障与内在层位和成分深度联系起来 [658]；视野长度研究进一步表明，较长的任务会加剧优化的不稳定性和 贡献归因（Credit Assignment） [279]。 AgentSynth 将易于生成的子任务组合成较长的计算机使用任务，而 TaskCraft 将原子任务扩展为多跳、多工具目标 [540, 713]。这里的视野是通过必要的中间依赖关系而不是仅仅通过交互长度来增长的。

图结构信息构建。当视野来自证据而不是行动时，任务被绘制为通过网页、文档、工具或证据图的路径，困难就转移到在分支、不完整或掩盖的观察下选择相关证据。 WebWalker 和 WebDancer 从遍历、爬行和向后问题演化构建多步骤信息查找任务 [689, 690]。 WebSailor 使用知识图行走和信息屏蔽，而 WebShaper 通过集合和知识投影制定信息搜索来控制证据结构、答案空间和可验证性 [327, 590]。 WebSailor-V2 和 WebWeaver 进一步将这条线扩展到更多开放式信息查找任务 [326, 356]。

国家过渡建设。第三种观点将任务视为初始状态和目标状态之间的差距，或者从错误状态中恢复。在软件设置中，SWE-Gym、R2E 和 SWE-smith 从存储库、问题、提交和测试中重建任务 [248, 467, 748]。逆向综合使这种状态转换视图变得明确：DIVE 从执行的证据跟踪中生成任务，TASTE 演化出有效的工具序列骨架，CLI-Gym 从异常的命令行状态和历史错误中派生修复任务 [53,271,369]。执行轨迹和状态间隙因此被重新塑造为具有可验证结果的任务。

课程驱动建设。上述路线预先确定了任务分配；相反，课程让困难跟踪当前的策略，减少静态任务集和不断发展的能力之间的不匹配。 WebRL 从失败的尝试中生成新的 Web 任务，而交互学习则从环境文档和交互历史中重建指令和监督 [484, 562]。 AgentFrontier 针对模型的最近发展区域，而 CuES 使用探索和好奇心信号来综合基于环境的 RL 任务 [75, 417]。自进化智能体论文将此类机制视为联合智能体-任务进化的一部分[165]。

这些路线对计划维护、证据选择和状态恢复施加了互补的能力压力，而课程则正交作用，调整难度增长的速度而不是测试的内容。强调哪种压力取决于目标部署，因为与策略不匹配的分配往往会导致训练饱和或不稳定。

5.2.2 环境综合

环境综合为长时程智能体（Long-Horizon Agents）积累静态语料库无法提供的交互经验提供了训练场。任务指定要解决的问题，而环境则决定如何执行、观察和奖励行动，从而形成行动-观察-反馈循环，将被动预测者转变为主动决策者。综合环境使该循环可重复、可重置、可并行且可验证，以便智能体可以大规模行动、安全失败并接收可检查的反馈。真实的存储库、浏览器、API、数据库、桌面系统和游戏模拟器提供保真度[145]，但只有受控合成才能使这种交互对于重复的 展开轨迹（Rollout） 来说足够便宜和可靠。我们根据环境合成的转变和反馈方式将其分为三个系列：符号环境、神经环境和神经符号混合体。

Symbolic Environments. Symbolic environments implement transitions and feedback through code, tests, databases, or business logic, which yields stable execution and verifiable hard signals. This family has developed along several technical routes, distinguished by the substrate they make executable. A first route builds *repository-level software environments*: SWE-Gym packages real repositories, issues, dependencies, and tests into executable tasks, and follow-up systems scale repository construction, test generation, and task expansion [248, 467, 748, 797, 856]. A second route targets *terminal and command-line environments*, where Terminal-Bench standardizes hard, realistic shell tasks and later work co-derives instructions, sandboxes, and teacher trajectories to scale terminal-agent training [82, 86, 369, 428]. A third route provides *GUI, web, and game execution environments*, represented by WebArena, OSWorld, and GameWorld and extended toward broader interface and generation settings [251, 461, 695, 716, 852, 885]. A fourth route constructs *tool- and domain-specific environments*, ranging from multi-step tool substrates such as AgentScaler to grounded scientific and medical settings [148, 538, 555, 626, 729]. Across these routes, explicit rules stabilize verification but remain less effective under ambiguity, dynamic interfaces, and cross-tool side effects.

Neural Environments and World Models. Neural environments approximate transitions, observations, or feedback with LLMs, visual generators, or latent dynamics, reducing the cost of repeated interaction. These methods treat the model as a *world action model*: conditioned on the interaction history and the agent's action, it simulates the next round of environmental feedback, so that rollout no longer touches the real system. Three technical routes differ mainly in the representation over which the next state is predicted. The first route learns *pixel-level world models*, represented by DIAMOND and GameNGen, which generate action-conditioned visual futures and have since been extended to embodied and open-world settings [2, 6, 89, 170, 196, 249, 619, 769, 839]. The second route builds *text-level simulators* better suited to web, GUI, tools, and code, where WebDreamer predicts future web states and later systems structure UI, webpage, or software states [47, 149, 188, 572, 661, 712, 870]; recent language world models push this route toward general agents, learning next-state prediction across many agent domains and coupling generative simulators with agents in difficulty-aligned co-evolution [195, 905]. The third route learns *latent world models* in the style of joint-embedding prediction, which compress dynamics for cheaper long-horizon prediction [24, 25, 30, 176]. Neural environments scale cheaply, but they may introduce hallucination, drift, unreliable feedback, and sim-to-real gaps.

Neural–Symbolic Hybrid Environments. Neural–symbolic hybrid environments keep hard state and final verification in databases, file systems, programs, or APIs, while using models for user simulation, language interfaces, or hard-to-specify observations. They can be grouped by where the model is inserted into an otherwise symbolic loop. One route places models at the *user and application interface*, as in τ-bench, which combines stateful applications, user simulation, and final-state checks [399, 614, 762]. A second route standardizes *tool-server backends* through MCP [299, 411, 668, 700]. A third route pairs *generated content with symbolic constraints*, coupling generated webpages with state-machine constraints and completion checks [50, 695, 852]. A fourth route grounds *domain environments* in real data, retrieval, or computation while using models to expand tasks and feedback [538, 626, 804, 894]. The design choice throughout is where to draw the boundary between model-based simulation and symbolic verification.

Across these families, feedback and verification quality remain central. Software tasks can rely on tests and execution; tool tasks use persistent database, file, or API states; web and GUI tasks often require scripts, page-state checks, visual-semantic matching, or expert rules; and open-ended tasks depend more heavily on rubrics, LLM judges, or hybrid verifiers. Dense step-level signals can improve credit assignment, but stronger automation also raises judge bias, verifier loopholes, and reward-hacking risks [238, 402]. Synthesized environments must therefore balance correctness, diversity, complexity, and fidelity.

5.2.3 Trajectory Synthesis

For long-horizon agents, trajectory synthesis internalizes interaction as trainable behavioral experience: it records how a policy acts, observes, errs, and recovers over an environment, and converts that history into supervision, preferences, rewards, and memory. What can be recorded is bounded by the environment's observability and verification, so trajectory quality inherits the strengths and gaps of the synthesized environment. A useful trajectory keeps the cross-step dependencies, delayed feedback, and recovery decisions

原文第 42 页 | 中文翻译初稿

符号环境。符号环境通过代码、测试、数据库或业务逻辑实现转换和反馈，从而产生稳定的执行和可验证的硬信号。该系列沿着多种技术路线发展，其特征在于它们可执行的基板。第一条路线构建_存储库级软件环境_：SWE-Gym 将真实存储库、问题、依赖项和测试打包为可执行任务，后续系统扩展存储库构建、测试生成和任务扩展[248、467、748、797、856]。第二条路线针对_终端和命令行环境_，其中 Terminal-Bench 标准化了困难、现实的 shell 任务，随后的工作共同导出指令、沙箱和教师轨迹以扩展终端智能体训练 [82,86,369,428]。第三条路线提供_GUI、Web和游戏执行环境_，以WebArena、OSWorld和GameWorld为代表，并向更广泛的界面和生成设置扩展[251、461、695、716、852、885]。第四条路线构建_工具和特定领域的环境_，范围从多步骤工具基础（例如 AgentScaler）到扎根的科学和医疗环境[148,538,555,626,729]。在这些途径中，明确的规则可以稳定验证，但在模糊性、动态接口和跨工具副作用的情况下仍然不太有效。

神经环境和 世界模型（World Models）。神经环境通过 LLM、视觉生成器或潜在动态来近似转换、观察或反馈，从而降低重复交互的成本。这些方法将模型视为_世界动作模型_：以交互历史和智能体动作为条件，模拟下一轮环境反馈，从而使 展开轨迹（Rollout） 不再接触真实系统。三种技术路线的主要区别在于预测下一个状态的表示方式。第一条路线学习_像素级 世界模型（World Models）_，以 DIAMOND 和 GameNGen 为代表，它生成动作条件的视觉未来，并已扩展到具体和开放世界设置 [2, 6, 89, 170, 196, 249, 619, 769, 839]。第二条路线构建更适合 Web、GUI、工具和代码的_文本级模拟器_，其中 WebDreamer 预测未来的 Web 状态和后来的系统结构 UI、网页或软件状态 [47、149、188、572、661、712、870]；最近的语言 世界模型（World Models） 将这条路线推向通用智能体，学习跨多个智能体领域的下一状态预测，并将生成模拟器与难度一致的共同进化中的智能体耦合起来 [195, 905]。第三条路线以联合嵌入预测的方式学习_latent 世界模型（World Models）_，它压缩动态以获得更便宜的长时程（Long-Horizon）预测[24,25,30,176]。神经环境的扩展成本较低，但它们可能会引入幻觉、漂移、不可靠的反馈和 仿真到现实（Sim-to-Real） 间隙。

神经-符号混合环境。神经符号混合环境在数据库、文件系统、程序或 API 中保留硬状态和最终验证，同时使用模型进行用户模拟、语言接口或难以指定的观察。它们可以根据模型插入符号循环的位置进行分组

。一种方法将模型放置在_用户和应用程序接口_上，如_τ_-bench中所示，它结合了有状态应用程序、用户模拟和最终状态检查[399,614,762]。第二条路线通过 MCP [299、411、668、700] 标准化_工具服务器后端_。第三条路线将生成的内容与符号约束配对，将生成的网页与状态机约束和完成检查耦合起来[50,695,852]。第四条路线将_领域环境_置于真实数据、检索或计算中，同时使用模型来扩展任务和反馈[538,626,804,894]。整个设计选择是在基于模型的仿真和符号验证之间划定界限。在这些系列中，反馈和验证质量仍然是核心。软件任务可以依赖测试和执行；工具任务使用持久数据库、文件或 API 状态；Web 和 GUI 任务通常需要脚本、页面状态检查、视觉语义匹配或专家规则；开放式任务更多地依赖于评分标准、法学硕士法官或混合验证者。密集的步骤级信号可以改善 贡献归因（Credit Assignment），但更强的自动化也会增加判断偏差、验证者漏洞和奖励黑客风险 [238, 402]。因此，合成环境必须平衡正确性、多样性、复杂性和保真度。

5.2.3 轨迹合成

对于 长时程智能体（Long-Horizon Agents），轨迹合成将交互内化为可训练的行为体验：它记录策略如何在环境中行动、观察、犯错和恢复，并将该历史转化为监督、偏好、奖励和记忆。可以记录的内容受到环境的可观察性和可验证性的限制，因此轨迹质量继承了合成环境的优点和差距。有用的轨迹保留了跨步骤依赖性、延迟反馈和恢复决策

42

that plain instruction–response data discard. Prior work develops this idea along four directions: grounding records in real environments, aligning rollouts with the trained policy, assigning step-level credit while reusing failures, and compressing long histories into reusable experience.

Environment-grounded Execution Records. Environment-grounded records capture state changes, tool side effects, and feedback produced during execution. Textual ReAct traces or function-call transcripts provide useful templates, but they often omit post-action states and failure signals. TOUCAN synthesizes large-scale tool-agentic trajectories in real MCP environments; OpenResearcher anchors deep-research trajectories in a reproducible offline corpus; and ASTRA constrains multi-turn tool trajectories with tool-call graphs [354, 604, 737]. WebSTAR, ANCHOR, and HATS further tie GUI and computer-use trajectories to interface states, branch points, action ambiguity, and step-level feedback [215, 531, 679].

Policy-aligned Rollouts. Policy-aligned rollouts reduce distribution shift between collected trajectories and the policy being trained. Expert, human, or strong-model trajectories provide useful cold-start priors, whereas on-policy or near-on-policy rollouts expose the current policy's own exploration, failures, and recovery states. AgentGym-RL, AgentRL, and ASearcher combine multi-turn rollout, horizon scheduling, and asynchronous sampling; DPEPO uses parallel trajectories for cross-path exploration; and Agent Lightning decomposes execution records into RL-consumable units [166, 409, 701, 811, 820].

Step-level Credit Assignment and Failure Reuse. Step-level credit assignment exposes local correctness, redundant actions, and critical errors that are hidden by final success labels. WebSTAR filters learnable GUI actions with a process reward model; ANCHOR expands and cleans post-branch trajectories; and HATS targets difficult segments under contextual and visual ambiguity [215, 531, 679]. CSO identifies outcome-changing critical steps, VPR constructs verifiable turn-level rewards, and GEAR reallocates trajectory-level advantages to segments [329, 333, 785]. The unit of supervision shifts from whole trajectories to steps, segments, and critical decisions. Failure trajectories provide complementary training signals for planning, evidence selection, tool use, state recovery, and harness interaction. CSO generates critical-step replacements; AgentHER relabels failed attempts under reachable alternative goals; and AOI and PIVOT use failed records for correction and trajectory refinement [125, 329, 751, 826]. HORIZON and HarnessFix further show that failures may stem from tools, context management, verifiers, or harnesses rather than model decisions alone [64, 658]. Useful trajectories therefore retain errors, control flow, state changes, and verification provenance.

History Compression and Experience Reuse. History compression addresses the context and credit-assignment difficulties caused by long interaction records. ReSum trains over summary-bounded segments; Context-Folding and AgentFold organize trajectories into foldable subprocesses; and Memory-as-Action and Memory-R2 treat memory edits as trajectory decisions with credit-assignment consequences [573, 693, 742, 768, 846]. Long-horizon trajectories can thus be segmented, summarized, folded, converted into memory states, and abstracted into reusable experience.

In summary, trajectory synthesis is moving from text-only demonstrations toward environment-grounded, verifiable, and reusable agentic experience. Task synthesis sets the capability pressure, environment synthesis supplies the interaction loop, and trajectory synthesis turns execution histories into trainable signals.

5.3 Pre-training and Mid-training

Pre-training and mid-training are the earliest stages for internalizing the capabilities required by long-horizon agents. Pre-training establishes broad linguistic, factual, coding, mathematical, and reasoning priors through large-scale self-supervised learning, while mid-training further reshapes these priors before instruction tuning or reinforcement learning. For long-horizon agents, these stages are important because later supervised fine-tuning (SFT), alignment, and policy optimization can only elicit and refine capabilities that are already sufficiently represented in the base model. We therefore discuss pre-training and mid-training from four aspects: reasoning capability, long-context state modeling, multimodal perception, and the data mixture and training strategies that support these capabilities.

那个简单的指令-响应数据被丢弃。之前的工作沿着四个方向发展了这个想法：落地关联（Grounding）在真实环境中记录，将展开轨迹（Rollouts）与经过训练的策略对齐，在重用失败的同时分配步骤级别的信用，以及将长期历史压缩为可重用的经验。

基于环境的执行记录。基于环境的记录捕获状态变化、工具副作用以及执行期间产生的反馈。文本 ReAct 跟踪或函数调用记录提供了有用的模板，但它们经常忽略操作后状态和失败信号。 TOUCAN在真实MCP环境中合成大规模工具智能体轨迹； OpenResearcher 将深度研究轨迹锚定在可复制的离线语料库中； ASTRA 通过工具调用图约束多回转工具轨迹 [354, 604, 737]。 WebSTAR、ANCHOR 和 HATS 进一步将 GUI 和计算机使用轨迹与界面状态、分支点、动作模糊性和步骤级反馈联系起来 [215, 531, 679]。

策略一致 展开轨迹（Rollouts）。策略一致的 展开轨迹（Rollouts） 减少了收集的轨迹和正在训练的策略之间的分布偏移。专家、人类或强模型轨迹提供有用的冷启动先验，而在策略或近策略 展开轨迹（Rollouts） 则暴露了当前策略自身的探索、失败和恢复状态。 AgentGym-RL、AgentRL和ASearcher结合了多轮展开轨迹（Rollout）、水平调度和异步采样； DPEPO 使用平行轨迹进行交叉路径探索； Agent Lightning 将执行记录分解为 RL 可消耗的单元 [166, 409, 701, 811, 820]。

步骤级 贡献归因（Credit Assignment） 和故障重用。步骤级 贡献归因（Credit Assignment） 暴露了最终成功标签隐藏的局部正确性、冗余操作和关键错误。 WebSTAR 使用过程奖励模型过滤可学习的 GUI 操作； ANCHOR 扩展并清理分支后轨迹； HATS 针对上下文和视觉模糊性下的困难部分 [215, 531, 679]。 CSO 确定改变结果的关键步骤，VPR 构建可验证的回合级别奖励，GEAR 将轨迹级别优势重新分配给细分[329,333,785]。监督单位从整体轨迹转向步骤、环节、关键决策。故障轨迹为规划、证据选择、工具使用、状态恢复和 运行支撑框架（Harness） 交互提供补充训练信号。 CSO 生成关键步骤替换； AgentHER 在可实现的替代目标下重新标记失败的尝试； AOI 和 PIVOT 使用失败的记录进行校正和轨迹细化 [125, 329, 751, 826]。 HORIZON 和 HarnessFix 进一步表明，失败可能源于工具、上下文管理、验证器或 运行支撑框架（Harnesses），而不仅仅是模型决策 [64, 658]。因此，有用的轨迹保留了错误、控制流、状态变化和验证来源。

历史压缩和经验重用。历史压缩解决了由长交互记录引起的上下文和信用分配困难。 ReSum 在摘要边界段上进行训练； Context-Folding 和 AgentFold 将轨迹组织成可折叠子流程

; Memory-as-Action 和 Memory-R2 将记忆编辑视为具有信用分配结果的轨迹决策[573,693,742,768,846]。因此，长时程（Long-Horizon） 轨迹可以被分段、总结、折叠、转换为记忆状态，并抽象为可重用的经验。

总之，轨迹合成正在从纯文本演示转向基于环境的、可验证的和可重用的智能体体验。任务综合设置能力压力，环境综合提供交互循环，轨迹综合将执行历史转化为可训练的信号。

5.3 预训练和训练中

预训练和训练中期是内化长时程智能体（Long-Horizon Agents）所需能力的最早阶段。预训练通过大规模自监督学习建立广泛的语言、事实、编码、数学和推理先验，而中期训练在指令调整或 强化学习（Reinforcement Learning） 之前进一步重塑这些先验。对于 长时程智能体（Long-Horizon Agents） 来说，这些阶段很重要，因为后来的监督微调（SFT）、对齐和策略优化只能引出和细化基础模型中已经充分体现的能力。因此，我们从四个方面讨论预训练和训练中：推理能力、长上下文状态建模、多模态感知以及支持这些能力的数据混合和训练策略。

5.3.1 Reasoning Priors

Reasoning is a foundation for long-horizon agency because agents must decompose goals, maintain intermediate hypotheses, evaluate partial results, and revise plans over multiple steps. Code and mathematics are representative sources for injecting such priors during pre-training and mid-training, since they expose models to structured dependencies, executable programs, multi-step derivations, and verifiable intermediate states. Recent model reports increasingly emphasize coding, mathematics, reasoning, and agentic capabilities as part of their training recipes, including Qwen2.5 [744], DeepSeek-V3 [110], GLM-4.5 and GLM-5 [178, 179], and Kimi K2 [597]. Qwen3-Coder-Next [46] makes this direction more explicit through targeted mid-training toward code reasoning, repository-level understanding, and agent-style interaction patterns. These structured domains make reasoning supervision closer to long-horizon agent workloads than ordinary short-form text, because they require cross-step consistency, error checking, and compositional problem solving.

5.3.2 Long-Context State

Long-context capability provides the substrate for maintaining task state across extended interactions. For a long-horizon agent, context is not merely a longer input window; it serves as a serialized working state that may contain goals, observations, retrieved evidence, tool outputs, failed attempts, intermediate plans, and revised decisions. A first line of work extends the usable window by rescaling rotary position embeddings: Position Interpolation linearly rescales positions to fit a target length [66], YaRN improves this with an NTK-aware, frequency-dependent schedule for cheaper extension [477], and LongLoRA makes such extension parameter-efficient through shifted sparse attention during fine-tuning [78]. A complementary line studies the training recipe rather than the positional scheme. ProLong shows that long-context capability depends on continued-training data composition, training sequence length, and instruction tuning [171]; UltraLong pushes context to the million-token regime through continued pre-training and instruction tuning [721]; and LongWriter shows that long input windows do not by themselves induce long-output behavior without long-output training examples [29]. Recent frontier models also operationalize this direction. GLM-5.2 couples a 1M-token context window with specialized training for long-horizon coding-agent scenarios [179]. What matters for agents is therefore whether extended context is actually used for state tracking and evidence integration, not the maximum length a model nominally accepts [133, 625].

5.3.3 Multimodal Perception

Long-horizon agents increasingly operate in environments where task state is distributed across webpages, screenshots, videos, and audio streams; their perceptual priors must therefore extend beyond linguistic shortcuts that can mask basic visual failures [61, 502, 608, 624]. Open vision–language models establish the basic substrate: Cambrian-1 explores vision-centric representations and visual grounding [607]; Qwen2-VL perceives images and long videos at varying scales through dynamic resolution and multimodal rotary position embedding [644]; and the InternVL3 series advances open pre-training for high-resolution, long-context visual understanding [654, 892]. Recent industrial reports push these priors toward agent-centric perception, with the common thread being that visual grounding and GUI control are trained as native capabilities rather than added after a text-only stage: Seed1.5-VL and the subsequent Seed1.6 series pre-train on large-scale interleaved vision, text, and interaction data to sustain long-video understanding and GUI control across evolving screens [42, 194], while MiMo-VL treats visual grounding and GUI localization as first-class targets on a reasoning-optimized backbone [393, 394]. On the omni-modal side, Qwen3.5-Omni and Kimi K2.5 extend joint text, vision, and audio pre-training toward multimodal agentic intelligence [498, 598]. The shared implication is that long-horizon capability depends on persistent multimodal state tracking and cross-modal evidence grounding, not textual reasoning alone.

5.3.4 Data Mixture and Training Strategies

The above capabilities are shaped by how training data are selected, mixed, and structured. Recent work shifts pre-training from generic web-scale language modeling toward capability-oriented data distribution design. DataComp-LM [316], FineWeb [475], Dolma [552], and OLMo [184] make pre-training data construction more reproducible. Beyond static corpus construction, DoReMi [715] and Data Mixing Laws [767] show that domain proportions can be optimized or predicted with proxy experiments, while OPUS [649] performs

原文第 44 页 | 中文翻译初稿

5.3.1 推理先验

推理是 长时程（Long-Horizon） 智能体的基础，因为智能体必须分解目标、维持中间假设、评估部分结果并通过多个步骤修改计划。代码和数学是在预训练和训练中期注入此类先验的代表性来源，因为它们将模型暴露给结构化依赖项、可执行程序、多步推导和可验证的中间状态。最近的模型报告越来越强调编码、数学、推理和智能体能力作为其训练方案的一部分，包括 Qwen2.5 [744]、DeepSeek-V3 [110]、GLM-4.5 和 GLM-5 [178, 179] 以及 Kimi K2 [597]。 Qwen3-Coder-Next [46] 通过针对代码推理、存储库级理解和智能体式交互模式的有针对性的中期训练，使这一方向更加明确。这些结构化域使推理监督比普通短格式文本更接近 长时程（Long-Horizon） 智能体工作负载，因为它们需要跨步骤一致性、错误检查和组合问题解决。

5.3.2 长上下文状态

长上下文功能为跨扩展交互维护任务状态提供了基础。对于 长时程（Long-Horizon） 智能体来说，上下文不仅仅是一个较长的输入窗口；它作为一个序列化的工作状态，可能包含目标、观察结果、检索到的证据、工具输出、失败的尝试、中间计划和修订的决定。第一项工作通过重新调整旋转位置嵌入来扩展可用窗口：位置插值线性地重新调整位置以适应目标长度[66]，YaRN 通过 NTK 感知、频率相关的调度来改进这一点，以实现更便宜的扩展[477]，而 LongLoRA 通过在微调期间转移稀疏注意力来使这种扩展参数高效[78]。补充线研究训练方案而不是位置方案。 ProLong 表明，长上下文能力取决于持续训练数据组成、训练序列长度和指令调整 [171]； UltraLong 通过持续的预训练和指令调整将上下文推向百万代币状态 [721]； LongWriter 表明，如果没有长输出训练示例，长输入窗口本身不会引起长输出行为 [29]。最近的前沿模型也实现了这个方向。 GLM-5.2 将 1M 令牌上下文窗口与 长时程（Long-Horizon） 编码智能体场景的专门训练结合起来 [179]。因此，对于智能体来说，重要的是扩展上下文是否实际上用于状态跟踪和证据集成，而不是模型名义上接受的最大长度[133, 625]。

5.3.3 多模态感知

长时程智能体（Long-Horizon Agents） 越来越多地在任务状态分布在网页、屏幕截图、视频和音频流中的环境中运行；因此，他们的感知先验必须超越可能掩盖基本视觉缺陷的语言捷径[61,502,608,624]。开放视觉语言模型建立了基本基础：Cambrian-1 探索以视觉为中心的表示和视觉 落地关联（Grounding） [607]； Qwen2-VL通过动态分辨率和多模态旋转位置嵌入感知不同尺度的图像和长视频[644]

; InternVL3 系列推进了高分辨率、长上下文视觉理解的开放式预训练 [654, 892]。最近的行业报告将这些先验推向以智能体为中心的感知，共同的线索是视觉 落地关联（Grounding） 和 GUI 控制作为本机功能进行训练，而不是在纯文本阶段之后添加：Seed1.5-VL 和后续的 Seed1.6 系列对大规模交错视觉、文本和交互数据进行预训练，以维持跨不断发展的屏幕的长视频理解和 GUI 控制 [42, 194]，而 MiMo-VL 则处理视觉落地关联（Grounding） 和 GUI 本地化作为推理优化主干上的一流目标 [393, 394]。在全模态方面，Qwen3.5-Omni 和 Kimi K2.5 将联合文本、视觉和音频预训练扩展到多模态智能体智能 [498, 598]。共同的含义是，长时程（Long-Horizon） 能力取决于持久的多模态状态跟踪和跨模态证据 落地关联（Grounding），而不仅仅是文本推理。

5.3.4 数据混合和训练策略

上述功能取决于训练数据的选择、混合和结构化方式。最近的工作将预训练从通用的网络规模语言建模转向面向能力的数据分布设计。 DataComp-LM [316]、FineWeb [475]、Dolma [552] 和 OLMo [184] 使预训练数据构建更具可重复性。除了静态语料库构建之外，DoReMi [715] 和数据混合定律 [767] 表明，可以通

过智能体实验来优化或预测域比例，而 OPUS [649] 则执行

44

principled data selection at each pre-training iteration by scoring sample utility in the optimizer-induced update space, shifting pre-training from more tokens to better tokens and improving both from-scratch and continued pre-training under a fixed compute budget. For long-horizon agents, such data strategies act as inductive biases for reasoning, state modeling, and compositional problem solving, rather than merely as mechanisms for scaling factual knowledge. A further strategy is to structure training data as trajectories rather than isolated text samples. AgentTuning [794], Agent-FLAN [79], and AgentBank [557] show that interaction trajectories can improve generalized agent abilities such as planning, memory, and tool use. Since high-quality long-horizon trajectories are scarce, synthetic and verifier-guided data are also used to control horizon length, task difficulty, feedback structure, and failure-recovery patterns [29, 46]. Here the decisive factor is not what the data contain but whether they preserve temporal dependencies, interaction feedback, and recovery processes.

Taken together, pre-training and mid-training form the first stage of internalizing long-horizon capability. Reasoning-oriented data provide structured problem-solving priors, long-context training supports state maintenance over extended interactions, multimodal training enables persistent observation across heterogeneous evidence, and data mixture strategies determine how these signals are balanced and selected. These latent priors are then elicited, sequenced, and optimized by SFT and RL into reliable agentic behavior.

5.4 Fine-tuning

Fine-tuning converts the latent priors acquired during pre-training and mid-training into controllable long-horizon behavior. Whereas pre-training shapes broad parameter-level capabilities, supervised fine-tuning (SFT) teaches the model how these capabilities should be expressed during interaction: how to decompose goals, call tools, interpret observations, update state, recover from errors, and terminate with a final answer. For long-horizon agents, SFT is therefore process-level imitation over trajectories that contain intermediate reasoning, actions, tool results, and state transitions, rather than instruction-following over isolated input–output pairs. It typically serves as a cold-start stage that constrains the model to valid, task-relevant interaction patterns before any online optimization.

The central issue is how to construct a supervised trajectory distribution that is diverse enough to teach general agent behavior, structured enough to ensure valid tool use, and progressive enough to prepare the model for later stages. We organize this section around three questions: which demonstrations to select and how to mix them, how to stage them through a curriculum, and how to obtain them through distillation.

5.4.1 Instruction Selection and Mixing

Trajectory Quality Over Quantity. Fine-tuning long-horizon agents requires data that expose not only correct answers, but also the intermediate decisions connecting goals, observations, actions, and recovery behaviors across many turns. AgentTuning [794] constructs AgentInstruct, a compact set of high-quality interaction trajectories, and mixes it with general instruction data to strengthen agent skills while preserving general ability. FireAct [54] shows that trajectories generated by stronger models improve agent behavior beyond prompt-only adaptation, and LIMI [711] argues that a small set of strategically curated demonstrations can elicit strong agency-oriented behavior. Rather than cloning full trajectories, ATLaS [81] identifies the *critical steps* of planning, subgoal reasoning, and strategic decision making, and fine-tunes only on them, mitigating expert bias and improving cross-environment generalization. These works reframe agent SFT as an instruction-selection problem: prioritizing trajectory quality, decision-point coverage, and task diversity over raw demonstration count.

Compact Reasoning Demonstrations. A complementary line explains why small trajectory sets can be effective. LIMO [770] elicits complex mathematical reasoning from a strong base model with only 817 curated demonstrations, treating them as cognitive templates over knowledge already encoded during pre-training; s1 [442] builds a 1K-example set under quality, difficulty, and diversity criteria; and LIMR [344] extends the less-is-more observation to reinforcement learning through learning-impact-based sample selection. Unified Data Selection [342] sharpens this idea by selecting samples around high-entropy tokens, which mark critical branching points in reasoning trajectories. This "less-is-more" view is not unchallenged: Ding et al. [122] report that, in an SFT-then-RL pipeline, data scale still governs the attainable performance ceiling while

原文第 45 页 | 中文翻译初稿

通过在优化器引起的更新空间中对样本效用进行评分，将预训练从更多标记转移到更好标记，并在固定计算预算下改进从头开始和持续预训练，在每次预训练迭代中进行有原则的数据选择。对于 长时程智能体（Long-Horizon Agents） 来说，此类数据策略充当推理、状态建模和组合问题解决的归纳偏差，而不仅仅是扩展事实知识的机制。进一步的策略是将训练数据构建为轨迹而不是孤立的文本样本。 AgentTuning [794]、Agent-FLAN [79] 和 AgentBank [557] 表明交互轨迹可以提高广义智能体能力，例如规划、记忆和工具使用。由于高质量的 长时程（Long-Horizon） 轨迹稀缺，因此合成和验证者引导的数据也用于控制范围长度、任务难度、反馈结构和故障恢复模式 [29, 46]。这里的决定性因素不是数据包含什么，而是它们是否保留时间依赖性、交互反馈和恢复过程。

预训练和训练中期共同构成长时程（Long-Horizon）能力内化的第一阶段。面向推理的数据提供结构化的解决问题的先验，长上下文训练支持扩展交互的状态维护，多模态训练支持跨异构证据的持续观察，数据混合策略决定如何平衡和选择这些信号。然后，通过 SFT 和 RL 引出、排序并优化这些潜在先验，形成可靠的智能体行为。

5.4 微调

微调将预训练和训练中期获得的潜在先验转化为可控的长期行为。预训练塑造了广泛的参数级功能，而监督微调（SFT）则教导模型如何在交互过程中表达这些功能：如何分解目标、调用工具、解释观察结果、更新状态、从错误中恢复以及以最终答案终止。因此，对于长视野智能体来说，SFT 是对包含中间推理、动作、工具结果和状态转换的轨迹的过程级模仿，而不是对孤立的输入输出对的指令跟踪。它通常充当冷启动阶段，在任何在线优化之前将模型限制为有效的、与任务相关的交互模式。

核心问题是如何构建一个有监督的轨迹分布，该轨迹分布足够多样化以教授一般智能体行为，足够结构化以确保有效的工具使用，并且足够渐进以为后续阶段准备模型。我们围绕三个问题组织本节：选择哪些演示以及如何混合它们，如何通过课程安排它们，以及如何通过蒸馏获得它们。

5.4.1 指令选择和混合

轨迹质量胜于数量。微调 长时程智能体（Long-Horizon Agents） 需要的数据不仅公开正确的答案，而且还公开连接多个回合的目标、观察、行动和恢复行为的中间决策

。 AgentTuning [794] 构建了 AgentInstruct，这是一组紧凑的高质量交互轨迹，并将其与通用指令数据混合，以增强智能体技能，同时保留通用能力。 FireAct [54] 表明，由更强大的模型生成的轨迹可以改善智能体行为，超越仅提示的适应，而 LIMI [711] 认为，一小部分策略性策划的演示可以引发强烈的智能体导向行为。 ATLaS [81] 不是克隆完整的轨迹，而是确定规划、子目标推理和战略决策的“关键步骤”，并仅对它们进行微调，从而减轻专家偏见并提高跨环境泛化能力。这些工作将智能体 SFT 重新构建为指令选择问题：优先考虑轨迹质量、决策点覆盖范围和任务多样性，而不是原始演示计数。

紧凑推理演示。一条互补线解释了为什么小轨迹组可以有效。 LIMO [770] 从一个强大的基础模型中引出了复杂的数学推理，只有 817 个精心策划的演示，将它们视为预训练期间已经编码的知识的认知模板；s1 [442] 根据质量、难度和多样性标准构建 1K 示例集； LIMR [344]通过基于学习影响的样本选择将少即是多的观察扩展到强化学习（Reinforcement Learning）。统一数据选择[342]通过选择高熵标记周围的样本来强化这一想法，这些标记标记了推理轨迹中的关键分支点。这种“少即是多”的观点并非没有受到挑战：Ding 等人。 [122] 报告称，在 SFT-then-RL 管道中，数据规模仍然决定着可达到的性能上限，而

trajectory difficulty acts as a multiplier, suggesting that selection should maximize decision coverage rather than merely shrink the dataset.

Executable Tool-use Data and Mixture. For tool-using agents, data value further depends on whether actions are syntactically valid, executable, and grounded in observations. CodeAct [656] represents actions as executable Python code, enabling code execution, observation feedback, and self-debugging; APIGen [392] generates function-calling data verified through format, execution, and semantic checks; ToolACE [385] scales synthetic tool-learning data with a self-evolving API pool and multi-agent dialogue generation; and TOUCAN [737] synthesizes large-scale multi-tool, multi-turn trajectories from real-world Model Context Protocol (MCP) environments. Selection also matters at scale: EDGE [844] picks informative multi-turn samples according to guideline effectiveness. Because such corpora remain fragmented across incompatible formats, the Agent Data Protocol [558] introduces a unified schema that standardizes heterogeneous agent datasets into a common trajectory representation; SFT on the resulting cross-domain mixture yields broad gains and avoids the negative transfer induced by single-domain tuning.

5.4.2 Curriculum Learning

Staging by Horizon. Long-horizon trajectories are difficult to imitate directly because errors compound over time and the model must maintain state across many steps before receiving any signal of success or failure. Curriculum learning addresses this by staging the training distribution rather than exposing the model to full-length tasks at once, with the interaction horizon as a natural first axis. ScalingInter-RL, introduced in AgentGym-RL [701], begins with short-horizon exploitation to establish stable foundational policies and then progressively enlarges the interaction budget, mitigating the collapse that plagues naive long-horizon optimization. Studying horizon in isolation, Kim et al. [279] construct controlled task suites that vary only the required action-sequence length and show that horizon itself is a training bottleneck: reducing the horizon first and then extending it stabilizes learning and induces *horizon generalization*, where competence acquired on short goals transfers to unseen longer ones at inference time.

Staging by Difficulty. A second axis orders tasks from easy to hard. E2H Reasoner [469] schedules problems probabilistically from easy to difficult, and shows both empirically and through approximate-policy-iteration bounds that gradually fading out easy tasks prevents overfitting and lets small models acquire reasoning skills that vanilla training cannot reach. Curricula for long-horizon agents can be graded along several dimensions at once, including horizon length, task difficulty, tool complexity, feedback density, and recovery demand, so that a staged cold-start distribution first instills basic interaction competence before later stages optimize decisions under delayed rewards.

5.4.3 Distillation

Because high-quality long-horizon demonstrations are scarce, many fine-tuning pipelines obtain supervision by distilling interaction processes from stronger models, scaffolded agent systems, or executable environments. This is better understood as trajectory distillation rather than answer distillation: the student learns not only final responses, but also planning steps, action formats, observation interpretation, and recovery strategies. We use *distillation* here in its offline sense, where supervision comes from fixed teacher or past-model trajectories consumed as static SFT data, in contrast to the on-policy distillation of Section 5.6 and the closed-loop self-evolution of Section 5.7.

Strong-to-weak Distillation. *Strong-to-weak distillation* transfers agentic behavior from powerful teacher models or expensive agent systems into smaller, deployable students, reusing costly execution as supervised traces. Because agent competence lives in full reason–act–observe trajectories, recent work supervises structure rather than final answers. Agent Distillation [269] transfers retrieval and code-execution trajectories from a large teacher into sub-3B students, using a first-thought prefix and self-consistent action generation to keep tiny agents competitive with far larger chain-of-thought-distilled baselines. Structured Agent Distillation [376] segments each trajectory into [REASON] and [ACT] spans with segment-specific losses, preserving both reasoning fidelity and action validity under compression. Pushing the teacher side further, AgentArk [410] distills a

原文第 46 页 | 中文翻译初稿

轨迹难度充当乘数，表明选择应该最大化决策覆盖范围，而不是仅仅缩小数据集。

可执行工具使用数据和混合物。对于使用工具的智能体来说，数据价值进一步取决于动作在语法上是否有效、可执行以及是否基于观察。 CodeAct [656]将动作表示为可执行的Python代码，支持代码执行、观察反馈和自调试； APIGen [392] 生成通过格式、执行和语义检查验证的函数调用数据； ToolACE [385] 通过自我进化的 API 池和多智能体对话生成来扩展合成工具学习数据； TOUCAN [737] 从真实的 世界模型（World Model） 上下文协议（MCP）环境中合成大规模多工具、多转弯轨迹。选择在规模上也很重要：EDGE [844] 根据指南有效性挑选信息丰富的多轮样本。由于此类语料库在不兼容的格式中仍然分散，智能体数据协议[558]引入了一个统一的模式，将异构智能体数据集标准化为通用轨迹表示；对所得跨域混合进行 SFT 会产生广泛的增益，并避免单域调谐引起的负迁移。

5.4.2 课程学习

由地平线分期。 长时程（Long-Horizon） 轨迹很难直接模仿，因为错误会随着时间的推移而复合，并且模型必须在接收任何成功或失败的信号之前在多个步骤中保持状态。课程学习通过分阶段训练分布来解决这个问题，而不是立即将模型暴露于完整的任务，并将交互范围作为自然的第一轴。 AgentGym-RL [701] 中引入的 ScalingInter-RL，从短期利用开始建立稳定的基础策略，然后逐步扩大交互预算，减轻困扰 长时程（Long-Horizon） 优化的崩溃。 Kim 等人孤立地研究地平线。 [279]构建仅改变所需的动作序列长度的受控任务套件，并表明视野本身是一个训练瓶颈：首先缩小视野，然后扩展它，稳定学习并诱导_视野泛化_，其中在短目标上获得的能力在推理时转移到看不见的较长目标。

按难度分级。第二个轴将任务从简单到困难排序。 E2H Reasoner [469] 按概率从易到难地安排问题，并通过经验和近似策略迭代界限表明，逐渐淡出简单的任务可以防止过度拟合，并让小模型获得普通训练无法达到的推理技能。 长时程智能体（Long-Horizon Agents）的课程可以同时从多个维度进行评分，包括时间长度、任务难度、工具复杂度、反馈密度和恢复需求，以便分阶段的冷启动分配首先灌输基本的交互能力，然后再在后期优化延迟奖励下的决策。

5.4.3 蒸馏

由于高质量的 长时程（Long-Horizon） 演示很稀缺，因此许多微调管道通过从更强大的模型、脚手架智能体系统或可执行环境中提取交互过程来获得监督

。这可以更好地理解为轨迹蒸馏而不是答案蒸馏：学生不仅学习最终反应，还学习计划步骤、行动格式、观察解释和恢复策略。我们在这里使用离线意义上的“蒸馏”，其中监督来自作为静态 SFT 数据消耗的固定教师或过去模型轨迹，这与第 5.6 节的在策略蒸馏和第 5.7 节的闭环自我演化形成鲜明对比。

由强到弱的蒸馏。 _从强到弱的蒸馏_将智能体行为从强大的教师模型或昂贵的智能体系统转移到更小的、可部署的学生中，重用昂贵的执行作为监督跟踪。由于智能体能力存在于完整的理性-行动-观察轨迹中，因此最近的工作监督结构而不是最终答案。智能体蒸馏[269]将检索和代码执行轨迹从大型教师转移到低于 3B 的学生，使用第一思想前缀和自洽的动作生成来保持小型智能体与更大的思想链蒸馏基线的竞争。结构化智能体蒸馏[376]将每个轨迹分割为[原因]和[行为]跨度，并具有特定于段的损失，在压缩下保持推理保真度和行动有效性。 AgentArk [410] 进一步推动教师方面的发展

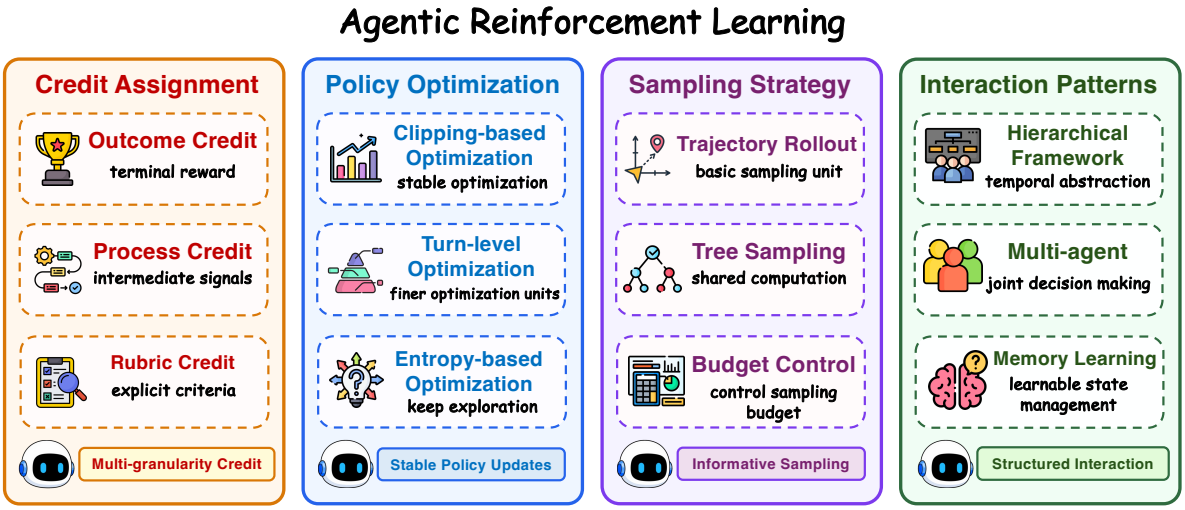


Figure 12 Agentic reinforcement learning through four key questions. These questions correspond to the four components discussed in Section 5.5: credit assignment, policy optimization, sampling strategy and interaction structure.

multi-agent debate system into a single model, turning expensive test-time interaction into implicit self-correction ability. Across these methods, effective distillation aligns supervision with trajectory structure rather than compressing outputs alone.

Self-distillation and Reflective Self-improvement. *Self-distillation* removes the external teacher: the model generates its own trajectories, filters or reflects on them, and trains back on the curated result. Since many long-horizon failures are local, such as choosing a wrong tool, misreading an observation, or failing to recover after an execution error, the central design choice is how to turn raw rollouts into clean, corrective supervision. Watch Every Step [719] explores along expert trajectories, collects step-level process feedback, and learns from contrastive action pairs. Agent-R [789] uses Monte Carlo Tree Search to locate the first erroneous step in a failed rollout and splices it onto an adjacent correct path, so that iterative fine-tuning teaches timely on-the-fly correction rather than end-of-episode fixes. Reflection enriches the signal further: RESD [845] converts failed rollouts into retrospective diagnoses and a reusable "playbook" of lessons, while HERO [374] turns each observation into a turn-level hindsight reflection used to re-score the student's own actions. These reflection-driven variants shade into the on-policy self-distillation of Section 5.6; used offline, their filtered traces still supply denser process supervision than final-answer imitation.

Both regimes share a lesson: distillation for long-horizon agents must preserve *process* structure, whether transferring capability from strong teachers or recycling an agent's own history into cleaner supervision. Both can also amplify biases and environment-specific habits, so execution-based verification, semantic filtering, and mixture with general instruction data remain essential for robust, transferable behavior.

Overall, fine-tuning determines how long-horizon priors surface as interactive behavior: instruction selection defines the supervised behavior distribution, curriculum learning stages it by horizon and difficulty, and distillation transfers complex interaction processes from stronger or self-generated sources. It improves controllability, tool-use validity, state tracking, and recovery, yet stays bounded by the coverage of supervised demonstrations, leaving decisions under delayed rewards and compounding errors to later online optimization.

5.5 Agentic Reinforcement Learning

Compared with supervised fine-tuning, Agentic Reinforcement Learning (Agentic RL) further optimizes the policy against feedback from the agent's own interaction. This extends the optimization targets of LLMs from single-turn responses to long-horizon interaction trajectories. As illustrated in Figure 12, the core challenges involve four aspects: sampling informative multi-turn trajectories, deriving multi-granularity credit signals

原文第 47 页 | 中文翻译初稿

智能体强化学习（Reinforcement Learning）

贡献归因（Credit Assignment） 策略优化 抽样策略 交互模式
基于结果信用裁剪的轨迹 展开轨迹（Rollout） 分层
优化框架
终端奖励基本抽样单元
稳定优化时间抽象
过程信用 轮级树抽样 多智能体
中间信号 优化共享计算联合决策
更精细的优化单元
Rubric Credit基于熵的预算控制记忆学习
显式准则优化控制采样可学习状态
不断探索预算管理
多粒度信用稳定策略更新信息采样结构化交互

图 12 Agentic 强化学习（Reinforcement Learning） 通过四个关键问题。这些问题对应于第 5.5 节中讨论的四个组成部分：贡献归因（Credit Assignment）、策略优化、抽样策略和交互结构。

将多智能体辩论系统转化为单一模型，将昂贵的测试时交互转化为隐含的自我纠正能力。在这些方法中，有效的蒸馏将监督与轨迹结构结合起来，而不是单独压缩输出。

自蒸馏和反射 自我改进（Self-Improvement）。 _自我蒸馏_消除了外部教师：模型生成自己的轨迹，过滤或反映它们，并根据策划的结果进行训练。由于许多 长时程（Long-Horizon） 故障都是局部的，例如选择错误的工具、误读观察或在执行错误后无法恢复，因此中心设计选择是如何将原始 展开轨迹（Rollouts） 转变为干净的、纠正性的监督。观察每一步 [719] 沿着专家轨迹进行探索，收集步骤级过程反馈，并从对比动作对中学习。 Agent-R [789] 使用蒙特卡罗树搜索来定位失败的 展开轨迹（Rollout） 中的第一个错误步骤，并将其拼接到相邻的正确路径上，以便迭代微调能够及时进行即时纠正，而不是在片段结束时进行修复。反思进一步丰富了信号：RESD [845] 将失败的 展开轨迹（Rollouts） 转化为回顾性诊断和可重复使用的课程“剧本”，而 HERO [374] 将每个观察结果转化为回合级事后反思，用于对学生自己的行为重新评分。这些反射驱动的变体逐渐融入第 5.6 节的 on-policy 自我蒸馏；离线使用时，它们的过滤痕迹仍然提供比最终答案模仿更密集的过程监督。

两种制度都有一个教训：长时程智能体（Long-Horizon Agents） 的蒸馏必须保留_流程_结构，无论是从强大的教师那里转移能力，还是将智能体人自己的历史回收到更清洁的监督中。两者都会放大偏见和特定于环境的习惯，因此基于执行的验证、语义过滤以及与一般指令数据的混合对于稳健、可转移的行为仍然至关重要。

总体而言，微调决定了 长时程（Long-Horizon） 先验如何呈现为交互行为：指令选择定义了监督行为分布，课程学习按范围和难度对其进行分级，蒸馏从更强或自生成的来源转移了复杂的交互过程。它提高了可控性、工具使用有效性、状态跟踪和恢复，但仍受到监督演示的覆盖范围的限制，导致延迟奖励决策和复合错误需要稍后在线优化。
5.5 智能体强化学习（Reinforcement Learning）

与监督微调相比，Agentic 强化学习（Reinforcement Learning）（Agentic RL）根据智能体自身交互的反馈进一步优化策略。这将 LLM 的优化目标从单轮响应扩展到 长时程（Long-Horizon） 交互轨迹。如图12所示，核心挑战涉及四个方面：采样信息丰富的多转弯轨迹、导出多粒度信用信号

47

from sparse feedback, integrating these signals into stable policy updates, and organizing training interaction patterns such as hierarchy, tool use, multi-agent collaboration, and memory. In the following subsections, we elaborate on credit assignment, policy optimization, and sampling methods in Agentic RL, as well as several unique interaction patterns of Agentic RL. Representative works for each part are shown in Table 6.

5.5.1 Credit Assignment

Credit assignment is one of the core challenges for Agentic RL. It concerns how to attribute credit to heterogeneous decision units when an agent completes long-horizon tasks. This section reviews credit assignment from three perspectives: outcome credit assignment, which scores only the final result; process credit assignment, which shapes the intermediate decision process; and rubric credit assignment, which expresses quality through explicit semantic criteria.

Outcome Credit Assignment. Outcome credit assignment is the most direct form of credit assignment, where the reward is determined only by the final result, such as exact match, F1, task success, final-state checks, unit-test results, or retrieved-answer correctness. Its appeal lies in being automatable, cheap to verify, and easy to scale to large online rollouts. This paradigm was first validated by verifiable terminal rewards in mathematical and code reasoning [112, 345, 533]. It was subsequently carried into interactive settings by a line of web and information-seeking agents led by Search-R1, which reuse task success, environment-state checks, or retrieved-answer correctness as terminal rewards across long-horizon application environments [60, 256, 261, 681, 871]. However, this form of credit is coarse-grained. A single terminal score is shared across the entire trajectory, conflating key exploration, redundant actions, and early mistakes into one learning signal. As a result, the policy struggles to discriminate the quality of individual local decisions.

Process Credit Assignment. Process supervision predates agentic RL in mathematical reasoning, where process reward models (PRMs) score intermediate steps rather than only final answers [363, 643]; reinforcement learning with verifiable rewards then scaled outcome-based optimization for reasoning [112, 533]. Agentic credit assignment inherits this lineage and extends it to multi-step, tool-using trajectories. Process credit assignment extends the evaluation target from the final answer to the intermediate process. Existing methods follow two complementary routes. The first decomposes the quality of an individual decision: a group of tool-use-oriented methods represented by ToolRL break tool invocation into separately scoreable dimensions, including output format, tool selection, and parameter names and values [129, 487, 834]. The second distributes denser reward across the interaction process rather than scoring a single decision, spanning a spectrum of granularity: some works such as AdaTIR and AgentPRM assign reward to intermediate units of the trajectory rather than to its endpoint alone [77, 150, 318, 554, 682, 702, 837]. Among them, some methods fold process signals such as interaction cost and query quality into trajectory-level rewards, while others attach explicit step- or turn-level credit via process reward models or advantage redistribution. Across both routes, the shared goal is not merely to make rewards denser, but to align credit signals with the true decision structure of the agent.

Rubric Credit Assignment. Rubric credit assignment expresses answer or trajectory quality through explicit, multi-dimensional semantic criteria (e.g., factuality, completeness, reasoning soundness, evidence grounding, and safety) when a single verifiable metric is insufficient; a recent paper separates such rubrics from reward models, verifiable rewards, and LLM-as-a-judge [386]. Methods differ along two axes: how rubrics are constructed, and how they are turned into credit.

On construction, rubrics are produced along an increasingly automated spectrum: (1) strong LLMs or human experts author checklists used directly as scoring criteria or as training rewards [534], as in Rubrics as Rewards and the rubric bank of Reinforcement Learning with Rubric Anchors [192, 240]; (2) rubrics are induced from preference contrasts, where OpenRubrics extracts rules and implicit principles filtered by preference-label consistency and Chasing the Tail targets the high-reward tail to curb over-optimization [384, 819]; and (3) rubrics are made context-grounded and online, as when Agentic Rubrics lets an expert agent inspect a repository before generating criteria and DR Tulu co-evolves positive and negative rubrics with the policy during deep-research training [500, 532].

On conversion, rubrics become credit signals (1) as direct RL rewards, aggregated by weights or an LLM judge and optionally decomposed into fine-grained signals such as answer correctness, step validity, and premise

原文第 48 页 | 中文翻译初稿

从稀疏的反馈中，将这些信号集成到稳定的策略更新中，并组织训练交互模式，例如层次结构、工具使用、多智能体协作和记忆。在下面的小节中，我们将详细介绍 Agentic RL 中的 贡献归因（Credit Assignment）、策略优化和采样方法，以及 Agentic RL 的几种独特的交互模式。各部分的代表作品如表6所示。

5.5.1 贡献归因（Credit Assignment）

贡献归因（Credit Assignment） 是 Agentic RL 的核心挑战之一。它涉及当智能体完成 长时程（Long-Horizon） 任务时如何将功劳归因于异构决策单元。本节从三个角度来回顾贡献归因（Credit Assignment）：结果贡献归因（Credit Assignment），只对最终结果进行评分；过程贡献归因（Credit Assignment），形成中间决策过程；以及标题 贡献归因（Credit Assignment），它通过明确的语义标准来表达质量。

结果 贡献归因（Credit Assignment）。结果 贡献归因（Credit Assignment） 是 贡献归因（Credit Assignment） 最直接的形式，其中奖励仅由最终结果决定，例如完全匹配、F1、任务成功、最终状态检查、单元测试结果或检索答案正确性。它的吸引力在于可自动化、验证成本低廉并且易于扩展到大型在线展开轨迹（Rollouts）。该范式首先通过数学和代码推理中可验证的终端奖励进行验证[112,345,533]。随后，由 Search-R1 领导的一系列网络和信息搜索智能体将其带入交互式设置，这些智能体在 长时程（Long-Horizon） 应用程序环境中重用任务成功、环境状态检查或检索到的答案正确性作为终端奖励[60,256,261,681,871]。然而，这种形式的信用是粗粒度的。整个轨迹共享一个最终分数，将关键探索、冗余动作和早期错误合并为一个学习信号。因此，该策略很难区分各个地方决策的质量。

进程 贡献归因（Credit Assignment）。过程监督早于数学推理中的智能体强化学习，其中过程奖励模型 (PRM) 对中间步骤进行评分，而不仅仅是最终答案 [363, 643]； 强化学习（Reinforcement Learning） 具有可验证的奖励，然后扩展基于结果的推理优化 [112, 533]。 Agentic 贡献归因（Credit Assignment） 继承了这一血统，并将其扩展到多步骤、使用工具的轨迹。过程贡献归因（Credit Assignment）将评估目标从最终答案延伸到中间过程。现有方法遵循两条互补的路线。第一个分解了单个决策的质量：以 ToolRL 为代表的一组面向工具使用的方法将工具调用分解为单独可评分的维度，包括输出格式、工具选择以及参数名称和值 [129,487,834]。第二个在整个交互过程中分配更密集的奖励，而不是对单个决策进行评分，跨越一系列粒度：AdaTIR 和 AgentPRM 等一些作品将奖励分配给轨迹的中间单元，而不是单独分配给其端点 [77, 150, 318, 554, 682, 702, 837]

。其中，一些方法将交互成本和查询质量等过程信号折叠到轨迹级奖励中，而其他方法则通过过程奖励模型或优势重新分配附加明确的步骤或回合级信用。在这两条路线中，共同的目标不仅是使奖励更加密集，而且使信用信号与智能体的真实决策结构保持一致。

标题 贡献归因（Credit Assignment）。当单个可验证指标不足时，Rubric 贡献归因（Credit Assignment）通过明确的多维语义标准（例如事实性、完整性、推理合理性、证据 落地关联（Grounding） 和安全性）来表达答案或轨迹质量；最近的一篇论文将此类评价标准与奖励模型、可验证奖励和法学硕士法官分开[386]。方法在两个方面有所不同：如何构建评分标准，以及如何将其转化为信用。

在构建过程中，Rubrics 是沿着日益自动化的范围产生的：(1) 强大的法学硕士或人类专家编写直接用作评分标准或训练奖励的清单 [534]，如 Rubrics as Rewards 和带有 Rubric Anchors 的 强化学习（Reinforcement Learning） 的 rubric 库 [192, 240]； （2）Rubrics是从偏好对比中归纳出来的，其中OpenRubrics提取通过偏好标签一致性过滤的规则和隐含原则，而Chasing the Tail则针对高奖励尾部以抑制过度优化[384, 819]； (3) 规则是基于上下文和在线的，就像 Agentic Rubrics 让专家智能体在生成标准

之前检查存储库一样，DR Tulu 在深度研究训练期间与策略共同发展积极和消极的规则 [500, 532]。

转换后，评分细则成为信用信号 (1) 作为直接 RL 奖励，通过权重或 LLM 法官进行聚合，并可选择分解为细粒度信号，例如答案正确性、步骤有效性和前提

48

Table 6 Representative works in agentic reinforcement learning. Rows are grouped by the four parts of Section 5.5: credit assignment, policy optimization, sampling strategy, and interaction patterns; **Type** names the mechanism family, **Key Mechanism** gives its one-line idea, and **Resource** links to a public code repository where available.

Method	Type	Key Mechanism	Resource
Credit Assignment			
DeepSeekMath [533]	Outcome Reward	Terminal reward for math reasoning	GitHub
Search-R1 [261]	Outcome Reward	Answer correctness as terminal reward	GitHub
DeepRetrieval [256]	Outcome Reward	Retrieval-metric terminal reward	GitHub
ToolRL [487]	Process Reward	Decomposes tool calls into scored aspects	GitHub
Tool-Star [129]	Process Reward	Monitors answer and tool usage	GitHub
CriticSearch [837]	Process Reward	Retrospective critic for fine-grained credit	–
RaR [192]	Rubric Reward	Expert checklists used directly as rewards	–
OpenRubrics [384]	Rubric Reward	Rubrics induced from preference contrasts	–
DR Tulu [532]	Rubric Reward	Co-evolving rubrics in deep research	GitHub
RuscaRL [887]	Rubric Reward	Rubrics as decaying exploration scaffolds	GitHub
Policy Optimization			
REINFORCE++ [222]	Ratio Clipping	Global advantage normalization	GitHub
DAPO [780]	Ratio Clipping	Decoupled clip-higher/lower	GitHub
Dr.GRPO [389]	Ratio Clipping	Removes length/std normalization bias	GitHub
GSPO [865]	Ratio Clipping	Sequence-level ratio and clipping	–
Turn-PPO [321]	Credit Granularity	Turn-level advantage estimation	–
StepPO [628]	Credit Granularity	Step-aligned advantage	GitHub
GiGPO [156]	Credit Granularity	Episode- and step-level advantage	GitHub
Clip-Cov [103]	Entropy Regularization	Covariance-gated updates control entropy	GitHub
CURE [330]	Entropy Regularization	Critical-token-guided re-concatenation	GitHub
EPO [732]	Entropy Regularization	Trajectory entropy regularization	GitHub
Sampling Strategy			
R1-searcher [553]	Trajectory Rollout	On-policy search-agent trajectory rollout	GitHub
WebDancer [689]	Trajectory Rollout	Web-search trajectory pipeline	GitHub
WebRL [484]	Trajectory Rollout	Self-evolving curriculum web trajectories	GitHub
Tree of Thoughts [760]	Tree Sampling	Organizes thoughts as a search tree	GitHub
RAP [204]	Tree Sampling	Planning as tree search with world model	GitHub
LATS [875]	Tree Sampling	Language-agent tree search over actions	GitHub
TreePO [348]	Tree Sampling	Reuses inference compute across tree paths	GitHub
LiteResearcher [334]	Budget-Aware Pruning	Difficulty/pass-rate-based pruning	GitHub
TCOD [636]	Budget-Aware Pruning	Trajectory-depth curriculum	GitHub
Interaction Patterns			
GLIDER [232]	Hierarchical Framework	Grounds LLMs as decision-making agents	GitHub
SkillRL [705]	Hierarchical Framework	Evolves a skill library from failures	GitHub
THOR [52]	Hierarchical Framework	Joint episode- and step-level optimization	GitHub
AT-GRPO [861]	Multi-agent Framework	Per-agent, per-turn advantage grouping	GitHub
MATPO [439]	Multi-agent Framework	Multiple roles in one LLM over joint rollouts	GitHub
SPIRAL [371]	Multi-agent Framework	Role-conditioned advantages	GitHub
Memory-R1 [743]	Memory-state Learning	Separates memory manager and answer agent	GitHub
Memory-as-Action [846]	Memory-state Learning	Working-memory editing as policy action	GitHub
Agent Workflow Memory [674]	Memory-state Learning	Induces reusable workflows	GitHub
Agent Lightning [409]	Memory-state Learning	Traces workflows into trainable samples	GitHub

原文第 49 页 | 中文翻译初稿

表 6 智能体强化学习中的代表性作品。行按第 5.5 节的四个部分进行分组：贡献归因（Credit Assignment）、策略优化、采样策略和交互模式；类型命名机制系列，关键机制给出其单行想法，资源链接到可用的公共代码存储库。

方法	类型	关键机制	资源
贡献归因（Credit Assignment）			
DeepSeekMath [533]	结果奖励	数学推理的终端奖励	GitHub
Search-R1 [261]	结果奖励	答案正确性作为最终奖励	GitHub
DeepRetrieval [256]	结果奖励	检索度量终端奖励	GitHub
ToolRL [487]	处理奖励	将工具调用分解为评分方面	GitHub
工具之星[129]	流程奖励	监控答案和工具使用情况	GitHub
CriticSearch [837]	过程奖励	细粒度信用的回顾性评论家	−
RaR [192]	Rubric 奖励	直接用作奖励的专家清单	−
OpenRubrics [384]	Rubric 奖励	从偏好对比中得出的Rubrics	−
DR Tulu [532]	Rubric 奖励	深度研究中的共同进化评估	GitHub
RuscaRL [887]	Rubric 奖励	Rubric 作为腐烂的探索脚手架	GitHub
策略优化			
REINFORCE++ [222]	比率裁剪	全球优势标准化	GitHub
DAPO [780]	比率裁剪	解耦裁剪-更高/更低	GitHub
Dr.GRPO [389]	比率裁剪	消除长度/标准标准化偏差	GitHub
GSPO [865]	比率剪辑	序列级比率和剪辑	−
回合-PPO [321]	信用粒度	回合级优势估计	−
StepPO [628]	信用粒度	步调一致的优势	GitHub
GiGPO [156]	信用粒度	情节级和步骤级优势	GitHub
Clip-Cov [103]	熵正则化	协方差门控更新控制熵	GitHub
CURE [330]	熵正则化	关键令牌引导的重新串联	GitHub
EPO [732]	熵正则化	轨迹熵正则化	GitHub
抽样策略			
R1-searcher [553]	轨迹 展开轨迹（Rollout）	策略搜索智能体轨迹 展开轨迹（Rollout）	GitHub
WebDancer [689]	轨迹 展开轨迹（Rollout）	网络搜索轨迹管道	GitHub
WebRL [484]	轨迹 展开轨迹（Rollout）	自我进化课程网络轨迹	GitHub
思想树 [760]	树采样	将思想组织为搜索树	GitHub
RAP [204]	树采样	使用 世界模型（World Model） 进行树搜索规划	GitHub
LATS [875]	树采样	针对操作的语言智能体树搜索	GitHub
TreePO [348]	树采样	跨树路径重用推理计算	GitHub
LiteResearcher [334]	预算感知修剪	基于难度/通过率的修剪	GitHub
TCOD [636]	预算感知修剪	轨迹深度课程	GitHub
交互模式			
GLIDER [232]	层次结构	法学硕士作为决策智能体	GitHub
SkillRL [705]	分层框架	从失败中发展技能库	GitHub

THOR [52]	分层框架	联合情节和步骤级优化	GitHub
AT-GRPO [861]	多智能体框架	每智能体、每轮优势分组	GitHub
MATPO [439]	多智能体框架	联合 LLM 中的多个角色 展开轨迹（Rollouts）	GitHub
SPIRAL [371]	多智能体框架	角色条件优势	GitHub
Memory-R1 [743]	内存状态学习	分离内存管理器和答案

智能体|GitHub|
|记忆即动作 [846]|记忆状态学习|工作记忆编辑作为策略动作|GitHub|
|智能体工作流程
内存状态学习	引入可重用的工作流程	GitHub
内存[674]			
智能体闪电 [409]	记忆状态学习	将工作流程跟踪到可训练样本	GitHub

49

utilization [164, 192]; (2) as supervision for reward models or verifiers, including Prometheus, the OpenRubrics reward model, and the interpretable scorer R3 [19, 278, 384]; and (3) as guidance for policy behavior rather than only final scoring, as in Rubric-Scaffolded RL, which injects rubrics as decaying exploration scaffolds during rollout and as judge references during exploitation [887]. Rubric credit is especially useful in Agentic RL, where context dependence, tool interaction, and side effects are easily obscured by a single composite reward; its open challenges remain rubric quality, judge bias, criterion weighting, and reward hacking.

5.5.2 Policy Optimization

Policy optimization is where credit signals are turned into parameter updates. In Agentic RL, this step is difficult because the updates must be computed from multi-turn trajectories that mix tool calls, environmental observations, GUI states, and delayed terminal rewards [261, 321, 628, 681]. We organize the discussion around three practical questions: how to make sparse-reward updates stable, which interaction unit should be optimized, and how to preserve exploration over long trajectories.

Clipping-based Policy Optimization. The first problem is how to make PPO- or GRPO-style RLVR reliable when trajectories are long and rewards are sparse or binary [460, 524, 525, 533]. The main difficulty is the mismatch between noisy group statistics and long sequence-level gradients. This mismatch appears as length bias, local normalization bias, homogeneous all-correct or all-wrong groups, unstable token-level importance ratios, and reward-scale drift. Existing methods fall into two broad categories. One category makes group-relative updates less biased through normalization, clipping, and sampling design; representative examples include REINFORCE++, DAPO, Dr.GRPO, CISPO, and GPPO [222, 389, 435, 565, 780]. GSPO defines the importance ratio and applies clipping at the sequence level rather than per token [865]. The other category changes the estimation unit or objective to reduce variance and constrain policy drift, with representative examples such as BNPO [95, 205, 391, 710].

Turn-level Policy Optimization. The second problem is the granularity of optimization. In agent-environment interaction, the meaningful decision unit is often a trajectory, turn, step, subgoal, or branch rather than a single response. Early agentic RL systems therefore lift optimization from response-level training to full interaction trajectories, as in web and search agents such as WebAgent-R1 [261, 681]. A more fine-grained line then aligns advantages with explicit segments, steps, or turns, represented by SegPO, StepPO, and Turn-PPO [321, 628, 901]. More recent methods make the comparison set itself more local and informative. Some anchor comparisons to shared states or trajectory structures, as in GiGPO and related methods [156, 212, 318, 349]. Others use shared branches, subgoal segments, or hindsight information to ask which local choice changed the downstream trajectory, as in Tree-GRPO and its extensions [250, 478, 582, 687].

Entropy-based Policy Optimization. The third problem is exploration. Long-horizon RL can quickly over-reinforce early successful patterns, causing entropy collapse and reducing the diversity of future trajectories. Recent work therefore treats entropy as a decision- or trajectory-level control variable rather than only a token-level regularizer. One group modifies the update rule to avoid excessive probability concentration: Clip-Cov and KL-Cov gate over-concentrating updates, and OPEFO balances entropy-increasing and entropy-decreasing gradients [103, 725]. Another group injects diversity at critical decision points, for example CURE, which re-concatenates rollouts around critical (high-entropy) tokens to prevent entropy collapse [330]. Other methods, such as AEnt and AER, attempt to stabilize policy entropy within a bounded interval to alleviate entropy collapse [536, 830]. For multi-turn agents, the same idea becomes trajectory-level exploration control: EPO regularizes trajectory entropy under sparse rewards [732]. ARPO and AEPO branch from high-entropy positions after tool feedback to trade off rollout diversity against gradient distortion [128, 130].

5.5.3 Sampling Strategy

In long-horizon agent tasks, the sampling target is multi-turn interaction trajectories encompassing tool calls, environmental observations, memory operations and final rewards [808, 821]. Unlike credit assignment and policy optimization, the sampling strategy decides which trajectories are generated in the first place. We organize this from three angles: trajectory-level rollout that fixes the basic sampling unit, tree-structured

利用 [164, 192]； (2) 作为奖励模型或验证者的监督，包括 Prometheus、OpenRubrics 奖励模型和可解释评分器 R3 [19,278,384]； （3）作为策略行为的指导，而不仅仅是最终评分，如Rubric-Scaffolded RL，它在展开轨迹（Rollout）期间注入Rubric作为腐烂的探索支架，并在开发期间作为判断参考[887]。Rubric Credit 在 Agentic RL 中特别有用，其中上下文依赖、工具交互和副作用很容易被单一的复合奖励所掩盖；它的公开挑战仍然是标题质量、法官偏见、标准权重和奖励黑客。

5.5.2 策略优化

策略优化是将信用信号转化为参数更新的地方。在 Agentic RL 中，这一步很困难，因为更新必须根据混合了工具调用、环境观察、GUI 状态和延迟终端奖励的多轮轨迹计算[261,321,628,681]。我们围绕三个实际问题组织讨论：如何使稀疏奖励更新稳定，应该优化哪个交互单元，以及如何保持长轨迹的探索。

基于裁剪的策略优化。第一个问题是当轨迹很长并且奖励稀疏或二元时如何使 PPO 或 GRPO 式的 RLVR 可靠 [460,524,525,533]。主要困难是嘈杂的组统计数据和长序列级梯度之间的不匹配。这种不匹配表现为长度偏差、局部标准化偏差、同质的全正确或全错误组、不稳定的代币级别重要性比率和奖励规模漂移。现有方法分为两大类。一类通过归一化、裁剪和采样设计减少组相关更新的偏差；代表性的例子包括 REINFORCE++、DAPO、Dr.GRPO、CISPO 和 GPPO [222、389、435、565、780]。 GSPO 定义了重要性比率，并在序列级别而不是每个标记应用剪辑[865]。另一类改变估计单位或目标以减少方差并限制策略漂移，代表性例子如 BNPO [95,205,391,710]。

轮次策略优化。第二个问题是优化的粒度。在主体环境交互中，有意义的决策单元通常是轨迹、转弯、步骤、子目标或分支，而不是单个响应。因此，早期的智能体 RL 系统将优化从响应级别训练提升到完全交互轨迹，如 WebAgent-R1 等网络和搜索智能体 [261, 681]。然后，一条更细粒度的线将优势与明确的段、步骤或转弯对齐，以 SegPO、StepPO 和 Turn-PPO 为代表[321、628、901]。最近的方法使比较集本身更加本地化和信息丰富。一些锚点与共享状态或轨迹结构的比较，如 GiGPO 和相关方法 [156,212,318,349]。其他人使用共享分支、子目标段或事后信息来询问哪个本地选择改变了下游轨迹，如 Tree-GRPO 及其扩展[250,478,582,687]。

基于熵的策略优化。第三个问题是探索

。 长时程（Long-Horizon） RL 可以快速过度强化早期的成功模式，导致熵崩溃并减少未来轨迹的多样性。因此，最近的工作将熵视为决策或轨迹级控制变量，而不仅仅是令牌级正则化器。一组修改更新规则以避免概率过度集中：Clip-Cov 和 KL-Cov 门过度集中更新，OPEFO 平衡熵增和熵减梯度 [103, 725]。另一组在关键决策点注入多样性，例如 CURE，它围绕关键（高熵）标记重新连接 展开轨迹（Rollouts） 以防止熵崩溃 [330]。其他方法，例如 AEnt 和 AER，试图将策略熵稳定在有界区间内，以减轻熵崩溃 [536, 830]。对于多轮智能体，同样的想法变成了轨迹级探索控制：EPO 在稀疏奖励下正则化轨迹熵[732]。ARPO 和 AEPO 在工具反馈后从高熵位置分支，以权衡 展开轨迹（Rollout） 多样性和梯度失真 [128, 130]。

5.5.3 抽样策略

在长时程（Long-Horizon）智能体任务中，采样目标是多轮交互轨迹，包括工具调用、环境观察、内存操作和最终奖励[808, 821]。与 贡献归因（Credit Assignment） 和策略优化不同，采样策略首先决定生成哪些轨迹。我们从三个角度进行组织：固定基本采样单元的轨迹级 展开轨迹（Rollout）、树形结构

sampling that shares computation and produces relative signals, and budget-aware pruning that keeps sampling informative under a fixed cost.

Trajectory-level Rollout. The basic sampling unit in Agentic RL is a multi-step agent-environment trajectory: at each step the model acts on observations, history, and task goals, and the environment returns new observations and reshapes the future state distribution. Early systems represented by ReAct established this reasoning-acting-observing loop, together with tool use, verbal reflection, and skill reuse, mostly through prompting rather than training [523, 544, 630, 761]. Subsequent RL work turned such loops into the rollout unit: a line of information-seeking agents led by Search-R1 folds query generation, evidence acquisition, and stopping decisions into on-policy trajectories [256, 261, 553, 871], while web and tool agents organize browser operations or executable tool composition as multi-turn online trajectories [484, 653, 681, 701, 834]. At larger scale, deep-research systems represented by WebDancer and the WebSailor family build end-to-end pipelines that combine synthetic data, tool-based web environments, and trajectory-level rollout [326, 327, 689]. On top of such pipelines, WebResearcher, ReSum, and WebWeaver further address long-context research, context summarization, and evidence structuring during sampling [356, 490, 693].

Tree-structured Sampling. Independent full-trajectory rollout is expensive and, when trajectories are homogeneous, yields weak relative learning signals. Tree-structured sampling addresses both problems by sharing prefixes across rollouts and contrasting sibling branches. The idea originates from organizing thoughts or actions as trees during reasoning and planning, as in Tree of Thoughts, RAP, and LATS [204, 760, 875], and is brought into RL training by Tree-GRPO, which compares sibling branches under a fixed token or tool-call budget [250]. A subsequent group refines where and how the tree expands: AT2PO, TreePO, and PORTool guide expansion toward uncertain turns, reuse inference computation, or organize shared and divergent tool-use paths [348, 687, 900], while branching methods such as BranPO and Tool-Light branch selectively from high-entropy positions, late critical decisions, or otherwise informative states to maximize rollout contrast [76, 860].

Budget-aware and Pruning Strategies. A third angle keeps sampling informative under a fixed rollout budget, either by pruning or reshaping the trajectory distribution or by allocating the branching budget adaptively. One line prunes or reshapes the trajectory distribution: representative mechanisms include experience reuse and confidence-based filtering in Tool-R1 and WebRL, difficulty- or pass-rate-based pruning in LiteResearcher, and trajectory-depth curricula in TCOD [334, 484, 636, 834]. These strategies discard low-signal rollouts, concentrate budget on contrastive or appropriately difficult cases, and bound trajectory length. They reduce the cost of producing each informative training sample in long-horizon settings. Another research approach makes branching itself budget-aware: ARPO triggers branch sampling only at high-entropy tool-call steps, reducing the tool-use budget of trajectory-level RL, while AEPO adaptively allocates the budget between global and branch sampling [128, 130].

5.5.4 Interaction Patterns

Interaction patterns concern how to organize optimizable decision-making structures during training. We discuss them from three perspectives: hierarchical interaction frameworks, multi-agent interaction frameworks, and memory-state learning.

Hierarchical Interaction Frameworks. Hierarchical frameworks exploit the temporal abstraction that naturally exists in long-horizon tasks, and split along two threads. The first is explicit planner-executor decomposition, where the policy is separated into a high-level module that sets subgoals and a low-level module that executes them, so that subgoal boundaries become new optimization units. Representative instances include HiPER, GLIDER, and STEP-HRL, which differ mainly in whether the low-level layer is a task-agnostic skill set or a progress-compressed executor [232, 478, 863]. The second thread pushes hierarchy toward experience abstraction and capability reuse: building on the observation that procedural skills in agentic exploration originally rely on external skill libraries and prompting. SkillRL evolves a skill library from validation failures during RL and Skill0 internalizes skills into model parameters via in-context agentic RL [405, 705]. Analogous high-low decompositions appear in tool-augmented reasoning, where THOR jointly optimizes episode-level problem solving and step-level tool or code operations [52].

原文第 51 页 | 中文翻译初稿

共享计算并产生相关信号的采样，以及在固定成本下保持采样信息的预算感知剪枝。

轨迹级 展开轨迹（Rollout）。 Agentic RL 中的基本采样单元是多步智能体-环境轨迹：模型在每一步都会作用于观测值、历史记录和任务目标，环境返回新的观测值并重塑未来的状态分布。以 ReAct 为代表的早期系统建立了这种推理-行动-观察循环，以及工具使用、言语反思和技能重用，主要是通过提示而不是训练[523,544,630,761]。随后的 RL 工作将此类循环转变为 展开轨迹（Rollout） 单元：由 Search-R1 领导的一系列信息搜索智能体将查询生成、证据获取和停止决策折叠为策略轨迹 [256, 261, 553, 871]，而 Web 和工具智能体将浏览器操作或可执行工具组合组织为多轮在线轨迹 [484, 653, 681、701、834]。在更大范围内，以 WebDancer 和 WebSailor 系列为代表的深度研究系统构建了端到端管道，结合了合成数据、基于工具的 Web 环境和轨迹级 展开轨迹（Rollout） [326,327,689]。在这些管道之上，WebResearcher、ReSum 和 WebWeaver 进一步解决了采样过程中的长上下文研究、上下文摘要和证据结构[356,490,693]。

树结构采样。独立的全轨迹 展开轨迹（Rollout） 价格昂贵，并且当轨迹均匀时，会产生较弱的相对学习信号。树结构采样通过在 展开轨迹（Rollouts） 之间共享前缀并对比同级分支来解决这两个问题。该想法源于在推理和规划过程中将思想或行为组织为树，如 Tree of Thoughts、RAP 和 LATS [204,760,875]，并由 Tree-GRPO 引入 RL 训练，该模型在固定令牌或工具调用预算下比较兄弟分支 [250]。随后的小组细化了树扩展的位置和方式：AT2PO、TreePO 和 PORTool 引导扩展向不确定的转弯、重用推理计算或组织共享和发散的工具使用路径 [348, 687, 900]，而分支方法（例如 BranPO 和 Tool-Light 分支）有选择地从高熵位置、后期关键决策或其他信息状态来最大化 展开轨迹（Rollout） 对比度 [76, 860]。

预算意识和修剪策略。第三个角度通过修剪或重塑轨迹分布或通过自适应分配分支预算，在固定的 展开轨迹（Rollout） 预算下保持采样信息。一行修剪或重塑轨迹分布：代表性机制包括 Tool-R1 和 WebRL 中的经验重用和基于置信度的过滤，LiteResearcher 中基于难度或通过率的修剪，以及 TCOD 中的轨迹深度课程 [334,484,636,834]。这些策略丢弃低信号 展开轨迹（Rollouts），将预算集中在对比或适当困难的情况上，并限制轨迹长度。它们降低了在 长时程（Long-Horizon） 设置中生成每个信息训练样本的成本

。另一种研究方法使分支本身具有预算意识：ARPO 仅在高熵工具调用步骤触发分支采样，减少轨迹级 RL 的工具使用预算，而 AEPO 自适应地在全局采样和分支采样之间分配预算 [128, 130]。

5.5.4 交互模式

交互模式涉及如何在训练期间组织可优化的决策结构。我们从三个角度讨论它们：分层交互框架、多智能体交互框架和记忆状态学习。

分层交互框架。分层框架利用 长时程（Long-Horizon） 任务中自然存在的时间抽象，并沿两个线程分割。第一个是显式规划器-执行器分解，其中策略被分为设置子目标的高级模块和执行子目标的低级模块，以便子目标边界成为新的优化单元。代表性实例包括 HiPER、GLIDER 和 STEP-HRL，它们的主要区别在于低层是任务无关的技能集还是进度压缩的执行器 [232,478,863]。第二条线索将层次结构推向体验抽象和功能重用：基于观察到主体探索中的程序技能最初依赖于外部技能库和提示。 SkillRL 从 RL 期间的验证失败中发展出一个技能库，而 Skill0 通过上下文智能体 RL 将技能内化为模型参数 [405, 705]。类似的高低分解出现在工具增强推理中，其中 THOR 联合优化情节级问题解决和步骤级工具或代码操作 [52]。

51

Multi-agent Frameworks. Multi-agent frameworks extend the training target from a single policy to a joint decision-making system. Their central difficulties are role non-stationarity, enlarged joint action spaces, and credit assignment under shared rewards. One thread formalizes LLM collaboration as cooperative MARL in Dec-POMDP settings and adapts group-relative training accordingly. MAGRPO uses centralized group-relative advantages with decentralized execution, while Stronger-MAS (AT-GRPO) restores variance reduction by grouping samples per agent and per turn [382, 861]. To assign credit across heterogeneous roles, M-GRPO computes separate group-relative advantages for a planner and its tool-executing sub-agents under decoupled cross-server training, whereas MATPO trains the planner and worker within a single LLM by normalizing over their joint rollouts [218, 439]. Dr.MAMR instead pushes credit to the step level with a Shapley-inspired causal-influence estimate to counter lazy-agent collapse [850]. A second thread covers self-play, debate, and role specialization, which inherit classical MARL principles such as centralized training with decentralized execution, counterfactual baselines, value decomposition, and MAPPO [396, 506, 776]. Within this thread, recent systems instantiate complementary System 1/System 2 policies for deep research, variance-reduced group-relative training with evolving memory, and role-conditioned advantage estimation in zero-sum language games [56, 201, 371]. Multi-agent debate adds a further variant, where competitive interaction provides additional preference, refutation, and verification signals for the final answer [139, 358].

Memory-state Learning. Memory-state learning turns runtime structures that are usually fixed by an external harness, including context, workflows, long-term memory, tool-use histories, and reusable experience, into learnable policies. This shifts Agentic RL from optimizing isolated LLM calls to training state management over long horizons. One thread makes memory operations themselves a learning target: Memory-R2 observes that memory writes place trajectories in different effective environments and uses LoGo-GRPO to combine a global objective with local comparisons from the same memory state, Memory-R1 separates a Memory Manager from an Answer Agent, and Memory as Action treats working-memory editing as an explicit policy action handled by Dynamic Context Policy Optimization [742, 743, 846]. A parallel thread lifts the same idea to the organization of the task process: Agent Workflow Memory induces reusable workflows from successful tasks, FlowSteer optimizes a workflow director with structural and task-level rewards, and Agent Lightning offers a system-level route that traces existing workflows as spans and transitions and converts them into trainable samples [409, 674, 823].

5.6 On-Policy Distillation

Alongside the emergence of Agentic RL, recent research has increasingly explored alternative training paradigms tailored to long-horizon trajectory optimization. As interaction horizons grow, agent training becomes increasingly challenged by distribution shift, error accumulation, and sparse, delayed supervision [377]. Existing training paradigms only partially address these challenges. Off-policy imitation, such as behavioral cloning and offline knowledge distillation [609], relies on static expert trajectories and therefore fails to expose agents to their own induced state distributions. Reinforcement learning [94], in contrast, optimizes on on-policy trajectories but often suffers from sparse rewards, costly exploration, and poor sample efficiency.

On-Policy Distillation (OPD), first formalized for language models by GKD [3], combines the high-quality supervision of distillation with the state coverage of on-policy learning [309, 635]. Instead of learning solely from fixed expert demonstrations or sparse outcome rewards, OPD trains the student on trajectories generated by its current policy and provides additional supervisory signals on these self-generated trajectories. As a result, OPD exposes the student to self-induced state distributions while providing richer supervision than sparse outcome rewards. This delimits OPD from its neighbors: unlike the offline trajectory distillation of Section 5.4, which learns from a fixed set of teacher or past-model traces, OPD draws supervision on the student's own current rollouts; unlike the self-evolution of Section 5.7, it keeps the task set and environment fixed and only densifies the supervision signal. These properties make OPD particularly attractive for long-horizon agents, where robustness to distribution shift, recovery from compounding errors, and access to dense supervisory signals are all critical for effective learning [697, 838].

5.6.1 Teacher-guided OPD

In teacher-guided OPD, one or more teacher models are queried on states visited by the student policy to provide supervisory signals for the generated trajectories [309, 312, 635, 697]. Unlike conventional offline

原文第 52 页 | 中文翻译初稿

多智能体框架。多智能体框架将训练目标从单一策略扩展到联合决策系统。他们的核心困难是角色非平稳性、扩大的联合行动空间以及共享奖励下的 贡献归因（Credit Assignment）。一个线程将 LLM 协作形式化为 Dec-POMDP 设置中的协作 MARL，并相应地调整与组相关的训练。 MAGRPO 使用集中式组相对优势和分散式执行，而 Stronger-MAS (AT-GRPO) 通过对每个智能体和每回合的样本进行分组来恢复方差减少 [382, 861]。为了跨异构角色分配学分，M-GRPO 在解耦的跨服务器训练下计算计划者及其工具执行子智能体的单独组相对优势，而 MATPO 通过对联合 展开轨迹（Rollouts） 进行规范化来在单个 LLM 中训练计划者和工作人员 [218, 439]。相反，MAMR 博士通过沙普利启发的因果影响估计将信用推向了阶梯水平，以应对懒惰智能体崩溃[850]。第二个线程涵盖自我游戏、辩论和角色专业化，它们继承了经典的 MARL 原则，例如带有分散执行的集中训练、反事实（Counterfactual） 基线、价值分解和 MAPPO [396,506,776]。在这个线程中，最近的系统实例化了系统 1/系统 2 的补充策略，用于深入研究、具有进化记忆的方差减少的群体相关训练，以及零和语言游戏中的角色条件优势估计 [56,201,371]。多智能体辩论增加了进一步的变体，其中竞争性交互为最终答案提供了额外的偏好、反驳和验证信号[139, 358]。

记忆状态学习。记忆状态学习将通常由外部 运行支撑框架（Harness） 固定的运行时结构（包括上下文、工作流程、长期记忆、工具使用历史和可重用经验）转变为可学习的策略。这将 Agentic RL 从优化孤立的 LLM 调用转变为长期训练状态管理。一个线程使内存操作本身成为学习目标：Memory-R2 观察到内存在不同的有效环境中写入位置轨迹，并使用 LoGo-GRPO 将全局目标与来自同一内存状态的局部比较相结合，Memory-R1 将内存管理器与应答智能体分开，而内存作为操作将工作内存编辑视为由动态上下文策略优化处理的显式策略操作[742,743,846]。并行线程将相同的想法提升到任务流程的组织中：Agent Workflow Memory 从成功的任务中引入可重用的工作流，FlowSteer 通过结构和任务级奖励优化工作流导向器，Agent Lightning 提供系统级路由，以跨度和转换的形式跟踪现有工作流并将其转换为可训练样本 [409,674,823]。

5.6 符合策略蒸馏

随着 Agentic RL 的出现，最近的研究越来越多地探索针对 长时程（Long-Horizon） 轨迹优化的替代训练范例。随着交互视野的增长，智能体训练越来越受到分布偏移、错误累积和稀疏、延迟监督的挑战[377]

。现有的训练范例只能部分解决这些挑战。离策略模仿，例如行为克隆和离线知识蒸馏[609]，依赖于静态专家轨迹，因此无法将智能体暴露给它们自己诱导的状态分布。相比之下，强化学习（Reinforcement Learning） [94] 对策略轨迹进行优化，但往往会遇到奖励稀疏、探索成本高昂和样本效率差的问题。

在策略蒸馏 (OPD) 最初由 GKD [3] 针对语言模型形式化，它将蒸馏的高质量监督与在策略学习的状态覆盖相结合 [309, 635]。 OPD 不是仅仅从固定的专家演示或稀疏的结果奖励中学习，而是根据当前策略生成的轨迹对学生进行训练，并针对这些自我生成的轨迹提供额外的监督信号。因此，OPD 让学生暴露于自我诱导的状态分布，同时提供比稀疏结果奖励更丰富的监督。这将 OPD 与其邻居区分开来：与第 5.4 节的离线轨迹蒸馏不同，OPD 从一组固定的教师或过去模型轨迹中学习，OPD 对学生自己当前的 展开轨迹（Rollouts） 进行监督；与第 5.7 节的自进化不同，它保持任务集和环境固定，仅密集化监督信号。这些特性使 OPD 对 长时程智能体（Long-Horizon Agents） 特别有吸引力，其中分布偏移的鲁棒性、复合错误的恢复以及对密集监督信号的访问对于有效学习都至关重要 [697, 838]。

5.6.1 教师指导的 OPD

在教师引导的 OPD 中，在学生策略访问的状态上查询一个或多个教师模型，以为生成的轨迹提供监督信号 [309,312,635,697]。与传统的离线方式不同

52

distillation that relies on fixed expert demonstrations, teacher-guided OPD generates supervision adaptively online according to the student's evolving trajectories, allowing teachers to provide guidance tailored to the student's current behaviors and mistakes. Existing methods can be characterized by two complementary design choices: how supervision is organized over trajectories and how teacher signals are constructed.

Supervision Scheduling. A first question is how teacher signals should be organized and delivered throughout long interaction trajectories. Representative methods such as DAgger-LLM introduce turn-level interpolation and step-wise supervision to provide dense corrections on states visited by the student policy [309, 873]. Other approaches further incorporate adaptive curricula and preference-based guidance, progressively adjusting the timing, granularity, and difficulty of supervision during training [313, 636, 835]. What emerges is that teacher-guided OPD hinges not only on the quality of teacher signals, but also on when and at what resolution they are injected during multi-turn interactions.

Teacher Signals. A second question is how to construct richer and more informative supervision for on-policy distillation. A series of methods represented by MAD-OPD leverage multiple agents or agent evolvers to aggregate complementary expertise and generate more reliable supervision than a single teacher policy [635, 809]. Along a related line, KAT-Coder-V2 [312] exploits specialist agents with complementary capabilities and distills their collective knowledge into a unified policy. Retrieval-augmented approaches such as LiteGUI [697] instead enrich teacher supervision with oracle trajectories and dynamically retrieved references, providing structured guidance in multi-solution interaction environments. Across these efforts, the recurring lesson is that supervision improves when teacher signals are made more diverse, complementary, and context-aware.

5.6.2 Self-improving OPD

Self-improving OPD removes direct reliance on external teacher models and instead constructs its own supervisory signals from auxiliary information available during training. Such information generally arises from two complementary sources: privileged information that is accessible only during training [838] and adaptive feedback generated through repeated interactions with the environment [216, 242, 845].

Privileged Information. One source of self-supervision is information that is accessible during training but unavailable at deployment time. Some methods leverage privileged annotations or modalities, such as outcome-aware information and fine-grained visual details, to provide dense supervision beyond what the agent's own observations reveal [788]. Others derive privileged information from the interaction trajectories themselves, treating intermediate construction processes, latent skills, or other trajectory-level abstractions as additional supervision that exposes the hidden structure of successful behaviors [631, 838]. Augmenting sparse interaction signals with training-only information in this way helps the student acquire more robust and reusable long-horizon behaviors.

Environment Feedback. The other source of self-supervision is the interaction outcomes and verifier signals returned by the environment. Some methods transform sparse environmental feedback, such as rewards, textual critiques, runtime errors, and verifier outputs, into dense self-supervision by conditioning the policy on feedback-informed predictions and distilling them back into the model [216, 242]. A complementary view treats failures as supervision, generating retrospective reflections and accumulating reusable lessons that guide future interactions [757, 845]. Beyond obtaining feedback, a further line studies how to use it well during distillation, for example by emphasizing informative signals, learning feedback-aware representations, and mitigating the instability caused by noisy or stale supervision [72, 161, 735, 782]. The decisive factor is therefore not whether environmental feedback is available, but how it is transformed, accumulated, and reused across long-horizon training.

5.7 Self-evolution

This subsection discusses self-evolution as a training mechanism for internalizing experience generated by the agent itself. The preceding stages mainly rely on prepared data, task sets, environment collections, or trajectories. Self-evolution adds a feedback loop: the current policy produces new attempts, failures,

原文第 53 页 | 中文翻译初稿

与依赖固定专家演示的蒸馏不同，教师引导的 OPD 可以根据学生的发展轨迹自适应地在线生成监督，从而使教师能够针对学生当前的行为和错误提供量身定制的指导。现有方法可以通过两种互补的设计选择来表征：如何组织轨迹监督以及如何构建教师信号。

监督调度。第一个问题是教师信号应如何在长互动轨迹中组织和传递。 DAgger-LLM 等代表性方法引入了轮级插值和逐步监督，以对学生策略访问的状态进行密集修正 [309, 873]。其他方法进一步结合适应性课程和基于偏好的指导，逐步调整训练期间监督的时间、粒度和难度[313,636,835]。由此可见，教师引导的 OPD 不仅取决于教师信号的质量，还取决于在多轮交互过程中注入信号的时间和分辨率。

老师发出信号。第二个问题是如何为在策略蒸馏构建更丰富、信息更丰富的监管。以 MAD-OPD 为代表的一系列方法利用多个智能体或智能体进化者来聚合互补的专业知识，并产生比单个教师策略更可靠的监督 [635, 809]。沿着相关路线，KAT-Coder-V2 [312]利用具有互补能力的专业智能体，并将他们的集体知识提炼成统一的策略。相反，诸如 LiteGUI [697] 之类的检索增强方法通过预言轨迹和动态检索参考文献来丰富教师监督，从而在多解决方案交互环境中提供结构化指导。在这些努力中，反复出现的教训是，当教师信号变得更加多样化、互补和情境感知时，监督就会得到改善。

5.6.2 自我改进（Self-Improving） OPD

自我改进（Self-Improving） OPD 消除了对外部教师模型的直接依赖，而是根据训练期间可用的辅助信息构建自己的监督信号。此类信息通常来自两个互补的来源：仅在训练期间才能访问的特权信息[838]和通过与环境重复交互而生成的自适应反馈[216,242,845]。

特权信息。自我监督的来源之一是在训练期间可访问但在部署时不可用的信息。一些方法利用特权注释或模式，例如结果感知信息和细粒度的视觉细节，来提供超出智能体自身观察所揭示的密集监督[788]。其他人从交互轨迹本身获取特权信息，将中间构建过程、潜在技能或其他轨迹级抽象视为暴露成功行为的隐藏结构的额外监督[631, 838]。以这种方式用仅训练信息增强稀疏交互信号有助于学生获得更稳健和可重用的 长时程（Long-Horizon） 行为。

环境反馈

。自我监督的另一个来源是环境返回的交互结果和验证者信号。一些方法通过根据反馈信息预测调整策略并将其提炼回模型，将稀疏的环境反馈（例如奖励、文本批评、运行时错误和验证者输出）转化为密集的自我监督[216, 242]。一种补充观点将失败视为监督，产生回顾性反思并积累可重复使用的经验教训，以指导未来的互动[757, 845]。除了获得反馈之外，另一条路线研究如何在蒸馏过程中很好地使用它，例如通过强调信息信号、学习反馈感知表示以及减轻由噪声或陈旧监督引起的不稳定性[72,161,735,782]。因此，决定性因素不是环境反馈是否可用，而是环境反馈如何在 长时程（Long-Horizon） 训练中转化、积累和重用。

5.7 自我进化

本小节讨论自我进化作为一种训练机制，用于内化智能体本身产生的经验。前面的阶段主要依赖于准备好的数据、任务集、环境集合或轨迹。自我进化增加了一个反馈循环：当前的策略会产生新的尝试、失败、

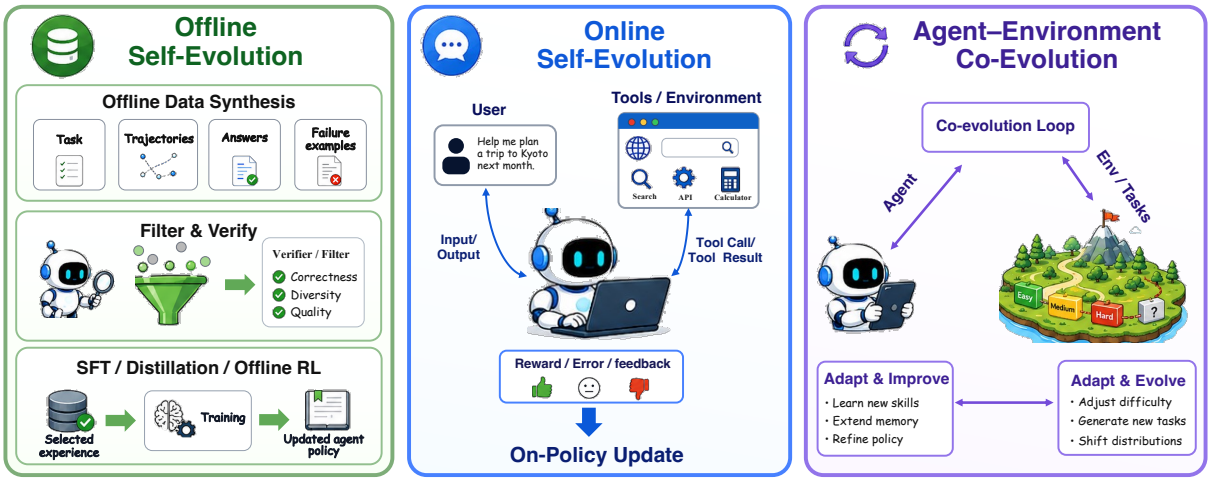


Figure 13 Three modes of self-evolution for long-horizon agents. *Offline self-evolution* learns from selected self-generated experience through a generate–filter–train pipeline; *online self-evolution* learns from live environment interaction on the agent's own induced state distribution; and *agent–environment co-evolution* jointly adapts the agent and its task/environment distribution to maintain a learnable capability boundary.

and feedback in its environment, and training writes selected records back into the model. This is what distinguishes it from the two distillation stages above: whereas fine-tuning distillation (Section 5.4) and on-policy distillation (Section 5.6) both consume a supervision source that is held fixed within a training round, self-evolution treats the data, tasks, and even the environment as objects that the loop itself updates. This has recently grown into a fast-moving research area, and dedicated papers now catalog how agents can autonomously evolve their models, memories, tools, and workflows toward lifelong adaptation [147, 165]. As illustrated in Figure 13, we organize this line of work into three forms: offline self-evolution, online self-evolution, and agent–environment co-evolution.

5.7.1 Offline Self-evolution

Offline bootstrapped self-evolution typically follows a generate–filter–train pattern. The model generates candidate tasks, answers, reasoning traces, code patches, tool calls, or interaction trajectories; verifiers, tests, reward models, teacher models, or self-evaluation mechanisms select successful samples, informative failures, and correctable fragments; training then turns these samples into SFT, distillation, preference optimization, or offline RL objectives. STaR bootstraps reasoning from successful rationales [793], and subsequent work extends the same idea to self-edits, tool-augmented reasoning, multi-agent demonstrations, and code-execution feedback [310, 517, 859, 906]. More recent work emphasizes how to turn search processes and failures into stable training assets. SOAR alternates evolutionary search with hindsight learning in program synthesis, converting search attempts into fine-tuning pairs; SAMULE synthesizes reflection data at the trajectory, task, and cross-task levels, and trains a retrospective model from failure analysis [173, 481]. A parallel search-based route runs the generate–filter–train loop for several rounds: rStar-Math uses Monte Carlo tree search to synthesize step-by-step verified reasoning trajectories and iteratively co-evolves a policy model together with a process preference model, so that each round's verified rollouts become the next round's supervision [191]. This route is attractive because data can be audited, denoised, and relabeled before an update, making it suitable for cold start and capability consolidation. Its main risk is that candidate data still largely comes from the existing model distribution, which can reinforce known behaviors while doing little to discover out-of-distribution strategies.

5.7.2 Online Self-evolution

Online interactive self-evolution interleaves experience collection with policy updates, training the model on state distributions induced by its own behavior. For long-horizon agents, difficult states are often not static

原文第 54 页 | 中文翻译初稿

离线在线智能体-环境
自我进化自我进化共同进化
离线数据合成用户工具/环境
任务轨迹答案示例失败帮助我计划下个月去京都共同进化循环
。
搜索API计算器
过滤和验证输入/工具调用/
验证器/过滤器输出工具结果
正确性
多样性
质量
SFT/蒸馏/离线强化学习奖励/错误/反馈
适应和改进适应和进化
•学习新技能•调整难度
训练•扩展记忆•生成新任务
选定的更新智能体•细化策略•轮班分配
经验策略按策略更新
智能体
Env /任务

图13 长时程智能体（Long-Horizon Agents）自我演化的三种模式。 _离线自我进化_通过生成-过滤-训练管道从选定的自生成经验中学习； _在线自我进化_从实时环境交互中学习智能体自身诱导的状态分布；和 _智能体-环境协同进化_共同调整智能体及其任务/环境分布，以维持可学习的能力边界。

及其环境中的反馈，训练将选定的记录写回到模型中。这就是它与上面两个蒸馏阶段的区别：微调蒸馏（第 5.4 节）和同策略蒸馏（第 5.6 节）都消耗在训练轮次中固定的监督源，而自进化将数据、任务甚至环境视为循环本身更新的对象。这最近已经发展成为一个快速发展的研究领域，现在有专门的论文对智能体如何自主进化其模型、记忆、工具和工作流程以实现终身适应进行分类[147, 165]。如图 13 所示，我们将这一工作线组织为三种形式：离线自我进化、在线自我进化和智能体-环境协同进化。

5.7.1 线下自我进化

离线引导的自我进化通常遵循生成-过滤-训练模式。模型生成候选任务、答案、推理轨迹、代码补丁、工具调用或交互轨迹；验证者、测试、奖励模型、教师模型或自我评估机制选择成功样本、信息性失败和可纠正片段；然后训练将这些样本转化为 SFT、蒸馏、偏好优化或离线 RL 目标。 STaR 从成功的基本原理中引导推理 [793]，后续工作将相同的想法扩展到自我编辑、工具增强推理、多智能体演示和代码执行反馈 [310、517、859、906]。最近的工作强调如何将搜索过程和失败转化为稳定的训练资产。 SOAR 在程序综合中交替使用进化搜索和事后学习，将搜索尝试转换为微调对； SAMULE 综合轨迹、任务和跨任务级别的反射数据，并根据故障分析训练回顾性模型 [173, 481]

。基于并行搜索的路线将生成-过滤-训练循环运行多轮：rStar-Math 使用蒙特卡罗树搜索来合成逐步验证的推理轨迹，并迭代地共同进化策略模型和过程偏好模型，以便每轮验证的 展开轨迹（Rollouts） 成为下一轮的监督[191]。这条路线很有吸引力，因为可以在更新之前对数据进行审计、去噪和重新标记，使其适

合冷启动和能力整合。其主要风险在于，候选数据仍然主要来自现有的模型分布，这可以强化已知的行为，但对发现分布外策略几乎没有作用。

5.7.2 在线自我进化

在线交互式自我进化将经验收集与策略更新交织在一起，根据其自身行为引起的状态分布来训练模型。对于长时程智能体（Long-Horizon Agents）来说，困难状态往往不是静态的

54

benchmark inputs; they arise from early decisions, tool side effects, recovery attempts, and environment state changes. Recent work therefore incorporates web, code, search, tool-use, and multi-agent game environments into training, using rollout, environment feedback, and reinforcement learning to continually update the policy [484, 672, 701, 799, 871]. A complementary line reduces external data seeds by using self-play, challenger–solver, or proposer–solver structures to create training pressure, so that the model obtains learning signals by proposing tasks, solving them, and verifying outcomes [233, 371, 854]. Recent agentic instantiations push this toward tool use and multi-step reasoning: Agent0 co-evolves a curriculum agent and a tool-using executor from the same base model with zero external data, so that a stronger executor pressures the curriculum agent to propose harder, tool-aware tasks [704]. Compared with offline bootstrapping, online self-evolution is more likely to move beyond the ceiling of existing data, but it is also more sensitive to the quality of environments and verifiers: delayed rewards, credit-assignment errors, policy collapse, verifier loopholes, and exploration cost are all amplified in long rollouts.

5.7.3 Agent–Environment Co-evolution

Agent–environment co-evolution further treats the training environment as an object of update. A fixed environment can quickly drift away from the model's capability boundary: if it is too easy, it only reinforces existing strategies; if it is too hard, it yields little usable learning signal. A first line makes the environment itself tunable to the policy: Environment Tuning uses curricula, environment augmentation, and progress feedback to mitigate cold start and sparse rewards in tool-use training, while RLVE and GenEnv align the difficulty of verifiable or generated environments with the policy's current capability, and Agent-World closes the loop by discovering environment–task pairs, exposing capability gaps through dynamic task synthesis, and improving the policy through multi-environment RL [131, 195, 401, 798]. In multi-agent settings, the environment may also be constituted by the policies of other agents; CoMAS uses interaction rewards to jointly update multiple agent policies, showing that peer interaction can serve as a training signal for self-evolution [738]. Socratic-Zero pushes this idea in a data-free direction, letting a Teacher, a Solver, and a Generator co-evolve so that the Teacher continually crafts questions targeting the Solver's weaknesses while the Generator distills its curriculum-design strategy, effectively turning the task distribution itself into a co-adapting counterpart [648]. From this perspective, the environment is not merely a source of samples, but a dynamic curriculum, feedback generator, and probe of the capability boundary: improvements in the agent change future reachable states and failure modes, while environment updates reshape the next round of exploration and optimization.

Overall, training self-evolution is not unconstrained self-training, but an experience loop jointly constrained by environments, verifiers, and optimization. External memories, skill libraries, and workflow traces can serve as intermediate forms of accumulated experience, but they constitute training self-evolution only when their stable patterns are absorbed through distillation, fine-tuning, reinforcement learning, or trainable modules [705]. The risks of this loop arise from its bootstrapping structure: self-generated data may shrink the distribution, weak verifiers may induce reward hacking, and dynamic curricula may reinforce flawed capabilities. Long-horizon agent self-evolution therefore requires strong verifiers, process-level feedback, and a combination of offline audit with online exploration.

6 Application: Long-Horizon Agents in Practice

The preceding sections examined the general ingredients of long-horizon agency, including persistent state, feedback-driven adaptation, and mechanisms for maintaining alignment over extended trajectories. We now turn to the settings in which these requirements arise in practice. Long-horizon agents operate across a wide range of environments, from code repositories and web documents to graphical interfaces, multimodal streams, and open-ended research workflows. These settings differ not only in task duration, but also in the interface through which an agent observes, acts, and receives feedback from the world, which in turn shapes what information is available during execution, what errors can be detected locally, and what capabilities the agent must add to compensate for what the environment does not provide. Therefore, we organize this section by the interface between the agent and its environment rather than by overall difficulty.

原文第 55 页 | 中文翻译初稿

基准输入；它们源于早期决策、工具副作用、恢复尝试和环境状态变化。因此，最近的工作将网络、代码、搜索、工具使用和多智能体游戏环境纳入训练中，使用 展开轨迹（Rollout）、环境反馈和 强化学习（Reinforcement Learning） 来不断更新策略[484、672、701、799、871]。互补线通过使用自我对弈、挑战者-求解器或提议者-求解器结构来产生训练压力来减少外部数据种子，以便模型通过提出任务、解决任务和验证结果来获得学习信号[233,371,854]。最近的智能体实例化将其推向工具使用和多步骤推理：Agent0 从相同的基础模型（零外部数据）共同进化出课程智能体和使用工具的执行器，因此更强大的执行器会迫使课程智能体提出更难的、工具感知的任务 [704]。与离线引导相比，在线自我进化更有可能超越现有数据的上限，但它对环境和验证者的质量也更加敏感：延迟奖励、信用分配错误、策略崩溃、验证者漏洞和探索成本都在长 展开轨迹（Rollouts） 中被放大。

5.7.3 智能体-环境协同进化

智能体-环境协同进化进一步将训练环境视为更新的对象。固定的环境会很快偏离模型的能力边界：如果太简单，它只会强化现有的策略；如果太简单，它只会强化现有的策略；如果太难，则几乎不会产生有用的学习信号。第一行使环境本身可以根据策略进行调整：环境调优使用课程、环境增强和进度反馈来减轻工具使用训练中的冷启动和稀疏奖励，而 RLVE 和 GenEnv 将可验证或生成的环境的难度与策略的当前能力相结合，Agent-World 通过发现环境-任务对、通过动态任务合成暴露能力差距以及通过多环境 RL 改进策略来闭合循环[131, 195, 401、798]。在多智能体设置中，环境也可能由其他智能体的策略构成； CoMAS 使用交互奖励来联合更新多个智能体策略，表明同伴交互可以作为自我进化的训练信号[738]。苏格拉底-零将这一想法推向无数据的方向，让教师、解决者和生成器共同进化，以便教师不断地针对解决者的弱点提出问题，而生成器提炼其课程设计策略，有效地将任务分配本身转变为相互适应的对应物[648]。从这个角度来看，环境不仅仅是样本的来源，而是动态的课程、反馈生成器和能力边界的探测：智能体的改进改变了未来的可达状态和故障模式，而环境的更新则重塑了下一轮的探索和优化。

总体而言，训练自进化并不是无约束的自训练，而是环境、验证者和优化共同约束的经验循环

。外部记忆、技能库和工作流程轨迹可以作为积累经验的中间形式，但只有当它们的稳定模式通过蒸馏、微调、强化学习（Reinforcement Learning）或可训练模块吸收时，它们才构成训练自我进化[705]。这个循环的风险来自于它的引导结构：自我生成的数据可能会缩小分布，弱验证者可能会导致奖励黑客攻击，动态课程可能会强化有缺陷的能力。因此，长时程（Long-Horizon） 智能体自我进化需要强大的验证者、过程级反馈以及离线审核与在线探索的结合。

6 应用：长时程智能体（Long-Horizon Agents） 实践

前面的部分研究了 长时程（Long-Horizon） 机构的一般要素，包括持久状态、反馈驱动的适应以及在扩展轨迹上保持对齐的机制。我们现在转向实践中出现这些要求的环境。 长时程智能体（Long-Horizon Agents） 可在广泛的环境中运行，从代码存储库和 Web 文档到图形界面、多模式流和开放式研究工作流程。这些设置不仅在任务持续时间上有所不同，而且在智能体观察、行动和接收来自世界的反馈的界面上也有所不同，这反过来又决定了执行期间可用的信息、可以在本地检测到的错误以及智能体必须添加哪些功能来补偿环境不提供的功能。因此，我们根据智能体与其环境之间的接口而不是总体难度来组织本节。

55

Figure 14 groups representative applications into five broad categories, distinguished by the interface through which each agent observes and acts:

Five Representative Applications of Long-Horizon Agents

1. **Software Engineering Agents** (Section 6.1): act over code repositories and execution environments, where edits can often be checked through tests, compilers, and runtime outputs.
2. **Information-seeking Agents** (Section 6.2): interact with web documents, search engines, and knowledge sources, where the challenge is to gather and synthesize evidence without a direct execution oracle.
3. **Computer-use Agents** (Section 6.3): operate graphical interfaces through screenshots, accessibility trees, and low-level actions, inferring and manipulating partially observable interface state.
4. **Multimodal Agents** (Section 6.4): reason over long visual, auditory, and textual streams, where relevant information is distributed across time and modalities.
5. **General-purpose Agents** (Section 6.5): combine several such interfaces within a single workflow, drawing on heterogeneous tools, permissions, and external services.

For each category, we ask three questions:

- What information and actions does the interface expose?
- What kinds of state and feedback must persist across a long trajectory?
- What capabilities must the agent or harness add to compensate for what the environment does not provide directly?

The following subsections address these questions by discussing the core capabilities required in each domain and representative systems that instantiate them.

6.1 Software Engineering

Software engineering agents are one of the most foundational and representative classes of long-horizon agents. The agent interacts with its environment through code: it acts by writing and running code, obtains feedback from how that code behaves, and uses this feedback to advance its task. A defining feature of this paradigm is unusually strong executable feedback: through tests, compilers, and runtime traces, the environment can often tell the agent automatically whether an action has succeeded, which is what distinguishes software engineering from many other long-horizon tasks. These signals nonetheless remain incomplete proxies for intended behavior: passing tests and clean compiles cannot always certify that the real issue is resolved, especially where hidden requirements, security, maintainability, and regression risk are concerned [499, 671, 801].

Turning this feedback into reliable long-horizon behavior, however, takes more than the signal itself. We organize long-horizon software engineering around three capabilities that an agent must add. The agent first has to work out *where* to act, building a usable picture of a codebase too large to fit in its context window; we call this *repository grounding*. It then has to keep the overall objective intact as it proceeds, ordering its edits and tracking progress rather than chasing one fix at a time, which is *workflow-level planning*. Finally, it has to use the feedback it receives to actually correct its mistakes, the task of *feedback-driven repair*. Together, these capabilities mark the step up from fixing isolated bugs to repository-scale work [124, 749], including implementation, migration, and pull-request preparation; commercial systems such as Devin[2] brought such end-to-end long-horizon coding agents into production. The next subsections examine each capability in turn.

6.1.1 Repository Grounding

Before any code is written, the agent must decide *where* to act. A natural-language issue typically says little about which files, dependencies, or project conventions are relevant, so long-horizon systems need a way to progressively construct and reuse a working model of the repository.

[2]Devin (Cognition): https://devin.ai.

原文第 56 页 | 中文翻译初稿

图 14 将代表性应用程序分为 5 大类，根据每个智能体观察和操作的界面进行区分：
长时程智能体（Long-Horizon Agents）的五个代表性应用

1. 软件工程智能体（第 6.1 节）：对代码存储库和执行环境进行操作，通常可以通过测试、编译器和运行时输出来检查编辑。

2. 信息寻求智能体（第 6.2 节）：与网络文档、搜索引擎和知识源交互，其中的挑战是在没有直接执行预言机的情况下收集和综合证据。

3.计算机使用智能体（第6.3节）：通过屏幕截图、可访问树和低级操作来操作图形界面，推断和操纵部分可观察的界面状态。

4. 多模式智能体（第 6.4 节）：对长视觉、听觉和文本流进行推理，其中相关信息跨时间和模式分布。

5. 通用智能体（第 6.5 节）：利用异构工具、权限和外部服务，将多个此类接口组合到一个工作流程中。

对于每个类别，我们提出三个问题：

- 界面公开哪些信息和操作？

- 什么样的状态和反馈必须在长期轨迹中持续存在？

- 智能体或 运行支撑框架（Harness） 必须添加哪些功能来弥补环境未直接提供的功能？

以下小节通过讨论每个领域所需的核心功能以及实例化它们的代表性系统来解决这些问题。
6.1 软件工程

软件工程智能体是长时程智能体（Long-Horizon Agents）中最基础、最具代表性的类别之一。智能体通过代码与其环境进行交互：它通过编写和运行代码来进行操作，从代码的行为方式中获取反馈，并使用该反馈来推进其任务。这种范例的一个定义特征是异常强大的可执行反馈：通过测试、编译器和运行时跟踪，环境通常可以自动告诉智能体某个操作是否成功，这就是软件工程与许多其他 长时程（Long-Horizon）任务的区别。尽管如此，这些信号仍然是预期行为的不完整智能体：通过测试和干净编译并不总是能证明真正的问题已得到解决，特别是在涉及隐藏需求、安全性、可维护性和回归风险的情况下[499,671,801]。

然而，将此反馈转化为可靠的 长时程（Long-Horizon） 行为需要的不仅仅是信号本身。我们围绕智能体必须添加的三个功能来组织 长时程（Long-Horizon） 软件工程。智能体首先必须弄清楚在哪里行动，构建一个可用的代码库图片，该代码库太大而无法适应其上下文窗口；我们称之为_存储库 落地关联（Grounding）_。然后，它必须在进行过程中保持整体目标的完整性，排序其编辑并跟踪进度，而不是一次追逐一个修复，这就是_工作流级规划_。最后，它必须使用收到的反馈来实际纠正错误，即“反馈驱动修复”的任务

。这些功能共同标志着从修复孤立的错误到存储库规模的工作的进步 [124, 749]，包括实施、迁移和拉取请求准备； Devin2 等商业系统将此类端到端 长时程（Long-Horizon） 编码智能体投入生产。接下来的小节将依次检查每项功能。
6.1.1 存储库 落地关联（Grounding）

在编写任何代码之前，智能体必须决定在哪里执行操作。自然语言问题通常很少说明哪些文件、依赖项或项目约定是相关的，因此 长时程（Long-Horizon） 系统需要一种方法来逐步构建和重用存储库的工作模型。

> 2Devin（认知）：https://devin.ai。

56

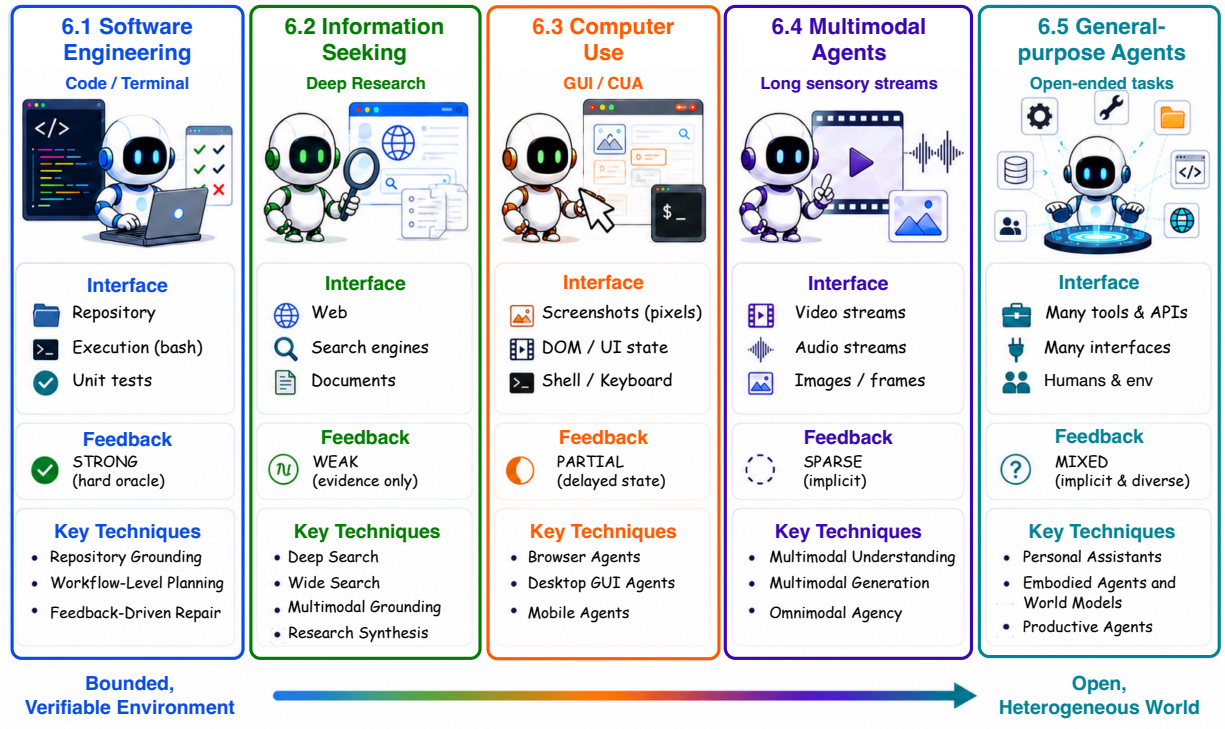


Figure 14 Representative long-horizon agent applications grouped by the structure of the agent–environment interface. Different interfaces expose different forms of observation, action, state, and feedback, leading to distinct capability requirements for long-horizon behavior.

Existing designs largely follow three strategies. The first is to improve *interactive navigation*: SWE-agent, for example, exposes search, navigation, and editing commands through an agent–computer interface tailored to LLM use [747], while structure-aware systems such as AutoCodeRover search over a program's class and method hierarchy rather than over raw text [843]. The second is to maintain a *persistent workspace*, so that file, shell, and browser state can accumulate across steps rather than being repeatedly reconstructed; OpenHands is representative of this approach [657]. The third is to externalize repository structure in compact, reusable artifacts, such as Aider's repository map,[3] IDE-level project rules,[4] and `AGENTS.md`-style instructions.[5] Although implemented differently, these approaches share a purpose: turning a large, partially observed codebase into stable task-relevant state that can guide later decisions. Related work extends this idea by constructing executable environments from arbitrary repositories [248], or by training models to internalize more repository-level structure directly [46].

6.1.2 Workflow-level Planning

Knowing where to act does not by itself determine how to sequence the work. Feature development, dependency migration, experiment reproduction, and pull-request preparation require the agent to decompose an objective, track intermediate progress, manage artifacts, and decide when available evidence is sufficient to stop [117, 142, 260, 317, 437]. The challenge here is to preserve the relationship between each local action and a global engineering objective.

Long-horizon systems increasingly externalize this relationship through explicit workflow state. One line of work structures execution around branches, plans, checkpoints, and approval gates, so that progress can be reviewed and resumed rather than remaining implicit in a conversation history.[6] A second line factors

[3] Aider: https://github.com/Aider-AI/aider.
[4] Cursor Rules: https://cursor.com/docs/rules.md.
[5] AGENTS.md: https://agents.md/.
[6] Codex, *Agent Approvals & Security*: https://developers.openai.com/codex/agent-approvals-security. Anthropic, *Claude*

原文第 57 页 | 中文翻译初稿

6.1 软件 6.2 信息 6.3 计算机 6.4 多模式 6.5 一般-
工程寻求 使用智能体目的 智能体
代码/终端 深入研究 GUI / CUA 长感官流 开放式任务
接口 接口 接口 接口 接口
存储库 网页截图（像素） 视频流 许多工具和 API
执行（bash） 搜索引擎 DOM / UI 状态 音频流 许多接口
单元测试 文档外壳/键盘 图像/帧 人类和环境
反馈 反馈 反馈 反馈 反馈
强弱部分稀疏混合
（硬预言）（仅证据）（延迟状态）（隐式）（隐式和多样化）
关键技术关键技术关键技术关键技术关键技术
存储库 落地关联（Grounding）深度搜索浏览器智能体 多模态理解 个人助理
工作流程级规划 广泛搜索桌面 GUI 智能体 多模态生成 体现智能体和
反馈驱动的修复 多模态 落地关联（Grounding） 移动智能体 全模态智能体 世界模型（World Models）
研究综合 高效智能体
有界、开放、
可验证的环境 异构世界

图 14 按智能体-环境界面的结构分组的代表性长期智能体应用程序。不同的接口暴露不同形式的观察、操作、状态和反馈，从而导致 长时程（Long-Horizon） 行为的不同能力要求。

现有的设计主要遵循三种策略。第一个是改进交互式导航：例如，SWE-agent 通过专为 LLM 使用而定制的智能体计算机界面公开搜索、导航和编辑命令 [747]，而 AutoCodeRover 等结构感知系统则通过程序的类和方法层次结构而不是原始文本进行搜索 [843]。第二是维护一个持久的工作空间，以便文件、shell 和浏览器状态可以跨步骤累积，而不是重复重建； OpenHands 是这种方法的代表[657]。第三种是将存储库结构外部化为紧凑、可重用的工件，例如 Aider 的存储库映射、3 IDE 级项目规则、4 和 AGENTS.md 样式指令。5 尽管实现方式不同，但这些方法都有一个目的：将大型的、部分观察到的代码库转变为稳定的任务相关状态，以指导后续决策。相关工作通过从任意存储库构建可执行环境[248]，或通过训练模型直接内化更多存储库级结构[46]来扩展这一想法。

6.1.2 工作流程级规划

知道在哪里行动本身并不决定如何安排工作顺序。功能开发、依赖项迁移、实验复制和拉取请求准备需要智能体分解目标、跟踪中间进度、管理工件，并决定何时可用证据足以停止[117、142、260、317、437]。这里的挑战是保持每个本地行动与全球工程目标之间的关系。

长时程（Long-Horizon） 系统通过显式工作流状态日益外化这种关系

。第一条工作线围绕分支、计划、检查点和审批关卡构建执行，以便可以审查和恢复进度，而不是隐含在对话历史记录中。6第二条线因素

> 3Aider：https://github.com/Aider-AI/aider。

> 4Cursor 规则：https://cursor.com/docs/rules.md。

> 5AGENTS.md：https://agents.md/。

> 6Codex，智能体审批和安全：https://developers.openai.com/codex/agent-approvals-security。人类，克劳德

57

recurring orchestration functions, such as planning, file-system state, tool invocation, and delegation, into reusable runtime components, as in OpenClaw and Deep Agents [293, 457]. A third line extends the workflow beyond a single coding loop by coordinating specialized agents across browsing, terminal use, file manipulation, context management and code editing [157, 381]. At deployment scale, managed execution systems further separate session-level reasoning from sandboxed execution, allowing long-running work to be supervised, audited, and resumed [18, 561]. Across these designs, the common goal is to make the task trajectory itself a persistent object of control rather than an unstructured sequence of model actions.

6.1.3 Feedback-driven Repair

Within each step of such a plan, the agent must turn execution feedback into revisions without mistaking local evidence for complete correctness. Tests and compiler errors make failures visible, but they do not fully specify the intended behavior, and an agent may repeatedly patch the most recent symptom, overfit to visible tests, or introduce a regression elsewhere in the repository.

Systems address this problem at both training and execution time. At training time, executable environments and repair trajectories teach agents to associate failures with plausible diagnosis-and-revision behavior, as in SWE-Gym and SWE-smith [467, 748]. At execution time, the central principle is to make repair *inspectable and reversible*: agents preserve diffs, logs, execution traces, and isolated workspaces so that failed attempts can be diagnosed rather than simply overwritten [657, 747]. Production-oriented systems extend this loop with branch isolation, hooks, approval boundaries, rollback, and final verification gates.[7] The result is more than iterative debugging: the agent accumulates evidence across failed attempts while retaining enough control to revise or discard earlier decisions. Not all of this requires an open-ended agent loop, however: Agentless shows that a fixed localize-then-repair pipeline can rival more elaborate agentic systems [703].

6.2 Information Seeking

Information-seeking agents are a prominent class of long-horizon agents, because answering complex information needs often requires progressively accumulating evidence over extended retrieval trajectories, where relevant information is distributed across heterogeneous sources and only partially observable at any individual search step. We organize information-seeking agents around four capabilities: *deep search* sustains a single line of inquiry across many retrieval hops; *wide search* coordinates numerous parallel investigations under a shared objective; *multimodal grounding* extends retrieval beyond text to evidence in documents, figures, and images; and *research synthesis* organizes dispersed evidence into coherent, well-attributed research reports.

Together, these capabilities define the progression from retrieval-centric systems to long-horizon information-seeking agents. Rather than locating isolated facts, such agents must continuously acquire, compress, verify, and integrate evidence across extended trajectories [284]. They coordinate multiple investigations, maintain evolving research state, synthesize findings into structured outputs, and ground conclusions in heterogeneous sources. The resulting challenge is no longer search alone, but the end-to-end management of evidence throughout the entire research lifecycle [239].

6.2.1 Deep Search

Deep search concerns sustained investigation along a single line of inquiry, where relevant evidence may only emerge after many rounds of retrieval, navigation, and reasoning [223, 883]. As search trajectories lengthen, maintaining useful state becomes increasingly important. Deep search agents can be broadly categorized into two paradigms according to how they optimize long-horizon search behaviors. The first paradigm consists of RL-based deep search agents, which explicitly optimize multi-step search policies through reinforcement learning [367]. Existing efforts mainly improve these systems from two complementary directions. One line focuses on constructing challenging long-horizon search tasks that encourage models to acquire complex search and reasoning behaviors [586, 689], while another develops more effective reinforcement learning algorithms to stabilize exploration, credit assignment, and long-term planning [120, 130, 682]. The second paradigm comprises workflow-based deep search agents, which rely on carefully designed reasoning and retrieval

Code documentation: https://code.claude.com/docs/en/overview.

[7] Anthropic, *Claude Code Hooks reference*: https://code.claude.com/docs/en/hooks.

原文第 58 页 | 中文翻译初稿

将重复编排功能（例如规划、文件系统状态、工具调用和委派）集成到可重用运行时组件中，如 OpenClaw 和 Deep Agents [293, 457] 中所示。第三条线通过协调跨浏览、终端使用、文件操作、上下文管理和代码编辑的专用智能体，将工作流程扩展到单个编码循环之外[157, 381]。在部署规模上，托管执行系统进一步将会话级推理与沙盒执行分开，从而允许监督、审核和恢复长时间运行的工作[18, 561]。在这些设计中，共同的目标是使任务轨迹本身成为持久的控制对象，而不是非结构化的模型操作序列。

6.1.3 反馈驱动修复

在此类计划的每一步中，智能体必须将执行反馈转化为修订，而不能将本地证据误认为是完全正确的。测试和编译器错误使失败可见，但它们没有完全指定预期的行为，并且智能体可能会重复修补最新的症状，过度适应可见的测试，或在存储库的其他位置引入回归。

系统在训练和执行时都解决了这个问题。在训练时，可执行环境和修复轨迹教导智能体将故障与合理的诊断和修改行为联系起来，如 SWE-Gym 和 SWE-smith [467, 748] 中所示。在执行时，中心原则是使修复“可检查和可逆”：智能体保留差异、日志、执行跟踪和隔离的工作空间，以便可以诊断失败的尝试，而不是简单地覆盖[657, 747]。面向生产的系统通过分支隔离、挂钩、批准边界、回滚和最终验证门来扩展此循环。7结果不仅仅是迭代调试：智能体在失败的尝试中积累证据，同时保留足够的控制权来修改或放弃早期决策。然而，并非所有这些都需要开放式智能体循环：无智能体表明固定的本地化然后修复管道可以与更复杂的智能体系统相媲美[703]。

6.2 信息查找

信息寻求智能体是 长时程智能体（Long-Horizon Agents） 中的一类重要的类别，因为回答复杂的信息需求通常需要在扩展的检索轨迹上逐步积累证据，其中相关信息分布在异构源中，并且在任何单个搜索步骤中只能部分观察到。我们围绕四种功能组织信息寻求智能体：_深度搜索_维持跨多个检索跳跃的单行查询；_广泛搜索_在共同目标下协调大量并行调查； 多模态 落地关联（Grounding） 将检索范围从文本扩展到文档、图形和图像中的证据； _研究综合_将分散的证据组织成连贯的、明确的研究报告。

这些功能共同定义了从以检索为中心的系统到 长时程（Long-Horizon） 信息搜索智能体的发展。这些智能体必须不断获取、压缩、验证和整合扩展轨迹中的证据，而不是定位孤立的事实[284]

。他们协调多项调查，维持不断发展的研究状态，将研究结果综合成结构化输出，并在异构来源中得出基本结论。由此产生的挑战不再是单独的搜索，而是整个研究生命周期中证据的端到端管理[239]。

6.2.1 深度搜索

深度搜索涉及沿着单一询问线进行持续调查，相关证据可能只有在经过多轮检索、导航和推理后才会出现[223, 883]。随着搜索轨迹的延长，保持有用状态变得越来越重要。根据深度搜索智能体如何优化 长时程（Long-Horizon） 搜索行为，深度搜索智能体可以大致分为两种范例。第一个范式由基于 RL 的深度搜索智能体组成，它通过 强化学习（Reinforcement Learning） [367] 显式优化多步搜索策略。现有的努力主要从两个互补的方向改进这些系统。其中一条线专注于构建具有挑战性的 长时程（Long-Horizon） 搜索任务，鼓励模型获得复杂的搜索和推理行为 [586, 689]，而另一条线开发更有效的 强化学习（Reinforcement Learning） 算法来稳定探索、贡献归因（Credit Assignment） 和长期规划 [120, 130, 682]。第二种范式包括基于工作流的深度搜索智能体，它依赖于精心设计的推理和检索

> Code 文档：https://code.claude.com/docs/en/overview。

> 7Anthropic，Claude Code Hooks 参考：https://code.claude.com/docs/en/hooks。

58

pipelines rather than policy optimization. A series of methods represented by ReSum therefore compress accumulated interaction histories into compact summaries or latent states that can be repeatedly reused across long trajectories [530, 693], while another class of approaches focus on condensing retrieved documents into task-relevant evidence representations for downstream reasoning [106, 336]. Long search chains also make retrieval failures harder to distinguish from reasoning failures [772] and amplify the risk of propagating incorrect assumptions or low-quality evidence [480]. Representative systems such as WebSeer continuously reassess intermediate conclusions, identify unsupported claims, and trigger additional retrieval when evidence remains incomplete or inconsistent [207, 332, 746]. Through these mechanisms, retrieval becomes a persistent evidence-accumulation process that supports deep investigation over extended horizons.

6.2.2 Wide Search

Wide search addresses research objectives whose scope exceeds a single search trajectory. Rather than pursuing one chain of evidence to completion, agents must coordinate many partially independent investigations while maintaining coverage of a large information space. A common strategy is task decomposition, where broad objectives are recursively broken into manageable research questions, evidence requirements, or execution plans. Explicit planning further enables agents to revise information needs and adapt research trajectories as new evidence emerges during execution [227, 292, 650, 746]. Increasingly, decomposition is coupled with parallel exploration. Different agents or search branches investigate complementary hypotheses, domains, or source collections before aggregating findings into a shared research state. Systems such as FlowSearch demonstrate that parallel search can substantially improve coverage and reduce the effective depth of individual trajectories [227, 241, 295, 556, 736]. Beyond parallel exploration, recent methods increasingly organize the search process as structured graphs, trees, or tables, which explicitly maintain search states and support dynamic expansion and revision as exploration proceeds [227, 289, 292]. The resulting challenge shifts from discovering isolated facts to coordinating large-scale evidence collection under a unified objective.

6.2.3 Multimodal Grounding

Multimodal grounding extends information seeking beyond textual retrieval. Many long-horizon research tasks require agents to gather evidence scattered across images, videos, and visually rich documents [114, 414, 714], where multilingual visual-text comprehension, document parsing, and OCR provide the perceptual access needed when text-based retrieval alone is insufficient [159, 290, 459, 584]. Recent work has therefore expanded information-seeking agents toward image search, enabling agents to retrieve, localize, and interpret visual evidence embedded in web-pages or documents, such as screenshots, diagrams, charts, and figures, while grounding subsequent reasoning in visual observations rather than textual descriptions alone [67, 138, 175, 346, 802, 825]. Another line of work focuses on video search, where agents actively explore long videos to identify answer-critical segments through iterative reasoning and interaction, replacing exhaustive frame-by-frame parsing with adaptive evidence seeking over temporal visual content [146, 361, 372]; the underlying active-perception and video-memory systems are treated in Section 6.4. These advances extend information seeking from text-centric retrieval toward unified search over heterogeneous sources.

6.2.4 Research Synthesis

Research synthesis focuses on transforming large collections of retrieved evidence into coherent research artifacts. In long-horizon settings, information gathering alone is insufficient; agents must organize findings accumulated across many searches, identify relationships among sources, and construct narratives that remain faithful to the underlying evidence. Its central application is deep-research report generation, where agents synthesize retrieved evidence into comprehensive reports for domains such as open-domain research, finance, and business analysis [136]. Beyond research prototypes, widely deployed commercial systems such as OpenAI Deep Research, Gemini Deep Research, and Perplexity have brought this capability to broad audiences. Representative systems progressively externalize intermediate artifacts, including outlines, section plans, research notes, and persistent workspaces, before composing final reports [337, 356, 827, 891]. They further maintain explicit evidence attribution so that readers can trace individual claims back to their sources, improving both faithfulness and auditability [292, 347, 621]. The core challenge is therefore not merely to summarize retrieved documents, but to preserve coverage, narrative coherence, factual support [434, 677], and source-level traceability across a long search trajectory.

原文第 59 页 | 中文翻译初稿

管道而不是策略优化。因此，以 ReSum 为代表的一系列方法将累积的交互历史压缩为紧凑的摘要或潜在状态，可以在长轨迹中重复重用 [530, 693]，而另一类方法则侧重于将检索到的文档压缩为与任务相关的证据表示，以进行下游推理 [106, 336]。长搜索链还使得检索失败与推理失败更难以区分[772]，并放大了传播错误假设或低质量证据的风险[480]。 WebSeer 等代表性系统不断重新评估中间结论，识别不受支持的主张，并在证据不完整或不一致时触发额外检索 [207, 332, 746]。通过这些机制，检索成为一个持久的证据积累过程，支持在更广泛的范围内进行深入调查。

6.2.2 广泛搜索

广泛的搜索解决了范围超出单一搜索轨迹的研究目标。特工们必须协调许多部分独立的调查，同时保持对大量信息空间的覆盖，而不是追求完成一系列证据。一个常见的策略是任务分解，其中广泛的目标被递归地分解为可管理的研究问题、证据要求或执行计划。明确的规划进一步使智能体人能够根据执行过程中出现的新证据来修改信息需求并调整研究轨迹[227,292,650,746]。分解越来越多地与并行探索相结合。不同的智能体或搜索分支在将发现汇总到共享研究状态之前调查互补的假设、领域或源集合。 FlowSearch 等系统证明并行搜索可以显着提高覆盖范围并减少各个轨迹的有效深度 [227,241,295,556,736]。除了并行探索之外，最近的方法越来越多地将搜索过程组织为结构化图、树或表，它们显式地维护搜索状态并支持随着探索的进行动态扩展和修订[227,289,292]。由此产生的挑战从发现孤立的事实转变为在统一目标下协调大规模证据收集。

6.2.3 多式联运 落地关联（Grounding）

多模式 落地关联（Grounding） 将信息搜索扩展到文本检索之外。许多 长时程（Long-Horizon） 研究任务要求智能体收集分散在图像、视频和视觉丰富的文档中的证据 [114,414,714]，其中多语言视觉文本理解、文档解析和 OCR 提供了当仅基于文本的检索不足时所需的感知访问[159,290,459,584]。因此，最近的工作将信息寻求智能体扩展到图像搜索，使智能体能够检索、本地化和解释嵌入网页或文档中的视觉证据，例如屏幕截图、图表、图表和图形，而 落地关联（Grounding） 在视觉观察中进行后续推理，而不仅仅是文本描述 [67,138,175,346,802,825]

。另一条工作重点是视频搜索，智能体积极探索长视频，通过迭代推理和交互来识别关键答案片段，用针对时间视觉内容的自适应证据搜索取代详尽的逐帧解析[146,361,372]； 6.4 节讨论了底层的主动感知和视频存储系统。这些进步将信息搜索从以文本为中心的检索扩展到异构源上的统一搜索。

6.2.4 研究综合

研究综合的重点是将大量检索到的证据转化为连贯的研究成果。在长时程（Long-Horizon）设置中，仅收集信息是不够的；特工必须组织多次搜索中积累的发现，识别来源之间的关系，并构建忠实于潜在证据的叙述。其核心应用是深度研究报告生成，智能体将检索到的证据合成为开放领域研究、金融和业务分析等领域的综合报告[136]。除了研究原型之外，广泛部署的商业系统（例如 OpenAI Deep Research、Gemini Deep Research 和 Perplexity）也为广大受众带来了这种功能。在撰写最终报告之前，代表性系统逐渐将中间工件具体化，包括大纲、剖面图、研究笔记和持久工作空间[337、356、827、891]。他们进一步维护明确的证据归属，以便读者可以追溯到个人主张的来源，从而提高可信度和可审计性[292,347,621]。因此，核心挑战不仅仅是总结检索到的文档，而是在较长的搜索轨迹中保持覆盖范围、叙述连贯性、事实支持 [434, 677] 以及源级可追溯性。

59

6.3 Computer Use

Computer-use agents interact with stateful software systems in which actions have persistent consequences and task completion requires sustained execution across extended interaction trajectories. These agents autonomously manipulate desktops, browsers, and applications through heterogeneous interfaces including screenshots, DOMs, accessibility trees, shells, executable code, and file systems [126, 588]. Recent systems such as Anthropic's Computer Use and OpenAI's Operator support planning, multi-tool coordination, and persistent execution state across long-horizon tasks [12, 452, 745].

We organize long-horizon computer use according to the execution environment: *browser agents*, *desktop GUI agents*, and *mobile agents*. Browser agents focus on sustained task execution within web environments through navigation and information extraction. Desktop GUI agents emphasize reliable visual grounding and interaction with operating systems and desktop applications. Mobile agents further extend long-horizon execution to smartphone applications, where agents must coordinate touch-based interaction across diverse mobile interfaces. Together, these directions characterize the evolution from browser automation to general-purpose computer use. Beyond executing actions within a single application, modern computer-use agents must coordinate perception, reasoning, and execution across heterogeneous software environments while maintaining persistent state and coherent progress toward long-running objectives.

6.3.1 Browser Agents

Browser agents investigate long-horizon interaction within web environments through navigation, information extraction, and form manipulation. Benchmarks such as WebArena, Mind2Web, and ClawBench establish browser interaction as a representative long-horizon computer-use setting, requiring agents to interpret page content, choose actions, and maintain progress across evolving interface states [116, 847, 885]. To support sustained execution, many systems expose browsers through reusable action primitives such as clicking, typing, scrolling, navigation, and form completion, allowing agents to reason over browser interaction as a structured action space rather than raw interface manipulation [210, 226, 404, 443, 781]. At the same time, accumulated observations, retrieved information, and intermediate decisions are increasingly externalized through memories, notes, summaries, or reusable knowledge stores that persist across long interaction trajectories [637, 781]. These developments progressively transform browser interaction from isolated page-level decisions into persistent task execution over extended horizons.

6.3.2 Desktop GUI Agents

Desktop GUI agents study how autonomous systems interact with operating systems and graphical desktop environments through multimodal perception and action. Benchmarks such as OSWorld and Workflow-GYM, together with representative systems including UI-TARS, have established desktop interaction as a realistic long-horizon setting, requiring agents to perceive interface state, execute multi-step workflows, and recover from execution failures across diverse applications [494, 526, 551, 716, 745]. Visual grounding plays a central role in desktop interaction by maintaining reliable correspondence between evolving interface observations and executable actions. Because graphical layouts can be both dense and dynamic across applications and execution steps, agents must identify relevant elements in high-resolution interfaces [324], interpret their functional roles, and update action targets throughout long interaction trajectories [181, 691]. Modern systems further complement multimodal perception with execution verification mechanisms based on screenshot comparison, visual change detection, semantic consistency checks, and artifact inspection [12, 259, 518], together with safety guardrails such as permission boundaries and confirmation policies [12, 455] to improve robustness under execution uncertainty.

6.3.3 Mobile Agents

Mobile agents extend long-horizon computer use to smartphones and tablets, where agents must coordinate interactions across multiple applications through touch-based interfaces [282, 373, 570]. Compared with desktop environments, mobile tasks typically involve long multi-app workflows, limited screen space, frequent context switching, and gesture-based operations, making persistent planning and state management particularly important [667, 766]. Recent systems increasingly adopt hierarchical agent architectures that separate high-level planning from low-level UI execution, allowing complex user requests to be decomposed into executable

原文第 60 页 | 中文翻译初稿

6.3 计算机使用

计算机使用的智能体与有状态的软件系统进行交互，其中的操作具有持久的后果，并且任务的完成需要在扩展的交互轨迹上持续执行。这些智能体通过异构接口（包括屏幕截图、DOM、可访问性树、shell、可执行代码和文件系统）自主地操作桌面、浏览器和应用程序 [126, 588]。最近的系统，例如 Anthropic 的 Computer Use 和 OpenAI 的 Operator，支持跨 长时程（Long-Horizon） 任务的规划、多工具协调和持久执行状态 [12, 452, 745]。

我们根据执行环境组织长时程（Long-Horizon）计算机使用：浏览器智能体、_桌面GUI智能体_和_移动智能体_。浏览器智能体专注于通过导航和信息提取在 Web 环境中持续执行任务。桌面 GUI 智能体强调可靠的视觉 落地关联（Grounding） 以及与操作系统和桌面应用程序的交互。移动智能体进一步将 长时程（Long-Horizon） 执行扩展到智能手机应用程序，其中智能体必须协调跨不同移动界面的基于触摸的交互。这些方向共同体现了从浏览器自动化到通用计算机使用的演变。除了在单个应用程序中执行操作之外，现代计算机使用智能体还必须协调异构软件环境中的感知、推理和执行，同时保持持久状态和一致的进展以实现长期运行的目标。

6.3.1 浏览器智能体

浏览器智能体通过导航、信息提取和表单操作来调查 Web 环境中的 长时程（Long-Horizon） 交互。WebArena、Mind2Web 和 ClawBench 等基准将浏览器交互建立为代表性的 长时程（Long-Horizon） 计算机使用设置，要求智能体解释页面内容、选择操作并在不断发展的界面状态中保持进度[116、847、885]。为了支持持续执行，许多系统通过可重用的操作原语（例如单击、打字、滚动、导航和表单完成）公开浏览器，允许智能体将浏览器交互作为结构化操作空间而不是原始界面操作进行推理[210,226,404,443,781]。与此同时，积累的观察结果、检索到的信息和中间决策越来越多地通过记忆、笔记、摘要或在长期交互轨迹中持续存在的可重用知识存储来外化[637, 781]。这些发展逐渐将浏览器交互从孤立的页面级决策转变为长期持续的任务执行。

6.3.2 桌面 GUI 智能体

桌面 GUI 智能体研究自主系统如何通过多模式感知和操作与操作系统和图形桌面环境交互。 OSWorld 和 Workflow-GYM 等基准测试以及包括 UI-TARS 在内的代表性系统，已将桌面交互建立为现实的 长时程（Long-Horizon） 设置，要求智能体感知界面状态，执行多步骤工作流，并从跨不同应用程序的执行失败中恢复[494、526、551、716、745]

。 Visual 落地关联（Grounding） 通过维护不断发展的界面观察和可执行操作之间的可靠对应关系，在桌面交互中发挥着核心作用。由于图形布局在应用程序和执行步骤中可能既密集又动态，因此智能体必须识别高分辨率界面中的相关元素[324]，解释其功能角色，并在整个长交互轨迹中更新操作目标[181, 691]。现代系统通过基于屏幕截图比较、视觉变化检测、语义一致性检查和工件检查 [12, 259, 518] 的执行验证机制进一步补充多模态感知，以及权限边界和确认策略等安全护栏 [12, 455]，以提高执行不确定性下的鲁棒性。

6.3.3 移动智能体

移动智能体将 长时程（Long-Horizon） 计算机的使用扩展到智能手机和平板电脑，其中智能体必须通过基于触摸的界面协调多个应用程序之间的交互[282,373,570]。与桌面环境相比，移动任务通常涉及较长的多应用工作流程、有限的屏幕空间、频繁的上下文切换和基于手势的操作，使得持久规划和状态管理尤为重要[667, 766]。最近的系统越来越多地采用分层智能体架构，将高级规划与低级 UI 执行分开，从而允许

将复杂的用户请求分解为可执行文件

60

subtasks while maintaining progress across extended interaction trajectories [543, 667, 723, 766]. At the same time, long-term memories, reusable interaction skills, and execution verification mechanisms are incorporated to improve robustness and enable continual improvement from previous experiences [105, 571, 723, 766]. These developments transform mobile agents from reactive UI operators into persistent assistants capable of accomplishing complex real-world workflows across diverse mobile applications.

6.4 Multimodal Agents

Multimodal agents are long-horizon agents that must process information from different modalities. In long-horizon scenarios, multimodal evidence may be distributed across long visual contexts or multiple videos [55, 320, 413, 666], requiring agents to locate and acquire relevant information, store it, reason over it, and integrate it into an answer or multimodal output. Recent benchmarks such as AgentVista and OmniGAIA mainly evaluate agents' ability to solve cross-modal tasks [339, 564, 589], rather than focusing only on isolated visual question answering or reasoning [45, 85, 190, 465].

We organize long-horizon multimodal agents into three categories according to their tasks and capabilities: *multimodal understanding*, *multimodal generation*, and *omnimodal agency*. Multimodal understanding focuses on how to locate useful information over extended timelines, including deciding which segments to inspect and which perception tools to invoke. Multimodal generation focuses on how to understand user intent and conduct multi-round planning, generation, and refinement, a process that relies on external evidence, verification mechanisms, and persistent memory. Omnimodal agency focuses on the coordination of different modalities so that they can jointly accomplish an end-to-end task. Overall, these three types of capabilities enable agents to handle richer multimodal information and interact more deeply with the real world.

6.4.1 Multimodal Understanding

For long-horizon multimodal understanding, agents must possess the ability of *active perception*. Under limited budgets, it is impossible to read all information streams at once; truly important information often appears only in a few frames of a long video, a small region of an image, or a single line of text in a document. Therefore, agents should be able to actively locate useful information within long information streams.

Current designs follow two main lines. The first line studies *tool-augmented active perception*. Works represented by VideoSeek treat long videos and images as explorable spaces rather than inputs that are fed into the model all at once [138, 365, 655, 825, 903]. The agent dynamically invokes browsing and retrieval tools according to the current question and existing clues, progressively locating key evidence in long videos. The second line studies *memory-augmented evidence organization*. Representative works such as DVD [510, 739, 829] construct structured video memory and then retrieve relevant evidence according to task demands. The former emphasizes active exploration strategies, while the latter emphasizes structured representations of long information streams and evidence preservation [796]. Their shared goal is to avoid blindly reading all multimodal inputs under limited budgets.

6.4.2 Multimodal Generation

In long-horizon scenarios, multimodal generation is not a one-step mapping from a prompt to an image or video clip. Agents usually need to understand user intent, plan the generation process, and conduct multiple rounds of generation and verification. Therefore, multimodal generation is a long-term task that must be completed through continuous iteration. Recent multimodal generation models, such as those in the Qwen-Image, Seedream, and SeedEdit series, have significantly improved intent understanding, multimodal creation, image editing, and video synthesis capabilities [73, 115, 172, 528, 600, 645, 855]. However, for long-horizon agents, beyond the generative capability of the model itself, the design of the surrounding harness is more important for ensuring that generated content remains consistent with user requirements.

Therefore, three mechanisms are crucial. *Evidence seeking* provides grounding for generation by retrieving images, documents, and videos. This is especially necessary in knowledge-intensive multimodal generation tasks, where sufficient task-relevant information must be collected. Recent generative agents autonomously invoke retrieval and visual analysis tools to gather task-relevant evidence before generation [68, 209, 255]. *Verification* checks generated results from multiple dimensions to ensure that they remain aligned with

原文第 61 页 | 中文翻译初稿

子任务，同时保持扩展交互轨迹的进展[543,667,723,766]。同时，结合了长期记忆、可重用的交互技能和执行验证机制，以提高鲁棒性并能够从以前的经验中不断改进[105,571,723,766]。这些发展将移动智能体从反应式 UI 操作员转变为持久的助手，能够跨不同的移动应用程序完成复杂的现实工作流程。

6.4 多模式智能体

多模式智能体是 长时程智能体（Long-Horizon Agents），必须处理来自不同模式的信息。在 长时程（Long-Horizon）场景中，多模态证据可能分布在长视觉上下文或多个视频中 [55,320,413,666]，要求智能体定位和获取相关信息，存储它，对其进行推理，并将其集成到答案或多模态输出中。最近的基准测试（例如 AgentVista 和 OmniGAIA）主要评估智能体解决跨模式任务的能力 [339,564,589]，而不是只关注孤立的视觉问题回答或推理 [45,85,190,465]。

我们根据 长时程（Long-Horizon）多模态智能体的任务和能力将其分为三类：多模态理解、_多模态生成_和_全模态智能体_。多模态理解侧重于如何在较长的时间内找到有用的信息，包括决定要检查哪些部分以及要调用哪些感知工具。多模态生成重点关注如何理解用户意图并进行多轮规划、生成和细化，这个过程依赖于外部证据、验证机制和持久记忆。全模式机构侧重于不同模式的协调，以便它们能够共同完成端到端的任务。总体而言，这三种类型的功能使智能体能够处理更丰富的多模式信息并与现实世界进行更深入的交互。

6.4.1 多模态理解

对于长时程（Long-Horizon）多模态理解，智能体必须具备_主动感知_的能力。在预算有限的情况下，不可能一次性读取所有信息流；真正重要的信息通常只出现在长视频的几帧、图像的一小部分或文档中的一行文本中。因此，智能体应该能够在长信息流中主动定位有用的信息。

当前的设计遵循两条主线。第一行研究_工具增强主动感知_。以VideoSeek为代表的作品将长视频和图像视为可探索的空间，而不是一次性输入模型的输入[138,365,655,825,903]。智能体根据当前问题和现有线索动态调用浏览和检索工具，逐步定位长视频中的关键证据。第二行研究_记忆增强证据组织_。代表性作品如DVD[510,739,829]构建结构化视频存储器，然后根据任务需求检索相关证据。前者强调主动探索策略，后者强调长信息流的结构化表示和证据保存[796]

。他们的共同目标是避免在有限的预算下盲目阅读所有多式联运投入。

6.4.2 多模态生成

在 长时程（Long-Horizon）场景中，多模式生成不是从提示到图像或视频剪辑的一步映射。智能体通常需要了解用户意图，规划生成过程，并进行多轮生成和验证。因此，多模态生成是一项长期任务，必须通过不断迭代来完成。最近的多模态生成模型，例如 Qwen-Image、Seedream 和 SeedEdit 系列中的模型，显着提高了意图理解、多模态创建、图像编辑和视频合成功能 [73、115、172、528、600、645、855]。然而，对于长时程智能体（Long-Horizon Agents）来说，除了模型本身的生成能力之外，周边运行支撑框架（Harness）的设计对于确保生成的内容保持与用户需求的一致更为重要。

因此，三个机制至关重要。 _取证_提供落地关联（Grounding），通过检索图像、文档和视频来生成。这在知识密集型多模式生成任务中尤其必要，因为必须收集足够的任务相关信息。最近的生成智能体在生成之前自主调用检索和视觉分析工具来收集与任务相关的证据[68,209,255]。 _验证_检查从多个维度生成的结果，以确保它们与

the original goal [574, 741, 831]. *Memory* stores user requirements and evidence at different granularities, enabling subsequent generation to remain consistent with earlier decisions [71, 733]. Recent multimodal generation systems increasingly organize planning, evidence collection, memory, verification, and rendering into unified agentic workflows. For example, PTAH coordinates specialized agents throughout the research and writing lifecycle, maintains structured visual working memory, and enforces cross-modal consistency through verifier-guided report generation [803]. Similarly, long-form audio-video systems such as MUSE employ planning-execution-verification-revision loops, transforming multimodal generation into a long-horizon iterative process [574].

6.4.3 Omnimodal Agency

Omnimodal tasks require agents to coordinate across different modalities, because a single modality or a single model is often insufficient to complete the task. This is closer to real-world scenarios, where inputs may include text, images, speech, audio, video, and other modalities. Therefore, agents need the ability of *cross-modal orchestration*. Specifically, an agent needs to decompose the task, route subtasks to appropriate specialized models, and then verify and integrate their outputs.

Recent work has developed along two complementary directions. One direction focuses on native omnimodal models, such as those in the Qwen-Omni series, which provide unified interfaces for multimodal understanding or generation and usually support streaming interaction and speech-centered multimodal dialogue [498, 726, 727]. The other direction focuses on orchestration-based agents, such as OmniAgent, Orchestra-o1, and OmniGAIA, which explicitly assign subtasks to different modalities and coordinate their execution processes and outputs [137, 339, 364, 718, 728, 806]. In long-horizon omnimodal tasks, information from different modalities may conflict with each other, so explicit verification is required. Omnimodal agent capabilities enable agents to interact more deeply with real-world environments and broaden their range of applications.

6.5 General-purpose Agents

General-purpose agents represent one of the broadest forms of long-horizon agency, because they must pursue open-ended goals across heterogeneous interfaces rather than operate within a single fixed environment. A user request may require an agent to interact with multiple interfaces and invoke different types of tools. The central challenge for such agents is therefore *sustained coordination*: an agent must conduct overall planning for the task, maintain and transfer state across different environments, and recover from failures during execution. Benchmarks such as GAIA and τ-bench reflect a recent shift toward evaluating whether agents can consistently complete tasks in heterogeneous environments [335, 387, 431, 762, 840]; we do not restate the individual environments here, and collect them together with the domain-specific benchmarks reused throughout this section in Table 7.

We organize general-purpose agents into three representative categories according to their deployment scenarios. *Personal assistants* deploy general-purpose agent capabilities in personal computing and office environments, where agents must operate across applications and interact continuously with users. *Embodied agents and world models* extend such capabilities to physical environments, where agents must perceive, reason, and predict the consequences of actions in the world. *Productive agents* apply general-purpose agent capabilities to professional workflows, where outputs must satisfy domain-specific standards and verification requirements.

6.5.1 Personal Assistants

Personal assistants are an increasingly important class of agents designed to operate across everyday digital environments. Unlike traditional conversational assistants, these agents are expected not only to answer user questions, but also to complete continuous tasks across applications and local resources. Personal assistants therefore mark a shift from conversational assistance to persistent computer agents: the agent no longer merely generates text, but can continuously carry out tasks through multi-turn interaction with both the user and the environment.

A capability underlying all personal assistants is *general tool use*: selecting and invoking the right tools to act beyond the model's parametric knowledge. Early benchmarks assess this ability *without environment state*, testing whether an agent can choose and call the correct function from large, scalable tool sets, as

原文第 62 页 | 中文翻译初稿

最初的目标 [574, 741, 831]。 _Memory_以不同的粒度存储用户需求和证据，使后续一代能够与早期决策保持一致[71, 733]。最近的多模式生成系统越来越多地将规划、证据收集、记忆、验证和渲染组织到统一的智能体工作流程中。例如，Ptah 在整个研究和写作生命周期中协调专门智能体，维护结构化视觉工作记忆，并通过验证者引导的报告生成来强制跨模式一致性[803]。类似地，MUSE 等长格式音视频系统采用规划-执行-验证-修订循环，将多模态生成转换为 长时程（Long-Horizon） 迭代过程 [574]。

6.4.3 全模式智能体

全模态任务需要智能体跨不同模态进行协调，因为单一模态或单一模型通常不足以完成任务。这更接近现实世界的场景，其中输入可能包括文本、图像、语音、音频、视频和其他模式。因此，智能体需要_跨模式编排_的能力。具体来说，智能体需要分解任务，将子任务路由到适当的专门模型，然后验证和集成其输出。

最近的工作沿着两个互补的方向发展。一个方向侧重于原生全模态模型，例如 Qwen-Omni 系列中的模型，它为多模态理解或生成提供统一的接口，通常支持流交互和以语音为中心的多模态对话[498,726,727]。另一个方向侧重于基于编排的智能体，例如 OmniAgent、Orchestra-o1 和 OmniGAIA，它们显式地将子任务分配给不同的模式并协调其执行过程和输出[137、339、364、718、728、806]。在长时程（Long-Horizon）全模态任务中，来自不同模态的信息可能会相互冲突，因此需要显式验证。全模式智能体功能使智能体能够与现实环境进行更深入的交互，并扩大其应用范围。

6.5 通用（General-Purpose） 智能体

通用（General-Purpose） 智能体代表了 长时程（Long-Horizon） 智能体的最广泛形式之一，因为它们必须跨异构接口追求开放式目标，而不是在单个固定环境中运行。用户请求可能需要智能体与多个接口交互并调用不同类型的工具。因此，此类智能体的核心挑战是_持续协调_：智能体必须对任务进行总体规划，在不同环境中维护和传输状态，并在执行过程中从故障中恢复。 GAIA 和 τ -bench 等基准反映了最近的转变，即评估智能体是否能够在异构环境中一致地完成任务 [335, 387, 431, 762, 840]；我们不会在这里重述各个环境，而是将它们与表 7 中本节中重复使用的特定于域的基准收集在一起。

我们根据部署场景将 通用（General-Purpose） 智能体分为三个有代表性的类别

。 _个人助理_在个人计算和办公环境中部署 通用（General-Purpose） 智能体功能，其中智能体必须跨应用程序操作并与用户持续交互。 具体智能体和 世界模型（World Models） 将此类功能扩展到物理环境，其中智能体必须感知、推理并预测世界上行为的后果。 _高效智能体_将 通用（General-Purpose） 智能体功能应用于专业工作流程，其中输出必须满足特定领域的标准和验证要求。

6.5.1 个人助理

个人助理是一类日益重要的智能体，旨在在日常数字环境中运行。与传统的会话助理不同，这些智能体不仅要回答用户问题，还要跨应用程序和本地资源完成连续任务。因此，个人助理标志着从会话辅助到持久计算机智能体的转变：智能体不再仅仅生成文本，而是可以通过与用户和环境的多轮交互来连续执行任务。

所有个人助理的基本功能是“一般工具使用”：选择和调用正确的工具来超越模型的参数知识。早期的基准测试在没有环境状态的情况下评估这种能力，测试智能体是否可以从大型、可扩展的工具集中选择并调用正确的函数，例如

in ToolBench [492] and the Berkeley Function-Calling Leaderboard [472]. τ-bench [762] then moves tool use into *stateful* environments, where each call mutates a persistent backend and success is judged against realistic multi-turn user interactions rather than isolated call correctness. More recent benchmarks push toward mixed, cross-application settings with increasingly long-tailed tool inventories: Toolathlon [322] spans diverse realistic applications, while MCP-Atlas [31] and MCP-Mark [700] stress-test tool-use competency over large, heterogeneous MCP tool ecosystems. Finally, Claw-Eval [764] and ClawMark [426] evaluate tool use mediated by reusable *skills*, enabling more flexible invocation that better reflects modern agent deployments.

Personal assistants can be broadly divided into four categories. The first targets *office scenarios*, focusing on whether agents can complete tasks in realistic workflows involving documents, emails, calendars, and related office applications. Benchmarks such as OfficeQA, APEX-Agents, PresentBench, and SpreadsheetBench examine capabilities in information retrieval, file and presentation generation, spreadsheet manipulation, and multi-step task execution in such office environments [74, 415, 458, 623]. The second targets *personal computing scenarios*, emphasizing agents' ability to operate real device interfaces, including browsers, desktop applications, mobile apps, and local file systems; this setting largely reuses the computer-use systems and benchmarks of Section 6.3 (e.g., UI-TARS and OSWorld [494, 716]) and is not detailed again here. The third targets *software development scenarios*, where agents assist developers with coding tasks that require code understanding, modification, and validation. Systems such as Claude Code and Codex are representative examples of this direction [10, 453]. The fourth targets *persistent personal assistant scenarios*, emphasizing agents that run continuously in user-controlled devices or environments and connect local resources, long-term memory, and external tools. Systems such as OpenClaw and Hermes illustrate this direction [420, 457, 513, 893].

6.5.2 Embodied Agents and World Models

Embodied agents extend agent capabilities from digital environments to the physical world, where they must continuously perceive the environment, reason about goals and constraints, and execute actions that alter the world state. Long-horizon embodied tasks therefore require a sustained loop of perception, planning, and action, together with reliable state tracking, safety constraints, and recovery from execution errors. Their capabilities can be broadly divided into *navigation* and *manipulation*. Navigation focuses on understanding spatial layouts, planning paths, and adapting to environmental changes [135, 755, 771], whereas manipulation requires fine-grained reasoning about spatial references, object states, physical properties, and interactions such as grasping, moving, and assembling [93, 684, 872, 878]. Recent vision-language-action models, including Gemini Robotics and Qwen's robotic VLA models, integrate language instructions, visual observations, and robotic control within a unified action interface, marking a shift from task-specific controllers toward more general-purpose embodied agents [35, 245, 594, 646, 784, 817].

World models provide embodied agents with predictive representations of environment dynamics, allowing them to evaluate possible action outcomes before acting. This capability is particularly important in physical environments, where exploration is costly and execution failures may be difficult to reverse. Recent world-model research can be broadly divided into two main directions. The first is *predictive representation models*, exemplified by V-JEPA 2 and DINO-WM, which compress the environment into abstract latent representations and predict how these representations evolve over time and in response to actions, rather than directly generating complete future images [25, 879]. By omitting visual details such as texture and illumination, these models can support more efficient multi-step prediction and planning. The second direction is *generative world models*, including Cosmos, Cosmos-Predict2.5, and OSCAR, which directly generate predicted outcomes following an agent's actions to support trajectory comparison, policy training, and risk assessment before real-world execution [2, 449, 699]. Building on these two directions, *world-action models*, such as DreamZero and τ_0-WM, have recently emerged to unify world prediction and action generation by jointly modeling future states and executable actions [769, 884].

6.5.3 Productive Agents

Productive agents apply general-purpose capabilities to workflows whose outputs must satisfy domain-specific standards. They retrieve authoritative sources, invoke specialized tools, and maintain traceable intermediate and final artifacts over extended trajectories. The central challenge is therefore not simply task automation, but the integration of planning, tool use, memory, and verification into established practice. We group these

原文第 63 页 | 中文翻译初稿

在 ToolBench [492] 和 Berkeley 函数调用排行榜 [472] 中。然后，τ -bench [762] 将工具使用转移到_有状态_环境中，其中每个调用都会改变持久后端，并且根据实际的多轮用户交互而不是孤立的调用正确性来判断成功。最近的基准测试通过越来越多的长尾工具清单推动了混合、跨应用程序设置：Toolathlon [322] 跨越了不同的实际应用程序，而 MCP-Atlas [31] 和 MCP-Mark [700] 在大型、异构 MCP 工具生态系统上压力测试工具使用能力。最后，Claw-Eval [764] 和 ClawMark [426] 评估由可重用_skills_介导的工具使用情况，从而实现更灵活的调用，更好地反映现代智能体部署。

个人助理大致可分为四类。第一个目标是_办公场景_，重点关注座席是否可以在涉及文档、电子邮件、日历和相关办公应用程序的现实工作流程中完成任务。 OfficeQA、APEX-Agents、PresentBench 和 SpreadsheetBench 等基准测试在此类办公环境中检查信息检索、文件和演示文稿生成、电子表格操作以及多步骤任务执行的功能 [74、415、458、623]。第二个目标是“个人计算场景”，强调智能体操作真实设备界面的能力，包括浏览器、桌面应用程序、移动应用程序和本地文件系统；该设置很大程度上重用了第 6.3 节的计算机使用系统和基准（例如，UI-TARS 和 OSWorld [494, 716]），这里不再详细说明。第三个目标是“软件开发场景”，其中智能体协助开发人员完成需要代码理解、修改和验证的编码任务。 Claude Code 和 Codex 等系统是这个方向的代表性例子 [10, 453]。第四个目标是_持久个人助理场景_，强调在用户控制的设备或环境中持续运行并连接本地资源、长期内存和外部工具的智能体。 OpenClaw 和 Hermes 等系统说明了这个方向[420,457,513,893]。

6.5.2 体现智能体和世界模型（World Models）

具身智能体将智能体的能力从数字环境扩展到物理世界，它们必须不断感知环境，推理目标和约束，并执行改变世界状态的行动。因此，长时程（Long-Horizon）体现的任务需要感知、规划和行动的持续循环，以及可靠的状态跟踪、安全约束和从执行错误中恢复。他们的能力大致可分为_导航_和_操纵_。导航侧重于理解空间布局、规划路径和适应环境变化[135,755,771]，而操作则需要对空间参考、对象状态、物理属性以及抓取、移动和组装等交互进行细粒度推理[93,684,872,878]

。最近的 视觉语言动作（Vision-Language-Action）模型，包括 Gemini Robotics 和 Qwen 的机器人 VLA 模型，将语言指令、视觉观察和机器人控制集成在统一的动作界面中，标志着从特定任务控制器到更多 通用（General-Purpose）体现智能体的转变 [35,245,594,646,784,817]。

世界模型（World Models）为具身智能体提供环境动态的预测表示，使他们能够在行动之前评估可能的行动结果。此功能在物理环境中尤其重要，因为在物理环境中探索成本高昂且执行失败可能难以逆转。最近的世界模型研究大致可以分为两个主要方向。第一个是_预测表示模型_，以 V-JEPA 2 和 DINO-WM 为例，它将环境压缩为抽象的潜在表示，并预测这些表示如何随着时间的推移以及响应动作而演变，而不是直接生成完整的未来图像 [25, 879]。通过省略纹理和照明等视觉细节，这些模型可以支持更有效的多步骤预测和规划。第二个方向是_generative 世界模型（World Models）_，包括Cosmos、Cosmos-Predict2.5和OSCAR，它们根据智能体的动作直接生成预测结果，以支持现实世界执行之前的轨迹比较、策略训练和风险评估[2, 449, 699]。基于这两个方向，最近出现了 世界动作模型（World-Action Models），例如 DreamZero 和 τ 0-WM，通过联合建模未来状态和可执行动作来统一世界预测和动作生成 [769, 884]。

6.5.3 生产性智能体

高效智能体将 通用（General-Purpose）功能应用于其输出必须满足特定于域的标准的工作流。他们检索权威来源，调用专门工具，并在扩展轨迹上维护可追踪的中间和最终工件。因此，核心挑战不仅仅是任务自动化，而是将规划、工具使用、记忆和验证整合到既定实践中。我们将这些分组

63

applications into two intuitive families: (1) *professional services*, where agents assist established work in finance, law, and medicine; and (2) *scientific discovery and reasoning*, where agents help generate and verify new knowledge.

Professional Services. In finance, agents integrate market data, financial statements, and related evidence to produce traceable analyses, as illustrated by FinRobot and FinSight [34, 263, 886]. Legal agents retrieve authoritative sources, connect facts to rules, and produce outputs that can be reviewed against legal doctrine and procedure; the Legal Agent Benchmarks, together with agents such as Parthenon Law and LawThinker, reflect the shift from static question answering toward long-horizon legal work [174, 185, 315, 756]. Medical agents must further combine patient records, images, and examination reports while respecting safety, privacy, and auditability requirements [21, 904]. MedAgentBench, MedAgentBoard, and PhysicianBench accordingly evaluate sustained clinical reasoning and workflow execution rather than isolated diagnosis [23, 257, 380, 895]. Across these domains, correctness cannot usually be reduced to a single executable oracle, so provenance, procedural compliance, and expert review remain central [474, 752].

Scientific Discovery and Reasoning. Scientific-discovery agents go beyond producing domain-compliant analyses: they generate new claims and test them against computational or empirical feedback. Their workflows connect literature review with expert-level and experimental scientific reasoning [642, 882], hypothesis generation, experimental design, execution, result analysis, and iterative refinement [505, 678, 869]. The AI Scientist and its successor organize machine-learning research into repeated cycles of ideation, experimentation, evaluation, and writing [397, 740]. AlphaEvolve and Kosmos extend this pattern to algorithmic and cross-domain discovery [438, 448], while NovelSeek closes the loop from hypothesis generation to verification across heterogeneous scientific tasks [800]. In empirical settings, Coscientist couples language-model planning with robotic chemical experimentation [37], and Robin and the Virtual Lab connect multi-agent reasoning to biomedical discovery and laboratory validation [177, 581]. Hypotheses, programs, experimental records, and manuscripts consequently become evolving research state that must be revised whenever new evidence arrives.

Formal mathematics represents a complementary form of discovery in which proof assistants can mechanically check each completed proof [615, 866]. Seed-Prover 1.5 combines natural-language reasoning, lemma-level sketches, recursive decomposition, and an agentic Lean prover to construct long proofs under iterative verification [59]. Prover Agent similarly coordinates informal reasoning, auxiliary-lemma generation, formal proof construction, and Lean diagnostics in a feedback-driven refinement loop [26]. Automated Proof Engineering broadens the objective beyond isolated theorem proving to repository-scale work, including feature addition, refactoring, and bug repair across evolving Lean libraries, with evaluation based on both compilation and semantic requirement satisfaction [717]. Together, empirical science and formal mathematics show how productive agents can move beyond summarizing existing knowledge to generating claims that survive domain-specific verification [629].

6.6 Benchmarks and Resources

Progress on long-horizon agents is measured and reproduced through a shared substrate of benchmarks and open resources. Because a reported score reflects the joint contribution of the model, the harness, and the training recipe, long-horizon evaluation is best read as *horizon-aware*, sensitive to how long a goal must be held, and *attribution-aware*, able to separate harness-induced from model-induced variance [123, 286, 468]. The same verifiable, long-horizon environments that serve as benchmarks are also frequently reused as training environments for the optimization methods of Section 5, so progress on evaluation and progress on internalization are tightly coupled. Beyond the domain benchmarks tabulated below, a growing family of suites deliberately stresses multi-hour autonomy. Examples range from machine-learning engineering in MLE-bench [51] to end-to-end replication of research papers in PaperBench [560], complementing the frontier time-horizon measurements of METR [286, 429]; these appear among the productive-agent entries of Table 7.

Table 7 consolidates representative resources for the application domains above, grouped by domain and, within each domain, ordered by the capability taxonomy of Section 6. Each row pairs an evaluation benchmark or a reusable system with the capability it targets and, where available, a link to its public repository or official release page (), offering a concrete entry point into the works discussed throughout this section. The multimodal-understanding entries (e.g., video and streaming comprehension benchmarks) are not long-horizon

原文第 64 页 | 中文翻译初稿

应用程序分为两个直观的系列：（1）专业服务，智能体人协助金融、法律和医学领域的既定工作；(2)科学发现和推理，智能体帮助生成和验证新知识。

专业服务。在金融领域，智能体整合市场数据、财务报表和相关证据来生成可追溯的分析，如 FinRobot 和 FinSight [34, 263, 886] 所示。法律智能体人检索权威来源，将事实与规则联系起来，并产生可根据法律原则和程序进行审查的输出；法律智能体基准以及诸如 Parthenon Law 和 LawThinker 之类的智能体反映了从静态问答向 长时程（Long-Horizon） 法律工作的转变 [174, 185, 315, 756]。医疗机构必须进一步结合患者记录、图像和检查报告，同时尊重安全、隐私和可审核性要求 [21, 904]。 MedAgentBench、MedAgentBoard 和 PhysicianBench 相应地评估持续的临床推理和工作流程执行，而不是孤立的诊断 [23, 257, 380, 895]。在这些领域中，正确性通常不能简化为单个可执行的预言，因此出处、程序合规性和专家审查仍然是核心[474, 752]。

科学发现和推理。科学发现智能体不仅仅产生符合领域的分析：它们生成新的主张，并根据计算或经验反馈对其进行测试。他们的工作流程将文献综述与专家级和实验科学推理 [642, 882]、假设生成、实验设计、执行、结果分析和迭代细化联系起来 [505, 678, 869]。 《人工智能科学家》及其继任者将机器学习研究组织成构思、实验、评估和写作的重复循环[397, 740]。 AlphaEvolve 和 Kosmos 将这种模式扩展到算法和跨域发现 [438, 448]，而 NovelSeek 则关闭了从假设生成到跨异构科学任务验证的循环 [800]。在实证环境中，Coscientist 将语言模型规划与机器人化学实验结合起来 [37]，Robin 和虚拟实验室将多智能体推理与生物医学发现和实验室验证联系起来 [177, 581]。因此，假设、程序、实验记录和手稿成为不断发展的研究状态，每当新证据出现时都必须进行修改。

形式数学代表了一种补充形式的发现，证明助理可以机械地检查每个已完成的证明 [615, 866]。Seed-Prover 1.5 结合了自然语言推理、引理级草图、递归分解和智能体精益证明器，在迭代验证下构造长证明 [59]。证明者智能体同样在反馈驱动的细化循环中协调非正式推理、辅助引理生成、形式证明构造和精益诊断[26]。自动证明工程将目标从孤立的定理证明扩展到存储库规模的工作，包括在不断发展的精益库中添加功能、重构和错误修复，并根据编译和语义需求满足情况进行评估[717]

。经验科学和形式数学共同表明，高效主体如何超越总结现有知识，生成能够在特定领域验证中幸存的主张[629]。

6.6 基准和资源

长时程智能体（Long-Horizon Agents） 的进展是通过基准测试和开放资源的共享基础来衡量和再现的。因为报告的分数反映了模型、运行支撑框架（Harness） 和训练配方的共同贡献，所以 长时程（Long-Horizon） 评估最好理解为 horizon-aware ，对目标必须保持多长时间敏感，以及 attribution-aware ，能够将 运行支撑框架（Harness） 引起的方差与模型引起的方差分开 [123, 286, 468]。用作基准的相同可验证 长时程（Long-Horizon） 环境也经常被重新用作第 5 节优化方法的训练环境，因此评估进展和内化进展紧密耦合。除了下表列出的领域基准之外，越来越多的套件家族刻意强调多小时的自主性。示例范围从 MLE-bench [51] 中的机器学习工程到 PaperBench [560] 中研究论文的端到端复制，补充了 METR [286, 429] 的前沿时间范围测量；这些出现在表 7 的生产智能体条目中。

表 7 整合了上述应用程序域的代表性资源，按域分组，并在每个域内按第 6 节的能力分类法排序。每一行将一个评估基准或可重用系统与其目标能力配对，并在可用时提供指向其公共存储库或官方发布页面（）的链接，为本节中讨论的工作提供具体的切入点。多模态理解条目（例如视频和流理解基准）不是 长时程（Long-Horizon）

64

agent tasks in the strict sense; we include them because they measure the foundational perception capabilities on which long-horizon multimodal agents build.

Table 7 Representative systems and benchmarks relevant to long-horizon agents, grouped by domain. One work per row, annotated with its capability, type, a short description, an (approximate) release date, and a code repository. *Type* distinguishes evaluation *Benchmark*s from reusable *System*s; within each domain, rows are grouped by capability (per Section 6) and ordered by release date. The list is representative rather than exhaustive.

Work	Capability	Type	Description	Date	GitHub Repo
Software Engineering					
AutoCodeRover [843]	Repository Grounding	System	Structure-aware fault localization and patching	2024.04	GitHub
SWE-agent [747]	Repository Grounding	System	Agent–computer interface for repositories	2024.05	GitHub
OpenHands [657]	Repository Grounding	System	Persistent coding workspace	2024.07	GitHub
NL2Repo-Bench [124]	Repository Grounding	Benchmark	From-scratch repository generation	2025.12	GitHub
OctoBench [123]	Repository Grounding	Benchmark	Scaffold-aware instruction following	2026.01	GitHub
SWE-bench [260]	Workflow-level Planning	Benchmark	Repository issue resolution	2023.10	GitHub
Claude Code [10]	Workflow-level Planning	System	Autonomous code editing	2025.02	GitHub
Trae Agent [169]	Workflow-level Planning	System	Ensemble agent for repository issue resolution	2025.07	GitHub
SWE-bench Pro [117]	Workflow-level Planning	Benchmark	Enterprise repository issue resolution	2025.09	GitHub
Terminal-Bench 2.0 [428]	Workflow-level Planning	Benchmark	Hard terminal-based tasks	2025.11	GitHub
Aider[8]	Feedback-driven Repair	System	Iterative pair-programming loop	2023.06	GitHub
DebugBench [602]	Feedback-driven Repair	Benchmark	Multi-language bug diagnosis	2024.01	GitHub
Agentless [703]	Feedback-driven Repair	System	Localize-then-repair pipeline	2024.07	GitHub
SWE-Gym [467]	Feedback-driven Repair	System	Executable training environments	2024.10	GitHub
SWE-smith [748]	Feedback-driven Repair	System	Repair-trajectory synthesis	2025.04	GitHub
Information Seeking					
Search-o1 [336]	Deep Search	System	Retrieval and document reading	2025.01	GitHub
BrowseComp [676]	Deep Search	Benchmark	Persistent web browsing	2025.04	GitHub
WebDancer [689]	Deep Search	System	Information-seeking policy	2025.05	GitHub
Tongyi DeepResearch [307]	Deep Search	System	Agentic deep-research model	2025.10	GitHub
MiroThinker [599]	Deep Search	System	Interactive-scaling research agent	2025.11	GitHub
Open Deep Research [292]	Wide Search	System	Evidence aggregation and synthesis	2025.06	GitHub
WideSearch [685]	Wide Search	Benchmark	Structured information collection	2025.08	GitHub
FlowSearch [227]	Wide Search	System	Knowledge-flow orchestration	2025.10	GitHub
GISA [896]	Wide Search	Benchmark	Deep reasoning and broad information aggregation	2026.02	GitHub
MMInA [603]	Multimodal Grounding	Benchmark	Multi-hop multimodal browsing	2024.04	GitHub
MM-BrowseComp [331]	Multimodal Grounding	Benchmark	Multimodal evidence seeking	2025.08	GitHub
WebWatcher [175]	Multimodal Grounding	System	Vision-language deep research	2025.08	GitHub
WebThinker [337]	Research Synthesis	System	Autonomous research workflow	2025.04	GitHub
DeepResearch Bench [136]	Research Synthesis	Benchmark	Research-report quality evaluation	2025.06	GitHub
WebWeaver [356]	Research Synthesis	System	Evidence organization and synthesis	2025.09	GitHub
DeepConsult [592]	Research Synthesis	Benchmark	Consulting and business deep research	~2025	GitHub
Computer Use					
Mind2Web [116]	Browser Agents	Benchmark	Cross-domain web demonstrations	2023.06	GitHub
WebArena [885]	Browser Agents	Benchmark	Realistic interactive websites	2023.07	GitHub
WebVoyager [208]	Browser Agents	Benchmark	End-to-end live-web agent evaluation	2024.01	GitHub
browser-use[9]	Browser Agents	System	Browser control through agent loop	2024.10	GitHub
BrowserAgent [781]	Browser Agents	System	Human-inspired native browsing	2025.10	GitHub
OSWorld [716]	Desktop GUI Agents	Benchmark	Desktop task environment	2024.04	GitHub
UI-TARS [494]	Desktop GUI Agents	System	Native visual GUI perception	2025.01	GitHub
ScreenSpot-Pro [324]	Desktop GUI Agents	Benchmark	High-resolution professional GUI grounding	2025.01	GitHub
PointArena [83]	Desktop GUI Agents	Benchmark	Language-guided visual pointing	2025.05	GitHub
Seed 2.0 [527]	Desktop GUI Agents	System	Frontier multimodal foundation model	2026.01	GitHub
AndroidWorld [508]	Mobile Agents	Benchmark	Dynamic Android environment	2024.05	GitHub
AndroidLab [734]	Mobile Agents	Benchmark	Android agent training framework	2024.10	GitHub
Mobile-Agent-v3 [766]	Mobile Agents	System	Fundamental GUI-automation agents	2025.08	GitHub

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原文第 65 页 | 中文翻译初稿

严格意义上的智能体任务；我们将它们纳入其中是因为它们衡量了 长时程（Long-Horizon） 多模式智能体构建的基础感知能力。

表 7 与 长时程智能体（Long-Horizon Agents） 相关的代表性系统和基准，按领域分组。每行一项工作，注释有其功能、类型、简短描述、（大致）发布日期和代码存储库。 Type 区分评估_Benchmark_ 和可重用_System_ ；在每个域中，行按功能分组（根据第 6 节）并按发布日期排序。该列表具有代表性，而非详尽无遗。

工作	能力	类型	描述	日期	GitHub 存储库
软件工程					
AutoCodeRover [843]	存储库 落地关联（Grounding）	系统	结构感知故障定位和修补	2024.04	GitHub
SWE-agent [747]	存储库 落地关联（Grounding）	系统	智能体-存储库的计算机接口	2024.05	GitHub
OpenHands [657]	存储库 落地关联（Grounding）	系统	持久编码工作区	2024.07	GitHub
NL2Repo-Bench [124]	存储库 落地关联（Grounding）	基准	从头开始生成存储库	2025.12	GitHub
OctoBench [123]	存储库 落地关联（Grounding）	基准	脚手架感知指令遵循	2026.01	GitHub
SWE-bench [260]	工作流程级规划	基准	存储库问题解决	2023.10	GitHub
Claude Code [10]	工作流级规划	系统	自主代码编辑	2025.02	GitHub
Trae Agent [169]	工作流级规划	系统	存储库问题的集成智能体解决方案	2025.07	GitHub
SWE-bench Pro [117]	工作流级规划	基准	企业存储库问题解决	2025.09	GitHub
终端-Bench 2.0 [428]	工作流级规划	基准	基于终端的硬任务	2025.11	GitHub
Aider8	反馈驱动修复	系统	迭代结对编程循环	2023.06	GitHub
DebugBench [602]	反馈驱动修复	Benchmark	多语言错误诊断	2024.01	GitHub
无智能体 [703]	反馈驱动的修复	系统	本地化然后修复管道	2024.07	GitHub
SWE-Gym [467]	反馈驱动的修复	系统	可执行的训练环境	2024.10	GitHub
SWE-smith [748]	反馈驱动的修复	系统	修复轨迹综合	2025.04	GitHub
信息查找					
Search-o1 [336]	深度搜索	系统	检索与文档阅读	2025.01	GitHub
BrowseComp [676]	深度搜索	基准测试	持久网页浏览	2025.04	GitHub
WebDancer [689]	深度搜索	系统	信息查找策略	2025.05	GitHub
统一DeepRe-搜索[307]	深度搜索	系统	智能体深度研究模型	2025.10	GitHub
MiroThinker [599]	深度搜索	系统	交互式扩展研究智能体	2025.11	GitHub
开放深度研究[292]	广泛搜索	系统	证据聚合与合成	2025.06	GitHub
WideSearch [685]	Wide Search	基准	结构化信息采集	2025.08	GitHub
FlowSearch [227]	广泛搜索	系统	知识流编排	2025.10	GitHub
GISA [896]	广泛搜索	基准	深度推理和广泛信息聚合	2026.02	GitHub
MMInA [603]	多模式 落地关联（Grounding）	基准	多跳多模式浏览	2024.04	GitHub

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|MM-BrowseComp [331]|多模态 落地关联（Grounding）|基准|多模态证据寻求|2025.08|GitHub|
|WebWatcher [175]|多式联运

落地关联（Grounding）|系统|视觉语言深度研究|2025.08|GitHub|
WebThinker [337]	研究综合	系统	自主研究工作流程	2025.04	GitHub
DeepResearch Bench [136]	研究综合	基准	研究报告质量评估	2025.06	GitHub
WebWeaver [356]	研究综合	系统	证据组织与综合	2025.09	GitHub
DeepConsult [592]	研究综合	基准	咨询和业务深度研究	~2025	GitHub
计算机使用					
Mind2Web [116]	浏览器智能体	基准测试	跨域 Web 演示	2023.06	GitHub
WebArena [885]	浏览器智能体	基准测试	真实的交互式网站	2023.07	GitHub
WebVoyager [208]	浏览器智能体	基准	端到端实时网络智能体评估	2024.01	GitHub
browser-use9	浏览器智能体	系统	通过智能体循环进行浏览器控制	2024.10	GitHub
BrowserAgent [781]	浏览器智能体	系统	受人启发的本机浏览	2025.10	GitHub
OSWorld [716]	桌面 GUI 智能体	基准测试	桌面任务环境	2024.04	GitHub
UI-TARS [494]	桌面 GUI 智能体	系统	原生视觉 GUI 感知	2025.01	GitHub
ScreenSpot-Pro [324]	桌面 GUI 智能体	基准测试	高分辨率专业 GUI
落地关联（Grounding）	2025.01	GitHub
PointArena [83]	桌面 GUI 智能体	基准	语言引导的视觉指向	2025.05	GitHub
Seed 2.0 [527]	桌面 GUI 智能体	系统	Frontier 多模式 基础模型（Foundation Model）	2026.01	GitHub
AndroidWorld [508]	移动智能体	基准测试	动态 Android 环境	2024.05	GitHub
AndroidLab [734]	移动智能体	基准测试	Android 智能体训练框架	2024.10	GitHub
Mobile-Agent-v3 [766]	移动智能体	系统	基本 GUI 自动化智能体	2025.08	GitHub

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Table 7 (continued)

Work	Capability	Type	Description	Date	GitHub Repo
Mobile-Agent-v3.5 [723]	Mobile Agents	System	Multi-platform fundamental GUI agents	2026.01	GitHub
Multimodal Agents					
EgoSchema [419]	Multimodal Understanding	Benchmark	Egocentric video question answering	2023.08	GitHub
MVBench [328]	Multimodal Understanding	Benchmark	Comprehensive video understanding	2023.11	GitHub
VideoAgent [655]	Multimodal Understanding	System	Agentic evidence selection over long video	2024.03	GitHub
Video-MME [158]	Multimodal Understanding	Benchmark	Long-video understanding	2024.05	GitHub
LongVideoBench [688]	Multimodal Understanding	Benchmark	Long-video and text understanding	2024.07	GitHub
StreamingBench [366]	Multimodal Understanding	Benchmark	Streaming-video understanding	2024.11	GitHub
DVD [829]	Multimodal Understanding	System	Tool-using video deep research	2025.05	GitHub
VideoSeek [365]	Multimodal Understanding	System	Tool-augmented active perception in long video	2026.03	GitHub
GenAI-Bench [306]	Multimodal Generation	Benchmark	Text-to-visual generation evaluation	2024.06	GitHub
MM-StoryAgent [733]	Multimodal Generation	System	Text–image–audio story generation	2025.03	GitHub
MUSE [574]	Multimodal Generation	System	Plan–execute–verify story-video generation	2026.02	GitHub
Ptah [803]	Multimodal Generation	System	Verifier-guided multimodal report generation	2026.05	GitHub
CrayOtter [741]	Multimodal Generation	System	Verification-driven multimodal generation	2026.06	GitHub
OmniVideoBench [308]	Omnimodal Agency	Benchmark	Audio-visual evaluation	2025.10	GitHub
Agent-Omni [364]	Omnimodal Agency	System	Omnimodal reasoning agent	2025.11	GitHub
OmniGAIA [339]	Omnimodal Agency	Benchmark	Cross-modal tool-use reasoning	2026.02	GitHub
AgentVista [564]	Omnimodal Agency	Benchmark	Multimodal agent evaluation	2026.02	GitHub
Orchestra-o1 [806]	Omnimodal Agency	System	Cross-modal orchestration agent	2026.06	GitHub
General-Purpose Agents					
ToolBench [492]	Personal Assistants	Benchmark	Large-scale tool use without env state	2023.07	GitHub
Open Interpreter[10]	Personal Assistants	System	Local computer and code execution	2023.07	GitHub
AgentBench [387]	Personal Assistants	Benchmark	Multi-environment agent evaluation	2023.08	GitHub
GAIA [431]	Personal Assistants	Benchmark	General assistant tasks	2023.11	GitHub
τ-bench [762]	Personal Assistants	Benchmark	Stateful tool–agent–user interaction	2024.06	GitHub
AppWorld [614]	Personal Assistants	Benchmark	Stateful app-based personal tasks	2024.07	GitHub
BFCL [472]	Personal Assistants	Benchmark	Function-calling leaderboard	2024.08	GitHub
OpenManus[11]	Personal Assistants	System	General-purpose agent workflows	2025.03	GitHub
MCP-Mark [700]	Personal Assistants	Benchmark	Realistic MCP tool-use stress test	2025.09	GitHub
Toolathlon [322]	Personal Assistants	Benchmark	Diverse realistic tool environments	2025.10	GitHub
MCP-Atlas [31]	Personal Assistants	Benchmark	Large-scale tool-use competency	2026.02	GitHub
Claw-Eval [764]	Personal Assistants	Benchmark	Trustworthy autonomous-agent evaluation	2026.04	GitHub
ClawMark [426]	Personal Assistants	Benchmark	Multi-day multimodal living-world eval	2026.04	GitHub
Agents' Last Exam [575]	Personal Assistants	Benchmark	Frontier professional-workflow evaluation	2026.06	GitHub
VLN-CE [283]	Embodied & World Models	Benchmark	Continuous vision-language navigation	2020.04	GitHub
OpenVLA [277]	Embodied & World Models	System	Open vision-language-action model	2024.06	GitHub
DINO-WM [879]	Embodied & World Models	System	Feature-space world model	2024.11	GitHub
EmbodiedBench [753]	Embodied & World Models	Benchmark	Embodied task evaluation suite	2025.02	GitHub
GR00T N1 [35]	Embodied & World Models	System	Humanoid vision-language-action model	2025.03	GitHub
$\pi_{0.5}$ [245]	Embodied & World Models	System	Open-world VLA model	2025.04	GitHub
V-JEPA 2 [25]	Embodied & World Models	System	Latent video-prediction world model	2025.06	GitHub
Cosmos-Predict2.5 [2]	Embodied & World Models	System	Video world simulation	2025.10	GitHub
ChemCrow [39]	Productive Agents	System	Tool-augmented chemistry reasoning	2023.04	GitHub
FinRobot [886]	Productive Agents	System	Financial research and analysis	2024.05	GitHub
MLE-bench [51]	Productive Agents	Benchmark	Machine-learning engineering tasks	2024.10	GitHub
LegalAgentBench [315]	Productive Agents	Benchmark	Multi-step legal tasks	2024.12	GitHub
MedAgentBench [257]	Productive Agents	Benchmark	Clinical EHR workflows	2025.01	GitHub
METR [286, 429]	Productive Agents	Benchmark	Frontier time-horizon measurement	2025.03	GitHub
AI Scientist-v2 [740]	Productive Agents	System	Autonomous scientific discovery	2025.04	GitHub
PaperBench [560]	Productive Agents	Benchmark	End-to-end research-paper replication	2025.04	GitHub
AlphaEvolve [448]	Productive Agents	System	Algorithmic discovery agent	2025.05	GitHub
Finance Agent Benchmark [34]	Productive Agents	Benchmark	Real-world financial analysis tasks	2025.08	GitHub
GDPval [474]	Productive Agents	Benchmark	Real-world economic task value	2025.10	GitHub

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原文第 66 页 | 中文翻译初稿

表 7 <u>（续）</u>

工作	能力	类型	描述	日期	GitHub 存储库
Mobile-Agent-v3.5 [723]	移动智能体	系统	多平台基本GUI
智能体	2026.01	GitHub
多式联运智能体					
EgoSchema [419]	多模态理解	基准	以自我为中心的视频问答	2023.08	GitHub
MVBench [328]	多模态理解	基准测试	全面的视频理解
2023.11	GitHub
VideoAgent [655]	多模态理解	系统	长视频智能体证据选择	2024.03	GitHub
视频-MME [158]	多模态理解	基准	长视频理解	2024.05	GitHub
LongVideoBench [688]	多模态理解	基准	长视频和文本理解	2024.07	GitHub
StreamingBench [366]	多模态理解	基准	流视频理解	2024.11	GitHub
DVD [829]	多模态理解	系统	工具使用视频深度研究
2025.05	GitHub
VideoSeek [365]	多模态理解	系统	长视频中工具增强的主动感知	2026.03	GitHub
GenAI-Bench [306]	多模态生成	基准	文本到视觉生成评估	2024.06	GitHub
MM-StoryAgent [733]	多模式生成	系统	文本-图像-音频故事生成
2025.03	GitHub
MUSE [574]	多模式生成	系统	计划-执行-验证故事视频
生成	2026.02	GitHub
Ptah [803]	多模式生成	系统	验证者引导的多模式报告
生成	2026.05	GitHub
CrayOtter [741]	多模式生成	系统	验证驱动的多模式
生成	2026.06	GitHub
OmniVideoBench [308]	Omnimodal Agency	Benchmark	视听评估	2025.10	GitHub
Agent-Omni [364]	全模式智能体	系统	全模式推理智能体	2025.11	GitHub
OmniGAIA [339]	Omnimodal Agency	Benchmark	跨模态工具使用推理	2026.02	GitHub
AgentVista[564]	OmnimodalAgency	基准	Multimodalagentevaluation	2026.02	GitHub
Orchestra-o1 [806]
全模式智能体	系统
跨模式编排智能体	2026.06	GitHub
通用（General-Purpose）智能体商					
ToolBench [492]	个人助理	Benchmark	无环境状态的大规模工具使用	2023.07	GitHub
打开解释器10	个人助理	系统	本地计算机和代码执行	2023.07	GitHub
AgentBench [387]	个人助理	Benchmark	多环境智能体评估	2023.08	GitHub
GAIA [431]	个人助理	基准	一般助理任务	2023.11	GitHub
τ-bench [762]	个人助理	基准	有状态工具-智能体-用户交互	2024.06	GitHub

|AppWorld [614]
BFCL [472]|个人助理
个人助理|基准测试
基准测试|基于应用程序的状态个人任务
函数调用排行榜|2024.07
2024.08|GitHub
GitHub|
|OpenManus11|个人助理|系统|通用（General-Purpose） 智能体工作流程|2025.03|GitHub
|
|MCP-Mark [700]|个人助理|基准|真实的 MCP 工具使用压力测试|2025.09|GitHub|
|工具马拉松 [322]|个人助理|基准测试|多样化的现实工具环境|2025.10|GitHub
|
|MCP-Atlas [31]|个人助理|基准|大规模工具使用能力|2026.02|GitHub|
|Claw-Eval [764]|私人助理|基准|值得信赖

自主智能体
评估|2026.04|GitHub|
|ClawMark [426]|个人助理|基准|多日多模式生活世界评估|2026.04|GitHub|
|智能体最后考试[575]|私人助理|基准|前沿专业工作流程
评估|2026.06|GitHub|
|VLN-CE [283]|体现 & 世界模型（World Models）|基准|连续视觉语言导航|2020.04|GitHub|
|OpenVLA [277]
|体现 & 世界模型（World Models）
|系统
|打开 视觉语言动作（Vision-Language-Action） 模型
|2024.06|GitHub
|
|DINO-WM [879]|体现 & 世界模型（World Models）|系统|特征空间 世界模型（World Model）|2024.11|
GitHub|
|EmbodiedBench [753]|Embodied & 世界模型（World Models）|基准|Embodied 任务评估套件|2025.02|
GitHub|
|GR00T N1 [35]|体现 & 世界模型（World Models）|系统|人形 视觉语言动作（Vision-Language-Action）
模型|2025.03|GitHub|
|_π_0_._5 [245]|体现 & 世界模型（World Models）|系统|开放世界VLA模型|2025.04|GitHub|
|V-JEPA 2 [25]|体现 & 世界模型（World Models）|系统|潜在视频预测 世界模型（World Model）
2025.06	GitHub
Cosmos-Predict2.5 [2]	Embodied & 世界模型（World Models）	系统	视频世界模拟	2025.10	GitHub
ChemCrow [39]	高效智能体	系统	工具增强化学推理	2023.04	GitHub
FinRobot [886]	高效智能体	系统	金融研究与分析	2024.05	GitHub
MLE-bench[51]	ProductiveAgents	基准	机器学习工程任务	202410	GitHub
LegalAgentBench [315]|
高效智能体|基准|

多步骤法律任务|。
2024.12|GitHub|
|MdAtBh[257]|PdtiAt|Bhk|CliilEHRkfl|202501|GitHb|
|egenenc
METR [286, 429]|roucve gens
高效智能体|encmar
基准|nca worlows
前沿时间范围测量|。
2025.03|u
GitHub|
|AI Scientist-v2 [740]|高效智能体|系统|自主科学发现|2025.04|GitHub
|
|PaperBench [560]|高效智能体|基准|端到端研究论文复制|2025.04|GitHub|
|AlphaEvolve [448]
|高效智能体|系统|算法发现智能体|2025.05|GitHub|
|财务智能体基准-
标记[34]|高效智能体|基准|现实世界的财务分析任务|2025.08|GitHub|
|
GDPval [474]|生产性智能体|基准|现实世界经济任务价值|2025.10|GitHub|

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66

Table 7 (continued)

Work	Capability	Type	Description	Date	GitHub Repo
Kosmos [438]	Productive Agents	System	Cross-domain discovery agent	2025.11	GitHub
APEX-Agents [623]	Productive Agents	Benchmark	Office multi-step task execution	2026.01	GitHub
OfficeQA Pro [458]	Productive Agents	Benchmark	Enterprise grounded reasoning	2026.03	GitHub
PresentBench [74]	Productive Agents	Benchmark	Rubric-based slide generation	2026.03	GitHub
OneMillion Bench [752]	Productive Agents	Benchmark	Human-level economic tasks	2026.03	GitHub
Workspace-Bench [588]	Productive Agents	Benchmark	Workspace file-dependency tasks	2026.05	GitHub
YC-Bench [211]	Productive Agents	Benchmark	Long-term planning consistency	2026.06	GitHub

Taken together, these resources show that long-horizon evaluation is consolidating around stateful, multi-hour, and economically grounded tasks, and that a reported score must be read as a joint property of the model, the harness, and the training recipe rather than of the model alone.

7 Frontier: Open Challenges and Outlooks

The previous sections have organized long-horizon agency as it currently stands, across both pillars and the experience loop that links them. We now present a set of *positions*: claims about the most consequential paradigm-level shifts that we expect to define the next several years of research, together with the open problems each shift entails. In our view, the frontiers that matter most lie in this *in-between* space: capability migrates bidirectionally between the harness and the weights. The harness itself turns from fixed scaffolding into a learnable, portable object. And in noisy, costly, real environments, long horizons invalidate the idealized premises that both pillars rely on. Our central diagnosis is that practical long-horizon progress is increasingly a property of the runtime around the model, not of the model alone. To make these pressures easier to compare, we group the frontiers into four higher-level axes: *evolution*, *effectiveness*, *efficiency*, and *trustworthiness*. Table 8 previews the nine positions under this structure. For each concrete frontier we first state where the field stands and what blocks it (*status and challenges*), then our position on where it should go and the open problems it leaves behind (*outlook*).

7.1 Evolution: Generalization and Learning Over Time

The first frontier axis is evolution: whether an agent's long-horizon competence can improve over time rather than remain locked to a hand-written scaffold and a frozen policy. This axis raises three linked questions. Can the harness itself self-evolve? Can the resulting competence transfer across harnesses? And can experience compound across episodes so that the agent keeps learning, and even learns for a lifetime, without destroying previously acquired capability?

7.1.1 Self-evolving Harness and Agents

Status and Challenges. The dominant paradigm of *hand-maintained* agents is a transitional artifact, and self-improvement offers a direct route to extending the effective horizon. We distinguish two levels as follow:

The first is the *self-evolving harness*: with little or no model training, the agent keeps rewriting the runtime structure around a fixed policy. This includes skill-driven fast adaptation, where new skills are distilled directly from failure and success trajectories [206, 705, 706]; recursive procedural-memory evolution, where memory and skills co-evolve from execution traces [813, 853]; and topology evolution, where sub-agents are spawned on demand and the multi-agent graph itself becomes a learnable object [101, 652, 670, 791]. Increasingly the harness is treated as a first-class artifact that can be automatically generated and optimized, as in Meta-Harness and AutoHarness [297, 395], and even hosted by agent operating systems such as AIOS that provide a unified, resource-managed runtime across agents and tasks [424]; industrial trace-based iteration treats run traces as the primary evolution signal [293, 765].

[8] Aider: https://github.com/Aider-AI/aider.
[9] browser-use: https://github.com/browser-use/browser-use.
[10] Open Interpreter: https://github.com/OpenInterpreter/open-interpreter.
[11] OpenManus: https://github.com/FoundationAgents/OpenManus.

原文第 67 页 | 中文翻译初稿

表 7 <u>（续）</u>

工作	能力	类型	描述	日期	GitHub 存储库
Kosmos [438]	高效智能体	系统	跨域发现智能体	2025.11	GitHub
APEX-Agents [623]	高效智能体	基准	Office 多步骤任务执行	2026.01	GitHub
OfficeQA Pro [458]	高效智能体	基准	企业接地推理	2026.03	GitHub
PresentBench [74]	高效智能体	基准	基于标题的幻灯片生成	2026.03	GitHub
百万工作台 [752]	高效智能体	基准	人类级别的经济任务	2026.03	GitHub
Workspace-Bench [588]	高效智能体	基准	工作区文件依赖项任务	2026.05	GitHub
YC-Bench [211]	高效智能体	基准	长期规划一致性	2026.06	GitHub

总而言之，这些资源表明，长时程（Long-Horizon）评估正在围绕有状态、多小时且经济基础的任务进行整合，并且报告的分数必须被视为模型、运行支撑框架（Harness）和训练配方的联合属性，而不是模型本身的联合属性。

7 前沿：开放的挑战和前景

前面的部分已经按目前的情况组织了 长时程（Long-Horizon）机构，跨越两个支柱以及连接它们的经验循环。我们现在提出一组_立场_：关于我们期望定义未来几年研究的最重要的范式级别转变的主张，以及每个转变所带来的开放问题。在我们看来，最重要的边界在于这个“中间”空间：能力在 运行支撑框架（Harness）和权重之间双向迁移。 运行支撑框架（Harness）本身从固定脚手架变成了可学习的便携式物体。在嘈杂、昂贵的真实环境中，长远的视野使这两个支柱所依赖的理想化前提失效。我们的核心诊断是，实际的 长时程（Long-Horizon）进展越来越成为模型周围运行时的属性，而不仅仅是模型的属性。为了使这些压力更容易比较，我们将前沿分为四个更高级别的轴：evolution、efficiency、_efficiency_和_trustworthiness_。表 8 预览了该结构下的九个位置。对于每个具体前沿，我们首先说明该领域的立场以及阻碍它的因素（现状和挑战），然后说明我们对它应该走向何方的立场以及它留下的未解决问题（展望）。

7.1 进化：随着时间的推移泛化和学习

第一个前沿轴是进化：智能体人的 长时程（Long-Horizon）能力是否可以随着时间的推移而提高，而不是仍然锁定在手写的脚手架和冻结的策略上。该轴提出了三个相互关联的问题。 运行支撑框架（Harness）本身可以自我进化吗？由此产生的能力可以在 运行支撑框架（Harnesses）之间转移吗？是否可以经历跨情节的复合，以便智能体不断学习，甚至终身学习，而不破坏以前获得的能力？

7.1.1 自进化运行支撑框架（Harness）和智能体

现状和挑战。手动维护智能体的主导范式是过渡工件，而 自我改进（Self-Improvement）提供了扩展有效范围的直接途径。我们区分两个级别如下：

第一个是_自我进化的运行支撑框架（Harness）_：在很少或没有模型训练的情况下，智能体不断围绕固定策略重写运行时结构。这包括技能驱动的快速适应，即直接从失败和成功轨迹中提取新技能[206,705,706]；递归程序记忆进化，记忆和技能从执行痕迹共同进化[813, 853]；和拓扑演化，其中子智能体按需生成，多智能体图本身成为可学习的对象[101,652,670,791]。 运行支撑框架（Harness）越来越多地被视为可以自动生成和优化的一流工件，如 Meta-运行支撑框架（Harness）和 AutoHarness [297, 395]，甚至由智能体操作系统（例如 AIOS）托管，这些操作系统提供跨智能体和任务的统一的、资源管

理的运行时 [424]；基于工业轨迹的迭代将运行轨迹视为主要演化信号 [293, 765]。

> 8Aider：https://github.com/Aider-AI/aider。

> 9浏览器使用：https://github.com/browser-use/browser-use。

> 10开放解释器：https://github.com/OpenInterpreter/open-interpreter。

> 11OpenManus：https://github.com/FoundationAgents/OpenManus。

67

Table 8 Four frontier axes and nine frontiers for long-horizon agents. For each frontier we summarize its *open challenges* and our *outlook*. A recurring thread: the harness is where much of the next advance must happen.

Frontier	Open challenges	Outlook
I. Evolution — generalize across runtimes and keep learning over time		
Self-evolving Harness & Agents	• Hand-set optimization metric • Limited gains on out-of-distribution tasks • Overfitting and drift over long runs	• Adaptively choose optimization objective and scope • Autonomous evolution guided by feedback • Lifelong, non-overfitting evolution
Harness Generalization and Transferability	• Models tied to specific harnesses • Performance varies across model–harness pairings • Portable standards still ad hoc	• Multi-harness (hybrid-harness) training • Portable skills / experience at inference time • A standard harness protocol
Continual & Lifelong Learning	• External experience is noisy and unverified • Catastrophic forgetting during weight updates	• Learning only from validated experience • Safe online learning from billion-user logs
II. Effectiveness — act reliably in realistic, verifiable environments		
Real-world Environment Interaction	• Slow, costly, and irreversible interactions with real-world systems • Synthesized Envs & world-models may be unfaithful	• Measure, audit, and optimize sim-to-real faithfulness • Fusion of symbolic and neural simulation environments
From Digital to Embodied Agents	• Deep planning vs. real-time control • Limited modeling of physical dynamics & laws • Coarse vs. fine-grained feedback	• Async, hierarchical harness (deliberation vs. reflex) • Predict physical outcomes with world models • Integrate feedback across timescales
III. Efficiency — spend compute, context, and modality budgets		
Cost- & Budget-aware Agency	• Model selection is poorly matched to task difficulty • No calibrated cost / infeasibility sense • No runtime enforcement of ceilings • No law linking budget to success	• Perceive cost, difficulty, infeasibility • Act mid-run: shorten, re-route, or stop • Harness-enforced ceilings & cost-aware routing • A budget-vs-success scaling law
Multimodal & Omni Harness	• Multimodality bolted on; no vision-native harness • Visual-token budgeting is heuristic • Cross-modal verification is unreliable	• A vision-native, unified omni harness substrate • Per-modality granularity control • Cross-modal verification & omni-modal routing
IV. Trustworthiness — stay robust and governable as the horizon grows		
Reflection & Error Robustness	• Failures detected late under noisy, delayed feedback • Intrinsic self-correction is unreliable • Errors compound into goal drift	• Early failure detection & path-switching • External-anchored, calibrated self-knowledge • Align to human course-correction
Safety & Governance	• Injected error & hazardous experience reuse • No unified safety-verification standard • Self-evolution erodes invariants; biased self-checks	• Untrusted external input; gate experience reuse • Score action safety beside effectiveness • Independent, invariant-preserving verification

The second, more ambitious level is the *self-evolving agent*, whose goal is to fold harness structure and skills back into the weights: the Darwin Gödel Machine rewrites its own code to improve a coding agent [814], and a small but growing line couples skill discovery with reinforcement learning or routes rich runtime trajectories into training, as in SkillRL and OpenClaw-style RL such as Claw-R1 [457, 627, 705].

The shared limitations are threefold, mirroring the outlook below: **(1)** the optimization objective is still a hand-set benchmark metric rather than something the agent chooses for itself; **(2)** gains remain largely within-distribution while the evolution history stays opaque about *what* changed and *why*; and **(3)** over long runs such systems risk overfitting a single benchmark or drifting.

Outlook. We expect self-evolution to mature into a complex *automated-optimization* problem pursued with minimal human supervision, and we highlight three requirements. **(1)** The agent should *autonomously discover its own optimization objective*. Today this is typically a single task's benchmark metric or a training-loss level; in the future it will more likely be a joint objective over many tasks. The agent should locate where its own competence is weak on each task and decide accordingly where to optimize. Sometimes it is enough to

原文第 68 页 | 中文翻译初稿

表 8 长时程智能体（Long-Horizon Agents）的四个前沿轴和九个前沿。对于每个前沿领域，我们总结其_开放挑战_和我们的_前景_。一个重复出现的线程：运行支撑框架（Harness）是接下来大部分进展必须发生的地方。

前沿	开放挑战	展望
我。进化--泛化	跨运行时并随着时间的推移不断学习	
自进化运行支撑框架（Harness）
和智能体	• 手持优化指标
• 分布外任务的收益有限
• 长期运行中的过度拟合和漂移	• 自适应选择优化目标和
范围
• 由反馈引导的自主进化
• 终身、非过度拟合进化
运行支撑框架（Harness）
泛化和
可移植性	• 与特定 运行支撑框架（Harnesses）相关的模型
• 性能因模型而异 - 运行支撑框架（Harness）配对
• 可移植标准仍然是临时的	• 多 运行支撑框架（Harness）（混合 运行支撑框架（Harness））训练
• 推理时的便携技能/经验
• 标准运行支撑框架（Harness）协议
持续和终身
学习	• 外部经验嘈杂且未经验证
• 体重更新期间的灾难性遗忘（Catastrophic Forgetting）	• 仅从经过验证的经验中学习
• 从十亿用户日志中安全在线学习
二．有效性--在现实、可验证的环境中可靠地行动	
现实世界
环境
交互	• 与
现实世界系统的缓慢、昂贵且不可逆的交互
• 合成的环境和世界模型可能
不忠实	• 测量、审核和优化仿真到现实（Sim-to-Real）
忠实性
• 符号和神经模拟的融合
环境
从数字到
具体智能体|• 深度规划与实时控制
• 物理动力学和规律的有限建模
• 粗粒度反馈与细粒度反馈|• 异步、分层运行支撑框架（Harness）（深思熟虑与
反射）
• 使用世界模型（World Models）预测身体结果

• 跨时间尺度集成反馈|
|三．效率 - 支出|计算、上下文和模式预算||
|成本和
预算意识机构|• 模型选择与任务难度不匹配
• 没有校准成本/不可行性意识
• 没有运行时执行上限
• 没有法律将预算与成功联系起来|• 感知成本、难度、不可行性
• 中途采取行动：缩短、重新安排路线或停止
• 运行支撑框架（Harness） 强制上限和成本感知路由
• 预算与成功缩放法则|
|多式联运和全方位
运行支撑框架（Harness）|• 多式联运固定；无视觉原生 运行支撑框架（Harness）
• 视觉代币预算是启发式的
• 跨模式验证不可靠|• 视觉原生、统一全向 运行支撑框架（Harness） 基底
• 每模态粒度控制
• 跨模态验证和全模态路由|
|四．值得信赖--随着视野的增长||保持稳健和可控
|反射与错误
稳健性|• 在嘈杂、延迟的反馈下较晚检测到故障
• 内在的自我校正不可靠
• 错误会导致目标漂移|• 早期故障检测和路径切换
• 外部锚定、校准的自我认知
• 与人类航向校正保持一致|
|安全与治理|• 注入错误和危险经验重用
• 没有统一的安全验证标准
• 自进化侵蚀不变量；有偏见的自检|• 不可信的外部输入

;登机口体验重用
• 除了有效性之外，还对操作安全性进行评分
• 独立、不变的验证|

第二个更雄心勃勃的级别是“自我进化智能体”，其目标是将 运行支撑框架（Harness） 结构和技能重新纳入权重：达尔文哥德尔机重写自己的代码以改进编码智能体[814]，并且一条虽小但不断增长的线路将技能发现与 强化学习（Reinforcement Learning） 结合起来，或将丰富的运行时轨迹路由到训练中，如 SkillRL 和 OpenClaw 式 RL 中那样爪-R1 [457, 627, 705]。

共同的局限性有三个，反映了以下前景：（1）优化目标仍然是手动设置的基准指标，而不是智能体为自己选择的东西； (2) 收益很大程度上保持在分布范围内，而演化历史对于改变的内容和原因仍然不透明； (3) 从长远来看，此类系统存在过度拟合单个基准或漂移的风险。

前景。我们期望自我进化成熟为一个复杂的“自动优化”问题，以最少的人类监督来解决，并且我们强调了三个要求。 (1) 智能体应该_自主发现自己的优化目标_。如今，这通常是单个任务的基准指标或训练损失水平；未来，它更有可能成为许多任务的共同目标。智能体应该在每项任务中找到自己的能力薄弱的地方，并相应地决定优化哪里。有时候就足够了

68

adjust one part of the harness (orchestration logic, a memory module, the tool set), while in other cases the harness and the model weights should be optimized together. **(2)** Self-evolution should proceed *autonomously*: the agent should acquire external information, distill reusable skills (for instance, high-value strategies for deep web search), and continuously update its optimization strategy in response to environmental feedback. **(3)** This process should be able to run *continually, even for a lifetime*: driven by a stream of sparse external signals, it should neither overfit to a single benchmark nor drift into random change, which makes stable long-run self-evolution a genuinely long-term goal. Crucially, if self-evolution is to be more than per-benchmark overfitting, the structure it produces must survive being moved to a different runtime. This is the transferability problem we turn to next.

7.1.2 Harness Generalization and Transferability

Status and Challenges. Harness paradigms keep multiplying, spanning early patterns such as ReAct and CodeAct, the Model Context Protocol, and today's workstation-style harnesses like OpenClaw, Hermes, and Claude Code. As they do, the field is drifting toward a regime in which each strong model is implicitly co-trained with, and bound to, a single harness. Yet what users expect from a capable model is the opposite: that they can drive it from their own agent, and even extend it to a harness they design themselves. We regard this binding as the central scalability risk of the field: a model competitive only inside the harness it was tuned for must be retrained for every new one. It cannot keep pace with a runtime ecosystem that turns over every few months.

Three concrete pressures make the risk tangible. **(1)** Harness-specific tuning effects are large: identical harnesses change rank when ported across providers, sometimes by tens of points, so a model's measured "ability" is partly an artifact of how well its idioms match the harness [468]. **(2)** Multi-model orchestration is already standard: large models plan while small models execute. Learned preference-based routing [451] and layered model composition [638] serve as building blocks, with systems such as EvoRoute, Chimera, and ORCH offering early systematic studies [447, 810, 880]; any single-harness assumption thus breaks immediately. **(3)** Portable cross-provider standards such as AGENTS.md and Agent Skills are emerging organically rather than from any single vendor [16, 98]. This development elevates *transferability*, the extent to which a harness configuration remains effective when the base model is swapped, to a first-class concern.

Outlook. We see three complementary routes toward harness-robust competence. **(1)** *External transfer*: package experience, and ideally skills, into portable artifacts so that a model can read externally mounted skills at inference time and gradually raise its competence without retraining. **(2)** *Internal generalization*, in our view the more fundamental route: it exposes the model to many harnesses during training by mixing multi-harness trajectories and even running hybrid-harness RL, so that harness-robust behavior is built into the weights rather than bolted on. **(3)** *Standardizing a harness protocol*: just as the Model Context Protocol connected the world's tools behind one discoverable, authorizable interface and thereby made tool use broadly transferable [100], a standard harness protocol that supports flexible design and registration could become the starting point for world-wide harness generalization.

The concrete open problems are: the minimal portable contract that preserves performance across providers; training recipes that explicitly optimize for cross-harness transfer rather than single-harness fit; robustness under harness drift, when a new server or tool-calling convention silently changes behavior; and portable observability, so that traces from different runtimes can be analyzed in one stack. Transferability concerns whether competence transfers across runtimes at a single point in time; the complementary question is whether competence *accumulates* across time, which is the problem of continual learning.

7.1.3 Continual and Lifelong Learning

Status and Challenges. Continual and lifelong learning are central to long-horizon agents because such agents operate across many tasks, sessions, environments, and users rather than within a single isolated episode. A capable agent should not only solve the current task, but also accumulate experience that improves later behavior while preserving previously acquired abilities. Existing work pursues this along two complementary routes. **(1)** *External* lifelong learning stores experience outside the parameters: Voyager maintains an expanding library of executable skills for open-ended exploration [630], ExpeL extracts natural-language lessons from past

原文第 69 页 | 中文翻译初稿

调整 运行支撑框架（Harness） 的一部分（编排逻辑、内存模块、工具集），而在其他情况下，运行支撑框架（Harness） 和模型权重应一起优化。（2）自我进化应该“自主地”进行：智能体应该获取外部信息，提炼可重用的技能（例如，深网搜索的高价值策略），并根据环境反馈不断更新其优化策略。(3) 这个过程应该能够持续运行，甚至终生运行：由稀疏的外部信号流驱动，它既不应该过度适应单个基准，也不应该漂移到随机变化，这使得稳定的长期自我进化成为真正的长期目标。至关重要的是，如果自我进化要超过每个基准的过度拟合，那么它产生的结构必须能够在移动到不同的运行时时幸存下来。这就是我们接下来要讨论的可迁移性问题。

7.1.2 运行支撑框架（Harness） 泛化和可移植性

现状和挑战。 运行支撑框架（Harness） 范例不断增加，涵盖早期模式，例如 ReAct 和 CodeAct、模型上下文协议，以及当今的工作站风格 运行支撑框架（Harnesses），例如 OpenClaw、Hermes 和 Claude Code。正如他们所做的那样，该领域正朝着这样的方向发展：每个强模型都隐式地与单个 运行支撑框架（Harness） 共同训练并绑定到单个 运行支撑框架（Harness）。然而，用户对功能模型的期望恰恰相反：他们可以从自己的智能体驱动它，甚至将其扩展到他们自己设计的 运行支撑框架（Harness）。我们认为这种绑定是该领域的核心可扩展性风险：仅在其调优的 运行支撑框架（Harness） 内具有竞争力的模型必须针对每个新模型进行重新训练。它无法跟上每隔几个月更新一次的运行时生态系统的步伐。

三个具体压力使风险变得切实可见。(1) 运行支撑框架（Harness） 特定的调整效果很大：相同的 运行支撑框架（Harnesses） 在跨提供者移植时会改变排名，有时会改变数十个点，因此模型的测量“能力”在一定程度上是其习语与 运行支撑框架（Harness） 匹配程度的产物[468]。（2）多模型编排已经成为标准：大模型规划，小模型执行。基于学习偏好的路由[451]和分层模型组合[638]作为构建块，EvoRoute、Chimera和ORCH等系统提供了早期的系统研究[447、810、880]；因此，任何单一的 运行支撑框架（Harness） 假设都会立即失效。(3) 可移植的跨提供商标准（例如 AGENTS.md 和 Agent Skills）正在有机地出现，而不是来自任何单一供应商 [16, 98]。这一发展将“可转移性”（即交换基本模型时 运行支撑框架（Harness） 配置保持有效的程度）提升为一流的关注点。

前景。我们看到了实现 运行支撑框架（Harness） 强大能力的三种互补途径。(1)外部转移：将经验和理想的技能打包到便携式工件中，以便模型可以在推理时读取外部安装的技能，并逐渐提高其能力，而无需重新训练。(2)内部泛化，在我们看来，更基本的路线：它在训练期间通过混合多个 运行支撑框架（Harness） 轨迹甚至运行混合 运行支撑框架（Harness） RL 来将模型暴露给许多 运行支撑框架（Harnesses），以便将 运行支撑框架（Harness） 的鲁棒行为内置到权重中而不是固定在权重上

。(3)标准化运行支撑框架（Harness）协议：正如模型上下文协议将世界上的工具连接到一个可发现的、可授权的接口后面，从而使工具的使用可以广泛转移[100]，支持灵活设计和注册的标准运行支撑框架（Harness）协议可以成为全球运行支撑框架（Harness）泛化的起点。

具体的开放问题是：保持跨提供商性能的最小可移植合约；明确优化跨 运行支撑框架（Harness） 传输而不是单 运行支撑框架（Harness） 拟合的训练配方；当新服务器或工具调用约定悄悄改变行为时，运行支撑框架（Harness） 漂移下的鲁棒性；和可移植的可观察性，以便可以在一个堆栈中分析来自不同运行时的跟踪。可转移性涉及能力是否在单个时间点跨运行时转移；补充问题是能力是否随着时间的推移而累积，这就是持续学习（Continual Learning）的问题。

7.1.3 持续和终身学习

现状和挑战。持续和终身学习是 长时程智能体（Long-Horizon Agents） 的核心，因为此类智能体跨多个任务、会话、环境和用户运行，而不是在单个孤立的事件中运行。一个有能力的智能体不仅应该解决当前的任务，还应该积累经验，改善以后的行为，同时保留以前获得的能力。现有的工作沿着两条互补的路线来实现这一目标。 (1)_外部_终身学习存储参数之外的经验：Voyager 维护一个可扩展的可执行技能库，用于开放式探索 [630]，ExpeL 从过去提取自然语言课程

69

trajectories [853], and Reflexion keeps verbal feedback in episodic memory to improve later trials without weight updates [544]; such methods are cheap, reversible, and quick to adapt. **(2)** *Internal* continual learning folds new knowledge or behavior into the model itself; this supports broader generalization, but frequent updates remain costly and risk catastrophic forgetting, especially under noisy, shifting deployment data [58, 641].

Outlook. The key challenge is to connect these two forms of learning into a stable lifelong-learning loop: most new experience should first be kept externally as memory, trajectories, or skills, where it can be inspected, revised, and tested across future tasks, and only experience that proves reliable and broadly useful should later be consolidated into weights. This staged process makes lifelong learning less about fine-tuning after every interaction and more about deciding what to remember, reuse, generalize, or forget.

A distinctive new opportunity is *online learning from deployment at scale*: as billion-user agent products such as Cursor[12], Claude Code[13], and Doubao[14] mature, they accumulate vast streams of user logs and feedback. Turning these traces into safe improvements is a difficult but consequential long-term research challenge. Doing so requires metrics that identify high-value episodes and online-learning algorithms that incorporate them into the agent with minimal human intervention and without regressions. The open problems include how to prevent forgetting during periodic updates, how to extract trustworthy learning signals from noisy trajectories and user feedback, how to transfer lessons across tasks and environments, and how to evaluate improvement over many sessions rather than within a single episode. In this sense, lifelong learning is a necessary condition for long-horizon agency: without it, an agent may act for a long time, yet never truly become better over time.

7.2 Effectiveness: Acting in Realistic Environments

Evolution matters only if the resulting agent remains effective when the environment becomes slow, noisy, partially irreversible, and hard to verify. The second axis concerns whether long-horizon agents can operate effectively in realistic environments. We first consider the digital environments used for training and evaluation, and then turn to the physical world, where challenges arise from real-time perception and control.

7.2.1 Real-world Environment Interaction

Status and Challenges. A great many agent applications are defined by sustained interaction with real-world environments: ordering food or booking a flight, drafting a Feishu or Notion document, running real-time financial analysis and forecasting, or communicating on platforms such as WeChat[15] and X[16]. For training long-horizon agents, however, we usually cannot act directly in these live environments because interaction is slow, costly, and often irreversible. The binding constraint becomes the *environment* rather than the model: where an agent can act, whether it can fail safely, and whether it can receive a checkable reward.

Two construction routes are emerging. **(1)** *Verifiable, executable synthesis* builds environments that behave like the real systems they stand in for and mechanically decide success and failure: realistic testbeds such as WebArena, OSWorld, and TheAgentCompany ground agents in real browsers, operating systems, and multi-day company workflows [716, 722, 885], and automated environment synthesis pushes this toward scale [131, 238, 555]. **(2)** *Model-simulated environments* treat a large model itself as an implicit world model, as in GenEnv, Qwen-AgentWorld, and reasoning-model simulators such as Simia-SFT [195, 905], trading mechanical guarantees for flexibility and coverage.

Both routes face the same test of *faithfulness*: an environment that is inexpensive to evaluate but unfaithful may train agents that perform well in controlled settings yet fail in deployment. Faithfulness is thus best treated not as one property but as several separately auditable axes: whether interfaces and available actions match the real system; whether identical actions produce the same state change; whether automatic validators agree with real acceptance criteria; whether simulated users behave like real ones; and whether a reset returns the system to a functionally equivalent state rather than one that merely looks similar.

[12] https://cursor.com
[13] https://www.anthropic.com/claude-code
[14] https://www.doubao.com
[15] https://www.wechat.com
[16] https://x.com

原文第 70 页 | 中文翻译初稿

轨迹[853]，反射将口头反馈保留在情景记忆中，以改善以后的试验，而无需更新体重[544]；这些方法成本低廉、可逆且适应速度快。 (2) Internal 持续学习（Continual Learning） 将新的知识或行为折叠到模型本身中；这支持更广泛的泛化，但频繁的更新成本仍然很高，并且存在 灾难性遗忘（Catastrophic Forgetting） 的风险，特别是在嘈杂、不断变化的部署数据下 [58, 641]。

前景。关键的挑战是将这两种形式的学习连接成一个稳定的终身学习循环：大多数新经验首先应作为记忆、轨迹或技能保存在外部，可以在未来的任务中进行检查、修改和测试，并且只有被证明可靠且广泛有用的经验才应在以后合并为权重。这种分阶段的过程使得终身学习不再是每次互动后的微调，而是更多地决定记住、重用、概括或忘记什么。

一个独特的新机会是“大规模部署在线学习”：随着 Cursor12、Claude Code13 和 Doubao14 等十亿用户智能体产品的成熟，它们积累了大量的用户日志和反馈。将这些痕迹转化为安全的改进是一项困难但重要的长期研究挑战。这样做需要识别高价值事件的指标和在线学习算法，以最少的人为干预且无需回归，将它们合并到智能体中。悬而未决的问题包括如何在定期更新期间防止遗忘，如何从嘈杂的轨迹和用户反馈中提取值得信赖的学习信号，如何跨任务和环境转移课程，以及如何评估多次会话而不是单个事件中的改进。从这个意义上说，终身学习是长时程（Long-Horizon）智能体的必要条件：没有它，智能体可能会行动很长时间，但随着时间的推移永远不会真正变得更好。

7.2 有效性：在现实环境中行动

只有当环境变得缓慢、嘈杂、部分不可逆且难以验证时，所产生的智能体仍然有效，进化才有意义。第二个轴涉及长时程智能体（Long-Horizon Agents）能否在现实环境中有效运行。我们首先考虑用于训练和评估的数字环境，然后转向物理世界，其中实时感知和控制带来挑战。

7.2.1 现实世界环境交互

现状和挑战。许多智能体应用程序都是通过与现实环境的持续交互来定义的：订餐或预订航班、起草飞书或概念文档、运行实时财务分析和预测，或者在微信15和X16等平台上进行交流。然而，对于训练 长时程智能体（Long-Horizon Agents），我们通常无法直接在这些实时环境中进行操作，因为交互缓慢、成本高昂且通常不可逆。绑定约束变成了_环境_而不是模型：智能体可以在哪里行动，它是否可以安全地失败，以及它是否可以收到可检查的奖励。

两条建设路线正在出现

。 (1)_可验证的、可执行的综合_构建的环境与它们所代表的真实系统相似，并机械地决定成功和失败：真实的测试平台，例如真实浏览器、操作系统和多日公司工作流程中的 WebArena、OSWorld 和 TheAgentCompany 地面智能体 [716, 722, 885]，而自动化环境综合将其推向规模 [131, 238, 555]。 (2)_模型模拟环境_将大型模型本身视为隐式 世界模型（World Model），如 GenEnv、Qwen-AgentWorld 和推理模型模拟器（例如 Simia-SFT [195, 905]），以机械保证换取灵活性和覆盖范围。

两条路线都面临着相同的“忠诚度”测试：评估成本低廉但不忠实的环境可能会训练出在受控设置中表现良好但部署失败的智能体。因此，最好不要将忠诚度视为一项属性，而是将其视为几个单独的可审核轴：界面和可用操作是否与真实系统匹配；相同的动作是否会产生相同的状态变化；自动验证器是否符合真实的验收标准；模拟用户的行为是否与真实用户一样；以及重置是否会将系统返回到功能等效的状态，而不是仅仅看起来相似的状态。

> 12https://cursor.com

> 13https://www.anthropic.com/claude-code

> 14https://www.doubao.com

> 15https://www.wechat.com

> 16https://x.com

70

Outlook. We hold that environment construction, not model scale, is the rate-limiting step for the next phase of agent capability, and that "faithfulness" deserves to be measured as explicitly as task success is today. The most promising direction is to *fuse* the two routes above by combining the verifiability of synthesized environments with the flexibility of model-simulated ones. This fusion would allow agents to train broadly while still receiving trustworthy rewards. On the learning side the binding constraint is sample efficiency: real-world rollouts are slow, costly, and sometimes irreversible, so the algorithms that matter extract the most signal per interaction through off-policy reuse, experience replay over returned traces, and curricula in which environment difficulty co-evolves with the policy.

The open problems are: automated construction of environments that are provably checkable and measurably faithful; metrics and audits for the sim-to-real faithfulness of synthetic and model-simulated environments; sample-efficient RL for slow, costly, partially irreversible interaction; and curricula that co-evolve environment difficulty with agent competence. The most demanding realistic environment of all, however, is the physical world rather than the digital one. In this setting, perception and control must run in real time, and many actions cannot be undone. These constraints define the frontier of embodiment.

7.2.2 From Digital to Embodied Agents

Status and Challenges. A natural extension of long-horizon digital agency is embodiment: equipping an agent with real-world perception and control so that it can provide high-level deliberation for an embodied system. The externalization logic of this paper extends directly to embodiment. The digital harness becomes a high-level controller that orchestrates low-level physical skills, recapitulating the harness pattern as a division of labor between the cerebrum and cerebellum. In this arrangement, a high-level LLM/VLM agent interprets goals, decomposes tasks, and replans based on feedback [243, 357], while Vision-Language-Action (VLA) models serve as callable real-time skill modules for atomic manipulation [35, 36, 134, 277, 899]. But embodiment sharpens three challenges specific to the physical world. **(1)** A *timescale conflict*: long-horizon planning is deep but slow, and its deliberation clashes with the millisecond reactive control an embodied system demands. **(2)** A *dimensionality and physics gap*: today's long-horizon agents reason mostly over language and, increasingly, other digital modalities, whereas the physical world has far higher effective dimensionality, so better modeling of physical laws and dynamics is a key hurdle to migrating a digital long-horizon agent into a body. **(3)** A *feedback-granularity gap*: real environments provide feedback at multiple spatial and temporal scales, from sparse task-level signals to dense control-level signals (pose deviations, contact forces, or tactile mismatches), and an embodied agent must integrate feedback across these scales rather than rely only on the coarse plan-level signals that text harnesses handle well.

Outlook. We expect embodied and digital agent architectures to converge on a shared engineering stack with the long-horizon harness as the deliberative core, and we read the three challenges above as its research agenda. **(1)** For the timescale conflict, the promising route is an asynchronous, hierarchical harness that decouples slow deliberation from a fast reactive control loop, letting the deliberative core replan out-of-band while cached skills and reflexes keep the body responsive. **(2)** For the physics gap, embodied harnesses will likely lean on action-conditioned world models and simulation, so that the agent can predict physical consequences and close part of the sim-to-real gap before acting. **(3)** For the feedback-granularity gap, memory, skill, and protocol designs must carry both coarse subgoals and fine-grained continuous corrections through one interface, with verifiers that operate at both scales.

The open problems are: real-time, safety-critical action verification under latency budgets text harnesses can ignore; memory, skill, and protocol designs for irreversible physical action; closing the sim-to-real gap between simulated and physical deployment; and a unified deliberative harness that drives both digital tools and physical effectors. Whether an agent acts through digital tools or physical effectors, each step also draws down a finite budget of computation, context, and time. Therefore, how these resources should be allocated becomes a first-class frontier.

7.3 Efficiency: Spending Compute, Context, and Modality Budgets

Long horizons turn resources into first-class design variables. An agent must decide not only what to do, but how much computation, context, modality bandwidth, and verification effort each part of the run deserves.

原文第 71 页 | 中文翻译初稿

前景。我们认为环境构建，而不是模型规模，是下一阶段智能体能力的限制步骤，并且“忠诚度”值得像今天的任务成功一样明确地衡量。最有前途的方向是通过将合成环境的可验证性与模型模拟环境的灵活性相结合来融合上述两条路线。这种融合将使特工能够进行广泛的训练，同时仍然获得值得信赖的奖励。在学习方面，绑定约束是样本效率：现实世界的 展开轨迹（Rollouts） 速度慢、成本高，有时是不可逆的，因此重要的算法通过策略外重用、返回跟踪的经验重放以及环境难度与策略共同演化的课程来提取每次交互的最多信号。

悬而未决的问题是：自动构建可证明可检查且可测量忠实的环境；综合和模型模拟环境的 仿真到现实（Sim-to-Real） 忠实度的指标和审计；样本高效的强化学习，用于缓慢、昂贵、部分不可逆的交互；以及环境困难与智能体能力共同发展的课程。然而，最苛刻的现实环境是物理世界，而不是数字世界。在这种情况下，感知和控制必须实时运行，并且许多操作无法撤消。这些约束定义了实施例的边界。
7.2.2 从数字智能体到具身智能体

现状和挑战。 长时程（Long-Horizon）数字智能体的自然延伸是体现：为智能体配备现实世界的感知和控制，以便它可以为体现的系统提供高层次的审议。本文的外化逻辑直接延伸到具体化。数字 运行支撑框架（Harness） 成为协调低级身体技能的高级控制器，将 运行支撑框架（Harness） 模式概括为大脑和小脑之间的分工。在这种安排中，高级 LLM/VLM 智能体解释目标、分解任务并根据反馈重新计划 [243, 357]，而 视觉语言动作（Vision-Language-Action）(VLA) 模型充当用于原子操作的可调用实时技能模块 [35, 36, 134, 277, 899]。但具体化加剧了物理世界特有的三个挑战。 (1)时间尺度冲突：长时程（Long-Horizon） 规划深入但缓慢，其深思熟虑与具体系统要求的毫秒反应控制相冲突。 (2) 维度和物理差距：今天的 长时程智能体（Long-Horizon Agents） 推理主要通过语言以及越来越多的其他数字形式进行，而物理世界的有效维度要高得多，因此更好的物理定律和动力学建模是将数字 长时程（Long-Horizon）智能体迁移到身体中的关键障碍。 (3) 反馈粒度差距：真实环境提供多个空间和时间尺度的反馈，从稀疏的任务级信号到密集的控制级信号（姿势偏差、接触力或触觉不匹配），并且具身智能体必须跨这些尺度集成反馈，而不是仅依赖文本 运行支撑框架（Harnesses） 处理良好的粗略计划级信号。

前景

。我们期望实体和数字智能体架构能够融合在一个共享工程堆栈上，并以 长时程（Long-Horizon） 运行支撑框架（Harness） 作为审议核心，并且我们将上述三个挑战作为其研究议程。 (1) 对于时间尺度冲突，有希望的路线是异步、分层的 运行支撑框架（Harness），它将缓慢的深思熟虑与快速反应控制循环解耦，让深思熟虑的核心在带外重新计划，同时缓存的技能和反应保持身体的响应。 (2) 对于物理差距，体现的 运行支撑框架（Harnesses） 可能会依赖于动作条件 世界模型（World Models） 和模拟，以便智能体可以在行动之前预测物理后果并关闭部分 仿真到现实（Sim-to-Real） 差距。 (3) 对于反馈粒度差距，记忆、技能和协议设计必须通过一个界面同时承载粗粒度子目标和细粒度连续校正，并具有在两种规模上运行的验证器。

未解决的问题是：延迟预算文本 运行支撑框架（Harnesses） 可以忽略的实时、安全关键的操作验证；不可逆身体动作的记忆、技能和协议设计；缩小模拟部署和物理部署之间的 仿真到现实（Sim-to-Real） 差距；以及驱动数字工具和物理效应器的统一审议 运行支撑框架（Harness）。无论智能体是通过数字工具还是物理效应器来行动，每个步骤都会消耗有限的计算、上下文和时间预算。因此，如何分配这些资源成为首要前沿问题。
7.3 效率：支出计算、上下文和模态预算

长远的视野将资源转化为一流的设计变量。智能体不仅必须决定要做什么，还必须决定运行的每个部分应有多少计算、上下文、模态带宽和验证工作量。

71

The efficiency frontier spans both classical compute budgeting and the newer problem of scheduling high-cost multimodal evidence.

7.3.1 Cost- and Budget-aware Agency

Status and Challenges. As long-horizon agents move from research demos into ordinary users' everyday workflows, such as hours-long coding sessions and personal-assistance workflows spanning an entire day, cost and budget cease to be implementation details and become first-class concerns for both usability and economics. Yet today's agents are largely *budget-blind*: they carry almost no working sense of what a token costs, no notion that context and compute are a finite budget to be spent deliberately, and no felt pressure to finish within a price or time ceiling. In practice they default to the most capable model at every step, re-read context they could have summarized, and keep spending on runs that are already doomed. This is merely wasteful at demo scale, but once such agents run continuously for millions of users it translates into real cost and a degraded day-to-day experience.

The mechanisms to spend more wisely already exist, including compute-optimal test-time allocation [550], adaptive reasoning budgets [442], tunable best-of-N and multi-agent aggregation [323], and cost-aware model routing and cascades [62, 237, 447, 451, 810, 880]. However, agents do not yet use these mechanisms with any awareness of their own spending. Budget-awareness is best seen not as a single capability but as a chain: an agent must perceive what a task will cost, act on that estimate, run inside a harness that can enforce the limits it would otherwise ignore, and ideally be guided by a law that predicts the trade-off.

Outlook. We frame the way forward as four open problems along this chain. **(1)** *Perceiving cost, difficulty, and infeasibility.* An agent should forecast, and keep re-forecasting as a run unfolds, how hard a task is, how much budget remains, and whether the task has already become infeasible; even a coarse predicted-infeasibility signal can save a large fraction of the tokens otherwise burned on doomed runs [370]. The quantity is multi-dimensional and coupled (tokens, money, wall-clock time, sometimes real resources), and the estimate must be *calibrated* rather than optimistic, since only a calibrated forecast tells the agent when and by how much to change course. **(2)** *Acting on the estimate.* A cost estimate is worthless unless it changes behavior: the agent should shorten reasoning on easy subtasks, avoid redundant re-reading, reserve depth for where it changes the outcome, and escalate, stop, or re-route when the forecast turns bad, while avoiding a measurable loss in quality. The difficulty is that these choices are made mid-trajectory under uncertainty, where a false economy early can cost far more later. **(3)** *Enforcing budgets from the harness.* Not all budget discipline should live in the model's head; the harness should enforce frugality through budget-aware routing and cascades, context compression and eviction, caching and reuse of intermediate results, and scheduling primitives that hold the agent to a hard price, token, or time ceiling even when the model would overspend. Because the harness sees the whole run, it is also the natural place to account for cost across a long, multi-tool trajectory. **(4)** *A scaling law for budget and success.* Further out, is there a predictable relationship linking compute budget, tool calls, and run length to success rate, analogous to a Chinchilla-style law for agency? Without one, budget allocation stays a craft; the first credible candidates should emerge from the long-running-task regime, where budgets are large enough that depth-versus-breadth trade-offs dominate.

In text-only agents these budgets are already difficult; in multimodal agents the dominant cost may be not reasoning depth but which sensory stream to keep, compress, or discard.

7.3.2 Multimodal and Omni Harnesses

Status and Challenges. Current harnesses are still designed primarily around textual inputs and command-line interaction, whereas future long-horizon agents will operate over far richer multimodal information, including images, video, audio, GUI states, and generated media. Current multimodal agents typically treat multimodality as an auxiliary capability rather than as a native substrate. Doubao's task-assistant mode, for example, uses a text-centric main agent that invokes external tools for image and video generation and perception. Recent work has begun to address this challenge. Examples include efforts toward on-policy data evolution for visual-native multimodal deep-search agents, as well as verifiable multimodal deep research through multi-agent harnesses for interleaved report generation. However, a unified definition and architecture for a multimodal or omni harness is still missing. The technical obstacles are concrete. **(1)** *The cost of visual*

原文第 72 页 | 中文翻译初稿

效率前沿涵盖了经典的计算预算和调度高成本多模式证据的新问题。

7.3.1 具有成本和预算意识的机构

现状和挑战。随着 长时程智能体（Long-Horizon Agents） 从研究演示转向普通用户的日常工作流程，例如长达数小时的编码会议和跨越一整天的个人协助工作流程，成本和预算不再是实施细节，而是成为可用性和经济性的首要问题。然而，今天的智能体在很大程度上是“预算盲”的：他们几乎没有对代币成本的工作意识，没有意识到上下文和计算是需要刻意花费的有限预算，也没有感受到在价格或时间上限内完成的压力。在实践中，他们在每一步都默认使用最有能力的模型，重新阅读他们可以总结的上下文，并继续在已经注定要失败的运行上投入资金。这在演示规模上只是浪费，但一旦此类智能体持续为数百万用户运行，它就会转化为实际成本和日常体验下降。

更明智地支出的机制已经存在，包括计算最优测试时间分配[550]、自适应推理预算[442]、可调整的 best-of-N_和多智能体聚合[323]以及成本感知模型路由和级联[62、237、447、451、810、880]。然而，智能体商在使用这些机制时还没有意识到自己的支出。预算意识最好不要被视为单一功能，而应被视为一条链：智能体必须了解任务的成本，根据该估计采取行动，在 运行支撑框架（Harness） 内运行，该 运行支撑框架（Harness） 可以强制执行它本来会忽略的限制，并且理想情况下受到预测权衡的法律的指导。

前景。我们将前进的道路定义为沿着这条链条的四个开放性问题。 (1)_感知成本、难度和不可行性。_智能体应该预测，并在运行过程中不断重新预测任务有多难，剩余预算有多少，以及任务是否已经变得不可行；即使是粗略的预测不可行性信号也可以节省很大一部分代币，否则在注定的运行中会被烧毁[370]。数量是多维且耦合的（代币、金钱、挂钟时间，有时还有实际资源），并且估计必须是“校准的”而不是乐观的，因为只有校准的预测才能告诉智能体何时以及改变多少。 (2)_根据估计采取行动。_除非改变行为，否则成本估计毫无价值：智能体应缩短对简单子任务的推理，避免多余的重读，为改变结果的地方保留深度，并在预测变坏时升级、停止或重新路由，同时避免可测量的质量损失。困难在于，这些选择是在不确定的情况下做出的，早期的错误经济可能会在以后付出更大的代价。 (3) _根据 运行支撑框架（Harness） 执行预算。_并非所有预算纪律都应该存在于模型的头脑中； 运行支撑框架（Harness） 应该通过预算感知路由和级联、上下文压缩和驱逐、中间结果的缓存和重用以及调度原语来强制节俭，这些原语使智能体保持硬价格、令牌或时间上限，即使模型超支

。由于 运行支撑框架（Harness） 可以看到整个运行过程，因此它也是计算长的多工具轨迹成本的自然场所。 (4) _预算和成功的缩放法则。_更进一步，是否存在将计算预算、工具调用和运行长度与成功率联系起来的可预测关系，类似于 Chinchilla 式的智能体法则？如果没有一个，预算分配仍然是一门手艺；第一批可信的候选人应该从长期任务制度中产生，该制度的预算足够大，以至于深度与广度的权衡占主导地位。

对于纯文本智能体来说，这些预算已经很困难了；在多模式智能体中，主要成本可能不是推理深度，而是保留、压缩或丢弃哪种感觉流。

7.3.2 多模式和全向 运行支撑框架（Harnesses）

现状和挑战。当前的 运行支撑框架（Harnesses） 仍然主要围绕文本输入和命令行交互进行设计，而未来的 长时程智能体（Long-Horizon Agents） 将在更丰富的多模式信息上运行，包括图像、视频、音频、GUI 状态和生成的媒体。当前的多模态智能体通常将多模态视为辅助能力而不是天然基质。例如，豆宝的任务辅助模式使用以文本为中心的主智能体，调用外部工具来生成和感知图像和视频。最近的工作已经开始应对这一挑战。例子包括针对视觉原生多模态深度搜索智能体的策略数据演化的努力，以及通过多智能体 运行支撑框架（Harnesses） 进行交错报告生成的可验证多模态深度研究。然而，多式联运或全向 运

行支撑框架（Harness）的统一定义和架构仍然缺失。技术障碍是具体的。 (1)视觉成本

72

tokens: video and image streams cannot be placed into context in full in the same way as text logs, so the agent must decide when to observe, what evidence to retain, and how to compress, retrieve, and memorize multimodal information; today this allocation across text, images, and video remains largely heuristic. **(2)** *Cross-modal verification*: multimodal evidence often must be transcribed, aligned, or abstracted into another representation before it can be judged, and these intermediate transformations can themselves fail or hallucinate, so adding more verification stages need not improve reliability linearly. **(3)** *A wider externalization surface*: multimodal externalization changes the design assumptions underlying skill specification, memory indexing, and protocol schemas, opening a substantially wider frontier for what can be externalized at all.

Outlook. The most consequential shift is that a unified, *vision-native* omni harness must emerge, and we frame the agenda as four requirements. **(1)** *A vision-native substrate.* It should play for real-world visual capability a role analogous to that of Claude Code and OpenClaw for code and the workstation, giving the whole research program a vision-native harness to build on rather than a text harness with modalities attached. **(2)** *Per-modality, budget-aware granularity.* Future omni harnesses must choose an appropriate information granularity per modality (raw frames preserve the most information at the highest cost, while sampling, clip summarization, or feature compression trade detail for savings) and decide autonomously at what granularity multimodal evidence is retained, compressed, and retrieved. **(3)** *Early error detection before rendering.* Because the outputs of image and video generators are no longer cheap textual artifacts, harnesses must detect errors and reduce rework *before* full artifacts are produced, through intermediate representations, low-fidelity previews, staged inspection, or constrained generation. **(4)** *Omni-modal routing.* The harness should decide not only which model and how deep to search, but which modality, temporal segment, and expert model deserve the budget [339, 364]. As agents extend perception and action beyond text, efficiency and trustworthiness become inseparable: every decision to retain, compress, discard, or regenerate multimodal evidence is at once a cost trade-off and a reliability risk.

7.4 Trustworthiness: Robust and Governed Autonomy

The final axis asks how autonomy remains trustworthy as the horizon grows. Long runs make failures cumulative rather than isolated: small early mistakes compound into goal drift, and the more authority a harness is granted, the larger the blast radius of a single wrong action. Accordingly, trustworthiness joins two problems that are usually studied apart: whether an agent can notice and recover from its own errors under noisy feedback, and whether its autonomy can be governed, contained, and audited as it acts on real systems.

7.4.1 Reflection and Error Robustness

Status and Challenges. Error robustness is the first pillar of trustworthy long-horizon autonomy: an agent that runs unattended over many steps can be trusted only insofar as it can tell when it is going wrong and change course, since an undetected early error is silently amplified over the rest of the run. Real tasks rarely succeed on the first attempt, and the feedback from a real environment is delayed, partial, and noisy, so robustness is a property of the *whole trajectory*, not of any single step: the decisive capability is to act sensibly under noisy feedback, detect mid-execution that a path is failing, and stop or switch reasoning paths early rather than committing to one track to the bitter end. The self-correction literature now provides a more qualified account of when these methods are effective. Reflexion and Self-Refine show that iterative critique-and-revise loops can raise reliability [416, 544], and CRITIC shows that tool-grounded critique, which verifies against an external signal rather than the model's own confidence, is markedly more trustworthy than ungrounded self-reflection [182]. The crucial caveat is that models frequently *cannot* reliably self-correct reasoning from intrinsic signals alone and may even degrade when they try [234], which is why external, harness-supplied verification must anchor reflection, and why over long horizons undetected errors compound into goal drift and inherited error from weaker prior trajectories [20, 427].

Outlook. We see three priorities. **(1)** *Early detection and path-switching*: under noisy, delayed environmental feedback, the harness should detect a failing trajectory as early and cheaply as possible and abandon it to try an alternative, rather than spending the budget before discovering the run was doomed. **(2)** *Aligning to human correction*: in practice, a human often steps in partway through a run to redirect the agent, and a central open problem is how to internalize that signal; possible routes include external, human-aligned validation

原文第 73 页 | 中文翻译初稿

tokens：视频和图像流无法像文本日志那样完全放入上下文中，因此智能体必须决定何时观察、保留哪些证据以及如何压缩、检索和记忆多模态信息；如今，这种跨文本、图像和视频的分配在很大程度上仍然是启发式的。 (2)跨模态验证：多模态证据通常必须被转录、对齐或抽象为另一种表示形式才能进行判断，而这些中间转换本身可能会失败或产生幻觉，因此添加更多验证阶段并不需要线性提高可靠性。 (3)更广泛的外化表面：多模式外化改变了技能规范、内存索引和协议模式背后的设计假设，为可外化的内容开辟了更广阔的领域。

前景。最重要的转变是必须出现一个统一的、vision-native 全方位 运行支撑框架（Harness），我们将议程框架为四个要求。 (1) _视觉原生基质。_对于现实世界的视觉能力，它应该发挥类似于代码和工作站的 Claude Code 和 OpenClaw 的作用，为整个研究计划提供一个视觉原生 运行支撑框架（Harness） 来构建，而不是附加模式的文本 运行支撑框架（Harness）

。 (2)_按模态、预算敏感的粒度。_未来的全能 运行支撑框架（Harnesses） 必须为每种模态选择适当的信息粒度（原始帧以最高成本保留最多信息，同时采样、剪辑摘要或特征压缩交换细节以节省费用），并自主决定保留、压缩和检索多模态证据的粒度。 (3) _渲染前的早期错误检测。_因为图像和视频生成器的输出不再是廉价的文本工件，运行支撑框架（Harnesses） 必须在生成完整工件之前通过中间表示、低保真度预览、分阶段检查或约束生成来检测错误并减少返工

。 (4) 全模态路由。 运行支撑框架（Harness） 不仅应该决定搜索哪个模型以及搜索深度，还应该决定哪个模态、时间段和专家模型值得预算 [339, 364]。随着智能体将感知和行动扩展到文本之外，效率和可信度变得密不可分：保留、压缩、丢弃或重新生成多模态证据的每一个决定都同时是成本权衡和可靠性风险。

7.4 可信度：强大且受治理的自治

The final axis asks how autonomy remains trustworthy as the horizon grows

。长时间的跑步会让失败累积起来，而不是孤立的：早期的小错误会加剧目标漂移，而安全带被授予的权力越大，单个错误动作的影响范围就越大

术语保留：运行支撑框架（Harness）

。因此，可信度连接了两个通常分开研究的问题：智能体是否能够在嘈杂的反馈下注意到并从自己的错误中恢复，以及当它在真实系统上运行时，它的自主性是否可以被治理、包含和审计。

7.4.1 反射和误差鲁棒性

现状与挑战

。错误鲁棒性是值得信赖的 长时程（Long-Horizon） 自治的第一个支柱：在无人值守的情况下运行多个步骤的智能体只有在它能够判断何时出错并改变方向时才值得信任，因为未检测到的早期错误会在运行的其余部分中悄悄放大。真正的任务很少会在第一次尝试时就成功，并且来自真实环境的反馈是延迟的、部分的和嘈杂的，因此鲁棒性是_整个轨迹_的属性，而不是任何单个步骤的属性：决定性的能力是在嘈杂的反馈下明智地采取行动，检测到执行中路径失败，并尽早停止或切换推理路径，而不是在一条轨道上坚持到最后。现在，自我校正文献对这些方法何时有效提供了更合格的说明。反思和自我完善表明，迭代批评和修改循环可以提高可靠性 [416, 544]，而 CRITIC 表明，基于工具的批评（根据外部信号而不是模型自身的信心进行验证）明显比无根据的自我反思更值得信赖 [182]。关键的警告是，模型经常_不能_仅根据内在信号可靠地自我纠正推理，甚至在尝试时可能会降级[234]，这就是为什么外部、运行支撑框架（

Harness ） 提供的验证必须锚定反射，以及为什么在很长一段时间内未检测到的错误会复合为目标漂移和来自较弱先验轨迹的继承错误 [20, 427]。

前景。我们看到三个优先事项。 (1) 早期检测和路径切换：在嘈杂、延迟的环境反馈下，运行支撑框架 (Harness ） 应该尽早且尽可能便宜地检测到失败的轨迹，并放弃它以尝试替代方案，而不是在发现运行注定失败之前花费预算。 (2) 与人类纠正保持一致：在实践中，人类经常在跑步过程中介入以重定向智能体，而一个核心的开放问题是如何内化该信号；可能的途径包括外部的、以人为本的验证

73

checkpoints and incorporating human course-correction trajectories into training so that the agent learns *when* and *how* people intervene. **(3)** *Calibrated self-knowledge*: reflection should be grounded in external verification rather than the model's own often-miscalibrated confidence, and the agent should carry a calibrated sense of what it does not know, so that it escalates or asks for help instead of confidently compounding an error. But detecting and recovering from one's own errors is only half of trustworthy autonomy; the other half is ensuring that an agent, however robust, stays governed, contained, and auditable as the authority it wields over real systems grows. This is the problem of safety and governance.

7.4.2 Safety and Governance

Status and Challenges. In many respects, safety is the highest-stakes frontier in long-horizon agency, and three pressures make this challenge particularly acute.

(1) *The environment itself can inject error and risk*: a real environment may return manipulated or adversarial observations. More subtly, an agent that summarizes its own experience can distill a hazardous heuristic and then reuse it; for example, having once observed that a successful refund elicits a grateful reply, an agent may over-generalize and become predisposed to issue refunds in later tasks. Such experience-reuse effects compound as harness artifacts (skills, servers, shared memory) are shared across teams, creating attack surfaces analogous to a software supply chain [463, 902]. **(2)** *No unified standard for safety verification*: harness-level verifiers and benchmarks still overwhelmingly treat task success as the primary measure of *effectiveness*, while largely ignoring whether the actions taken along the way were *safe*, so local verifiers can all pass even when the run violates real-world constraints. **(3)** *Safety under self-evolution*: self-modifying harnesses make it hard to preserve initial safety invariants, and because a self-evolving agent that audits its own modifications is prone to the same biases that produced them, self-checking alone is not a trustworthy guarantee.

Underlying all three is a structural fact: most deployed safety mechanisms (least privilege, approval gates, capability scoping, audit, sandbox isolation) are *harness-level* controls living above the model and outside its weights, so treating safety as a model-only property miscounts the true attack surface.

Outlook. We sketch a technical path for each pressure. **(1)** Against injected error and hazardous experience reuse, the harness should treat external observations as untrusted input by sandboxing them and tracking their provenance, and should gate which distilled experiences may be reused, so that a heuristic must clear a safety review before it can shape future behavior. **(2)** Against the verification gap, the community needs a *unified safety-verification standard* that scores action safety as a first-class axis alongside effectiveness, with external acceptance tests and red-team suites that certify what an agent did, not just whether it succeeded. **(3)** Against self-evolution risk, invariant-preserving evolution should be enforced by *independent* verifiers rather than the evolving agent's own judgment, ideally with provable guarantees that a modification cannot weaken a fixed set of safety invariants.

More broadly, each safety layer adds latency, token cost, and friction, so the recurring engineering challenge is to quantify the *alignment tax* of a harness design and minimize it without compromising safety [1, 41, 303, 368, 520, 547, 617, 783]. Our position is that the harness, not the model, is the principal alignment and privilege surface for production agents: safety properties that hold robustly in production are almost always harness-imposed, while model-side alignment is necessary but rarely sufficient. This argument only sharpens once the harness leaves the screen: as embodied agents turn digital decisions into physical action, governance stops being a runtime policy and becomes a real-time safety constraint, which makes the harness, not the model, the decisive locus of trust for long-horizon autonomy.

8 Conclusion

In this paper, we review long-horizon AI agents through a central thesis: long-horizon agency emerges from the *co-evolution of externalized harness engineering and internalized model optimization*. From this view, we develop the discussion through six aspects. **Foundation** formalizes the agent as a coupling of a base policy and a surrounding harness, organizes long-horizon tasks into three nested levels (H1 to H3) together with the capabilities they demand (C1 to C3), and characterizes the scaling of the execution time horizon.

原文第 74 页 | 中文翻译初稿

检查点并将人类路线修正轨迹纳入训练中，以便智能体了解人们何时以及如何进行干预。 (3) 校准的自我知识：反思应该基于外部验证，而不是模型自己经常错误校准的信心，并且智能体应该对它不知道的事情保持校准的感觉，以便它升级或寻求帮助，而不是自信地复合错误。但发现自己的错误并从中恢复只是值得信赖的自主权的一半；另一半是确保智能体，无论多么强大，随着其对真实系统的权力的增长，保持受治理、遏制和可审计。这就是安全和治理的问题。

7.4.2 安全与治理

现状和挑战。在许多方面，安全是 长时程（Long-Horizon） 机构最重要的前沿领域，三种压力使这一挑战尤为严峻。

(1) 环境本身可以注入错误和风险：真实环境可能会返回受操纵或对抗性的观察结果。更巧妙的是，一个总结自己经验的智能体可以提炼出危险的启发式方法，然后重复使用它；例如，一旦观察到成功退款会引起感激的答复，智能体可能会过度概括并倾向于在以后的任务中发放退款。像 运行支撑框架（Harness）工件（技能、服务器、共享内存）这样的经验重用效果复合体在团队之间共享，创建了类似于软件供应链的攻击面 [463, 902]。 （2）安全验证没有统一的标准：运行支撑框架（Harness）级别的验证器和基准测试仍然绝大多数将任务成功作为_有效性_的主要衡量标准_，而很大程度上忽略了一路上采取的操作是否_安全_，因此即使运行违反了现实世界的约束，本地验证器也可以全部通过。 (3)自我进化下的安全：自我修改 运行支撑框架（Harnesses） 使得很难保留初始的安全不变量，并且因为审计自己的修改的自我进化智能体很容易产生与产生它们相同的偏差，所以单独的自我检查并不是一个值得信赖的保证。

这三者背后是一个结构性事实：大多数部署的安全机制（最小特权、批准门、能力范围、审计、沙箱隔离）都是位于模型之上且超出其权重的_运行支撑框架（Harness）-level_控制，因此将安全视为仅模型属性会错误地计算真正的攻击面。

前景。我们为每种压力绘制了一条技术路径。 (1) 针对注入错误和危险经验重用，运行支撑框架（Harness） 应通过沙箱处理外部观察并跟踪其来源，将外部观察视为不可信输入，并应控制哪些蒸馏经验可以重用，以便启发式必须先通过安全审查，然后才能塑造未来的行为。 (2) 针对验证差距，社区需要一个_统一的安全验证标准_，将操作安全性与有效性一起作为一流的轴，通过外部验收测试和红队套件来证明智能体做了什么，而不仅仅是它是否成功

。 （3）针对自我进化风险，保持不变的进化应该由独立验证者而不是进化智能体自己的判断来强制执行，理想情况下可以保证修改不会削弱一组固定的安全不变量。

更广泛地说，每个安全层都会增加延迟、令牌成本和摩擦，因此经常出现的工程挑战是量化 运行支撑框架（Harness） 设计的“对齐税”并在不影响安全性的情况下将其最小化[1, 41, 303, 368, 520, 547, 617, 783]。我们的立场是，运行支撑框架（Harness），而不是模型，是生产智能体的主要对齐和特权表面：在生产中稳健保持的安全属性几乎总是由 运行支撑框架（Harness） 强加的，而模型侧对齐是必要的，但很少是充分的。一旦 运行支撑框架（Harness） 离开屏幕，这一论点只会变得更加尖锐：随着具身智能体将数字决策转化为物理行动，治理不再是运行时策略，而是成为实时安全约束，这使得 运行支撑框架（Harness）（而不是模型）成为 长时程（Long-Horizon） 自治的决定性信任源。

8 结论

在本文中，我们通过一个中心论点回顾了 长时程（Long-Horizon） 人工智能智能体：长时程（Long-Horizon） 智能体是从外部 运行支撑框架（Harness） 工程和内部模型优化的共同进化中产生的。以此为出发点，我们从六个方面展开讨论。 Foundation 将智能体形式化为基本策略和周围 运行支撑框架

（Harness）的耦合，将 长时程（Long-Horizon）任务及其所需的功能（C1 到 C3）组织为三个嵌套级别（H1 到 H3），并表征执行时间范围的扩展。

74

Evolution traces how the control surface has progressively widened across three stages: prompt engineering, context engineering, and runtime harnesses. **Harness** takes a full lifecycle view, organizing the runtime by component: the agent loop and workflow, context and memory, tools and skills, orchestration, hooks, and verification. **Optimization** lays out a complete training recipe for internalizing competence, proceeding from an architectural and data–environment substrate through pre-/mid-training, fine-tuning, reinforcement learning, and on-policy distillation to self-evolution. **Application** organizes practice by the interface between the agent and its environment, spanning software engineering, information seeking, computer use, multimodal, and general-purpose agents, paired with horizon-aware evaluation. textbfFrontier consolidates the open problems into four axes: evolution, effectiveness, efficiency, and trustworthiness.

Beyond organizing prior work, we distill several emerging directions that define the next stage of long-horizon agent research. We hope this paper lays a solid foundation for future work and serves as a useful reference for researchers and practitioners alike. As agentic systems continue to mature, the long-horizon agent will remain a central and open problem, one likely to play a decisive role in building robust, general, and enduring artificial intelligence.

原文第 75 页 | 中文翻译初稿

进化追踪了控制面如何逐步扩大到三个阶段：提示工程、上下文工程和运行时 运行支撑框架（Harnesses）。 运行支撑框架（Harness） 采用完整的生命周期视图，按组件组织运行时：智能体循环和工作流程、上下文和内存、工具和技能、编排、挂钩和验证。优化为内化能力制定了完整的训练方案，从架构和数据环境基础开始，通过预/中期训练、微调、强化学习（Reinforcement Learning）、策略蒸馏到自我进化。应用程序通过智能体与其环境之间的接口来组织实践，涵盖软件工程、信息搜索、计算机使用、多模式和 通用（General-Purpose） 智能体，并与水平感知评估相结合。 textbfFrontier 将开放问题整合为四个轴：演化、有效性、效率和可信度。

除了组织先前的工作之外，我们还提炼了几个新兴方向，定义了 长时程（Long-Horizon） 智能体研究的下一阶段。我们希望本文为今后的工作奠定坚实的基础，并为研究人员和实践者提供有益的参考。随着智能体系统的不断成熟，长时程（Long-Horizon） 智能体仍将是一个核心且开放的问题，它可能在构建强大、通用和持久的人工智能方面发挥决定性作用。

75

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原文第 76 页 | 中文翻译初稿

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78

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原文第 82 页 | 中文翻译初稿

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82

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原文第 83 页 | 中文翻译初稿

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85

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原文第 86 页 | 中文翻译初稿

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86

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原文第 88 页 | 中文翻译初稿

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原文第 89 页 | 中文翻译初稿

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89

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91

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94

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101

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原文第 102 页 | 中文翻译初稿

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103

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105

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原文第 106 页 | 中文翻译初稿

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106

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109

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110

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原文第 111 页 | 中文翻译初稿

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原文第 115 页 | 中文翻译初稿

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原文第 116 页 | 中文翻译初稿

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121

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原文第 123 页 | 中文翻译初稿

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126

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134

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135

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原文第 137 页 | 中文翻译初稿

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137

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138

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原文第 139 页 | 中文翻译初稿

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140

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原文第 141 页 | 中文翻译初稿

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141

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142

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原文第 143 页 | 中文翻译初稿

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143

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原文第 144 页 | 中文翻译初稿

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144

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148

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原文第 149 页 | 中文翻译初稿

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149